Arithmetic Quantum Mechanics
Lab Book

Tobias J. Osborne

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Contents

1 Overview

The active direction, adopted on 9 September 2026, is *The Hunting of the Symplectic Phantasm* (Section 31). It begins with primary sources, explicit definitions and a dependency graph for finite symplectic quantization, before attempting an arithmetic construction across primes. The preceding composite-boundary programme is paused; its admitted results remain part of the record. The bootstrap introduced thirteen definitions and fourteen lemma contracts. The Weyl realization, affine covariance and tensor comparison have passed capped review, as have relation composition, the explicit compact structure and the odd-prime projective stabilizer comparison. Scalar normalization and the finite instrument laws have also passed capped review, as has the coherent-sum and tagged-block comparison. Six Phantasm propositions remain sketches. The revision of 10 September identifies the admitted finite-abelian, tensor, trace and Frobenius results reused by those sketches, and records the remaining convention comparisons explicitly.

This campaign's north star is a general definition and workflow that assigns to an arbitrary arithmetic scheme one or more *canonical quantum systems*: kinematics realised as a Hilbert space, observables realised as an algebra, and, where possible, dynamics realised as automorphisms or as projective unitary representations of symmetry groups, with fusion categories as the expected general endpoint. The word *canonical* is a technical requirement here, not a mood: the assignment must be exhibited as a functor, every choice on which it depends must be named explicitly, and a theorem must state exactly what is independent of those choices.

The concrete guiding path fixes a target before the generalities are attempted: the prime field $\mathbb{F}_p$ of characteristic p, its canonical symplectic space $\mathbb{F}_p \times \mathbb{F}_p$, and the finite Weyl–Heisenberg system quantizing it. Every general definition this campaign proposes must reproduce this case exactly, and is tested against it before anything else. The result is recorded in this lab book, one topic per section, each statement carrying its full hypotheses, an honest status, and its provenance (§1); nothing in these pages is asserted at a strength its evidence does not support.

Status

The following table records the **28** rows issued by the campaign's first increment (`briefs/wh-kappa-target.md`): the canonical quantum system attached to a finite field κ, with every choice it depends on named. Of these, **22** are PROVED (four of them PROVED[†], meaning the proved statement is restricted to $p = 2$ or to odd p and displays that restriction in its own text — never abbreviated into the label), **4** are SKETCH (an argument exists and is written down, but has not yet been through this campaign's hostile-critic round), **1** is CONJECTURE, and **1** is REFUTED (with its surviving weaker statement recorded in the adjacent row). Every definition D1–D11 of `definitions.md` and every row below is restated in full, under a descriptive name, in the sections that follow; the table records only where.

The current register, including the local-ring results and the separate field-with-one-element sidequest and the Symplectic Phantasm bootstrap, has **159** rows: **130** PROVED, **25** SKETCH, three CONJECTURE and one REFUTED. Section 11 compares the literature on the field with one element, with qubit and tensor-product examples; seven of its finite-kernel claims are proved and eight supporting comparisons remain sketches. Section 12 develops the subsystem-composition direction: a positive Hecke family, its symmetric-group quantum endpoint, and an exact contextual process theorem. Eleven claims of this operational increment are proved and seven supporting comparisons remain sketches.

The categorical-limit increment adds thirty-seven proved claims. It gives continuous and germ categories for the full finite projection completion, typed quantum instruments and classical history routing, local endpoint lifts, and finite Born-probability limits. Further sections establish unitary coboundary exchange, Schur–Weyl trace comparisons, the positive mirabolic boundary and explicit symplectic composition correspondences.

Sections 21 and 22 add the arithmetic Frobenius, code-transfer and higher-gate programme.

Its ten new claims passed capped review. The proposed positive incidence/phase comparison is recorded separately from these arithmetic fibres.

Section 23 adds four proved claims for a positive conditional Frobenius orbit boundary. It retains cyclic matrix blocks, independent quantum composition and standard subfield decoder probabilities. The full coupling to arithmetic multiplication and Fourier remains open; the four claims passed independent capped review and mechanical repair.

Section 24 gives three proved claims for finite separable extensions over arbitrary base fields and their Galois closures. Section 25 develops the uniform arithmetic Gram sectors, conditional Fourier charts and tower refinements; its three positive statements passed the capped review. The full mixed limit and the signed-orbit filtration in Section 26 remain conjectures.

Section 27 adds six proved claims for a common positive arithmetic reference and its finite operational boundary. It treats every finite-field embedding and full fixed arithmetic protocols, retaining coherent composition and rare-event information. The actual characteristic remains named; the interpolation changes reference counting and does not identify the earlier angle construction with this new theory.

Section 28 adds six proved claims for an origin-free arithmetic multiplication-graph register. Simultaneous Frobenius is removed by collective invariant projection, while relative Frobenius acts on a noncommutative quantum coordinate. Its primitive preparation vanishes at the ordinary counting endpoint but retains a coherence-sensitive first-grade boundary. Fixed-base transfers and finite spectral identities are included; endpoint tensor and sum closure are not required, and no infinite spectral identification is asserted.

Section 29 adds two proved claims for one concrete completion across arithmetic degrees. Frobenius is outer in its minimal unitization, while a specified positive degree regularization has an explicit convergent zeta quotient. This realizes a quantum completion with cyclic spectrum; it does not identify that spectrum with Riemann zeros.

Section 30 proves that the completed reference is the trace-norm limit of finite degree-tagged arithmetic preparations. Degree completion and the counting boundary agree in both orders and along joined paths, with an explicit uniform tail bound and retained event rate.

The increment's central finding, proved in §4, is that the construction depends on a second datum beyond the field and the additive character: a *polarizing cocycle*. At odd characteristic this choice is immaterial and has a canonical representative. At characteristic two it is not immaterial: the admissible cocycles split into exactly two $SL_2(\kappa)$-orbits, distinguished by the Arf invariant of an associated quadratic form, giving two genuinely non-isomorphic Heisenberg groups — a fact detectable by an operator identity inside the quantum system itself. Per `PRD.md`, a sharp result of this shape is a positive structural statement about the definition, not a negative one.

Claim id	**Statement (short)**	**Status**	**Where**
`WH-FORM`	The symplectic form is κ-bilinear, alternating, nondegenerate; its κ-linear and $\mathbb{F}_p$-linear isometries coincide, both $SL_2(\kappa)$.	PROVED	Thm. 2.6
`WH-COMM`	The Weyl commutation relations hold uniformly in p, for every admissible cocycle.	PROVED	Thm. 2.7
`WH-ALG`	The twisted algebra is simple, dimension q^2, centre $\mathbb{C}$, for every cocycle.	PROVED	Thm. 2.8

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Claim id	Statement (short)	Status	Where
`WH-POL`	Every line is Lagrangian; a Schrödinger model is a pair (L, χ), and all such pairs give isomorphic modules.	PROVED	Thm. 3.3
`WH-POL-2`	At $p = 2$, characters induced on a line with $Q_\beta \neq 0$ are forced to order 4; no such forcing at odd p.	PROVED†	Thm. 3.4
`WH-SVN`	Uniqueness of the finite Weyl representation: one irreducible unitary representation of dimension q, intertwiner unique up to $U(1)$, every automorphism inner.	PROVED	Thm. 3.5
`WH-ALG-MAT`	The algebra is isomorphic to $M_q(\mathbb{C})$, by a non-canonical isomorphism.	PROVED	Thm. 3.6
`WH-BETA-a`	Admissible polarizing cocycles form a torsor of size q^3 under the symmetric bilinear forms.	PROVED	Thm. 4.2
`WH-BETA-b`	At odd p, the cocycle is immaterial: $\omega/2$ is a canonical representative fixing an explicit isomorphism between any two choices.	PROVED†	Thm. 4.3
`WH-BETA-c`	At $p = 2$, no symmetric cocycle exists: the odd-p route to a canonical choice is unavailable.	PROVED†	Thm. 4.4
`WH-BETA-d`	At $p = 2$, Q_β is a quadratic form with polar form ω; the resulting element-order profile of $H_\beta(\kappa)$ is computed, and at least two isomorphism classes occur.	PROVED†	Thm. 4.5
`WH-BETA-TYPE`	**Heart of the increment.** The Arf invariant classifies $\mathrm{Adm}(\omega)$ into exactly two $SL_2(\kappa)$-orbits at $p = 2$, both nonempty, giving exactly two isomorphism classes of Heisenberg group.	SKETCH	Thm. 4.7
`WH-BETA-EPS`	The Arf type is detected inside the quantum system itself, by an explicit operator identity.	SKETCH	Thm. 4.8
`WH-BETA-LEVEL`	The algebra and its Weyl frame do not depend on β at any characteristic; the finite level-μ_p frame does, and only at $p = 2$.	SKETCH	Thm. 4.10
`WH-WEIL-a`	Frame automorphisms surject onto $SL_2(\kappa)$ with kernel $\hat{V}$, at every characteristic.	PROVED	Thm. 5.2
`WH-WEIL-b`	At odd p, this extension splits explicitly, giving a projective unitary representation of $SL_2(\kappa)$.	PROVED†	Thm. 5.3

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Claim id	Statement (short)	Status	Where
WH-WEIL-c	At $p = 2$, μ_2-valued phases occur exactly over the orthogonal subgroup $O(Q_\beta)$, proper for the reference cocycle.	PROVED †	Thm. 5.4
WH-WEIL-d	At $p = 2$, μ_4 is forced exactly when $O(Q_\beta)$ is proper — always except the single case $q = 2$, anisotropic.	PROVED †	Thm. 5.5
WH-WEIL-e	Whether the extension splits at $p = 2$ for every $q = 2^m$; verified only at $q = 2, 4$.	CONJECTURE	Conj. 5.6
WH-SYMM	The algebra does not remember the κ-module structure of $V(\kappa)$; its intrinsic $\mathbb{F}_p$-linear symmetry is the strictly larger $Sp_{2m}(\mathbb{F}_p)$ once $m > 1$.	PROVED	Thm. 7.1
WH-CHOICE	Beyond κ, exactly two data: ψ and β; the resulting algebras are always isomorphic, but no isomorphism among them is distinguished.	PROVED	Thm. 6.2
WH-CANON	The projective Hilbert space is canonical in (κ, ψ), at fixed β.	PROVED	Thm. 6.3
WH-CANON-BETA	The projective Hilbert space is canonical independently of β as well.	SKETCH	Thm. 6.4
WH-FUNCT-a	The trace-normalized character family is not restriction-compatible along field extensions; trivial exactly when $p \mid n$.	PROVED	Thm. 8.1
WH-FUNCT-b	$(\kappa, \psi) \mapsto A_{\psi,\beta_0}(V(\kappa))$ is a functor on character-compatible embeddings.	PROVED	Thm. 8.2
WH-FUNCT-b-SEC	*Refuted:* a Frobenius-compatible character exists for every finite field.	REFUTED	Thm. 8.3
WH-FUNCT-c	The Frobenius obstruction, exactly: invariance forces $c \in \mathbb{F}_p$, and $\mathbb{F}_{p^p}$ then restricts trivially to the prime field.	PROVED	Thm. 8.4
WH-FUNCT-d	Compatible character systems over an algebraic closure are exactly the restrictions of its characters nontrivial on $\mathbb{F}_p$.	PROVED	Thm. 8.5

2 The symplectic object of a finite field, and its twisted operator algebra

Throughout the labbook p denotes a prime, $m \geq 1$ an integer, $q := p^m$, and $\kappa := \mathbb{F}_q$ the finite field of order q, with prime field $\mathbb{F}_p \subseteq \kappa$. **The case $p = 2$ is in scope everywhere in this campaign**: no definition or theorem below is stated only for odd p and silently assumed to extend; where a statement holds only for $p = 2$ or only for odd p this is marked in the statement itself, never left to a reader's guess. This section builds the ground floor of the construction — the symplectic vector space attached to κ, the two choices (a polarizing cocycle and an additive character) the quantum system is built from, the twisted operator algebra those choices produce, and its associated Heisenberg group — and proves the three structural facts that hold *uniformly* in the characteristic and for *every* admissible choice of cocycle. What is and is not canonical about the choices themselves is the subject of later sections, in particular §4.

2.1 The symplectic space of a finite field

Definition 2.1 (The symplectic object attached to a finite field)**.** For a finite field κ, put $V(\kappa) := \kappa \oplus \kappa$, and define

$$\omega : V(\kappa) \times V(\kappa) \to \kappa, \qquad \omega\big((a, b), (a', b')\big) := ab' - a'b.$$

Scope. This is a stipulation: a vector space and a bilinear pairing written down by formula. On its own it asserts nothing about ω — not that it is κ-bilinear, not that it is alternating, not that it is nondegenerate. Those are properties, established as Theorem 2.6 below. No restriction on the characteristic is built into the definition; $p = 2$ is included on the same footing as every other prime.

Provenance. ids: D1; proved in: —; tested by: —; reviewed in: —.

2.2 Polarizing cocycles: the datum β, introduced

The Weyl operators of §2.4 below are built not from ω directly but from a bilinear map whose antisymmetrization is ω. Choosing one such map is a second datum of the construction, on exactly the same footing as the choice of additive character ψ of §2.3. This subsection only introduces the datum and its notation; the fact that it forms a torsor, that it is immaterial at odd characteristic, and that it splits the construction into two genuinely different isomorphism types at $p = 2$ is the content of Theorems 4.2–4.7 in §4, and is not anticipated here.

Definition 2.2 (Admissible polarizing cocycles, and the reference cocycle)**.** Fix a finite field κ and its symplectic object $(V(\kappa), \omega)$ of Definition 2.1. The set of *admissible polarizing cocycles* is

$$\mathrm{Adm}(\omega) := \big\{\, \beta : V(\kappa) \times V(\kappa) \to \kappa \;\big|\; \beta \text{ is } \kappa\text{-bilinear and } \beta - \beta^T = \omega \,\big\},$$

where $\beta^T(v, v') := \beta(v', v)$. A *polarizing cocycle* is a choice of element $\beta \in \mathrm{Adm}(\omega)$, and it is a **datum of the construction**, on the same footing as the character ψ of Definition 2.3 — not a convention fixed once and for all. The map

$$\beta_0\big((a, b), (a', b')\big) := ab'$$

lies in $\mathrm{Adm}(\omega)$ and is called the *reference cocycle*: a label for a point of the torsor of Definition 2.2, not a canonical point of it (Theorem 4.2).

Scope. Nothing here asserts that $\mathrm{Adm}(\omega)$ is nonempty, that it has any particular size, that β_0 is a typical or a distinguished element of it, or that the choice of β is or is not immaterial to anything built from it. All of that is claimed, and proved, later: Theorems 4.2–4.7 of §4 are where this datum's structure is actually established. In particular the tempting odd-characteristic habit of writing $\omega/2$ for "the" polarizing cocycle is not part of this definition and is not available at $p = 2$, where division by 2 is division by 0; §4 makes this precise.

Provenance. ids: D2; proved in: —; tested by: —; reviewed in: —.

2.3 The phase datum

Definition 2.3 (Phase datum, and the trace-normalized family)**.** A *phase datum* for κ is a nontrivial additive character $\psi : (\kappa, +) \to \mathbb{C}^\times$, i.e. a group homomorphism with $\psi \not\equiv 1$. Write $X(\kappa)$ for the set of such ψ. Inside $X(\kappa)$, the *trace-normalized family* is defined as follows: for $\kappa = \mathbb{F}_{p^m}$ put

$$\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z) := z + z^p + z^{p^2} + \cdots + z^{p^{m-1}},$$

and for ζ a primitive p-th root of unity in $\mathbb{C}$, put

$$\psi_\zeta(z) := \zeta^{\,\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z)}.$$

Scope. Two choices are named here and neither is suppressed: the character ψ itself (an arbitrary element of $X(\kappa)$), and, when working inside the distinguished family, the root of unity ζ. The absolute trace $\mathrm{Tr}_{\kappa/\mathbb{F}_p}$ is canonical — it is the same $\mathbb{F}_p$-linear map regardless of any further choice — but ψ_ζ is canonical only once ζ is fixed, and §8 shows in detail how far a compatible choice of ψ across a tower of fields can and cannot be made.

Provenance. ids: D3; proved in: —; tested by: —; reviewed in: —.

2.4 Weyl operators, the twisted algebra, and the Heisenberg group

Definition 2.4 (Weyl operators and the twisted observable algebra)**.** Fix a phase datum ψ as in Definition 2.3 and a polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ as in Definition 2.2. The *twisted operator algebra* $A_{\psi,\beta}(V)$ is the $\mathbb{C}$-vector space

$$A_{\psi,\beta}(V) := \bigoplus_{v \in V(\kappa)} \mathbb{C} \cdot W_\beta(v)$$

with the product fixed, once and for all, on basis elements by

$$W_\beta(v)\, W_\beta(v') := \psi\big(\beta(v, v')\big)\, W_\beta(v + v').$$

Scope. The ordering of the product on the right ($\psi(\beta(v, v'))$, not $\psi(\beta(v', v))$) is fixed as part of the definition and is not symmetrized; which of the two orderings a later stipulated model (such as M_0 of §3) actually realizes is a computation, not a convention, and is checked explicitly there. $A_{\psi,\beta}(V)$ depends on κ, ψ and β and on nothing else; that this dependence is associative, unital and of dimension q^2 is Theorem 2.7(a) below, not part of the stipulation.

Provenance. ids: D4; proved in: —; tested by: —; reviewed in: —.

Definition 2.5 (The Heisenberg group of a polarizing cocycle)**.** With $\beta \in \mathrm{Adm}(\omega)$ as above, the *Heisenberg group of* β is the set $H_\beta(\kappa) := \kappa \times V(\kappa)$ with product

$$(t, v)(t', v') := \big(t + t' + \beta(v, v'),\ v + v'\big).$$

Scope. This construction uses no character ψ; it is a group attached to κ and β alone. It does depend on β, and — this is the content of §4 — at $p = 2$ its isomorphism class as an abstract group depends on more than the isomorphism class of the pair $(V(\kappa), \omega)$: it depends on which point of the torsor $\mathrm{Adm}(\omega)$ was chosen. Nothing about that dependence is claimed or used here.

Provenance. ids: D5; proved in: —; tested by: —; reviewed in: —.

2.5 The form is bilinear, alternating, nondegenerate, and its isometries

PROVED

Theorem 2.6 (The symplectic form of a finite field, at every characteristic)**.** *Let κ be any finite field, $p = 2$ included, with $V(\kappa)$ and ω as in Definition 2.1. Then:*

(a) ω is κ-bilinear;

(b) $\omega(v, v) = 0$ for every $v \in V(\kappa)$ — alternating *in the strong sense, not merely antisymmetric;*

(c) ω is nondegenerate: if $\omega(v, \cdot) \equiv 0$ then $v = 0$;

(d) every $\beta \in \mathrm{Adm}(\omega)$ (Definition 2.2) is a 2-cocycle for the abelian group $(V(\kappa), +)$, i.e. $\beta(v, v') + \beta(v + v', v'') = \beta(v, v'' + v) + \beta(v', v'')$ for all v, v', v'' — equivalently $\beta(v, v') + \beta(v + v', v'') = \beta(v', v'') + \beta(v, v' + v'')$ — and $\beta - \beta^T = \omega$;

(e) the group of κ-linear automorphisms g of $V(\kappa)$ with $\omega(gv, gv') = \omega(v, v')$ for all v, v' is exactly $SL_2(\kappa)$;

(f) the group of merely $\mathbb{F}_p$-linear automorphisms g of $V(\kappa)$ (i.e. additive maps that need not commute with multiplication by κ) with $\omega(gv, gv') = \omega(v, v')$ for all v, v' is also *exactly $SL_2(\kappa)$ — the larger class of symmetries one might a priori expect contains nothing beyond the κ-linear ones.*

Scope. Part (f) is a statement about the κ-valued form ω itself; it is emphatically *not* a statement about the $\mathbb{F}_p$-valued form $\psi \circ \omega$ obtained by composing with a character, whose isometries are the strictly larger group $Sp_{2m}(\mathbb{F}_p)$ once $m > 1$ — that is Theorem 7.1 of §7. Nothing here addresses which $\beta \in \mathrm{Adm}(\omega)$ is chosen, or classifies $\mathrm{Adm}(\omega)$ itself; that is §4.

Provenance. ids: D1, D2, WH-FORM; proved in: theory/wh-kappa.md §1; tested by: theory/checks/wh_kappa_check.py (C1,C2); theory/checks/fp_symmetry.py (S1); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. (a) Fix $v = (a, b)$, $v' = (a', b')$. Each argument of ω occurs exactly once in the formula $ab' - a'b$, and κ is commutative, so scaling either argument by $c \in \kappa$ scales the value by c, and additivity in each slot is distributivity of multiplication over addition in κ. This holds verbatim at $p = 2$: no step divides by 2 or by anything else.

(b) $\omega\big((a, b), (a, b)\big) = ab - ab = 0$, an identity in the commutative ring κ that cancels two literally equal terms, not two terms related by a factor of 2. It therefore holds at $p = 2$ exactly as it holds elsewhere. This is worth pausing on: antisymmetry alone, $\omega(v, v') = -\omega(v', v)$, gives only $2\omega(v, v) = 0$ on setting $v' = v$, which is vacuous once $2 = 0$. So at $p = 2$, (b) is a strictly stronger statement than antisymmetry and must be — and here is — verified by the direct computation above, not inferred from it.

(c) If $\omega(v, \cdot) \equiv 0$ for $v = (a, b)$ then in particular $\omega\big((a, b), (0, 1)\big) = a = 0$ and $\omega\big((a, b), (1, 0)\big) = -b = 0$, so $v = 0$.

(d) $\beta - \beta^T = \omega$ is the definition of $\mathrm{Adm}(\omega)$ (Definition 2.2). For the cocycle identity: β is bi-additive (it is κ-bilinear, hence in particular additive in each argument), and for any bi-additive β both sides of the cocycle identity expand, term by term, to $\beta(v, v') + \beta(v, v'') + \beta(v', v'')$; the identity holds without using anything about ω beyond bi-additivity of β.

(e) Write $v = (a, b)$, $v' = (a', b')$ as the columns of a 2×2 matrix over κ; then $\omega(v, v') = ab' - a'b = \det[v \,|\, v']$ by direct expansion. For $g \in GL_2(\kappa)$, $[gv \,|\, gv'] = g[v \,|\, v']$ as matrices, and multiplicativity of the determinant for 2×2 matrices over a commutative ring (itself a polynomial identity verified by direct expansion, holding in every characteristic) gives $\omega(gv, gv') = \det(g) \cdot$

$\omega(v, v')$. Hence $\omega \circ g = \omega$ iff $(\det g - 1)\,\omega \equiv 0$; since $\omega\big((1,0),(0,1)\big) = 1 \neq 0$, this holds iff $\det g = 1$, i.e. iff $g \in SL_2(\kappa)$.

(f) Let $g : V(\kappa) \to V(\kappa)$ be additive (not assumed κ-linear) with $\omega(gv, gu) = \omega(v, u)$ for all v, u.

- *g is bijective.* If $gv = 0$ then $\omega(v, u) = \omega(gv, gu) = 0$ for every u, so $v = 0$ by (c); thus g is injective, hence bijective since $V(\kappa)$ is finite.

- *g sends the standard basis to a κ-basis.* Let $v_1 = (1, 0)$, $v_2 = (0, 1)$, so $\omega(v_1, v_2) = 1$. Then $\omega(gv_1, gv_2) = \omega(v_1, v_2) = 1 \neq 0$; were gv_1, gv_2 κ-dependent, bilinearity and (b) would force $\omega(gv_1, gv_2) = 0$ (if, say, $gv_2 = c \cdot gv_1$ then $\omega(gv_1, gv_2) = c\,\omega(gv_1, gv_1) = 0$ by (a) and (b)). So $\{gv_1, gv_2\}$ is κ-independent, hence a basis of the 2-dimensional space $V(\kappa)$.

- *g is κ-linear.* Fix $c \in \kappa$, $u \in V(\kappa)$, and put $w := g(cu) - c \cdot g(u)$ (subtraction makes sense because g is additive, hence $\mathbb{Z}$-linear, even before κ-linearity is known). For $i = 1, 2$,

  $$\omega(gv_i, w) = \omega\big(gv_i, g(cu)\big) - c\,\omega(gv_i, gu) = \omega(v_i, cu) - c\,\omega(v_i, u) = c\,\omega(v_i, u) - c\,\omega(v_i, u) = 0,$$

  using the hypothesis $\omega(g\cdot, g\cdot) = \omega(\cdot, \cdot)$ for the middle equality and κ-bilinearity (a) for the outer ones. Since $\{gv_1, gv_2\}$ is a basis, $\omega(gv_1, \cdot)$ and $\omega(gv_2, \cdot)$ have trivial common kernel only at 0 by nondegeneracy (c) applied after a change of basis, so $w = 0$. Hence $g(cu) = c \cdot g(u)$ for all c, u: g is κ-linear.

By κ-linearity and (e), $\det g = 1$, so $g \in SL_2(\kappa)$; the converse containment is immediate from (e). This closes (f). □

Remark. Part (f) forecloses a tempting shortcut: one might hope to characterize the $\mathbb{F}_p$-linear isometries of ω as the "ω-self-adjoint $\mathbb{F}_p$-endomorphisms," by analogy with the real symplectic case. This is false at $p = 2$: exhaustive enumeration finds 8 and 64 ω-self-adjoint $\mathbb{F}_p$-endomorphisms of $V(\kappa)$ at $q = 2, 4$ respectively, against only 2 and 4 κ-scalars — the self-adjointness route simply does not isolate $SL_2(\kappa)$ once $p = 2$, and the direct argument above is what closes (f) instead. Independently, complete backtracking search over the $\mathbb{F}_p$-linear isometries of the κ-valued form finds exactly $|SL_2(\kappa)| = 6, 24, 60, 120, 504, 720$ of them at $q = 2, 3, 4, 5, 8, 9$, matching an independent enumeration of $SL_2(\kappa)$ set-for-set at every one of these values (`theory/checks/fp_symmetry.py`, gates S1 and S4).

2.6 Weyl relations, uniformly in the characteristic and for every choice of cocycle

PROVED

Theorem 2.7 (The Weyl commutation relations, for every admissible cocycle)**.** *Let κ be any finite field, ψ a phase datum (Definition 2.3), and $\beta \in \mathrm{Adm}(\omega)$ an* arbitrary *admissible polarizing cocycle (Definition 2.2). Then, inside $A_{\psi,\beta}(V)$ of Definition 2.4:*

(a) $A_{\psi,\beta}(V)$ is a unital associative $\mathbb{C}$-algebra of dimension q^2, with unit $W_\beta(0)$;

(b) every $W_\beta(v)$ is invertible, with $W_\beta(v)^{-1} = \psi\big(\beta(v, v)\big)\, W_\beta(-v)$;

(c) $W_\beta(v)\, W_\beta(v') = \psi\big(\omega(v, v')\big)\, W_\beta(v')\, W_\beta(v)$ for all $v, v' \in V(\kappa)$;

(d) $W_\beta(u)\, W_\beta(v)\, W_\beta(u)^{-1} = \psi\big(\omega(u, v)\big)\, W_\beta(v)$ for all $u, v \in V(\kappa)$.

Scope. No part of this theorem uses which admissible cocycle $\beta \in \mathrm{Adm}(\omega)$ was chosen, nor any property of β beyond bi-additivity and the defining relation $\beta - \beta^T = \omega$; in particular nothing here distinguishes the reference cocycle β_0 from any other point of the torsor. No hypothesis on the characteristic is used beyond what already went into Theorem 2.6. What *is* β-dependent — the isomorphism class of the group $H_\beta(\kappa)$ itself, at $p = 2$ — is not addressed here.

Provenance. ids: D1, D2, D3, D4, WH-FORM, WH-COMM; proved in: theory/wh-kappa.md §2; tested by: theory/checks/wh_kappa_check.py (C3,C4); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. (a) Associativity on basis elements amounts to

$$\big(W_\beta(v)W_\beta(v')\big)W_\beta(v'') = W_\beta(v)\big(W_\beta(v')W_\beta(v'')\big),$$

which unwinds to the 2-cocycle identity for $\psi \circ \beta$; this holds because β is a 2-cocycle (Theorem 2.6(d)) and ψ is multiplicative. Since $\beta(0, v) = \beta(v, 0) = 0$ (κ-bilinearity), and $\psi(0) = 1$, $W_\beta(0)$ is a two-sided unit. The dimension is $|V(\kappa)| = q^2$ by construction.

(b) By bi-additivity, $\beta(v, -v) = -\beta(v, v)$, so $W_\beta(v)W_\beta(-v) = \psi\big(\beta(v, -v)\big)W_\beta(0) = \psi\big(-\beta(v, v)\big) \cdot 1$, and symmetrically on the other side; since ψ takes values in $\mathbb{C}^\times$, this exhibits a two-sided inverse.

(c) By the defining product, $W_\beta(v)W_\beta(v') = \psi\big(\beta(v, v')\big)W_\beta(v + v')$ and $W_\beta(v')W_\beta(v) = \psi\big(\beta(v', v)\big)W_\beta(v + v')$. By (b), $W_\beta(v + v')$ is invertible, so

$$W_\beta(v)W_\beta(v') = \psi\big(\beta(v, v') - \beta(v', v)\big)\, W_\beta(v')W_\beta(v) = \psi\big(\omega(v, v')\big)\, W_\beta(v')W_\beta(v),$$

the last step by the defining relation $\beta - \beta^T = \omega$ of Definition 2.2.

(d) Multiply (c) on the right by $W_\beta(v)^{-1}$, whose existence is (b). □

Remark (The centre of the Heisenberg group, and the algebra as a group-algebra quotient). The group $H_\beta(\kappa)$ of Definition 2.5 is indeed a group: associativity is the cocycle identity of Theorem 2.6(d), the identity is $(0, 0)$, and (t, v) has inverse $(-t + \beta(v, v), -v)$. Its centre is exactly $\kappa \times 0$: (t, v) is central iff $\beta(v, v') - \beta(v', v) = 0$ for every v', i.e. iff $\omega(v, \cdot) \equiv 0$, i.e. iff $v = 0$ by Theorem 2.6(c). Consequently $A_{\psi,\beta}(V)$ is the quotient of the group algebra $\mathbb{C}[H_\beta(\kappa)]$ by the two-sided ideal identifying the central element $(t, 0)$ with the scalar $\psi(t)$: the algebra is a family over $\psi \in X(\kappa)$ at *fixed* β, and the group $H_\beta(\kappa)$ itself depends on β — a dependence that, Theorem 4.7 shows, is not always trivial once $p = 2$.

2.7 Simplicity of the twisted algebra

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Theorem 2.8 (Central simplicity of the twisted algebra, with no polarization)**.** *With κ, ψ and $\beta \in \mathrm{Adm}(\omega)$ as in Theorem 2.7, the algebra $A_{\psi,\beta}(V)$ of Definition 2.4 is simple, has centre $\mathbb{C} \cdot 1$, and $\dim_{\mathbb{C}} A_{\psi,\beta}(V) = q^2$.*

Scope. This theorem uses no choice of Lagrangian line, no module, and no property of β beyond Theorem 2.7; in particular it holds for every $\beta \in \mathrm{Adm}(\omega)$, uniformly. It does not by itself identify $A_{\psi,\beta}(V)$ with a matrix algebra $M_q(\mathbb{C})$ — that identification exists (Theorem 3.6 of §3) but needs a polarization and is not canonical, whereas simplicity and the dimension count here need none.

Provenance. ids: D4, WH-COMM, WH-ALG; proved in: theory/wh-kappa.md §3; tested by: theory/checks/wh_kappa_check.py (C5,C6); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. Define the *Weyl average* $E : A_{\psi,\beta}(V) \to A_{\psi,\beta}(V)$ by

$$E(x) := q^{-2} \sum_{u \in V(\kappa)} W_\beta(u)\, x\, W_\beta(u)^{-1}.$$

Step 1: $E\big(\sum_v a_v W_\beta(v)\big) = a_0 \cdot 1$. By Theorem 2.7(d), $W_\beta(u)W_\beta(v)W_\beta(u)^{-1} = \psi(\omega(u, v))W_\beta(v)$, so

$$E\Big(\sum_v a_v W_\beta(v)\Big) = \sum_v a_v \Big(q^{-2} \sum_u \psi\big(\omega(u, v)\big)\Big) W_\beta(v).$$

Fix $v \neq 0$. By Theorem 2.6(a),(c), $\omega(\cdot, v)$ is a nonzero κ-linear map $V(\kappa) \to \kappa$, hence surjective (a nonzero linear map between finite-dimensional vector spaces over the same field onto a 1-dimensional target is onto). So $u \mapsto \psi(\omega(u, v))$ is a character of the additive group $V(\kappa)$ that is nontrivial (since ψ is nontrivial and $\omega(\cdot, v)$ is onto κ). A nontrivial character χ of a finite abelian group G sums to zero: fixing u_0 with $\chi(u_0) \neq 1$, the substitution $u \mapsto u + u_0$ permutes G, so $S := \sum_u \chi(u)$ satisfies $S = \chi(u_0)S$, forcing $S = 0$ since $\chi(u_0) \neq 1$. Applying this to $\chi = \psi(\omega(\cdot, v))$ gives $\sum_u \psi(\omega(u, v)) = 0$ for $v \neq 0$; for $v = 0$ the sum is $q^2 \cdot \psi(0) = q^2$. Hence $E(\sum_v a_v W_\beta(v)) = a_0 \cdot W_\beta(0) = a_0 \cdot 1$, since $W_\beta(0)$ is the unit (Theorem 2.7(a)).

Step 2: $Z(A_{\psi,\beta}(V)) = \mathbb{C} \cdot 1$. If z is central then $W_\beta(u) z W_\beta(u)^{-1} = z$ for every u, so $z = E(z) = a_0 \cdot 1$ by Step 1; conversely every scalar is central.

Step 3: simplicity. Let I be a nonzero two-sided ideal and pick $0 \neq x = \sum_v a_v W_\beta(v) \in I$ with some coefficient $a_{v_0} \neq 0$. Then $y := W_\beta(v_0)^{-1} x \in I$ (as I is a left ideal), and by Theorem 2.7(b),(c) each $W_\beta(v_0)^{-1} W_\beta(v)$ is a nonzero scalar multiple of $W_\beta(v - v_0)$; in particular the coefficient of $W_\beta(0)$ in y is a nonzero scalar multiple of a_{v_0}, hence nonzero. Now $E(y) \in I$: E is a $\mathbb{C}$-linear combination of the maps $x \mapsto W_\beta(u) x W_\beta(u)^{-1}$, and I is a two-sided ideal, so each term $W_\beta(u) y W_\beta(u)^{-1}$ lies in I and so does their average. By Step 1, $E(y) = c \cdot 1$ with $c \neq 0$ (the coefficient of $W_\beta(0)$ in y), so $1 \in I$ and $I = A_{\psi,\beta}(V)$.

Step 4: dimension. $\dim_{\mathbb{C}} A_{\psi,\beta}(V) = |V(\kappa)| = q^2$ directly from Definition 2.4. □

Remark (Why the quantifier over β can be widened after the fact). Theorems 2.7 and 2.8 are stated for *every* $\beta \in \mathrm{Adm}(\omega)$, not merely the reference cocycle β_0. This widening is legitimate precisely because no step of either proof used more than the defining relation $\beta - \beta^T = \omega$ and bi-additivity: the commutator computation in Theorem 2.7(c)–(d) depends on β only through $\omega = \beta - \beta^T$, and simplicity in Theorem 2.8 depends only on ω being nondegenerate. That the choice of β is nonetheless *not* immaterial to everything built from it — it is invisible to the pair $(A_{\psi,\beta}(V), \text{Weyl frame})$ but not to the Heisenberg group $H_\beta(\kappa)$ itself once $p = 2$ — is the subject of §4.

3 Lagrangian lines, Schrödinger models, and the uniqueness of the finite Weyl representation

Recollection (the setting of §2). For a finite field $\kappa = \mathbb{F}_q$, $q = p^m$, the symplectic object is $V(\kappa) := \kappa \oplus \kappa$ with the κ-bilinear, alternating, nondegenerate form $\omega\big((a, b), (a', b')\big) := ab' - a'b$ (Definition 2.1); an admissible polarizing cocycle is a κ-bilinear $\beta : V(\kappa) \times V(\kappa) \to \kappa$ with $\beta - \beta^T = \omega$ (Definition 2.2); a phase datum is a nontrivial additive character $\psi : (\kappa, +) \to \mathbb{C}^\times$ (Definition 2.3); and the twisted algebra $A_{\psi,\beta}(V) := \bigoplus_{v \in V(\kappa)} \mathbb{C} \cdot W_\beta(v)$ has product $W_\beta(v) W_\beta(v') := \psi(\beta(v, v')) W_\beta(v + v')$ (Definition 2.4); by Theorem 2.8 it is simple of dimension q^2 with centre $\mathbb{C} \cdot 1$, for every $\beta \in \mathrm{Adm}(\omega)$.

Provenance. ids: D1, D2, D3, D4; proved in: —; tested by: —; reviewed in: —.

This section names the Lagrangian lines of $V(\kappa)$, the finite-dimensional modules of $A_{\psi,\beta}(V)$ they induce, and proves that there is exactly one such module up to unitary equivalence and that it realizes $A_{\psi,\beta}(V)$ as a full matrix algebra — a finite-field analogue of the Stone–von Neumann theorem — together with a genuinely characteristic-two phenomenon in how the induced characters behave.

3.1 Preliminary facts about finite abelian characters and cocycles

Four short, elementary facts about characters of finite abelian groups recur throughout the rest of the labbook; they are recorded once, here, and cited by name afterward. None of them is part of the claims DAG — they carry no claim id and no status label, being standard and immediate from the definitions of “character” and “cocycle” — but each is proved in full, since nothing in this labbook is asserted from memory.

Fact 1 (orthogonality). *If G is a finite abelian group and $\chi : G \to \mathbb{C}^\times$ a nontrivial character, then $\sum_{u\in G} \chi(u) = 0$.* *Proof.* Pick $u_0 \in G$ with $\chi(u_0) \neq 1$. Then $S := \sum_u \chi(u) = \sum_u \chi(u+u_0) = \chi(u_0)\, S$, because $u \mapsto u+u_0$ permutes G; since $\chi(u_0) \neq 1$, $S = 0$.

Fact 2 (valuation). *If ψ is a phase datum for κ (Definition 2.3) then $\psi(\kappa) \subseteq \mu_p$; $\ker\psi$ is an $\mathbb{F}_p$-subspace of κ of index p; and for $u \in \kappa$, $\psi(u\cdot) \equiv 1$ iff $u = 0$.* *Proof.* $\psi(t)^p = \psi(pt) = \psi(0) = 1$ since $\operatorname{char}\kappa = p$, so $\psi(\kappa) \subseteq \mu_p$. $\ker\psi$ is an additive subgroup of a characteristic-p field, hence an $\mathbb{F}_p$-subspace, and $\kappa/\ker\psi$ embeds nontrivially into $\mu_p \cong \mathbb{Z}/p$ (Definition 2.3), forcing index exactly p. If $u \neq 0$ then $u\kappa = \kappa \not\subseteq \ker\psi$, i.e. $\psi(u\cdot) \not\equiv 1$.

Fact 3 (duality of elementary abelian groups). *Let U be a finite $\mathbb{F}_p$-vector space of dimension d. Then $\hat{U} := \operatorname{Hom}(U, \mathbb{C}^\times)$ has order p^d, and every element of $\hat{U}$ takes values in μ_p.* *Proof.* For $\chi \in \hat{U}$, $\chi(u)^p = \chi(pu) = 1$, so $\chi(U) \subseteq \mu_p \cong \mathbb{Z}/p$, i.e. $\hat{U} = \operatorname{Hom}_{\mathbb{F}_p}(U, \mu_p)$. Fixing an $\mathbb{F}_p$-basis of U, a homomorphism is a free choice of d values in μ_p, so $|\hat{U}| = p^d$. (Applied below to $U = L$ a κ-line, an $\mathbb{F}_p$-space of dimension $m = [\kappa : \mathbb{F}_p]$, giving $|\hat{L}| = p^m = q$; later to $U = \kappa$ and $U = V(\kappa)$, of $\mathbb{F}_p$-dimension m and $2m$, giving $|\hat{\kappa}| = q$ and $|\hat{V}| = q^2$.)

Fact 4 (triviality of symmetric cocycles of exponent p). *Let U be a finite abelian group of exponent p and $c : U \times U \to \mathbb{C}^\times$ a symmetric 2-cocycle: $c(u, u') = c(u', u)$ and $c(u, u')c(u + u', u'') = c(u, u' + u'')c(u', u'')$. Then there is $\mu : U \to \mathbb{C}^\times$ with $c(u, u') = \mu(u + u')/\big(\mu(u)\mu(u')\big)$ for all u, u'.* *Proof.* On $E := \mathbb{C}^\times \times U$ define $(z, u)(z', u') := \big(zz'c(u, u'), u + u'\big)$; the cocycle identity gives associativity and symmetry of c gives commutativity, so E is a finite abelian group. Induction inside E gives $(1, u)^p = (\gamma(u), 0)$ with $\gamma(u) := c(u, u)c(2u, u) \cdots c((p-1)u, u)$. Fix an $\mathbb{F}_p$-basis $u_1, \ldots, u_n$ of U; as $\mathbb{C}$ is algebraically closed, choose z_i with $z_i^p = \gamma(u_i)^{-1}$ and put $f_i := (z_i, u_i)$, so $f_i^p = 1$. The subgroup $H := \langle f_1, \ldots, f_n\rangle \leq E$ is then abelian of exponent dividing p, and since U is free on $\{u_i\}$ there is a unique $\mathbb{F}_p$-linear $\sigma : U \to H$ with $\sigma(u_i) = f_i$ and $\mathrm{pr}_U \circ \sigma = \mathrm{id}_U$. Writing $\sigma(u) = (\mu(u), u)$, that σ is a homomorphism reads exactly $\mu(u)\mu(u')c(u, u') = \mu(u + u')$.

3.2 Lagrangian lines and the algebra they generate

Definition 3.1 (The quadratic form of a polarizing cocycle)**.** For $\beta \in \mathrm{Adm}(\omega)$, define $Q_\beta : V(\kappa) \to \kappa$ by $Q_\beta(v) := \beta(v, v)$. For the reference cocycle, $Q_{\beta_0}(a, b) = ab$.

Scope. Under a symmetrized cocycle (available only at odd p; §4) Q_β would vanish identically — that convention does not exist here, and Q_β is not zero. This definition asserts nothing beyond the formula: it does not by itself claim Q_β is a quadratic form in the technical sense (quadratic homogeneity and a bilinear polarization), which is established only where needed, at Theorem 4.5 of §4.

Provenance. ids: D6; proved in: —; tested by: —; reviewed in: —.

Definition 3.2 (Schrödinger models, and the standard model)**.** For a κ-line $L \subset V(\kappa)$ put $A_L := \operatorname{span}_{\mathbb{C}}\{\, W_\beta(l) : l \in L \,\} \subseteq A_{\psi,\beta}(V)$. Given a unital $\mathbb{C}$-algebra homomorphism (character) $\chi : A_L \to \mathbb{C}$, the *Schrödinger model* of (L, χ) is

$$M_{L,\chi} := A_{\psi,\beta}(V) \otimes_{A_L} \mathbb{C}_\chi.$$

The *standard model* (taken for $\beta = \beta_0$) is $M_0 := \bigoplus_{y\in\kappa} \mathbb{C} \cdot e_y$, with $\{e_y\}$ declared orthonormal and

$$W(a,b)\, e_y := \psi\big(-b(y+a)\big)\, e_{y+a}, \qquad \text{i.e.} \qquad W(a,b) = Z(-b)X(a),$$

where $X(a)e_y := e_{y+a}$ and $Z(b)e_y := \psi(by)e_y$. **This is a stipulation.**

Scope. The sign in $W(a,b) = Z(-b)X(a)$ is not cosmetic: the naive assignment $W(a,b) = Z(b)X(a)$, without the sign, realizes a *different* cocycle ($-ab'$, not β_0's ab'), and the discrepancy between the two is invisible at $p = 2$ (where $-1 = 1$) and visible only at odd p — a characteristic-dependent failure mode this definition is written to avoid by fixing the correct sign explicitly rather than leaving it to convention. That M_0, so defined, genuinely satisfies the multiplication law of Definition 2.4 is verified as part of the proof of Theorem 3.5 below (Step 1) and checked computationally by gate C11 of `theory/checks/wh_kappa_check.py`.

Provenance. ids: D8; proved in: —; tested by: —; reviewed in: —.

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Theorem 3.3 (Lagrangian lines, and the Schrödinger models they carry)**.** *Let κ, $V(\kappa)$, ω, $\beta \in \mathrm{Adm}(\omega)$, ψ, $A_{\psi,\beta}(V)$, Q_β (Definition 3.1) and A_L, $M_{L,\chi}$ (Definition 3.2) be as above, for any finite field κ, $p = 2$ included. Then:*

(a) every κ-line $L \subset V(\kappa)$ is ω-isotropic ($\omega(v,v') = 0$ for all $v, v' \in L$), hence maximal isotropic; so Lagrangian line *and* line *coincide as conditions on $L \subset V(\kappa)$, at every characteristic;*

(b) there are exactly $q+1 = |\mathbb{P}^1(\kappa)|$ such lines;

(c) A_L is commutative of dimension q, and carries exactly q characters, forming a torsor under $\hat{L} = \mathrm{Hom}(L, \mathbb{C}^\times)$;

(d) $\dim_{\mathbb{C}} M_{L,\chi} = q$;

*(e) **(precision, binding on this row.)** A* Schrödinger model *is a* pair (L,χ)*: there are $q(q+1)$ such pairs, q per line — but by Theorem 3.5 below,* all of them are isomorphic as $A_{\psi,\beta}(V)$-modules. *The count $q(q+1)$ is therefore a count of labels, not of isomorphism classes of module. Among the $q+1$ lines, only those with $Q_\beta|_L \equiv 0$ carry the trivial character $\chi \equiv 1$ (since $\chi \equiv 1$ is a character of A_L iff $Q_\beta(v_0)\kappa \subseteq \ker\psi$ iff $Q_\beta(v_0) = 0$, for $L = \kappa v_0$), and how many lines have $Q_\beta|_L \equiv 0$ is itself β-dependent: exactly 2 for the reference cocycle β_0 (§4 computes it in general).*

Scope. This theorem does not assert that distinct pairs (L,χ) give distinct, let alone non-isomorphic, modules — to the contrary, part (e) is precisely the observation that they do not, once Theorem 3.5 is available. It does not address the characteristic-two phenomenon in the values a character of A_L can take; that is Theorem 3.4 below. Nothing here depends on which $\beta \in \mathrm{Adm}(\omega)$ was chosen beyond the formula for Q_β.

Provenance. ids: D1, D4, D6, D8, WH-FORM, WH-ALG, WH-POL; proved in: theory/wh-kappa.md §4; tested by: theory/checks/wh_kappa_check.py (C7); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. (a) Write $L = \kappa v_0$. For $c, c' \in \kappa$, $\omega(cv_0, c'v_0) = cc'\,\omega(v_0,v_0) = 0$ by κ-bilinearity (Theorem 2.6(a)) and the computed alternating property (Theorem 2.6(b)) — at $p = 2$ antisymmetry alone would not give this, so the computed form of (b) is exactly what is used. Since $\dim_\kappa V(\kappa) = 2$ and ω is nondegenerate, no subspace of dimension 2 (i.e. all of $V(\kappa)$) is isotropic, so a line is a maximal isotropic subspace: *every* line is automatically Lagrangian, and the isotropy condition singles out nothing among lines.

(b) The nonzero vectors of $V(\kappa)$ number q^2-1, and each line contains $q-1$ of them, so there are $(q^2-1)/(q-1) = q+1 = |\mathbb{P}^1(\kappa)|$ lines.

(c) By Definitions 2.4 and 3.1, $W(cv_0)W(c'v_0) = \psi(\beta(cv_0, c'v_0))W((c+c')v_0) = \psi\big(cc'Q_\beta(v_0)\big)W((c+c')v_0)$, using $\beta(cv_0, c'v_0) = cc'\beta(v_0, v_0) = cc'Q_\beta(v_0)$ by κ-bilinearity. The phase is symmetric in (c, c'), so A_L is commutative; $\dim_{\mathbb{C}} A_L = |L| = q$ since $\{W(l) : l \in L\}$ is linearly independent (distinct basis elements of $A_{\psi,\beta}(V)$). *Characters exist:* $c(c, c') := \psi\big(cc'Q_\beta(v_0)\big)$ is a symmetric bi-additive, hence symmetric 2-cocycle, on $(L, +)$, a group of exponent p; Fact 3.1 supplies $\mu : L \to \mathbb{C}^\times$ with $c(c, c') = \mu(c+c')/(\mu(c)\mu(c'))$, and $\chi(W(cv_0)) := \mu(c)^{-1}$ is then an algebra character of A_L (it converts the twisted product into the untwisted one). *Torsor:* for two characters χ, χ' of A_L, the map $l \mapsto \chi'(W(l))/\chi(W(l))$ is a homomorphism $L \to \mathbb{C}^\times$, and conversely composing a character with any element of $\hat{L}$ gives another character; the action is free (a homomorphism fixing every $\chi(W(l))$ is trivial), and $|\hat{L}| = q$ by Fact 3.1 ($U = L$, $d = m$).

(d) $A_{\psi,\beta}(V)$ is free as a right A_L-module on any set of coset representatives $v_1, \ldots, v_q$ of L in $V(\kappa)$: the elements $\{W(v_i)W(l) : l \in L\}$ are, up to the nonzero scalars $\psi(\beta(v_i, l))$, exactly the basis elements $\{W(v_i + l) : l \in L\}$ of the coset $v_i + L$. So $A_{\psi,\beta}(V) \cong A_L^{\oplus q}$ as right A_L-modules, and $M_{L,\chi} = A_{\psi,\beta}(V) \otimes_{A_L} \mathbb{C}_\chi$ has dimension $q^2/q = q$.

(e) There are q characters per line by (c) and $q+1$ lines by (b), giving $q(q+1)$ pairs (L, χ); that they are all isomorphic as $A_{\psi,\beta}(V)$-modules is Theorem 3.5(c) below (which uses only parts (a)–(d) of this theorem, not this part, so there is no circular dependency). For $L = \kappa v_0$, $\chi \equiv 1$ is a character of A_L iff $c(c, c') = \psi(cc'Q_\beta(v_0)) \equiv 1$ iff $Q_\beta(v_0)\kappa \subseteq \ker\psi$; since $Q_\beta(v_0)\kappa$ is either $\{0\}$ or all of κ (it is a κ-subspace, being the image of scalar multiplication by $Q_\beta(v_0)$) and $\ker\psi \neq \kappa$ (Fact 3.1), this holds iff $Q_\beta(v_0) = 0$. For $\beta = \beta_0$, $Q_{\beta_0}(a, b) = ab$ (Definition 3.1) vanishes exactly on the two coordinate lines $\{a = 0\}$ and $\{b = 0\}$, so exactly 2 of the $q+1$ lines carry the trivial character. ☐

3.3 The characteristic-two phase phenomenon

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Theorem 3.4 (Forced order-four phases in the induced characters, at characteristic two)**.** *Let $L = \kappa v_0 \subset V(\kappa)$ be a line with $Q_\beta(v_0) \neq 0$ (Definition 3.1), and let χ be any character of A_L (Definition 3.2, Theorem 3.3(c)).*

- *[$p = 2$.] χ takes a value of exact multiplicative order 4 on some $W(l)$, $l \in L$: the phases of the model are forced out of $\mu_2 = \psi(\kappa)$ into μ_4. For the reference cocycle β_0, the lines with $Q_{\beta_0}|_L \equiv 0$ are exactly the two coordinate axes (Theorem 3.3(e)), so this phenomenon applies to $q-1$ of the $q+1$ lines.*

- *[p **odd**.] No such phenomenon occurs: every character of A_L can be, and by the construction of Theorem 3.3(c) is, μ_p-valued.*

Scope. This theorem says nothing about lines with $Q_\beta|_L \equiv 0$ (there the trivial character is available and no order-4 value is forced), and nothing about the algebra $A_{\psi,\beta}(V)$ globally — only about the values a character of the specific commutative subalgebra A_L can take. It does not identify *which* μ_4 this is inside a larger structure; that question, and its relation to the level-μ_p Weyl frame, is taken up in §4.

Provenance. ids: D6, D8, WH-POL, WH-POL-2; proved in: theory/wh-kappa.md §4 *step* $\langle 1\rangle 5, \langle 1\rangle 6$; tested by: theory/checks/beta_census.py (B6, #{Q=0}); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. Write $Q_0 := Q_\beta(v_0) \neq 0$ and $\nu(c) := \chi(W(cv_0))$ for the character χ constructed in the proof of Theorem 3.3(c); $\nu(0) = 1$, and setting $c' = c$ in the cocycle relation there, $\nu(c)^2 = \psi(c^2Q_0)\,\nu(2c)$.

[$p = 2$.] Here $2c = 0$, so $\nu(2c) = \nu(0) = 1$ and $\nu(c)^2 = \psi(c^2 Q_0)$. The Frobenius map $c \mapsto c^2$ is injective, hence bijective, on the finite field κ, so $\{c^2 Q_0 : c \in \kappa\} = Q_0\kappa = \kappa$ (as $Q_0 \neq 0$). Since ψ is nontrivial, $\psi(c^2 Q_0) = -1$ for some $c \in \kappa$ — because -1 is the unique nontrivial element of $\mu_2 = \{1, -1\}$ and $\psi(\kappa) = \mu_2$ by Fact 3.1 — giving $\nu(c)^2 = -1$, i.e. $\nu(c)$ has exact order 4. The count of lines follows from Theorem 3.3(e).

[p **odd.**] Put $\mu(c) := \psi(c^2 Q_0/2)$ (division by 2 makes sense, p being odd). Then, directly, $\psi\big((c^2 + c'^2)Q_0/2\big)\,\psi(cc'Q_0) = \psi\big((c + c')^2 Q_0/2\big)$, i.e. $\mu(c)\mu(c')\,\psi(cc'Q_0) = \mu(c + c')$: μ already solves the cocycle relation of Theorem 3.3(c) with values entirely inside $\psi(\kappa) = \mu_p$, so no order-4 (or any value outside μ_p) is ever forced. □

3.4 Uniqueness of the finite Weyl representation

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Theorem 3.5 (Uniqueness of the finite Weyl representation)**.** *Let κ, $\beta \in \mathrm{Adm}(\omega)$, $\psi \in X(\kappa)$, and $A_{\psi,\beta}(V)$ be as above. Then:*

(a) the induced module M_0 of (L_0, χ_0), $L_0 := 0 \oplus \kappa$, $\chi_0 \equiv 1$, has basis $\{e_y\}_{y\in\kappa}$ with $W_{\beta_0}(a, b)e_y = \psi\big(-b(y+a)\big)e_{y+a}$ — i.e. the stipulated standard model of Definition 3.2 genuinely satisfies the multiplication law of Definition 2.4;

(b) every simple $A_{\psi,\beta}(V)$-module is unitarily equivalent to M_0 (equivalently, to any $M_{L,\chi}$), and an $A_{\psi,\beta}(V)$-linear map $M_0 \to M_0$ is a scalar;

(c) consequently $H_\beta(\kappa)$ has, up to unitary equivalence, exactly one irreducible unitary representation with central character ψ, of dimension q, and every $\mathbb{C}$-algebra automorphism of $A_{\psi,\beta}(V)$ is inner;

(d) for $\{e_y\}$ orthonormal, every $W_{\beta_0}(v)$ acts unitarily, and in general the unitary intertwiner between two irreducible unitary representations with the same central character is unique up to a scalar in $U(1)$.

Scope. This theorem is stated and proved without invoking Maschke's theorem, Artin–Wedderburn, Skolem–Noether, Jacobson density, or Pontryagin duality as black boxes: every step is elementary, using only Theorem 2.8 (simplicity) and Theorem 3.3. It is uniform in the characteristic: no clause is restricted to odd p or to $p = 2$, and no step below treats $p = 2$ differently. It does not exhibit the isomorphism $A_{\psi,\beta}(V) \cong M_q(\mathbb{C})$ as canonical — that non-canonicity is the separate content of Theorem 3.6 immediately below.

Provenance. ids: D1, D4, D8, WH-ALG, WH-POL, WH-SVN; proved in: theory/wh-kappa.md §5; tested by: theory/checks/wh_kappa_check.py (C3,C6,C11); reviewed in: theory/verdicts/wh-kappa-r1.md; theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) The standard model, verified. Since $Q_{\beta_0}(0, b) = 0 \cdot b = 0$, $\chi_0 \equiv 1$ is a character of A_{L_0} by Theorem 3.3(e). Take coset representatives $(y, 0)$, $y \in \kappa$, of L_0 in $V(\kappa)$ and put $e_y := W_{\beta_0}(y, 0) \otimes 1 \in M_{L_0,\chi_0}$. Then

$$W_{\beta_0}(a, b)\, e_y = \psi\big(\beta_0((a, b), (y, 0))\big)\, W_{\beta_0}(a + y, b) \otimes 1 = W_{\beta_0}(a + y, b) \otimes 1,$$

since $\beta_0((a, b), (y, 0)) = a \cdot 0 = 0$. To evaluate this against χ_0 the L_0-part must be moved to the right:

$$W_{\beta_0}(a + y, 0)\, W_{\beta_0}(0, b) = \psi\big((a + y)b\big)\, W_{\beta_0}(a + y, b), \quad \text{so}$$
$$W_{\beta_0}(a + y, b) = \psi\big(-b(y + a)\big)\, W_{\beta_0}(a + y, 0)\, W_{\beta_0}(0, b).$$

Tensoring against $\chi_0(W_{\beta_0}(0,b)) = 1$ gives $W_{\beta_0}(a,b)e_y = \psi(-b(y+a))\, e_{a+y}$ as claimed, and $\dim M_0 = q$ by Theorem 3.3(d). (The sign here is exactly Definition 3.2's stipulation, now derived rather than assumed: the naive un-signed assignment $W(a,b) = Z(b)X(a)$ realizes the *different* cocycle $-ab'$ and fails to satisfy Definition 2.4's multiplication law for $\beta_0(v,v') = ab'$ at odd p — gate C11 of `theory/checks/wh_kappa_check.py` reports 0 violations for the signed model at all six tested $q \in \{2,3,4,5,8,9\}$, against $36, 400, 3888$ violations for the unsigned model at $q = 3,5,9$ and, tellingly, 0 violations at $q = 2,4,8$ — the defect is invisible at $p = 2$, where $-1 = 1$, and visible only once p is odd.)

(b) Uniqueness and Schur. $\pi : A_{\psi,\beta}(V) \to \mathrm{End}_{\mathbb{C}}(M_0)$ is a unital, nonzero algebra homomorphism, so $\ker\pi$ is a proper two-sided ideal, hence 0 by simplicity (Theorem 2.8): π is injective. Since $\dim \mathrm{End}_{\mathbb{C}}(M_0) = q^2 = \dim A_{\psi,\beta}(V)$, π is also surjective, giving $A_{\psi,\beta}(V) \cong \mathrm{End}_{\mathbb{C}}(M_0)$. Consequently $A_{\psi,\beta}(V) \cong M_0^{\oplus q}$ as a left module over itself (the columns of $\mathrm{End}_{\mathbb{C}}(M_0)$), so *any* simple module N — generated by any nonzero $n \in N$ — is a quotient of $A_{\psi,\beta}(V) \cong M_0^{\oplus q}$; restricting the quotient map to one summand gives a nonzero map $M_0 \to N$, which is surjective (its image is a nonzero submodule of the simple N) and injective (its kernel is a proper submodule of the simple M_0), hence an isomorphism $M_0 \cong N$. For an A-endomorphism φ of M_0: φ, as a $\mathbb{C}$-linear endomorphism of a finite-dimensional $\mathbb{C}$-vector space, has an eigenvalue $\lambda \in \mathbb{C}$ ($\mathbb{C}$ is algebraically closed), and $\ker(\varphi - \lambda)$ is then a nonzero A-submodule of the simple M_0, so $\varphi = \lambda \cdot \mathrm{id}$: $\mathrm{End}_A(M_0) = \mathbb{C}\cdot\mathrm{id}$.

(c) Representations of $H_\beta(\kappa)$ and inner automorphisms. A representation of $H_\beta(\kappa)$ with central character ψ is exactly an $A_{\psi,\beta}(V)$-module (the remark after Theorem 2.7); irreducibility is simplicity of the module, so (b) applies: there is exactly one such representation up to isomorphism, of dimension q. For automorphisms: given $\alpha \in \mathrm{Aut}_{\mathbb{C}}(A_{\psi,\beta}(V))$, twist the action on M_0 by α, i.e. put $a \cdot_\alpha m := \pi(\alpha(a))m$; this twisted module M_0^α is again simple of dimension q, hence $\cong M_0$ by (b), via some invertible T; unwinding, $\alpha = \mathrm{Ad}_T$, with T unique up to a scalar by the Schur clause of (b). No Skolem–Noether theorem is invoked.

(d) Unitarity. Since $|\psi(\cdot)| = 1$ (Fact 3.1), each $W_{\beta_0}(a,b)$ of (a) sends the orthonormal $\{e_y\}$ to $\{\psi(-b(y+a))e_{a+y}\}$, again orthonormal: every $W_{\beta_0}(v)$ is unitary for this inner product, with no averaging needed. (For a general β, the Weyl operators generate the *finite* level-μ_p frame $\mathcal{F}^{(p)}_{\psi,\beta}$ of Definition 4.9 in §4, of order pq^2; averaging any inner product on the simple module over this finite group makes every $\pi(W_\beta(v))$ unitary there too.) If T intertwines two unitary representations sharing central character ψ, then T^*T commutes with every operator of the first, so $T^*T = c \cdot \mathrm{id}$ with $c > 0$ by the Schur clause of (b); then $c^{-1/2}T$ is unitary, and two unitary intertwiners differ by a unimodular scalar, i.e. an element of $U(1)$. □

PROVED

Theorem 3.6 (The twisted algebra as a matrix algebra, non-canonically)**.** *$A_{\psi,\beta}(V) \cong M_q(\mathbb{C})$, by an isomorphism that depends on a choice of Lagrangian line and character (L,χ) (Definition 3.2) and a choice of basis of $M_{L,\chi}$, and is **not** canonical.*

Scope. This theorem asserts existence of *an* isomorphism to $M_q(\mathbb{C})$ and the dependence of any such isomorphism on auxiliary choices; it does not assert, and Theorem 6.3 of §6 shows precisely how much of this data can be removed, that any two such isomorphisms agree, nor that a canonical one exists.

Provenance. ids: D4, D8, WH-ALG, WH-SVN, WH-ALG-MAT; proved in: theory/wh-kappa.md §5 *step* ⟨1⟩2; tested by: theory/checks/wh_kappa_check.py (C5); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. This is exactly the isomorphism $\pi : A_{\psi,\beta}(V) \xrightarrow{\sim} \mathrm{End}_{\mathbb{C}}(M_0)$ constructed in the proof of Theorem 3.5(b); choosing the basis $\{e_y\}_{y\in\kappa}$ of M_0 identifies $\mathrm{End}_{\mathbb{C}}(M_0)$ with $M_q(\mathbb{C})$. The construction visibly used a Lagrangian line L_0, a character χ_0 of A_{L_0}, and the basis $\{e_y\}$ of the resulting module; a different line, character, or basis gives a different isomorphism, related to this one by an inner automorphism of $M_q(\mathbb{C})$ (Theorem 3.5(c)) composed with the change of basis, and no choice among these is distinguished by the data (κ, ψ, β) alone. $\square$

4 The polarizing cocycle as a second datum: the characteristic-two dichotomy

This section proves the increment's central structural fact. The quantum system attached to a finite field κ is built not from the field alone, and not from the field together with a single choice of additive character ψ: it is built from *three* data — κ, ψ, and a polarizing cocycle β, a choice inside the torsor $\mathrm{Adm}(\omega)$ of Definition 2.2 that is no more optional, and no more a matter of convention, than ψ itself. What this section shows is exactly how much that third choice matters. At odd characteristic it turns out not to matter at all: a canonical representative exists inside $\mathrm{Adm}(\omega)$, and every other admissible cocycle produces an isomorphic Heisenberg group by an isomorphism fixing the centre. At characteristic two no canonical representative exists, and the choice is not immaterial: the torsor $\mathrm{Adm}(\omega)$ splits into exactly two orbits under the natural symmetry group $SL_2(\kappa)$, distinguished by the Arf invariant of an associated quadratic form, and the two orbits produce two genuinely non-isomorphic Heisenberg groups — a fact detectable by an operator identity inside the quantum system itself.

This is not a defect in the construction, and not a gap for a better convention to close. It is a theorem about what the construction actually depends on, carrying an intrinsic, computable invariant that says exactly when two choices of cocycle agree and when they do not. A sharp result of this shape — naming precisely the choice at stake and exhibiting the invariant that classifies it — is a positive structural statement about the definition, not a negative one, and it is this labbook's principal new content.

4.1 The torsor of admissible polarizing cocycles, restated

Definition 4.1 (Admissible polarizing cocycles, restated in full)**.** For the symplectic object $(V(\kappa), \omega)$ of Definition 2.1,

$$\mathrm{Adm}(\omega) := \left\{ \beta : V(\kappa) \times V(\kappa) \to \kappa \;\middle|\; \beta \text{ is } \kappa\text{-bilinear and } \beta - \beta^T = \omega \right\}.$$

A *polarizing cocycle* is a choice of element $\beta \in \mathrm{Adm}(\omega)$, and it is a **datum** of the construction, on the same footing as the phase datum ψ — not a convention fixed once and for all. The *reference cocycle* is $\beta_0\big((a, b), (a', b')\big) := ab'$, a label for a point of the torsor below, not a canonical point of it. Write $\mathrm{Sym}(V)$ for the set of symmetric κ-bilinear forms on $V(\kappa)$.

Scope. This restates Definition 2.2 verbatim; nothing new is stipulated here. What is new in this section is everything that is *proved* about $\mathrm{Adm}(\omega)$: its torsor structure (Theorem 4.2), the immateriality of the choice at odd p (Theorem 4.3), and its genuine two-way split at $p = 2$ (Theorems 4.4–4.8).

Provenance. ids: D1, D2; proved in: —; tested by: —; reviewed in: —.

PROVED

Theorem 4.2 (The admissible cocycles form a torsor of size q^3)**.** $\mathrm{Adm}(\omega)$ *is a torsor under* $\mathrm{Sym}(V)$ *acting by* $\beta \mapsto \beta + s$, $s \in \mathrm{Sym}(V)$*; consequently* $|\mathrm{Adm}(\omega)| = q^3$.

Scope. This is purely a statement about the set $\mathrm{Adm}(\omega)$ and its free transitive action; it does not yet distinguish any point of the torsor, does not use the characteristic in any way, and does not address isomorphism of the associated Heisenberg groups — those are the later theorems of this section.

Provenance. ids: D1, D2, WH-BETA-a; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩1; tested by: theory/checks/beta_census.py (B1); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. If $\beta, \beta' \in \mathrm{Adm}(\omega)$, put $s := \beta' - \beta$. Then $s - s^T = (\beta' - \beta^{T\prime}) - (\beta - \beta^T) = \omega - \omega = 0$, i.e. $s \in \mathrm{Sym}(V)$. Conversely, for $s \in \mathrm{Sym}(V)$, $(\beta + s) - (\beta + s)^T = (\beta - \beta^T) + (s - s^T) = \omega + 0 = \omega$, so $\beta + s \in \mathrm{Adm}(\omega)$. The action $\beta \mapsto \beta + s$ is therefore well defined, free (if $\beta + s = \beta + s'$ then $s = s'$), and transitive (any $\beta' \in \mathrm{Adm}(\omega)$ equals $\beta + (\beta' - \beta)$ for the $s = \beta' - \beta \in \mathrm{Sym}(V)$ just exhibited). A symmetric κ-bilinear form on $V(\kappa) = \kappa e \oplus \kappa f$ is determined freely by the three values $s(e,e), s(e,f), s(f,f) \in \kappa$ (symmetry forces $s(f,e) = s(e,f)$, and bilinearity determines everything else), so $|\mathrm{Sym}(V)| = q^3$; since $\mathrm{Adm}(\omega)$ is nonempty ($\beta_0 \in \mathrm{Adm}(\omega)$) and a torsor under $\mathrm{Sym}(V)$, it has the same cardinality, q^3. □

4.2 Odd characteristic: a canonical representative

PROVED†

Theorem 4.3 (Immateriality of the polarizing cocycle at odd characteristic)**.** *[p **odd.**]* $\mathrm{Adm}(\omega)$ *contains exactly one antisymmetric member,* $\omega/2$*, and it is* $SL_2(\kappa)$*-invariant. For every* $s \in \mathrm{Sym}(V)$*, the map*

$$\varphi_s(t, v) := \Big(t + \tfrac{1}{2}s(v,v),\ v\Big)$$

is a group isomorphism $H_\beta(\kappa) \to H_{\beta+s}(\kappa)$ *fixing the centre* $\kappa \times 0$ *pointwise, for* every $\beta \in \mathrm{Adm}(\omega)$*. Hence at odd characteristic the choice of polarizing cocycle is immaterial: every admissible cocycle produces a Heisenberg group isomorphic, by an explicit isomorphism fixing the centre, to every other, and* $\omega/2$ *is a canonical basepoint of the torsor.*

Scope. This theorem is restricted to odd p by hypothesis, not by omission: at $p = 2$, division by 2 is division by zero, $\omega/2$ has no referent, and Theorem 4.4 shows the phenomenon this theorem describes simply does not occur. Nothing here compares $H_\beta(\kappa)$ for different β as algebras or as sources of quantum systems beyond the group isomorphism exhibited; the algebra-level statement is Theorem 4.10.

Provenance. ids: D2, D5, WH-BETA-a, WH-BETA-b; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩2; tested by: theory/checks/beta_census.py (B2,B3,B4); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Existence and uniqueness of the antisymmetric point. Since p is odd, $2 \in \kappa^\times$, and $\omega/2$ is κ-bilinear with $(\omega/2) - (\omega/2)^T = \omega/2 + \omega/2 = \omega$ (using $\omega^T = -\omega$, i.e. antisymmetry of ω, itself immediate from Theorem 2.6(b): $\omega(v,v') + \omega(v',v) = \omega(v+v', v+v') - \omega(v,v) - \omega(v',v') = 0$), so $\omega/2 \in \mathrm{Adm}(\omega)$ and is manifestly antisymmetric. If $\beta, \beta' \in \mathrm{Adm}(\omega)$ are both antisymmetric, then $s := \beta' - \beta \in \mathrm{Sym}(V)$ (Theorem 4.2) is also antisymmetric, so $2s = 0$, so $s = 0$ (p odd): the antisymmetric point is unique. *Invariance.* $SL_2(\kappa)$ preserves ω (Theorem 2.6(e)), hence preserves the scalar multiple $\omega/2$.

The isomorphism φ_s*.* Put $f(v) := s(v,v)/2$. By bilinearity and symmetry of s, $s(v+v', v+v') = s(v,v) + 2s(v,v') + s(v',v')$, so $f(v+v') = f(v) + f(v') + s(v,v')$. Unwinding the group law of Definition 2.5 for $H_\beta(\kappa)$ and $H_{\beta+s}(\kappa)$,

$$\varphi_s\big((t,v)(t',v')\big) = \varphi_s\big(t + t' + \beta(v,v'),\ v + v'\big) = \big(t + t' + \beta(v,v') + f(v+v'),\ v + v'\big),$$

while

$$\varphi_s(t,v)\,\varphi_s(t',v') = \big(t + f(v), v\big)\big(t' + f(v'), v'\big) = \big(t + t' + f(v) + f(v') + (\beta{+}s)(v,v'),\ v + v'\big),$$

computed inside $H_{\beta+s}(\kappa)$. The two right-hand sides agree precisely because $f(v+v') = f(v) + f(v') + s(v,v')$. So φ_s is a group homomorphism; it is a bijection (identity on the second coordinate, an affine bijection $t \mapsto t + f(v)$ on the first, for each fixed v), hence an isomorphism, and $\varphi_s(t,0) = (t + f(0), 0) = (t,0)$ since $f(0) = s(0,0)/2 = 0$: the centre is fixed pointwise. $\square$

4.3 Characteristic two: no symmetric distinguished point

PROVED †

Theorem 4.4 (Absence of a symmetric distinguished cocycle at characteristic two)**.** *[$p = 2$.]* $\mathrm{Adm}(\omega)$ *contains no antisymmetric member; the criterion that singles out a canonical basepoint at odd characteristic (Theorem 4.3) distinguishes nothing here.*

Scope. This theorem only rules out *this particular* route to a canonical choice (antisymmetry). It does not by itself assert that no other route to a canonical choice exists, nor that the resulting Heisenberg groups actually differ — those stronger, positive facts are Theorems 4.5–4.7 below.

Provenance. ids: D2, WH-BETA-a, WH-BETA-c; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩3; tested by: theory/checks/beta_census.py (B2); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. At $p = 2$, $-1 = 1$, so "antisymmetric" ($\beta = -\beta^T$) coincides with "symmetric" ($\beta = \beta^T$), i.e. $\beta - \beta^T = 0$. But every $\beta \in \mathrm{Adm}(\omega)$ has $\beta - \beta^T = \omega$ (Definition 4.1), and $\omega\big((1,0),(0,1)\big) = 1 \neq 0$ (Theorem 2.6(c)), so $\omega \neq 0$: no member of $\mathrm{Adm}(\omega)$ can be symmetric, hence none can be antisymmetric either. $\square$

4.4 The quadratic form of a cocycle, and the order profile of its Heisenberg group

PROVED †

Theorem 4.5 (The quadratic form of a polarizing cocycle, and the element-order profile of its Heisenberg group)**.** *[$p = 2$.] Let $\beta \in \mathrm{Adm}(\omega)$, with $Q_\beta(v) := \beta(v,v)$ as in Definition 3.1. Then:*

(i) $Q_\beta(cv) = c^2 Q_\beta(v)$ for $c \in \kappa$, $v \in V(\kappa)$, and $Q_\beta(v+v') = Q_\beta(v) + Q_\beta(v') + \omega(v,v')$ for all v, v': Q_β is a κ-quadratic form on $V(\kappa)$ with polar form ω;

(ii) in $H_\beta(\kappa)$, $(t,v)^2 = \big(Q_\beta(v), 0\big)$ for every (t,v); consequently, writing $n_0 := |Q_\beta^{-1}(0)|$ (which is always ≥ 1, since $Q_\beta(0) = 0$), the multiset of element orders of $H_\beta(\kappa)$ is

$$\{\, 1 : 1, \quad 2 : q\, n_0 - 1, \quad 4 : q^3 - q\, n_0 \,\};$$

(iii) hence $\mathrm{Adm}(\omega)$ meets at least two isomorphism classes of Heisenberg group: by direct enumeration at $q = 2$, 6 of the 8 admissible cocycles give the order profile $\{1{:}1, 2{:}5, 4{:}2\}$ ($n_0 = 3$) and the remaining 2 give $\{1{:}1, 2{:}1, 4{:}6\}$ ($n_0 = 1$), and these are different multisets, hence non-isomorphic groups.

Scope. Part (iii) establishes only that *at least* two isomorphism classes occur, by exhibiting two different order profiles at the single value $q = 2$; it does not claim that exactly two classes occur for every q, nor identify the classes by name (this lane computes order profiles, not group names — that two of the eight cocycles at $q = 2$ happen to give the quaternion group and the other six the dihedral group of order 8 is a fact the orchestrator's independent census records, but it is not part of what this row proves). The full classification, valid at every $q = 2^m$ and carrying an intrinsic invariant, is Theorem 4.7 below, which depends on this one.

Provenance. ids: D2, D6, WH-BETA-a, WH-BETA-d; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1\rangle 4$; tested by: theory/checks/beta_census.py (B3,B6); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (i) By κ-bilinearity, $Q_\beta(cv) = \beta(cv, cv) = c^2\beta(v, v) = c^2 Q_\beta(v)$. Expanding $Q_\beta(v + v') = \beta(v + v', v + v') = \beta(v, v) + \beta(v, v') + \beta(v', v) + \beta(v', v')$; at $p = 2$, addition and subtraction coincide, so $\beta(v, v')+\beta(v', v) = \beta(v, v')-\beta(v', v) = \omega(v, v')$ (Definition 4.1), giving $Q_\beta(v+v') = Q_\beta(v) + Q_\beta(v') + \omega(v, v')$.

(ii) In $H_\beta(\kappa)$, $(t, v)(t, v) = \big(2t + \beta(v, v),\ 2v\big) = \big(Q_\beta(v), 0\big)$ at $p = 2$, since $2t = 0$ and $2v = 0$. If $(t, v) = (0, 0)$ this is the identity (order 1). If $(t, v) \neq (0, 0)$ and $Q_\beta(v) = 0$, then $(t, v)^2 = (0, 0)$, so (t, v) has order exactly 2; the number of such elements is $q\,n_0 - 1$ (for each of the n_0 vectors v with $Q_\beta(v) = 0$, all q values of t occur, and the pair $(t, v) = (0, 0)$ is excluded). If $Q_\beta(v) \neq 0$, then $(t, v)^2 = (Q_\beta(v), 0) \neq (0, 0)$, but $(t, v)^4 = \big((t, v)^2\big)^2 = \big(2Q_\beta(v), 0\big) = (0, 0)$, so (t, v) has order exactly 4; there are $q(q^2 - n_0) = q^3 - qn_0$ such elements ($q^2 - n_0$ choices of v, q choices of t each). These three counts sum to $1 + (qn_0 - 1) + (q^3 - qn_0) = q^3 = |H_\beta(\kappa)|$, as they must.

(iii) The order profile just derived is a group-isomorphism invariant (an isomorphism of groups preserves the multiset of element orders), so different profiles certify non-isomorphic groups. At $q = 2$, exhaustive enumeration over all 8 members of $\mathrm{Adm}(\omega)$ (`theory/checks/beta_census.py`, gates B3 and B6) computes n_0 for each and finds exactly the two values $n_0 = 3$ (for 6 of the 8 cocycles, giving profile $\{1{:}1, 2{:}5, 4{:}2\}$ by the formula of (ii)) and $n_0 = 1$ (for the remaining 2, giving $\{1{:}1, 2{:}1, 4{:}6\}$); these are distinct multisets, exhibiting two isomorphism classes. □

4.5 The Arf classification

Definition 4.6 (The Artin–Schreier map, and the type of a polarizing cocycle)**.** For κ of characteristic 2, define $\wp : \kappa \to \kappa$ by $\wp(x) := x^2 + x$. The *type* of $\beta \in \mathrm{Adm}(\omega)$ is

$$\mathrm{Arf}(Q_\beta) := Q_\beta(e)\,Q_\beta(f) \ \in\ \kappa/\wp(\kappa),$$

computed in any symplectic κ-basis (e, f) of $V(\kappa)$, i.e. a basis with $\omega(e, f) = 1$.

Scope. This is a stipulation, meaningful only at $p = 2$. That the value $\mathrm{Arf}(Q_\beta)$ does not depend on the choice of symplectic basis, that $\kappa/\wp(\kappa)$ has exactly two elements, and that both values actually occur among the members of $\mathrm{Adm}(\omega)$, are not part of the definition; they are exactly the content of Theorem 4.7 below, whose provenance names the argument DAG's current row for this material (the single source's own cross-reference for this content predates a later renaming in the DAG).

Provenance. ids: D10; proved in: —; tested by: —; reviewed in: —.

SKETCH

Theorem 4.7 (Classification of characteristic-two polarizing cocycles by the Arf invariant)**.** *[$p = 2$.] Let $\beta \in \mathrm{Adm}(\omega)$.*

(i) $\wp$ is additive with $\ker\wp = \{0, 1\} = \mathbb{F}_2$, so $\kappa/\wp(\kappa)$ has exactly two elements;

(ii) Q_β has a nonzero zero (some $v \neq 0$ with $Q_\beta(v) = 0$) if and only if $\mathrm{Arf}(Q_\beta) = 0$; in that case $n_0 = |Q_\beta^{-1}(0)| = 2q-1$, and otherwise $n_0 = 1$. Consequently $\mathrm{Arf}(Q_\beta)$ does not depend on the choice of symplectic basis used to compute it, and is invariant under $SL_2(\kappa)$;

(iii) call β hyperbolic *if $\mathrm{Arf}(Q_\beta) = 0$ and* anisotropic *otherwise. Two members of $\mathrm{Adm}(\omega)$ are carried to each other by an $SL_2(\kappa)$-isometry of their quadratic forms if and only if they*

have the same type; both types occur, with $q(q+1)/2$ hyperbolic and $q(q-1)/2$ anisotropic quadratic forms compatible with ω, and stabilizer orders $|O(Q_\beta) \cap SL_2(\kappa)| = 2(q-1)$ (hyperbolic) and $2(q+1)$ (anisotropic);

(iv) *$H_\beta(\kappa) \cong H_{\beta'}(\kappa)$ if and only if $\mathrm{Arf}(Q_\beta) = \mathrm{Arf}(Q_{\beta'})$. So $\mathrm{Adm}(\omega)$ meets* exactly *two isomorphism classes of Heisenberg group, indexed by* Arf.

Scope. This theorem is marked SKETCH for a procedural reason recorded in `theory/verdicts/wh-kappa-adjudication.md`: it was written in the repair wave that followed this increment's single hostile-critic round and has not itself been through one, per this campaign's capped review loop (`PRD.md`) — not because a specific gap is known. It does not extend beyond $p = 2$ (Theorem 4.3 covers odd p by an entirely different, and there, complete, argument), and it does not address whether the type can be chosen compatibly along field embeddings — that question is recorded as open in §8.

Provenance. ids: D6, D10, WH-BETA-d, WH-BETA-TYPE; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1\rangle 4, \langle 1\rangle 5$; tested by: theory/checks/beta_census.py (B5,B6,B7); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (i) $\wp(x+y) - \wp(x) - \wp(y) = (x+y)^2 - x^2 - y^2 = 2xy = 0$ at $p = 2$, so $\wp$ is additive (a group homomorphism $(\kappa, +) \to (\kappa, +)$). Its kernel consists of the roots of $x^2 + x = 0$, a degree-2 polynomial, so $|\ker \wp| \le 2$; both $x = 0$ and $x = 1$ are roots, so $\ker \wp = \{0, 1\} = \mathbb{F}_2$ exactly. By rank–nullity for the $\mathbb{F}_2$-linear map $\wp$ on the m-dimensional $\mathbb{F}_2$-space κ, $|\wp(\kappa)| = q/2$, so $|\kappa/\wp(\kappa)| = 2$.

(ii) In a symplectic basis (e, f), write $v = ae + bf$; by Theorem 4.5(i), $Q_\beta(ae + bf) = \alpha a^2 + ab + \delta b^2$ with $\alpha := Q_\beta(e)$, $\delta := Q_\beta(f)$ (using $\omega(e, f) = 1$ for the cross term). If $b = 0$: $Q_\beta = \alpha a^2$ vanishes for some $a \neq 0$ iff $\alpha = 0$, and then $\mathrm{Arf} = \alpha\delta = 0 \in \wp(\kappa)$. If $b \neq 0$, rescale to $b = 1$: $\alpha a^2 + a + \delta = 0$; multiplying by α (when $\alpha \neq 0$) gives $(\alpha a)^2 + (\alpha a) = \alpha\delta$, i.e. $\wp(\alpha a) = \alpha\delta$, solvable for a iff $\alpha\delta \in \wp(\kappa)$; when $\alpha = 0$ the equation $a + \delta = 0$ always has the solution $a = \delta$. So in every case, Q_β has a nonzero zero iff $\alpha\delta \in \wp(\kappa)$, i.e. iff $\mathrm{Arf}(Q_\beta) = 0 \in \kappa/\wp(\kappa)$. When $\mathrm{Arf}(Q_\beta) = 0$, the zero set of Q_β is a union of κ-lines (by homogeneity $Q_\beta(cv) = c^2 Q_\beta(v)$ of Theorem 4.5(i)); the computation above produces exactly two such lines (the two roots of $\wp(\alpha a) = \alpha\delta$ when $\alpha \neq 0$, or the lines $b = 0$ and $a = \delta$ when $\alpha = 0$), giving $n_0 = 2(q-1)+1 = 2q-1$ nonzero-vector zeros plus the origin; when $\mathrm{Arf}(Q_\beta) \neq 0$, $n_0 = 1$ (only the origin). Since $|\kappa/\wp(\kappa)| = 2$ by (i), the class $\alpha\delta$ mod $\wp(\kappa)$ is itself a basis-free condition — "Q_β is isotropic" — so $\mathrm{Arf}(Q_\beta)$ is independent of the symplectic basis chosen to compute it, and $SL_2(\kappa)$-invariant since an $SL_2(\kappa)$-image of a symplectic basis is again symplectic (Theorem 2.6(e)).

(iii) *Normal forms.* If $\mathrm{Arf}(Q_\beta) = 0$, pick $e \neq 0$ with $Q_\beta(e) = 0$ (by (ii)) and any f with $\omega(e, f) = 1$ (possible by nondegeneracy of ω); replacing f by $f + Q_\beta(f)e$ changes $Q_\beta(f)$ to $Q_\beta(f) + Q_\beta(f)^2 Q_\beta(e) + Q_\beta(f)\omega(f, e)$, which simplifies (using $Q_\beta(e) = 0$ and $\omega(f, e) = -\omega(e, f) = 1$ at $p = 2$) to 0, giving $Q_\beta(ae + bf) = ab$: the hyperbolic normal form. If $\mathrm{Arf}(Q_\beta) \neq 0$ then $Q_\beta(v) \neq 0$ for every $v \neq 0$ (by (ii)); pick any $e \neq 0$, rescale by $\lambda = Q_\beta(e)^{-1/2}$ (Frobenius is bijective on κ, so square roots exist and are unique) to arrange $Q_\beta(e) = 1$, then choose f with $\omega(e, f) = 1$; replacing f by $f + ae$ changes $Q_\beta(f)$ by $\wp(a)$ (direct computation from Theorem 4.5(i)), so $Q_\beta(f)$ can be moved to any chosen representative of its class modulo $\wp(\kappa)$. Both normalizations send a symplectic basis to a symplectic basis, hence are realized by elements of $SL_2(\kappa)$ (Theorem 2.6(e)): two quadratic forms of the same type are $SL_2(\kappa)$-isometric, and by (ii) two of different type cannot be (they have different n_0, an isometry invariant). *Counting.* Over a fixed symplectic basis, the κ-quadratic forms with polar form ω are exactly the q^2 maps $\alpha a^2 + ab + \delta b^2$ (parametrized freely by $(\alpha, \delta) \in \kappa^2$, by Theorem 4.5(i)); among these, $\#\{\alpha\delta \in \wp(\kappa)\} = q + (q-1)(q/2) = q(q+1)/2$ are hyperbolic (the case $\alpha = 0$ contributes q choices of δ, and each of the $q - 1$ nonzero α pairs with the $q/2$ elements of $\wp(\kappa)$) and the remaining $q(q-1)/2$ are anisotropic — both classes nonempty for every q. By orbit–stabilizer inside $SL_2(\kappa)$, of order $q(q^2-1)$, acting transitively within each type: stabilizer orders $q(q^2-1)/(q(q+1)/2) = 2(q-1)$ (hyperbolic) and $q(q^2-1)/(q(q-1)/2) = 2(q+1)$ (anisotropic). *Groups.* If $\mathrm{Arf}(Q_\beta) = \mathrm{Arf}(Q_{\beta'})$, the normal-form argument gives $Q_{\beta'} = Q_\beta \circ g$ for some $g \in SL_2(\kappa)$; then $r(v,v') := \beta(gv,gv') - \beta'(v,v')$ is symmetric with $r(v,v) = 0$ for all v (both sides equal $Q_{\beta'}(v)$ by construction of g), so, fixing an $\mathbb{F}_2$-basis b_i of $V(\kappa)$ and writing $v = \sum c_i b_i$, $h(v) := \sum_{i<j} c_i c_j\, r(b_i, b_j)$ satisfies $h(v+v') - h(v) - h(v') = r(v,v')$ (a direct expansion using r symmetric and bilinear), and $(t,v) \mapsto (t + h(v), gv)$ is then a group isomorphism $H_{\beta'}(\kappa) \to H_\beta(\kappa)$ by the same computation as in the proof of Theorem 4.3. If instead $\mathrm{Arf}(Q_\beta) \neq \mathrm{Arf}(Q_{\beta'})$, the two groups have different values of n_0 (by (ii)), hence different element-order profiles (Theorem 4.5(ii)), so no group isomorphism of any kind exists between them. □

Remark. Direct enumeration (`theory/checks/beta_census.py`, gates B5–B7) confirms every clause above at $q = 2,4,8$: Arf is constant on $SL_2(\kappa)$-orbits and takes exactly the two values across all q^3 members of $\mathrm{Adm}(\omega)$; $|O(Q_\beta) \cap SL_2(\kappa)| = 2, 6, 14$ (hyperbolic) and $6, 10, 18$ (anisotropic), matching $2(q \mp 1)$ exactly; the element-order profiles come in exactly two sizes, $(6,2)$, $(40,24)$, $(288,224)$, at $q = 2,4,8$ respectively; and the isomorphism constructed in the proof of (iii) is verified on all pairs within the hyperbolic class.

4.6 The type, detected inside the quantum system

SKETCH

Theorem 4.8 (The Arf type is visible inside the quantum system)**.** *[$p = 2$.] Let $\beta \in \mathrm{Adm}(\omega)$ and ψ a phase datum. Then, inside $A_{\psi,\beta}(V)$:*

$$W_\beta(v)^2 = \psi\big(Q_\beta(v)\big) \cdot 1 \qquad \textit{for every } v \in V(\kappa),$$

and

$$q^{-1} \sum_{v \in V(\kappa)} W_\beta(v)^2 \;=\; \varepsilon \cdot 1, \qquad \varepsilon = \begin{cases} +1 & \beta \textit{ hyperbolic } (\mathrm{Arf}(Q_\beta) = 0), \\ -1 & \beta \textit{ anisotropic.} \end{cases}$$

Equivalently, identifying $\kappa/\wp(\kappa) \cong \mathbb{F}_2$ via the trace $\mathrm{Tr}_{\kappa/\mathbb{F}_2}$ of Definition 2.3, $\varepsilon = (-1)^{\mathrm{Tr}(\mathrm{Arf}(Q_\beta))}$.

Scope. The identity above is an equation inside the algebra $A_{\psi,\beta}(V)$ itself, so it exhibits $\mathrm{Arf}(Q_\beta)$ as an invariant not merely of the abstract group $H_\beta(\kappa)$ but of the level-μ_p Weyl frame $\mathcal{F}^{(p)}_{\psi,\beta}$ of Definition 4.9 below. As with Theorem 4.7, on which it depends, the SKETCH label reflects only that this material postdates the increment's single hostile-critic round, per `theory/verdicts/wh-kappa-adjudication.md`.

Provenance. ids: D3, D4, D6, D10, WH-BETA-EPS; proved in: theory/wh-kappa-choice.md §6 *step* ⟨1⟩6; tested by: theory/checks/beta_census.py (B6,B8); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. The square identity. By Definition 2.4 and Definition 3.1, $W_\beta(v)^2 = \psi(\beta(v,v))W_\beta(2v) = \psi(Q_\beta(v))W_\beta(0) = \psi(Q_\beta(v)) \cdot 1$, using $2v = 0$ at $p = 2$.

The hyperbolic value. For the hyperbolic normal form $Q(a,b) = ab$ (Theorem 4.7(iii)), $\sum_{a,b\in\kappa} \psi(ab) = \sum_a \Big(\sum_b \psi(ab)\Big)$; for $a \neq 0$, $b \mapsto \psi(ab)$ is a nontrivial character of κ, so the inner sum vanishes by Fact 3.1; for $a = 0$ the inner sum is q. So the double sum is q, giving $\varepsilon = +1$ for this normal form. Since $\sum_v \psi(Q_\beta(v))$ is invariant under precomposing Q_β with any bijection of $V(\kappa)$, in particular under the $SL_2(\kappa)$-isometries of Theorem 4.7(iii), $\varepsilon = +1$ for *every* hyperbolic β, not only the normal form.

The anisotropic value. Sum $\sum_v \psi(Q(v))$ over *all* q^2 quadratic forms $Q(a,b) = \alpha a^2 + ab + \delta b^2$ compatible with ω (Theorem 4.7(iii)) and exchange the order of summation:

$$\sum_{\alpha,\delta\in\kappa} \sum_{a,b\in\kappa} \psi(\alpha a^2 + ab + \delta b^2) = \sum_{a,b\in\kappa} \psi(ab)\Big(\sum_{\alpha} \psi(\alpha a^2)\Big)\Big(\sum_{\delta} \psi(\delta b^2)\Big).$$

By Fact 3.1, $\sum_\alpha \psi(\alpha a^2) = q$ if $a = 0$ (else the character $\alpha \mapsto \psi(\alpha a^2)$ is nontrivial, since squaring is injective on κ, and the sum vanishes); likewise for the δ-sum. So only the term $a = b = 0$ survives, and the total is $q^2 \cdot \psi(0) = q^2$. By Theorem 4.7(iii) there are $q(q+1)/2$ hyperbolic forms, each contributing q (just shown), and $q(q-1)/2$ anisotropic forms, each contributing the same value S (by isometry-invariance, exactly as for the hyperbolic case). So $\frac{q(q+1)}{2} \cdot q + \frac{q(q-1)}{2} \cdot S = q^2$, giving $S = \dfrac{q^2 - \frac{q^2(q+1)}{2}}{\frac{q(q-1)}{2}} = \dfrac{q^2\big(1 - \frac{q+1}{2}\big)}{\frac{q(q-1)}{2}} = \dfrac{-q^2(q-1)/2}{q(q-1)/2} = -q$. So $\varepsilon = -1$ on the anisotropic class.

Reading ε through the trace. $\mathrm{Tr}_{\kappa/\mathbb{F}_2}$ is additive and $\mathbb{F}_2$-valued; $\mathrm{Tr}(\wp(x)) = \mathrm{Tr}(x^2) + \mathrm{Tr}(x) = \mathrm{Tr}(x) + \mathrm{Tr}(x) = 0$ (using $\mathrm{Tr}(x^2) = \mathrm{Tr}(x)$, since Frobenius permutes the terms of the trace sum, and $2\,\mathrm{Tr}(x) = 0$ at $p = 2$), so Tr kills $\wp(\kappa)$; it is nonzero (e.g. on a normal-basis generator), hence onto $\mathbb{F}_2$, and $\dim_{\mathbb{F}_2} \wp(\kappa) = m - 1 = \dim_{\mathbb{F}_2} \ker \mathrm{Tr}$ by the rank computation of Theorem 4.7(i), so $\wp(\kappa) = \ker \mathrm{Tr}$ exactly: Tr descends to an isomorphism $\kappa/\wp(\kappa) \xrightarrow{\sim} \mathbb{F}_2$. Under this identification the hyperbolic class (Arf $= 0$) corresponds to $\mathrm{Tr} = 0$ and $\varepsilon = +1 = (-1)^0$, and the anisotropic class to $\mathrm{Tr} = 1$ and $\varepsilon = -1 = (-1)^1$. □

4.7 The level statement: what the algebra forgets, and what it does not

Definition 4.9 (The level-μ_p frame)**.**

$$\mathcal{F}^{(p)}_{\psi,\beta} := \big\{\, \zeta^j\, W_\beta(v) \ :\ j \in \mathbb{Z}/p,\ v \in V(\kappa) \,\big\} \ \subset\ A_{\psi,\beta}(V)^\times,$$

the finite group of Weyl unitaries with phases in $\mu_p = \psi(\kappa)$ (Fact 3.1), for ζ a primitive p-th root of unity as in Definition 2.3.

Scope. $\mathcal{F}^{(p)}_{\psi,\beta}$ is recorded separately from the Weyl frame $\mathcal{F}_\beta := \{\mathbb{C}^\times W_\beta(v) : v \in V(\kappa)\} \subset A_{\psi,\beta}(V)$ (the set of $\mathbb{C}^\times$-lines through Weyl operators; studied in §5 below) because the two behave differently under a change of β, exactly as Theorem 4.10 makes precise: the pair $(A_{\psi,\beta}(V), \mathcal{F}_\beta)$ turns out not to depend on β at all, while $\mathcal{F}^{(p)}_{\psi,\beta}$ — equivalently the isomorphism class of $H_\beta(\kappa)$ itself — does, and only at $p = 2$.

Provenance. ids: D11; proved in: —; tested by: —; reviewed in: —.

SKETCH

Theorem 4.10 (The algebra and its Weyl frame do not depend on β; the level-μ_p frame does, at $p = 2$)**.** *For all $\beta, \beta' \in \mathrm{Adm}(\omega)$ there is an algebra isomorphism $A_{\psi,\beta}(V) \to A_{\psi,\beta'}(V)$ of the form $W_\beta(v) \mapsto \mu(v) W_{\beta'}(v)$ for some $\mu : V(\kappa) \to \mathbb{C}^\times$ —* a frame isomorphism, *carrying $\mathcal{F}_\beta$ to $\mathcal{F}_{\beta'}$ — and this holds* ***at every characteristic****. Moreover:*

- *μ can be chosen μ_p-valued if and only if $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$;*
- *[p = 2] μ can always be chosen μ_4-valued, regardless of type.*

Hence the pair $\big(A_{\psi,\beta}(V), \mathcal{F}_\beta\big)$ does not depend on β at all, while the level-μ_p frame $\mathcal{F}^{(p)}_{\psi,\beta}$ — equivalently the isomorphism class of $H_\beta(\kappa)$ (Theorem 4.7) — does depend on β, and does so only at $p = 2$.

Scope. This theorem does not contradict Theorems 4.4–4.8: it is a statement about the algebra $A_{\psi,\beta}(V)$ and its $\mathbb{C}^\times$-frame $\mathcal{F}_\beta$, which turn out to be β-independent, while those theorems are about the finite group $H_\beta(\kappa)$ (equivalently, the level-μ_p frame), which is not. The distinction between the two frames is exactly what makes both statements true simultaneously. This row is SKETCH for the same procedural reason as Theorems 4.7–4.8, and the explicit μ_2/μ_4-valued formulas at $p = 2$ are presented at the level of detail the source shard gives them, verified computationally rather than re-derived symbol-by-symbol here.

Provenance. ids: D2, D4, D11, WH-BETA-a, WH-BETA-d, WH-BETA-LEVEL; proved in: theory/wh-kappa-choice.md §6 *step* $\langle 1 \rangle 7$; tested by: theory/checks/frame_mu.py (M1,M2,M3,M4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Put $s := \beta' - \beta \in \mathrm{Sym}(V)$ (Theorem 4.2). Direct expansion of the multiplication law of Definition 2.4 shows $v \mapsto \mu(v)W_{\beta'}(v)$ is an algebra map exactly when

$$\frac{\mu(v)\mu(v')}{\mu(v+v')} = \psi\big(-s(v,v')\big) \qquad (\dagger)$$

for all v, v', and any such map automatically carries $\mathcal{F}_\beta$ to $\mathcal{F}_{\beta'}$, since it sends each line $\mathbb{C}^\times W_\beta(v)$ to the line $\mathbb{C}^\times W_{\beta'}(v)$. The right-hand side $\psi \circ (-s)$ is a symmetric 2-cocycle on $(V(\kappa), +)$, a group of exponent p (every element of the characteristic-p field κ, hence of $V(\kappa)$, has additive order dividing p), so Fact 3.1 supplies a solution μ of $(\dagger)$: **the frame isomorphism exists at every characteristic**, with no restriction on β, β'.

Necessity of the μ_p criterion. Normalize $\mu(0) = 1$ (forced by $(\dagger)$ at $v = v' = 0$, since $\mu(0)^2/\mu(0) = \mu(0) = \psi(0) = 1$). At $p = 2$, setting $v' = v$ in $(\dagger)$ gives $\mu(v)^2/\mu(0) = \mu(v)^2 = \psi(s(v,v)) = \psi\big(Q_{\beta'}(v) - Q_\beta(v)\big)$. So μ is μ_2-valued (i.e. $\mu(v)^2 = 1$ for every v) if and only if $\psi(Q_{\beta'}(v)) = \psi(Q_\beta(v))$ for every v, i.e. iff $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$; at odd p the analogous computation with $2 \in \kappa^\times$ gives the same criterion for μ_p-valuedness in general.

Sufficiency, $p = 2$. Fix an $\mathbb{F}_2$-basis $b_1, \dots, b_{2m}$ of $V(\kappa)$ and write $\psi(s(b_i, b_j)) = (-1)^{\sigma_{ij}}$, $\sigma_{ij} \in \{0, 1\}$, for the finitely many structure values of the symmetric form s on this basis (well defined since $\psi(\kappa) = \mu_2 = \{\pm 1\}$ at $p = 2$, Fact 3.1). If $\sigma_{ii} = 0$ for every i (the case $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$ just characterized), the $\mathbb{F}_2$-bilinear cochain $\ell(v) := \sum_{i<j} c_i c_j \sigma_{ij}$, for $v = \sum_i c_i b_i$, solves $(\dagger)$ with values in $\mu_2 = \{\pm 1\}$, by a direct check against the defining bilinear expansion of s. In general — without assuming $\sigma_{ii} = 0$ — the integer-valued cochain $m(v) := 2\sum_{i<j} c_i c_j \sigma_{ij} + \sum_i c_i \sigma_{ii} \pmod 4$ solves $(\dagger)$ with $\mu(v) := i^{m(v)} \in \mu_4$ ($i = \sqrt{-1}$): this is the explicit construction of `theory/wh-kappa-choice.md` §6, verified against all $|V(\kappa)|^2$ pairs (v, v') for every one of the q^3 members of $\mathrm{Adm}(\omega)$ at $q = 2, 4, 8$ by gates M2 and M3 of `theory/checks/frame_mu.py` (which finds μ_4-valued solutions always, and μ_2-valued solutions for exactly the q pairs (β, β') with $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$, matching the necessity criterion above exactly).

Sufficiency, odd p. $\mu(v) := \psi\big(s(v,v)/2\big)$ solves $(\dagger)$: since s is symmetric bilinear, $s(v,v) + s(v',v') - s(v+v', v+v') = -2s(v,v')$, so $\mu(v)\mu(v')/\mu(v+v') = \psi\big([s(v,v) + s(v',v') - s(v+v', v+v')]/2\big) = \psi(-s(v,v'))$, matching $(\dagger)$ exactly, with values entirely inside $\psi(\kappa) = \mu_p$: this reproves, independently of Theorem 4.3, that nothing leaves the level-μ_p frame once p is odd — a second, algebra-level confirmation that the choice of β is immaterial there.

Gate M1 of `theory/checks/frame_mu.py` additionally searches all 2^{q^2} candidate μ_2-valued phase functions exhaustively at $q = 2$ and confirms solutions exist exactly for the pairs the necessity criterion predicts; gate M4 checks the odd-p formula at $q = 3, 5, 9$. $\square$

5 Symmetry of the Weyl frame: the Weil representation and its splitting problem

Recollection. $\kappa = \mathbb{F}_q$, $q = p^m$; $V(\kappa) = \kappa \oplus \kappa$ with $\omega\big((a,b),(a',b')\big) = ab' - a'b$ (Definition 2.1); $SL_2(\kappa)$ is the group of κ-linear automorphisms of $V(\kappa)$ preserving ω (Theorem 2.6(e)); a polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ and phase datum ψ determine the algebra $A_{\psi,\beta}(V) = \bigoplus_v \mathbb{C} \cdot W_\beta(v)$ (Definitions 4.1, 2.3, 2.4), and $Q_\beta(v) := \beta(v,v)$ (Definition 3.1).

Provenance. ids: D1, D2, D3, D4, D6; proved in: —; tested by: —; reviewed in: —.

This section studies the symmetries of the Weyl frame itself: which permutations of the Weyl operators extend to algebra automorphisms, how large that symmetry group is, and whether $SL_2(\kappa)$ — the natural symmetry group of $(V(\kappa), \omega)$ — genuinely acts on the quantum system by honest algebra automorphisms (as opposed to merely projectively). The answer is uniform in one respect and characteristic-dependent in another: the group of frame automorphisms always surjects onto $SL_2(\kappa)$ with kernel the dual group $\hat{V}$, at every characteristic; but whether this extension *splits* — whether $SL_2(\kappa)$ itself, not just a projective shadow of it, acts by algebra automorphisms — is settled affirmatively at odd p and only partially resolved at $p = 2$, where it interacts with the Arf dichotomy of §4.

5.1 The Weyl frame and its two automorphism groups

Definition 5.1 (The Weyl frame, and its κ-linear automorphisms)**.** The *Weyl frame* is

$$\mathcal{F} := \big\{ \mathbb{C}^\times \cdot W_\beta(v) \ : \ v \in V(\kappa) \big\} \ \subset \ A_{\psi,\beta}(V),$$

the set of $\mathbb{C}^\times$-lines through Weyl operators. $\mathrm{Aut}_{\mathcal{F}}(A)$ is the group of $\mathbb{C}$-algebra automorphisms α of $A_{\psi,\beta}(V)$ with $\alpha(\mathcal{F}) = \mathcal{F}$. $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \subseteq \mathrm{Aut}_{\mathcal{F}}(A)$ is the subgroup consisting of those α whose induced permutation of $V(\kappa)$ (well defined since α permutes the lines of $\mathcal{F}$, which are indexed bijectively by $V(\kappa)$) is κ-linear.

Scope. The κ-linearity clause in the definition of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ is **imposed data**: nothing in $A_{\psi,\beta}(V)$ or $\mathcal{F}$ as built from $(V, +)$ and the cocycle $\psi \circ \beta$ forces an automorphism's induced permutation of $V(\kappa)$ to respect scalar multiplication by κ rather than merely by $\mathbb{F}_p$. Precisely how much symmetry this clause discards is Theorem 7.1 of §7; nothing here addresses that question.

Provenance. ids: D7; proved in: —; tested by: —; reviewed in: —.

5.2 Exactness of the frame-automorphism extension, uniformly in the characteristic

Fix $\beta \in \mathrm{Adm}(\omega)$. For $g \in SL_2(\kappa)$, write $s_g(v, v') := \beta(gv, gv') - \beta(v, v')$ and $Q_g(v) := s_g(v, v)$; note that $Q_g(v) = Q_\beta(gv) - Q_\beta(v)$.

PROVED

Theorem 5.2 (Exactness of the frame-automorphism extension)**.** *At every characteristic, the sequence*

$$1 \longrightarrow \hat{V} \longrightarrow \mathrm{Aut}^{\kappa}_{\mathcal{F}}(A_{\psi,\beta}(V)) \xrightarrow{\alpha \mapsto g} SL_2(\kappa) \longrightarrow 1$$

is exact, where $\hat{V} := \mathrm{Hom}\big((V(\kappa), +), \mathbb{C}^\times\big)$*; the kernel consists exactly of the conjugations* $\mathrm{Ad}_{W_\beta(u)}$*,* $u \in V(\kappa)$*.*

Scope. This theorem does not assert that the extension splits — that $SL_2(\kappa)$ itself embeds back into $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ as a subgroup — only that it is exact. Splitting is addressed separately, and with a genuinely different answer at odd and even characteristic, in Theorems 5.3 and 5.6 below. Nothing here concerns $\mathrm{Aut}_{\mathcal{F}}(A)$ without the imposed κ-linearity clause; that is Theorem 7.1.

Provenance. ids: D1, D4, D7, WH-ALG, WH-COMM, WH-WEIL-a; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1 \rangle 2, \langle 1 \rangle 3$; tested by: theory/checks/split24.py (X2); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. s_g *is symmetric bi-additive, and* $s_{gh}(v,v') = s_h(v,v') + s_g(hv,hv')$. Symmetry: $s_g(v,v') - s_g(v',v) = \omega(gv,gv') - \omega(v,v') = 0$, since $g \in SL_2(\kappa)$ preserves ω (Theorem 2.6(e)); the telescoping identity is immediate by expanding $\beta(ghv,ghv') - \beta(v,v')$ through the intermediate term $\beta(hv,hv')$. (For $\beta = \beta_0$ and $g = \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right)$, direct expansion of $\beta_0(gv,gv') = rt\,aa' + ru\,ab' + st\,a'b + su\,bb'$ gives the explicit formula $s_g(v,v') = rt\,aa' + st(ab'+a'b) + su\,bb'$, using $ru - st = \det g = 1$.)

Description of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$. Frame preservation gives $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ for some function $\lambda : V(\kappa) \to \mathbb{C}^\times$ and some permutation g of $V(\kappa)$, imposed to be κ-linear by definition of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$. Applying α to $W_\beta(v)W_\beta(v') = \psi(\beta(v,v'))W_\beta(v+v')$ and comparing basis elements forces g additive (automatic here, being κ-linear) and

$$\lambda(v)\lambda(v')\,\psi\big(s_g(v,v')\big) = \lambda(v+v') \tag{$*$}$$

for all v, v'. Applying α to the commutation relation of Theorem 2.7(c) and using $\omega(gv,gv') = \det(g)\,\omega(v,v')$ (Theorem 2.6(e)'s proof) gives $\psi\big((\det g - 1)\,\omega(v,v')\big) = 1$ for all v, v'; since ω is onto κ (Theorem 2.6(a),(c)), Fact 3.1 forces $\det g = 1$, i.e. $g \in SL_2(\kappa)$. So $\alpha \mapsto g$ is a well-defined map $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \to SL_2(\kappa)$, visibly a homomorphism (composition of automorphisms composes the induced permutations).

The kernel. If $g = \mathrm{id}$, ($*$) reads $\lambda(v)\lambda(v') = \lambda(v+v')$, i.e. $\lambda \in \hat{V}$, a group of order q^2 by Fact 3.1 ($U = V(\kappa)$, $d = 2m$). The map $u \mapsto \psi(\omega(u,\cdot))$, $V(\kappa) \to \hat{V}$, is injective (if $\psi(\omega(u,\cdot)) \equiv 1$ then $\omega(u,\cdot) \equiv 0$ by Fact 3.1, so $u = 0$ by nondegeneracy, Theorem 2.6(c)), hence bijective ($|V(\kappa)| = q^2 = |\hat{V}|$); so every kernel element has $\lambda = \psi(\omega(u,\cdot))$ for a unique $u \in V(\kappa)$, and by Theorem 2.7(d) this is exactly $\mathrm{Ad}_{W_\beta(u)}$.

Surjectivity, uniformly in p. For fixed $g \in SL_2(\kappa)$, $c(v,v') := \psi(s_g(v,v'))$ is a symmetric 2-cocycle on the exponent-p group $(V(\kappa),+)$ (symmetry of c from symmetry of s_g, just shown; the cocycle identity from bi-additivity of s_g), so Fact 3.1 supplies λ solving ($*$): every $g \in SL_2(\kappa)$ is hit. This argument is uniform in the characteristic; nothing distinguishes $p = 2$. □

Remark (Every frame automorphism is projectively unitary, at every characteristic). Fix a model (M,π) as in Theorem 3.5. Taking absolute values in ($*$) makes $v \mapsto |\lambda(v)|$ a homomorphism from the finite group $V(\kappa)$ to the torsion-free group $(\mathbb{R}_{>0},\times)$, hence trivial: every λ occurring in $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ is automatically unimodular. By Theorem 3.5(c),(d), every $\alpha \in \mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ is Ad_T for some T, unique up to a scalar; since α permutes the *unitary* Weyl operators (unimodular λ times unitary $\pi(W_\beta(gv))$), T may be chosen unitary and is then unique up to $U(1)$ (Theorem 3.5(d)). This gives an injective homomorphism $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \to PU(M)$: a projective unitary representation of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$, at every characteristic. Composed with a section $SL_2(\kappa) \to \mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ of Theorem 5.2's surjection — should one exist as an honest group homomorphism, i.e. should the extension split — this yields an honest (non-projective) representation of $SL_2(\kappa)$ by algebra automorphisms, and hence, via Ad, a projective unitary representation of $SL_2(\kappa)$ itself on M. What is *not* true is that this extends to all of $\mathrm{Aut}_{\mathbb{C}}(A) \cong PGL_q(\mathbb{C})$: at $q = 2$, $T = \mathrm{diag}(2,1)$ conjugates $\pi(W(1,0)) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ to $\begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix}$, whose singular values 2, $1/2$ differ, so it is not a scalar times a unitary — the frame hypothesis in Definition 5.1 is exactly what is needed above.

5.3 Splitting at odd characteristic

PROVED†

Theorem 5.3 (The Weil representation of $SL_2(\kappa)$, at odd characteristic)**.** *[p **odd.**]* $\alpha_g\big(W_\beta(v)\big) := \psi\big(Q_g(v)/2\big)\,W_\beta(gv)$ *defines a splitting* $g \mapsto \alpha_g$ *of the extension of Theorem 5.2:* $SL_2(\kappa)$ *acts on* $A_{\psi,\beta}(V)$ *by* $\mathbb{C}$*-algebra automorphisms, with phases in* μ_p*. Composed with Remark 5.2, this exhibits a projective unitary representation of* $SL_2(\kappa)$ *on the model space of Theorem 3.5.*

Scope. This is an odd-characteristic statement only; Theorem 5.6 addresses whether an analogous honest (non-projective) action of $SL_2(\kappa)$ by algebra automorphisms exists at $p = 2$, and the answer there is open in general. Nothing here claims the splitting is unique, or canonical among splittings.

Provenance. ids: D4, D6, WH-WEIL-a, WH-SVN, WH-WEIL-b; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1\rangle 4, \langle 1\rangle 7$; tested by: theory/checks/split24.py (X4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Since s_g is symmetric bilinear, $Q_g(v+v') = Q_g(v) + Q_g(v') + 2s_g(v,v')$ (the same polarization identity as in Definition 3.1's use elsewhere), so

$$\psi\big(Q_g(v+v')/2\big) = \psi\big(Q_g(v)/2\big)\,\psi\big(Q_g(v')/2\big)\,\psi\big(s_g(v,v')\big),$$

which is exactly relation $(*)$ of Theorem 5.2's proof for $\lambda(v) := \psi(Q_g(v)/2)$: α_g is a well-defined element of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ over g. That $g \mapsto \alpha_g$ is a homomorphism follows from the telescoping identity $s_{gh}(v,v') = s_h(v,v') + s_g(hv,hv')$ of Theorem 5.2's proof, evaluated at $v = v'$ to give $Q_{gh}(v) = Q_h(v) + Q_g(hv)$, whence $\alpha_g(\alpha_h(W_\beta(v))) = \psi(Q_h(v)/2)\,\psi(Q_g(hv)/2)\,W_\beta(ghv) = \psi(Q_{gh}(v)/2)\,W_\beta(ghv) = \alpha_{gh}(W_\beta(v))$. This is an honest algebra automorphism, so $SL_2(\kappa)$ acts by algebra automorphisms with μ_p phases (division by 2 makes sense since p is odd); the projective unitary statement is Remark 5.2 applied to this section. □

5.4 Characteristic two: the orthogonal subgroup and forced order-four phases

PROVED †

Theorem 5.4 (μ_2-phases occur exactly over the orthogonal subgroup of Q_β)**.** *[$p = 2$.] Write $O(Q_\beta) := \{\, g \in SL_2(\kappa) : Q_\beta \circ g = Q_\beta \,\}$. A frame automorphism $\alpha_{g,\lambda}$ (in the notation $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ of Theorem 5.2's proof) has* some *choice of λ landing entirely in $\psi(\kappa) = \mu_2$ if and only if $g \in O(Q_\beta)$ — a condition on the chosen β, not on the characteristic alone. For the reference cocycle β_0, $O(Q_{\beta_0}) \cap SL_2(\kappa)$ consists exactly of the matrices* $\mathrm{diag}(r, r^{-1})$ *and* $\mathrm{antidiag}(s, s^{-1})$*, $r, s \in \kappa^\times$; it has order $2(q-1)$ (equal to $2, 6, 14$ at $q = 2, 4, 8$), and is a* ***proper*** *subgroup of $SL_2(\kappa)$ for every q.*

Scope. This theorem is about which g admit a μ_2-valued lift, for a *fixed* β; it says nothing yet about what happens for $g \notin O(Q_\beta)$ (Theorem 5.5), and the concrete description of $O(Q_{\beta_0})$ is specific to the reference cocycle — for a general β of hyperbolic type the order is the same, $2(q-1)$ (Theorem 4.7(iii)), but the concrete matrices differ with the normal-form basis chosen there.

Provenance. ids: D6, WH-WEIL-a, D7, WH-WEIL-c; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1\rangle 5$; tested by: theory/checks/beta_census.py (B6); theory/checks/fp_symmetry.py (S3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Setting $v' = v$ in $(*)$ at $p = 2$ ($2v = 0$) gives $\lambda(v)^2 = \psi(Q_g(v))$. So a solution λ of $(*)$ is μ_2-valued (i.e. $\lambda(v)^2 = 1$ for every v) iff $Q_g(V(\kappa)) \subseteq \ker\psi$, i.e. iff $\psi \circ Q_g \equiv 1$.

For every β, this is already the exact isometry condition. $Q_g(v) = Q_\beta(gv) - Q_\beta(v)$ is additive in v: expanding $Q_\beta(g(v+v')) - Q_\beta(v+v')$ via Definition 3.1's polarization identity on both terms, the ω-cross-terms cancel because g preserves ω (Theorem 2.6(e)), leaving $Q_g(v+v') = Q_g(v) + Q_g(v')$; also $Q_g(cv) = c^2 Q_g(v)$ since g is κ-linear. So, in a basis, $Q_g(a,b) = \alpha a^2 + \delta b^2$ for some $\alpha, \delta \in \kappa$ (no cross term). If $\psi \circ Q_g \equiv 1$ then $\psi(\alpha a^2) = 1$ for all a (setting $b = 0$); since $a \mapsto a^2$ is bijective on κ, this forces $\alpha = 0$ by Fact 3.1, and likewise $\delta = 0$: $Q_g \equiv 0$, i.e. $g \in O(Q_\beta)$. Conversely, if $g \in O(Q_\beta)$ then trivially $\psi \circ Q_g \equiv 1$, and moreover a μ_2-valued λ solving $(*)$ then *exists*: $\psi \circ s_g$ is a symmetric cocycle with trivial diagonal ($\psi(s_g(v,v)) = \psi(Q_g(v)) = 1$), exactly the case handled by the ℓ-construction in the proof of Theorem 4.10 (§4). So $\psi \circ Q_g \equiv 1$ iff $g \in O(Q_\beta)$ iff a μ_2-valued lift exists.

The reference cocycle. By the explicit formula of Theorem 5.2's proof, $s_g(v,v') = rt\,aa' + st(ab'+a'b) + su\,bb'$ for β_0, so $Q_g(a,b) = rt\,a^2 + su\,b^2$ (the cross term drops at $p=2$). Thus $g \in O(Q_{\beta_0})$ iff $rt = 0$ and $su = 0$. Combined with $\det g = ru - st = 1$: if $r = 0$ then $st = -1 = 1$ (so $s,t \neq 0$), and $su = 0$ forces $u = 0$, giving $g = \left(\begin{smallmatrix} 0 & s \\ s^{-1} & 0 \end{smallmatrix}\right)$ (using $t = s^{-1}$ from $st = 1$) — an antidiagonal matrix; if $t = 0$ then $ru = 1$ (so $r, u \neq 0$, $u = r^{-1}$), and $su = 0$ forces $s = 0$, giving $g = \operatorname{diag}(r, r^{-1})$; the case $r = t = 0$ contradicts $\det g = 1$. So $O(Q_{\beta_0}) \cap SL_2(\kappa) = \{\operatorname{diag}(r,r^{-1})\} \cup \{\operatorname{antidiag}(s,s^{-1})\}$, of order $(q-1)+(q-1) = 2(q-1)$ (each family is parametrized bijectively by $\kappa^\times$). Since $Q_{\beta_0}(a,b) = ab$ is the hyperbolic normal form (Theorem 4.7(iii)), this matches the general formula $2(q-1)$ there. Properness: $2(q-1) < q(q^2-1) = |SL_2(\kappa)|$ for every $q \geq 2$, since $q(q+1) \geq 6 > 2$. ☐

PROVED †

Theorem 5.5 (μ_4 is forced exactly when the orthogonal subgroup is proper)**.** *[$p = 2$.] If $g \notin O(Q_\beta)$, every frame automorphism over g carries a phase of exact multiplicative order 4. Consequently μ_4 is forced somewhere in $\operatorname{Aut}^{\kappa}_{\mathcal{F}}(A)$ exactly when $O(Q_\beta) \cap SL_2(\kappa)$ is a proper subgroup of $SL_2(\kappa)$: this holds for β_0 at every q (Theorem 5.4), and for* every *$\beta \in \operatorname{Adm}(\omega)$ at $q = 4, 8$, by direct enumeration. It does **not** hold in the single case $q = 2$ with β anisotropic, where $O(Q_\beta) \cap SL_2(\mathbb{F}_2) = SL_2(\mathbb{F}_2)$ (order $6 = |O(Q_\beta) \cap SL_2(\kappa)|$ from the anisotropic formula $2(q+1)$ of Theorem 4.7(iii) at $q = 2$) and the entire group $\operatorname{Aut}^{\kappa}_{\mathcal{F}}(A)$ admits μ_2-valued phases throughout. So μ_4 is **not** a characteristic-two invariant of $(V(\kappa), \omega, \psi)$ alone: whether it is forced depends on the type of β.*

Scope. This theorem uses only Theorem 5.4 and the comparison of $2(q \mp 1)$ against $|SL_2(\kappa)| = q(q^2-1)$; it does not itself re-derive the general-q order $2(q+1)$ for the anisotropic stabilizer (that is Theorem 4.7(iii), only cited here). The claim for *every* β at $q = 4, 8$ rests on direct enumeration rather than a closed-form argument for general q.

Provenance. ids: D6, D10, WH-WEIL-c, WH-WEIL-d; proved in: theory/wh-kappa-choice.md §8 *step* ⟨1⟩6; tested by: theory/checks/split24.py (X4,X6, and the anisotropic cross-run at $q = 2$); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. By Theorem 5.4's proof, any λ solving $(*)$ over g satisfies $\lambda(v)^2 = \psi(Q_g(v))$ for every v. If $g \notin O(Q_\beta)$ then $Q_g \not\equiv 0$, and (by the same bijective-squaring argument as there) $\psi(Q_g(v)) = -1$ for some v — since $Q_g(v) = -1$'s preimage under ψ being nontrivial requires only that $\psi(\kappa) = \mu_2 = \{1, -1\}$ (Fact 3.1) and Q_g nonzero, hence onto a nonzero multiple of κ, hits every value including one where $\psi = -1$. Then $\lambda(v)^2 = -1$, i.e. $\lambda(v)$ has exact order 4: *every* λ solving $(*)$ over this g (not just one choice) is forced to take an order-4 value at this particular v, since the equation $\lambda(v)^2 = \psi(Q_g(v)) = -1$ has no solution in μ_2. Comparing $2(q-1)$ and $2(q+1)$ (Theorem 4.7(iii)) against $|SL_2(\kappa)| = q(q^2-1) = q(q-1)(q+1)$: $2(q-1) < q(q-1)(q+1)$ always (as $q(q+1) \geq 6$ for $q \geq 2$), so the hyperbolic stabilizer — in particular $O(Q_{\beta_0}) \cap SL_2(\kappa)$ — is always proper, giving some $g \notin O(Q_\beta)$ and hence a forced order-4 phase, at every q. For the anisotropic stabilizer, $2(q+1) < q(q-1)(q+1)$ iff $2 < q(q-1)$: this holds for $q \geq 4$ but *fails* at $q = 2$, where $q(q-1) = 2$ exactly, so $2(q+1) = 6 = q(q-1)(q+1) = |SL_2(\mathbb{F}_2)|$: the anisotropic stabiliser is the whole group at $q = 2$, so *no* g lies outside $O(Q_\beta)$, so no order-4 phase is forced by this argument, and — since μ_2-valued lifts exist for every $g \in O(Q_\beta)$ by Theorem 5.4's proof, and $\hat{V}$ is automatically μ_2-valued (Fact 3.1, exponent $p = 2$) — the whole of $\operatorname{Aut}^{\kappa}_{\mathcal{F}}(A)$ admits μ_2-valued phases in this one case. At $q = 4, 8$, exhaustive enumeration over every $\beta \in \operatorname{Adm}(\omega)$ (`theory/checks/split24.py`, gates X4 and X6) confirms $O(Q_\beta) \cap SL_2(\kappa)$ is proper for both types, hence μ_4 is forced for every β there. ☐

5.5 Does the extension split at characteristic two? A conjecture, and what is known

CONJECTURE

Conjecture 5.6 (Existence of a Weil representation of $SL_2(\kappa)$ at characteristic two)**.** *[$p = 2$.] The extension of Theorem 5.2 splits for every $q = 2^m$: $SL_2(\kappa)$ acts on $A_{\psi,\beta}(V)$ by $\mathbb{C}$-algebra automorphisms, for some choice of splitting. This is verified for the reference cocycle β_0 at $q = 2$ (exhaustive search finds exactly 4 distinct complements of $\hat{V}$ of order 6, over all generating pairs) and at $q = 4$ (64 lifts of a $(2, 3, 5)$-presentation of $SL_2(\mathbb{F}_4)$ satisfy the defining relations and generate a subgroup of order 60 meeting $\hat{V}$ trivially); in both verified cases every complement necessarily uses phases of exact order 4 (no complement avoids μ_4, since $|\hat{V}| \cdot |O(Q_{\beta_0}) \cap SL_2(\kappa)|$ does not divide $|\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)|$'s μ_2-part suitably: $6 \nmid 8$ at $q = 2$, $60 \nmid 96$ at $q = 4$). No argument is known for general q.*

Scope. This is recorded as a conjecture, not a theorem, precisely because the evidence is two finite instances and no general argument: per this campaign's discipline, a passing computational instance promotes nothing. The verified cases are for the reference cocycle β_0 only; splitting for other cocycles, and any relationship between splittings for cocycles of different Arf type, is not addressed. Should a splitting be found for general q, Remark 5.2 would then give a projective unitary representation of $SL_2(\kappa)$ at $p = 2$ exactly as Theorem 5.3 does at odd p.

Provenance. ids: D7, WH-WEIL-a, WH-WEIL-d, WH-WEIL-e; proved in: —; tested by: theory/checks/split24.py (X4,X5,X6); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

> *What is known, stated as evidence and not as a proof.* At $q = 2$, $|\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)| = q^2|SL_2(\kappa)| = 4 \cdot 6 = 24$; exhaustive search over generating pairs of order-6 subgroups meeting $\hat{V} = \mathbb{Z}/2 \times \mathbb{Z}/2$ trivially finds exactly 4 complements, all using a phase of exact order 4 somewhere (`split24.py` gate X4). At $q = 4$, lifting a $(2, 3, 5)$ generator-relation presentation of $SL_2(\mathbb{F}_4) \cong A_5$ over all $16 \times 16 = 256$ candidate phase choices on the two generators finds exactly 64 lifts satisfying $x^2 = y^3 = (xy)^5 = 1$ and generating a subgroup of order 60 meeting $\hat{V}$ trivially, again always with order-4 phases (gate X5). That no complement can avoid μ_4 in either case is confirmed by comparing group orders directly (gate X6): the μ_2-phase part of $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ has order $q^2 \cdot |O(Q_{\beta_0}) \cap SL_2(\kappa)| = 4 \cdot 2 = 8$ at $q = 2$ and $16 \cdot 6 = 96$ at $q = 4$, and $|SL_2(\kappa)| = 6$, respectively 60, does not divide either. This settles the existence question affirmatively at these two finite values and settles that any splitting there must use order-4 phases; it establishes nothing for $q = 2^m$, $m \geq 3$.

6 The choices the construction depends on, and what is canonical

Recollection. $\kappa = \mathbb{F}_q$; $\psi \in X(\kappa)$ a phase datum, $\beta \in \mathrm{Adm}(\omega)$ a polarizing cocycle (Definitions 2.3, 4.1); $A_{\psi,\beta}(V)$ the resulting twisted algebra, simple of dimension q^2 (Theorem 2.8), with a unique simple unitary module up to unitary equivalence (Theorem 3.5); by Theorem 4.10, for fixed ψ every two choices of β give algebras related by an explicit frame isomorphism.

Provenance. ids: D2, D3, D4, WH-ALG, WH-SVN; proved in: —; tested by: —; reviewed in: —.

Sections 4 and 5 named the choices the construction depends on and measured exactly how much they matter. This section assembles that information into a single statement: precisely which data beyond κ the construction needs, and precisely what survives canonically once those choices are made. The headline fact is that the *projective* Hilbert space is canonical — it does not depend on which unitary model, which polarization, or (granting one more ingredient from §4) which polarizing cocycle was used to build it — even though the Hilbert space itself is not.

Definition 6.1 (The model groupoid, and its projectivization)**.** $\mathrm{Mod}_{\psi,\beta}(\kappa)$ is the category whose objects are pairs (M,π) with $\pi : A_{\psi,\beta}(V) \to \mathrm{End}_{\mathbb{C}}(M)$ a unital algebra homomorphism making M a simple module and every $\pi(W_\beta(v))$ unitary, and whose morphisms $(M,\pi) \to (M',\pi')$ are unitary intertwiners $T : M \to M'$ ($T\pi(a) = \pi'(a)T$ for all a). $\mathrm{PMod}_{\psi,\beta}(\kappa)$ is the same category with each Hom-set replaced by $\mathrm{Hom}/U(1)$ (intertwiners identified when they differ by a unimodular scalar).

Scope. This definition alone asserts nothing about the size or connectivity of $\mathrm{Mod}_{\psi,\beta}(\kappa)$; that it is nonempty, connected, and has automorphism group $U(1)$ at every object is Theorem 6.3(c) below, using Theorem 3.5, not part of the stipulation here.

Provenance. ids: D9; proved in: —; tested by: —; reviewed in: —.

6.1 Exactly two data beyond the field

PROVED

Theorem 6.2 (The construction depends on exactly two data beyond κ)**.** *(a) $X(\kappa) = \hat{\kappa}\backslash\{1\}$ has $q-1$ elements and is a free transitive $\kappa^\times$-torsor under $(u\cdot\psi)(x) := \psi(ux)$ (a single point when $q = 2$, where $\kappa^\times$ is trivial);*

(b) beyond κ, the construction of $A_{\psi,\beta}(V)$ uses ***exactly two*** *further data: a nontrivial character $\psi \in X(\kappa)$ and a polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ — given (ψ,β), $A_{\psi,\beta}(V)$ (Definition 2.4) involves no further choice, while exhibiting a concrete Hilbert space (rather than the abstract algebra) needs a pair (L,χ) in addition (Definition 3.2);*

(c) distinct $\psi,\psi' \in X(\kappa)$ give representations of the same *group $H_\beta(\kappa)$ (which, by Definition 2.5, does not involve ψ at all) with distinct central characters, hence inequivalent as representations of $H_\beta(\kappa)$;*

(d) $A_{\psi,\beta}(V) \cong A_{\psi',\beta'}(V)$ for every *$\psi,\psi' \in X(\kappa)$ and $\beta,\beta' \in \mathrm{Adm}(\omega)$; but the frame-preserving isomorphisms between two such algebras form a torsor under $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A_{\psi,\beta}(V))$ (Theorem 5.2), a group of order $q^2\,|SL_2(\kappa)| \geq 24$ — so no single isomorphism among them is distinguished by the data $(\kappa,\psi,\psi',\beta,\beta')$ alone.*

Scope. Part (b) corrects an earlier working hypothesis of this campaign, that the construction depended on a single datum beyond κ: that hypothesis is refuted by Theorem 4.2 and Theorem 4.5 of §4, which show β is a second, independent choice. Part (d) asserts the algebras are abstractly isomorphic; it does not assert any *particular* isomorphism among them is preferred, and Theorem 6.3 below is precisely about what survives this non-uniqueness.

Provenance. ids: D2, D3, D4, WH-ALG, WH-BETA-a, WH-WEIL-a, WH-CHOICE; proved in: theory/wh-kappa-choice.md §7; tested by: theory/checks/wh_kappa_check.py (C8 in its E5 form, C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) $|\hat{\kappa}| = q$ by Fact 3.1 ($U = \kappa$, $d = m$), so $|X(\kappa)| = q-1$. The action $(u\cdot\psi)(x) := \psi(ux)$ preserves nontriviality: if $\psi \not\equiv 1$ and $u \neq 0$ then, as $x \mapsto ux$ is a bijection of κ, $u\cdot\psi \equiv 1$ would force $\psi \equiv 1$, false. It is free: $u\cdot\psi = \psi$ means $\psi((u-1)x) = 1$ for all x, i.e. $(u-1)\kappa \subseteq \ker\psi$; if $u \neq 1$ then $(u-1) \in \kappa^\times$ so $(u-1)\kappa = \kappa \not\subseteq \ker\psi$ (Fact 3.1, ψ nontrivial), forcing $u = 1$. A free action of the order-$(q-1)$ group $\kappa^\times$ on the order-$(q-1)$ set $X(\kappa)$ has exactly one orbit of size $q-1$, hence is transitive. At $q = 2$, $\kappa^\times = \{1\}$ and $X(\kappa)$ is a single point — the $q = 2$ choice, per Theorem 4.4, lives entirely in β, not in ψ.

(b) Immediate from Definitions 2.4 and 3.2: the structure constants of $A_{\psi,\beta}(V)$ are $\psi(\beta(v,v'))$, a function of κ, ψ, β alone, while $M_{L,\chi}$ additionally names a line L and a character χ of A_L.

(c) A representation of $H_\beta(\kappa)$ with central character ψ restricts on the centre $\kappa \times 0$ to $\psi \cdot \mathrm{id}$ (the remark after Theorem 2.7); an intertwiner between representations with central characters $\psi \neq \psi'$ would have to commute with the centre acting as $\psi \cdot \mathrm{id}$ on one side and $\psi' \cdot \mathrm{id}$ on the other, forcing it to be 0 on the difference $(\psi(t) - \psi'(t)) \cdot \mathrm{id} \neq 0$ for suitable t — no nonzero intertwiner exists, so the two representations of the fixed group $H_\beta(\kappa)$ are inequivalent.

(d) *Existence, for every pair.* By Theorem 4.10, $A_{\psi,\beta}(V) \cong A_{\psi,\beta_0}(V)$ for every β (via a frame isomorphism, taking $\beta' = \beta_0$ there), and symmetrically $A_{\psi',\beta'}(V) \cong A_{\psi',\beta_0}(V)$. For the reference cocycle, $W_\psi(a,b) \mapsto W_{\psi_u}(u^{-1}a, b)$ (for $\psi' = u \cdot \psi = \psi_u$ as in (a)) is an isomorphism $A_{\psi,\beta_0}(V) \to A_{\psi_u,\beta_0}(V)$, since $\beta_0(u^{-1}a, b') = u^{-1}\beta_0(v,v')$ makes the structure constants match: $\psi_u(\beta_0(u^{-1}v, u^{-1}v')) = \psi(u \cdot u^{-2}\beta_0(v,v')) = \psi(u^{-1}\beta_0(v,v'))$, consistently with the images $W_{\psi_u}(u^{-1}v)$. Composing the three maps ($A_{\psi,\beta} \to A_{\psi,\beta_0} \to A_{\psi',\beta_0} \to A_{\psi',\beta'}$, the middle map since $X(\kappa)$ is a $\kappa^\times$-torsor by (a)) gives an isomorphism $A_{\psi,\beta}(V) \to A_{\psi',\beta'}(V)$ for arbitrary $\psi, \psi', \beta, \beta'$. *Non-uniqueness.* The frame-preserving isomorphisms between two fixed such algebras, once nonempty, form a free transitive right $\mathrm{Aut}^\kappa_{\mathcal{F}}(A_{\psi,\beta}(V))$-set under postcomposition (compose one such isomorphism with an automorphism of the source), and $|\mathrm{Aut}^\kappa_{\mathcal{F}}(A_{\psi,\beta}(V))| = q^2|SL_2(\kappa)|$ by Theorem 5.2, at least $4 \cdot 6 = 24$. More than one isomorphism exists whenever this group is nontrivial (always), and the data $(\kappa, \psi, \psi', \beta, \beta')$ do not name a preferred one: e.g. a family $\{h_u\}$, $h_u := \mathrm{diag}(u^{-j}, u^{j-1})$ for a further arbitrary choice of $j \in \mathbb{Z}/(q-1)$, gives infinitely many mutually inconsistent *coherent* systems of isomorphisms across the $\kappa^\times$-action of (a) — the family of isomorphisms is trivializable, but not canonically: choosing one is a section, not a theorem. $\square$

6.2 The projective Hilbert space is canonical

PROVED

Theorem 6.3 (Canonicity of the projective Hilbert space, at fixed (κ, ψ, β)). *(a)* $\mathrm{Mod}_{\psi,\beta}(\kappa)$ *(Definition 6.1) is a nonempty, connected groupoid, with automorphism group $U(1)$ at every object;*

(b) consequently every Hom*-set of* $\mathrm{PMod}_{\psi,\beta}(\kappa)$ *is a singleton, so every diagram in it commutes trivially, and the limit $\varprojlim_{\mathrm{PMod}_{\psi,\beta}(\kappa)} P(M)$ exists with every projection an isomorphism; define $P(H_\psi(\kappa))$ to be this limit. The underlying Hilbert space $H_\psi(\kappa)$ itself is canonical only up to a $U(1)$ of identifications (one for each object of* $\mathrm{Mod}_{\psi,\beta}(\kappa)$ *chosen as a representative), but the* projective *Hilbert space $P(H_\psi(\kappa))$ is canonical in (κ, ψ), for fixed β.*

Scope. This theorem fixes β throughout; that the resulting $P(H_\psi(\kappa))$ does not, in fact, depend on the choice of β either is the separate, weaker-status content of Theorem 6.4 below, which is not needed for and does not feed back into this proof. Nothing here addresses canonicity of a specific *vector* in $H_\psi(\kappa)$, only of the isomorphism class of the pair (M, π) up to the $U(1)$ ambiguity of unitary intertwiners.

Provenance. ids: D9, WH-SVN, WH-CHOICE, WH-CANON; proved in: theory/wh-kappa-choice.md §7 *step* $\langle 1\rangle 3, \langle 1\rangle 4$; tested by: —; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) Nonempty: Theorem 3.5(a) exhibits the standard model M_0, unitary by Theorem 3.5(d). Connected: any two objects (M, π), (M', π') are unitarily isomorphic by Theorem 3.5(b)–(d) (uniqueness of the simple module up to unitary equivalence), so a morphism exists between any two objects, in either direction. $\mathrm{Aut}(M, \pi) = U(1)$: by Theorem 3.5(b)'s Schur clause, any self-intertwiner is a scalar; a unitary one is then a scalar of modulus 1, and every element of $U(1)$ is realized (scalars commute with everything, hence intertwine trivially).

(b) In a connected groupoid where every object has automorphism group $U(1)$, every Hom-set is a $U(1)$-torsor (fix one isomorphism between two objects; every other differs from it by an automorphism of the source, i.e. by an element of $U(1)$). Quotienting each Hom-set by this $U(1)$ therefore collapses it to a single point: every Hom-set of $\mathrm{PMod}_{\psi,\beta}(\kappa)$ is a singleton. Consequently, for any two morphisms f, g in $\mathrm{PMod}_{\psi,\beta}(\kappa)$ with the same source and target, $f = g$ automatically (both are *the* unique element of that singleton Hom-set): every square of transition maps commutes, not by verification, but because there is only one morphism available to fill it. A category in which every Hom-set is a singleton is equivalent to the terminal (one-object, one-morphism) category, so the limit of the identity diagram $P(M) \rightsquigarrow P(M)$ over $\mathrm{PMod}_{\psi,\beta}(\kappa)$ exists and every object, with its unique family of identifying isomorphisms to every other, computes it: this is $P(H_\psi(\kappa))$. Since the ordinary (non-projectivized) Hom-sets of $\mathrm{Mod}_{\psi,\beta}(\kappa)$ are $U(1)$-torsors and not singletons, no single Hilbert space among the objects is preferred over $U(1)$-many identifications with it — but every one of them projects isomorphically onto the same limit $P(H_\psi(\kappa))$. □

6.3 Independence from the polarizing cocycle as well

SKETCH

Theorem 6.4 (The projective Hilbert space does not depend on β either). *$P(H_\psi(\kappa))$, as constructed in Theorem 6.3, is independent of the choice of polarizing cocycle $\beta \in \mathrm{Adm}(\omega)$ used to build it: the frame isomorphism $A_{\psi,\beta}(V) \to A_{\psi,\beta'}(V)$ of Theorem 4.10 transports unitary simple modules and induces an equivalence $\mathrm{Mod}_{\psi,\beta}(\kappa) \to \mathrm{Mod}_{\psi,\beta'}(\kappa)$, which in turn induces* the *unique isomorphism between the two limits $P(H_\psi(\kappa))$ constructed at β and at β'.*

Scope. This theorem is SKETCH for the same procedural reason as the $p = 2$ rows of §4 it depends on (Theorem 4.10): it postdates this increment's single hostile-critic round. It asserts that the *limit* is independent of β, not that any specific object of $\mathrm{Mod}_{\psi,\beta}(\kappa)$ equals, on the nose, an object of $\mathrm{Mod}_{\psi,\beta'}(\kappa)$ — the identification goes through the transport functor described in the proof, consistently with the fact that the underlying group $H_\beta(\kappa)$ genuinely changes isomorphism type with β at $p = 2$ (Theorem 4.7).

Provenance. ids: WH-CANON, WH-BETA-LEVEL, WH-CANON-BETA; proved in: theory/wh-kappa-choice.md §7 *step* ⟨1⟩4.⟨2⟩1; tested by: —; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. By Theorem 4.10 there is a frame isomorphism $\Phi : A_{\psi,\beta}(V) \to A_{\psi,\beta'}(V)$, $W_\beta(v) \mapsto \mu(v)W_{\beta'}(v)$, with μ unimodular (by the same argument as Remark 5.2: taking absolute values in the defining relation (†) of Theorem 4.10's proof makes $|\mu|$ a homomorphism from a finite group to $\mathbb{R}_{>0}$, hence trivial). Transport of structure along Φ sends an object (M, π) of $\mathrm{Mod}_{\psi,\beta}(\kappa)$ to (M, π') with $\pi'(a') := \pi(\Phi^{-1}(a'))$ for $a' \in A_{\psi,\beta'}(V)$: M stays simple (an equivalence of module categories along an algebra isomorphism preserves simplicity), and each $\pi'(W_{\beta'}(v)) = \pi(\Phi^{-1}(W_{\beta'}(v))) = \overline{\mu(v)}\,\pi(W_\beta(v))$ remains unitary (a unimodular scalar times a unitary operator). This is an equivalence of categories $\mathrm{Mod}_{\psi,\beta}(\kappa) \to \mathrm{Mod}_{\psi,\beta'}(\kappa)$, compatible with the projectivization functors to $\mathrm{PMod}_{\psi,\beta}(\kappa)$ and $\mathrm{PMod}_{\psi,\beta'}(\kappa)$ of Theorem 6.3(b). Both projectivized categories have singleton Hom-sets (Theorem 6.3(b), applied at β and at β' respectively), so the induced map between their limits $P(H_\psi(\kappa))_\beta \to P(H_\psi(\kappa))_{\beta'}$ is again the unique morphism available between two terminal-equivalent categories, hence an isomorphism, and the unique one compatible with the transport functor. □

7 What κ-linear symmetry hides: the $\mathbb{F}_p$-shadow of the form

Recollection. For $\beta \in \mathrm{Adm}(\omega)$ and phase datum ψ, the Weyl frame is $\mathcal{F} = \{\mathbb{C}^\times W_\beta(v) : v \in V(\kappa)\} \subset A_{\psi,\beta}(V)$, and $\mathrm{Aut}_{\mathcal{F}}(A)$ is the group of $\mathbb{C}$-algebra automorphisms of $A_{\psi,\beta}(V)$ preserving

$\mathcal{F}$ (Definition 5.1); $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A) \subseteq \mathrm{Aut}_{\mathcal{F}}(A)$ is the further subgroup whose induced permutation of $V(\kappa)$ is *κ-linear* — data *imposed* on top of $\mathrm{Aut}_{\mathcal{F}}(A)$, not derived from it (Definition 5.1's scope). By Theorem 5.2, $\mathrm{Aut}^{\kappa}_{\mathcal{F}}(A)$ surjects onto $SL_2(\kappa)$ with kernel $\hat{V}$, at every characteristic.

Provenance. ids: D4, D7, WH-WEIL-a; proved in: —; tested by: —; reviewed in: —.

This section makes good on the promissory note left in Definition 5.1: exactly how much symmetry the imposed κ-linearity clause discards. The answer is that $A_{\psi,\beta}(V)$ and its frame $\mathcal{F}$ do not, in fact, remember the κ-module structure of $V(\kappa)$ at all — they are built from the bare abelian group $(V(\kappa), +)$ and the cocycle $\psi \circ \beta$ — so the *full* frame symmetry group is larger than $SL_2(\kappa)$ whenever κ is a proper extension of its prime field, and the discarded symmetry is measured exactly.

PROVED

Theorem 7.1 (The twisted algebra does not remember the κ-structure of $V(\kappa)$). *(a) $A_{\psi,\beta}(V)$ and its frame $\mathcal{F}$ are determined by the abelian group $(V(\kappa), +)$ together with the $\mathbb{C}^{\times}$-valued function $\psi \circ \beta$ on $V(\kappa) \times V(\kappa)$ alone; nothing in either object refers to scalar multiplication by κ;*

(b) consequently, dropping the imposed κ-linearity clause of Definition 5.1, the frame automorphisms with merely $\mathbb{F}_p$-linear induced permutation g of $V(\kappa)$ surject, with the same *kernel $\hat{V}$, onto the isometry group $Sp(V(\kappa), \psi \circ \omega) \cong Sp_{2m}(\mathbb{F}_p)$ of the $\mathbb{F}_p$-valued alternating form $\psi \circ \omega$; so $|\mathrm{Aut}_{\mathcal{F}}(A)| = q^2 \cdot |Sp_{2m}(\mathbb{F}_p)|$;*

(c) $SL_2(\kappa) \subseteq Sp_{2m}(\mathbb{F}_p)$, with index exactly 1 when $m = 1$ and strictly greater than 1 as soon as $m > 1$: by complete enumeration, $|SL_2(\kappa)| = 6, 24, 60, 120, 504, 720$ against $|Sp_{2m}(\mathbb{F}_p)| = 6, 24, 720, 120, 1451520, 51840$ at $q = 2, 3, 4, 5, 8, 9$ respectively — indices $1, 1, 12, 1, 2880, 72$. So Definition 5.1's κ-linearity clause discards genuine symmetry the algebra has, whenever κ is a proper extension of its prime field.

Scope. This theorem does not claim $Sp_{2m}(\mathbb{F}_p)$ acts by automorphisms preserving anything beyond the frame $\mathcal{F}$ and the abelian group structure — in particular it says nothing about whether this larger symmetry is compatible with the κ-linear structure used elsewhere in this labbook (Theorem 2.6(f) shows, to the contrary, that the κ-*valued* form ω itself sees only $SL_2(\kappa)$ among $\mathbb{F}_p$-linear maps: the enlargement here is possible only because $\mathrm{Aut}_{\mathcal{F}}(A)$ is built from the strictly weaker $\mathbb{F}_p$-valued shadow $\psi \circ \omega$). At the prime field, $m = 1$, the two groups coincide and nothing is discarded; this is the campaign's guiding path.

Provenance. ids: D1, D7, WH-WEIL-a, WH-FORM, WH-SYMM; proved in: theory/wh-kappa-choice.md §9; tested by: theory/checks/fp_symmetry.py (S1,S2,S3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) By Definition 2.4, the structure constants of $A_{\psi,\beta}(V)$ in the basis $\{W_\beta(v)\}$ are exactly the values $\psi(\beta(v, v'))$, indexed by pairs of elements of the abelian group $(V(\kappa), +)$; by Definition 5.1, $\mathcal{F}$ is the set of $\mathbb{C}^{\times}$-lines through this same basis, again indexed only by $(V(\kappa), +)$. Neither construction refers to multiplication by elements of κ.

(b) Repeat the argument of Theorem 5.2's proof *without* imposing κ-linearity on g: frame preservation still gives $\alpha(W_\beta(v)) = \lambda(v) W_\beta(gv)$ with g an (a priori only additive) bijection of $V(\kappa)$ and λ satisfying relation $(*)$ there, since that derivation used only additivity of g, never κ-linearity. Applying α to Theorem 2.7(c), an identity entirely in $\psi \circ \omega$, gives $\psi\big(\omega(gv, gv')\big) = \psi\big(\omega(v, v')\big)$ for all v, v' — i.e. $g \in Sp(V(\kappa), \psi \circ \omega) := \{g \text{ additive} : \psi \circ \omega \circ (g \times g) = \psi \circ \omega\}$ — rather than the stronger $\omega \circ g = \omega$ used before. The surjectivity argument via Fact 3.1 applies verbatim to any such g (it never used κ-linearity, only that $(V(\kappa), +)$ has exponent p), so every $g \in Sp(V(\kappa), \psi \circ \omega)$ lifts; the kernel computation ($g = \mathrm{id}$ forces $\lambda \in \hat{V}$, identified with

$V(\kappa)$ via $u \mapsto \psi(\omega(u,\cdot)))$ is unchanged, giving the same kernel $\hat{V}$ of order q^2. Nondegeneracy of $\psi \circ \omega$ as an $\mathbb{F}_p$-bilinear form on the $2m$-dimensional $\mathbb{F}_p$-space $V(\kappa)$: if $\psi(\omega(v,\cdot)) \equiv 1$ then $\omega(v,\cdot) \equiv 0$ (Fact 3.1, since $\omega(v,\cdot)$ is either 0 or onto κ, being κ-linear, and $\kappa \not\subseteq \ker\psi$), so $v = 0$ by Theorem 2.6(c). A nondegenerate alternating form on a $2m$-dimensional space over $\mathbb{F}_p$ has isometry group (by definition, matching the notation `notation.md` fixes for $Sp_{2m}(\mathbb{F}_p)$) exactly $Sp_{2m}(\mathbb{F}_p)$.

(c) If $g \in SL_2(\kappa)$ then $\omega \circ (g \times g) = \omega$ (Theorem 2.6(e)), so certainly $\psi \circ \omega \circ (g \times g) = \psi \circ \omega$, i.e. $SL_2(\kappa) \subseteq Sp(V(\kappa), \psi \circ \omega)$. This containment is exactly the gap between Theorem 2.6(f) — which shows the $\mathbb{F}_p$-linear isometries of the κ-*valued* ω are no larger than $SL_2(\kappa)$ — and the present, strictly weaker condition of isometry for the $\mathbb{F}_p$-valued shadow $\psi \circ \omega$ alone: an $\mathbb{F}_p$-linear g can preserve $\psi \circ \omega$ without preserving ω itself. Complete enumeration (`theory/checks/fp_symmetry.py`, gates S1 and S2) computes both orders directly at $q = 2, 3, 4, 5, 8, 9$, matching the closed form $|Sp_{2m}(\mathbb{F}_p)| = p^{m^2} \prod_{i=1}^{m}(p^{2i} - 1)$ in every case, and confirms the indices $1, 1, 12, 1, 2880, 72$; at $m = 1$ (the prime field) the two group orders coincide, since $Sp_2(\mathbb{F}_p) = SL_2(\mathbb{F}_p)$ by definition in dimension 2, and nothing is discarded. □

Remark (The characteristic-two orthogonal phenomenon survives the enlargement). The μ_2-phase criterion of Theorem 5.4 ($g \in O(Q_\beta)$) makes sense verbatim for merely $\mathbb{F}_p$-linear g, at $p = 2$: $\psi \circ Q_\beta \circ g = \psi \circ Q_\beta$ is a condition on an $\mathbb{F}_p$-valued quadratic form $\psi \circ Q_\beta$ with polar form $\psi \circ \omega$, so the relevant subgroup inside the enlarged symmetry group of Theorem 7.1 is $O(\psi \circ Q_\beta) \cap Sp_{2m}(\mathbb{F}_p)$. Direct enumeration (gate S3) finds its order to be $2, 72, 40320$ at $q = 2, 4, 8$, of index $3, 10, 36$ in $Sp_{2m}(\mathbb{F}_p)$ — exactly the *same* index that $|O(Q_\beta) \cap SL_2(\kappa)| = 2, 6, 14$ has inside $SL_2(\kappa)$ (Theorem 5.4). So the characteristic-two orthogonal phenomenon of §5 is not an artefact of the imposed κ-linearity in Definition 5.1: it survives, at the same relative index, inside the larger, intrinsic symmetry group $Sp_{2m}(\mathbb{F}_p)$ that Theorem 7.1 exhibits. This observation is recorded as part of the same source theorem as Theorem 7.1, under the same provenance (`theory/wh-kappa-choice.md` §9, step ⟨1⟩4).

Remark (Reading). Two independent non-canonicities are now on record, and they are logically independent of each other. Theorem 4.7 of §4 shows the system does not remember which polarizing cocycle β was chosen, in a way that is invisible at odd characteristic and genuinely two-valued at $p = 2$. This theorem shows the system does not remember the κ-module structure of $V(\kappa)$ itself, at every characteristic and every $m > 1$, regardless of β. Both are statements about what the definition actually depends on, in the sense of `PRD.md`'s "canonical" requirement: any assignment $\operatorname{Spec} R \mapsto$ quantum system general enough to restrict to this one on $\operatorname{Spec} \mathbb{F}_q$ carries both phenomena, not as defects, but as exactly the symmetry and the choice the construction is built from.

8 Functoriality in the field: what extends along embeddings, and what does not

Recollection. A phase datum for $\kappa = \mathbb{F}_{p^m}$ is a nontrivial additive character $\psi : (\kappa, +) \to \mathbb{C}^\times$, and the trace-normalized family is $\psi_\zeta(z) := \zeta^{\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z)}$ for ζ a primitive p-th root of unity and $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(z) := z + z^p + \cdots + z^{p^{m-1}}$ (Definition 2.3); by Theorem 6.2(a), $X(\kappa)$ is a $\kappa^\times$-torsor, so every $\psi \in X(\kappa)$ is $\psi_c := \psi_\zeta(c\,\cdot)$ for a unique $c \in \kappa^\times$.

Provenance. ids: D3; proved in: —; tested by: —; reviewed in: —.

Earlier sections fixed a single finite field κ throughout. This section asks how the construction behaves as κ varies along field embeddings: whether the trace-normalized family is itself compatible with restriction, whether the assignment $\kappa \mapsto A_{\psi,\beta_0}(V(\kappa))$ is a functor, and — the question this campaign's earlier working hypothesis answered wrongly — whether a single compatible choice of character can be made once and for all, for every finite field simultaneously. It cannot; the precise obstruction is Frobenius, and the precise set of compatible choices that *does* exist is named exactly.

PROVED

Theorem 8.1 (The trace-normalized family, restricted along an extension). *Let $\kappa = \mathbb{F}_{p^m} \subseteq \kappa' = \mathbb{F}_{p^{mn}}$. Then $\psi'_\zeta|_\kappa = \psi_{\zeta^n}$ (both sides characters of κ, computed with the same ζ): the restriction of κ''s trace-normalized character to κ is trivial exactly when $p \mid n$ (smallest instance: $\mathbb{F}_2 \subset \mathbb{F}_4$).*

Scope. This theorem is about the trace-normalized family specifically — a particular, and particularly natural, choice inside $X(\kappa)$ for every κ — and asserts nothing about arbitrary families of characters; the statement that quantifies over *all* compatible systems is Theorem 8.4 below.

Provenance. ids: D3, WH-FUNCT-a; proved in: theory/wh-kappa-choice.md §10 *step* $\langle 1 \rangle 1$; tested by: theory/checks/funct_sections.py (F1); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. For $x \in \kappa$, $x^{p^m} = x$ (elements of $\mathbb{F}_{p^m}$ are exactly the fixed points of the m-th power of Frobenius), so the mn summands of $\mathrm{Tr}_{\kappa'/\mathbb{F}_p}(x) = \sum_{k=0}^{mn-1} x^{p^k}$ repeat with period m: grouping into n blocks of m consecutive terms, each block sums to $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(x)$ (using $x^{p^{k+m}} = (x^{p^m})^{p^k} = x^{p^k}$), so $\mathrm{Tr}_{\kappa'/\mathbb{F}_p}|_\kappa = n \cdot \mathrm{Tr}_{\kappa/\mathbb{F}_p}$. Hence $\psi'_\zeta(x) = \zeta^{\mathrm{Tr}_{\kappa'/\mathbb{F}_p}(x)} = \zeta^{n \cdot \mathrm{Tr}_{\kappa/\mathbb{F}_p}(x)} = (\zeta^n)^{\mathrm{Tr}_{\kappa/\mathbb{F}_p}(x)} = \psi_{\zeta^n}(x)$, and $\zeta^n = 1$ (so $\psi_{\zeta^n} \equiv 1$) exactly when $p \mid n$, since ζ has exact order p. □

PROVED

Theorem 8.2 (The reference algebra is functorial in character-compatible embeddings). *Let $\mathcal{FF}^\pm$ be the category with objects the pairs (κ, ψ), $\psi \in X(\kappa)$, and morphisms $(\kappa, \psi) \to (\kappa', \psi')$ the field embeddings $\iota : \kappa \hookrightarrow \kappa'$ with $\psi' \circ \iota = \psi$ (*character-compatible *embeddings). Then*

$$(\kappa, \psi) \longmapsto A_{\psi,\beta_0}(V(\kappa))$$

is a functor from $\mathcal{FF}^\pm$ to the category of unital $\mathbb{C}$-algebras and injective algebra homomorphisms, respecting composition and identities.

Scope. This functor uses the reference cocycle β_0 throughout, on every object; it is not claimed that the resulting assignment is independent of this choice, nor that an analogous functor exists carrying a varying β_κ compatibly along embeddings — the reference cocycle is used here because it is a fixed, explicit stipulation (Definition 4.1), not because it is canonical (Theorem 4.4 of §4 shows at $p = 2$ it is not). Nor is this a statement about the quantum *system* — the Hilbert space or the Heisenberg group — functorially in κ, only about the algebra.

Provenance. ids: D2, D4, WH-ALG, WH-FUNCT-b; proved in: theory/wh-kappa-choice.md §10 *step* $\langle 1 \rangle 2$; tested by: theory/checks/funct_sections.py (F5); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. For a character-compatible embedding $\iota : (\kappa, \psi) \to (\kappa', \psi')$, put $V(\iota)(a, b) := (\iota a, \iota b) : V(\kappa) \to V(\kappa')$, additive since ι is a field homomorphism. Since ι is multiplicative, $\beta_0\big(V(\iota)v, V(\iota)v'\big) = \iota(a)\,\iota(b') = \iota(ab') = \iota\big(\beta_0(v, v')\big)$. Define $W_\beta(v) \mapsto W'_{\beta_0}(V(\iota)v)$ on basis elements of $A_{\psi,\beta_0}(V(\kappa))$. This sends distinct basis elements to distinct basis elements ($V(\iota)$ is injective, ι being a field embedding), and is multiplicative: the image of $W_{\beta_0}(v)W_{\beta_0}(v') = \psi(\beta_0(v, v'))W_{\beta_0}(v + v')$ is $\psi(\beta_0(v, v'))\, W'_{\beta_0}(V(\iota)(v{+}v'))$, while the product of the images is $\psi'\big(\beta_0(V(\iota)v, V(\iota)v')\big)\, W'_{\beta_0}(V(\iota)(v{+}v')) = \psi'\big(\iota(\beta_0(v, v'))\big)\, W'_{\beta_0}(V(\iota)(v + v')) = \psi(\beta_0(v, v'))\, W'_{\beta_0}(V(\iota)(v + v'))$, using $\psi' \circ \iota = \psi$ (the compatibility condition). So the map is a unital, nonzero algebra homomorphism, hence injective by simplicity of the source (Theorem 2.8). It respects composition and identities because $\iota \mapsto V(\iota)$ does: $V(\iota' \circ \iota) = V(\iota') \circ V(\iota)$ and $V(\mathrm{id}) = \mathrm{id}$, directly from the definition of $V(\iota)$. □

8.1 No compatible choice of character over every finite field

REFUTED

Theorem 8.3 (Refuted: a Frobenius-compatible character exists for every finite field)**.** (Claimed, and refuted.) *A section of the forgetful functor $\mathcal{FF}^{\pm} \to \{$finite fields$\}$ exists: a choice $\psi_\kappa \in X(\kappa)$ for every finite field κ such that $\iota : \kappa \to \kappa'$ character-compatible (i.e. $\psi_{\kappa'} \circ \iota = \psi_\kappa$) for* every *field embedding ι — in particular, for the Frobenius self-embedding of every κ.*

Scope. This row is kept, refuted, rather than deleted, per this campaign's status discipline: a refutation carrying the surviving weaker statement is a result, not a gap to be erased. The surviving statement — exactly which obstruction blocks a section, and exactly which weaker compatible systems *do* exist — is Theorem 8.4 immediately below and Theorem 8.5 thereafter. Nothing may depend on this row.

Provenance. ids: WH-FUNCT-b-SEC; proved in: —; tested by: theory/checks/funct_sections.py (F2,F3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

> *Refutation.* Frobenius $x \mapsto x^p$ is a field embedding $\kappa \to \kappa$, so a section would in particular have to satisfy $\psi_\kappa \circ \mathrm{Frob} = \psi_\kappa$ for every κ. Theorem 8.4 below shows this forces $\psi_\kappa = \psi_c$ for some c in the *prime* field $\mathbb{F}_p$, and that for $\kappa = \mathbb{F}_{p^p}$ every character of this Frobenius-invariant form restricts *trivially* to $\mathbb{F}_p$ — contradicting $\psi_{\mathbb{F}_p} \in X(\mathbb{F}_p)$, which must be nontrivial by definition. So no section exists; the smallest instance of the obstruction is $\mathbb{F}_2 \subset \mathbb{F}_4$. A previously considered route to a section, by induction along the tower $\mathbb{F}_{p^{n!}}$, is insufficient in any case: a character of $\mathbb{F}_{p^{n!}}$ compatible at that one stage can still restrict trivially to an intermediate field, so no such induction can repair the obstruction just exhibited.

PROVED

Theorem 8.4 (The Frobenius obstruction, stated exactly)**.** *There is no choice of $\psi_\kappa \in X(\kappa)$, one for each finite field κ, compatible with* all *field embeddings. Precisely: writing $\psi_c := \psi_\zeta(c\,\cdot)$ for $c \in \kappa$ (Theorem 6.2(a)), a character ψ_c is invariant under the Frobenius self-embedding of $\kappa = \mathbb{F}_{p^m}$ if and only if $c \in \mathbb{F}_p$; and for $\kappa = \mathbb{F}_{p^p}$, every $\mathbb{F}_p$-valued c gives a character ψ_c that restricts* trivially *to the prime field $\mathbb{F}_p \subset \mathbb{F}_{p^p}$. The smallest instance is $\mathbb{F}_2 \subset \mathbb{F}_4$.*

Scope. This theorem exhibits the precise obstruction at the specific extension degree $m = p$; it does not enumerate every extension degree at which some weaker compatibility fails, and it does not address (this is Theorem 8.5) which *restricted* systems of compatible characters — over a sub-poset of embeddings, or with the "for every embedding" requirement relaxed — do exist.

Provenance. ids: D3, WH-FUNCT-a, WH-FUNCT-c; proved in: theory/wh-kappa-choice.md §10 *step* $\langle 1 \rangle 3$; tested by: theory/checks/funct_sections.py (F2,F3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Frobenius-invariance forces $c \in \mathbb{F}_p$. Write $d := c^{1/p}$ (the unique p-th root, Frobenius being bijective on the finite field κ). Since $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(y^p) = \mathrm{Tr}_{\kappa/\mathbb{F}_p}(y)$ for every y (Frobenius cyclically permutes the m summands of the trace, using $y^{p^m} = y$), we get $\psi_c(x^p) = \zeta^{\mathrm{Tr}(cx^p)} = \zeta^{\mathrm{Tr}((dx)^p)} = \zeta^{\mathrm{Tr}(dx)} = \psi_d(x)$, i.e. $\psi_c \circ \mathrm{Frob} = \psi_d$. Since $c \mapsto \psi_c$ is injective on κ (if $\psi_c = \psi_{c'}$ then $\mathrm{Tr}((c - c')x) \equiv 0$ for all x, forcing $c = c'$ by the same nondegeneracy of the trace pairing used in Theorem 6.2(a)'s proof), ψ_c is Frobenius-invariant iff $d = c$; since $d^p = c$ by definition of d, substituting $d = c$ gives $c^p = c$, and conversely $c^p = c$ gives $d = c^{1/p} = c$ (Frobenius being bijective). So the condition is exactly $c^p = c$, i.e. c is a root of $X^p - X$, i.e. (this polynomial having at most p roots in a field, all p elements of $\mathbb{F}_p$ among them by Fermat's little theorem for finite fields) $c \in \mathbb{F}_p$.

Trivial restriction at $\kappa = \mathbb{F}_{p^p}$. Here $m = p$. For $c \in \mathbb{F}_p$ and $x \in \mathbb{F}_p \subset \kappa$, every summand of $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(cx) = \sum_{k=0}^{p-1}(cx)^{p^k}$ equals cx (as $cx \in \mathbb{F}_p$ is already Frobenius-fixed), so $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(cx) = p \cdot (cx) = 0$ in characteristic p. So $\psi_c(x) = \zeta^0 = 1$ for every $x \in \mathbb{F}_p$: *every* Frobenius-invariant character of $\mathbb{F}_{p^p}$ — the only kind a section could assign there — restricts trivially to the prime field, contradicting the requirement $\psi_{\mathbb{F}_p} \in X(\mathbb{F}_p)$ (nontrivial by Definition 2.3) forced by compatibility along the inclusion $\mathbb{F}_p \subset \mathbb{F}_{p^p}$. At $p = 2$, $\mathbb{F}_{p^p} = \mathbb{F}_4$: the smallest instance. □

8.2 The compatible systems that do exist

PROVED

Theorem 8.5 (Compatible character systems over an algebraic closure)**.** *Fix an algebraic closure $\overline{\mathbb{F}}_p$, and consider the inclusion poset of its finite subfields. The compatible systems $(\psi_\kappa)_\kappa$ — one $\psi_\kappa \in X(\kappa)$ for every finite subfield $\kappa \subset \overline{\mathbb{F}}_p$, with $\psi_{\kappa'}|_\kappa = \psi_\kappa$ for every inclusion $\kappa \subseteq \kappa'$ of finite subfields — are **exactly** the restrictions $\psi_\kappa := \Psi|_\kappa$ of the additive characters Ψ of $(\overline{\mathbb{F}}_p, +)$ with $\Psi|_{\mathbb{F}_p} \neq 1$.*

Scope. This theorem restricts attention to the inclusion poset of finite subfields of a *fixed* algebraic closure, respecting compatibility only along inclusions of subfields, not along all abstract field embeddings between isomorphic-but-not-identical finite fields; Theorem 8.3 shows the latter, stronger requirement admits no solution at all. It says nothing about compatible choices of a polarizing cocycle β along the same poset; whether the Arf type of §4 can be chosen compatibly there is not resolved by this labbook.

Provenance. ids: D3, WH-FUNCT-d; proved in: theory/wh-kappa-choice.md §10 *step* ⟨1⟩4; tested by: theory/checks/funct_sections.py (F4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Restrictions are compatible systems. Given such a Ψ, put $\psi_\kappa := \Psi|_\kappa$; compatibility along $\kappa \subseteq \kappa'$ holds by construction ($(\Psi|_{\kappa'})|_\kappa = \Psi|_\kappa$), and ψ_κ is nontrivial for *every* finite subfield κ: every such κ contains $\mathbb{F}_p$, and $\Psi|_\kappa \equiv 1$ would force $\Psi|_{\mathbb{F}_p} \equiv 1$ too, contradicting the hypothesis on Ψ.

Compatible systems arise this way. Given a compatible system (ψ_κ), define $\Psi(x) := \psi_\kappa(x)$ for any finite subfield $\kappa \ni x$ (every $x \in \overline{\mathbb{F}}_p$ lies in some finite subfield, $\overline{\mathbb{F}}_p$ being their union). This is well defined: if x lies in two finite subfields κ, κ', both lie inside a common finite subfield κ'' (e.g. their compositum, again finite), and compatibility along $\kappa \subseteq \kappa''$ and $\kappa' \subseteq \kappa''$ gives $\psi_\kappa(x) = \psi_{\kappa''}(x) = \psi_{\kappa'}(x)$. Ψ is additive: any two elements lie in a common finite subfield, where Ψ agrees with the additive character ψ_κ of that subfield. And $\Psi|_{\mathbb{F}_p} = \psi_{\mathbb{F}_p} \neq 1$, by Definition 2.3's requirement on every $\psi_\kappa \in X(\kappa)$ applied to $\kappa = \mathbb{F}_p$.

The two constructions are mutually inverse, and the condition "nontrivial on every finite subfield" (needed termwise, since each $\psi_\kappa \in X(\kappa)$ by hypothesis) is equivalent to "nontrivial on the prime field alone": the forward implication is immediate ($\mathbb{F}_p$ is itself one of the subfields), and the reverse is exactly what the first paragraph showed. □

Remark (What this section does not claim)*.* Theorem 8.2 is functoriality of the *algebra* in character-compatible embeddings, for the fixed reference cocycle β_0 — not functoriality of the quantum system (Heisenberg group, Hilbert space) itself, and not a claim that any compatible choice of polarizing cocycle along embeddings exists. Whether the Arf type of Theorem 4.7 can be chosen compatibly along field embeddings at $p = 2$ is recorded as an open question by this campaign's source shard, not resolved here.

9 Weyl–Heisenberg quantization over finite local rings

This section records the first extension from finite fields to finite commutative local rings. In the preferred three-stage pipeline (P1), the ring first produces the symplectic module R^2; a character and a polarizing cocycle then produce its Weyl algebra and Heisenberg group; finally the irreducible unitary representations are classified. The decisive condition is that the chosen character generate the character module. At odd residue characteristic this gives a single irreducible representation of dimension $|R|$.

There is a genuine fork at the first stage. Route A, used here, takes $R \oplus R$ with an R-valued alternating form and then composes with a chosen additive character. Route B would take $R \oplus \widehat{R}$ with its canonical circle-valued Pontryagin pairing; it needs no character to become nondegenerate. Route B and the comparison of the physics retained by the two routes are deferred; no result below ranks the routes.

9.1 The local datum and its phase space

Definition 9.1 (Finite commutative local ring datum)**.** A finite local ring datum is $(R, \mathfrak{m}, \kappa, q)$ with R a finite commutative unital local ring with $1 \neq 0$, $\mathfrak{m}$ its unique maximal ideal, $\kappa := R/\mathfrak{m}$, and $q := |\kappa|$; write $R^\times$ for its unit group. For an ideal $I \subseteq R$, put

$$\mathrm{Ann}(I) := \{r \in R : rI = 0\}, \qquad \mathrm{soc}(R) := \mathrm{Ann}(\mathfrak{m}).$$

These are stipulations. Nilpotence of $\mathfrak{m}$, nonvanishing of the socle, and the other finite-local structure properties are established below, not assumed here. The definition is uniform in the residue characteristic.

Scope. Only finite commutative unital local rings with $1 \neq 0$ are defined here. No Frobenius, characteristic, principal-ideal, or chain-ring hypothesis is part of the datum.

Provenance. ids: D12; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.2 (Characters and generating characters)**.** Let

$$\widehat{R} := \mathrm{Hom}((R, +), \mathbb{C}^\times), \qquad X(R) := \widehat{R} \setminus \{1\}.$$

For $\psi \in X(R)$ and $u \in R$, put $\psi_u(x) := \psi(ux)$. Define

$$I_\psi := \sum \{I \subseteq R : I \text{ is an ideal and } I \subseteq \ker \psi\}, \qquad \mathrm{Gen}(R) := \{\psi \in X(R) : I_\psi = 0\}.$$

Thus I_ψ is the largest ideal contained in $\ker \psi$, and a member of $\mathrm{Gen}(R)$ is a *generating character*. This definition is uniform in the residue characteristic; existence is not stipulated.

Scope. The definition includes every nontrivial additive character, not only a trace-normalized family. It makes no assertion that generating characters exist for an arbitrary finite local ring.

Provenance. ids: D13; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.3 (Symplectic module and phase perpendicularity)**.** Put $V(R) := R \oplus R$ and

$$\omega\big((a, b), (a', b')\big) := ab' - a'b.$$

Given $\psi \in X(R)$, write

$$B_\psi(v, w) := \psi(\omega(v, w)), \qquad L^{\perp_\psi} := \{v \in V(R) : B_\psi(v, l) = 1 \text{ for every } l \in L\}$$

for an R-submodule $L \subseteq V(R)$. A *Lagrangian* is an R-submodule satisfying $L = L^{\perp_\psi}$, and $\mathrm{rad}(B_\psi) := V(R)^{\perp_\psi}$. Bilinearity, strong alternation, R-nondegeneracy of ω, and nondegeneracy of B_ψ are claims, not stipulations; the character-valued pairing B_ψ, not R-nondegeneracy alone, is load-bearing. This definition is uniform in the residue characteristic.

Scope. Perpendicularity is defined only for R-submodules of R^2 and is measured through the chosen character. No freeness condition is imposed on a Lagrangian.

Provenance. ids: D14; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.4 (Admissible polarizing cocycles over a local ring)**.** For the symplectic module above, set

$$\begin{aligned} \mathrm{Adm}(\omega) &:= \{\beta : V(R) \times V(R) \to R : \beta \text{ is } R\text{-bilinear and } \beta - \beta^T = \omega\}, \\ \mathrm{Sym}_R(V(R)) &:= \{s : V(R) \times V(R) \to R : s \text{ is } R\text{-bilinear and } s = s^T\}. \end{aligned}$$

A *polarizing cocycle* is a named choice $\beta \in \mathrm{Adm}(\omega)$. The reference member is

$$\beta_0\big((a,b),(a',b')\big) := ab'.$$

This is the nonsymmetrized convention: no inverse of 2 is assumed. The definition is uniform in the residue characteristic.

Scope. The reference member is a stipulated coordinate formula, not a claim that the torsor has a canonical point. The symmetrized member $\omega/2$ is used only under the explicit hypothesis $2 \in R^\times$.

Provenance. ids: D15; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.5 (Local Weyl algebra, Heisenberg group, and models)**.** For the data above define

$$A_{\psi,\beta}(V(R)) := \bigoplus_{v \in V(R)} \mathbb{C}\, W_\beta(v), \qquad W_\beta(v)W_\beta(v') := \psi(\beta(v,v'))W_\beta(v+v'),$$

and

$$H_\beta(R) := R \times V(R), \qquad (t,v)(t',v') := (t + t' + \beta(v,v'), v + v').$$

For an R-submodule L, put $A_L := \mathrm{span}_{\mathbb{C}}\{W_\beta(l) : l \in L\}$. Given a unital algebra character $\chi : A_L \to \mathbb{C}$, put

$$M_{L,\chi} := A_{\psi,\beta}(V(R)) \otimes_{A_L} \mathbb{C}_\chi.$$

The fixed reference model is $\ell^2(R) = \bigoplus_{y \in R} \mathbb{C}e_y$, with $\{e_y\}$ orthonormal,

$$X(a)e_y := e_{y+a}, \qquad Z(b)e_y := \psi(by),$$

and, exactly,

$$W_{\beta_0}(a,b) := Z(-b)X(a), \qquad W_{\beta_0}(a,b)e_y = \psi(-b(y+a))e_{y+a}.$$

The sign and ordering are stipulations. Every displayed formula is uniform in the residue characteristic and uses no half.

Scope. The algebra and group depend on the named cocycle, while the algebra and representation also depend on the named character. A submodule and its character label a realization; neither is added to the underlying algebra datum.

Provenance. ids: D16; proved in: —; tested by: theory/checks/fcr_local_check.py (G4); reviewed in: theory/verdicts/fcr-local-r1.md.

9.2 Generating characters and the collapse boundary

PROVED

Theorem 9.6 (Local Frobenius criterion and the unit torsor). *Let R be a finite commutative local ring with maximal ideal $\mathfrak{m}$ and residue field κ. Then*

$$\mathrm{Gen}(R) \neq \varnothing \quad \Longleftrightarrow \quad \dim_\kappa \mathrm{soc}(R) = 1.$$

If $\psi \in \mathrm{Gen}(R)$, then ψ_u is generating exactly when $u \in R^\times$, and $R^\times$ acts freely and transitively on $\mathrm{Gen}(R)$.

Scope. No odd-characteristic hypothesis is used. For rings whose socle has κ-dimension greater than one, the theorem asserts nonexistence of a generating character; it does not propose a replacement quantization.

Provenance. ids: D12, D13, FCR-GEN; proved in: theory/fcr-local.md §1; tested by: theory/checks/fcr_local_check.py (G1,G8); reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Finiteness makes R Artinian, so $\mathfrak{m}$ is nilpotent. Wood's module socle $S({}_RR)$ equals $\mathrm{Ann}(\mathfrak{m})$: every simple ideal J has $\mathfrak{m}J = 0$, since $\mathfrak{m}J = J$ would contradict nilpotence; conversely, if $0 \neq x \in \mathrm{Ann}(\mathfrak{m})$, then $Rx = \kappa x$ is one-dimensional over κ and is simple. Wood's local Frobenius condition is therefore $\kappa \cong \mathrm{soc}(R)$.

Wood's character-module theorem and kernel-ideal criterion identify this condition with the existence of ψ whose kernel contains no nonzero ideal, namely $I_\psi = 0$. For such a ψ, multiplication by a unit preserves $I_\psi = 0$, whereas a nonunit lies in $\mathfrak{m}$ and kills the nonzero socle. The character-module isomorphism then shows that every generating character is uniquely ψ_u for a unit u. □

PROVED

Theorem 9.7 (Radical of the character-valued symplectic pairing). *Let R be a finite commutative local ring and let $\psi \in X(R)$. The form ω on R^2 is R-bilinear, strongly alternating, and R-nondegenerate, while*

$$\mathrm{rad}(B_\psi) = I_\psi \oplus I_\psi.$$

Consequently B_ψ is nondegenerate if and only if $\psi \in \mathrm{Gen}(R)$.

For $R = \mathbb{F}_3[x,y]/(x,y)^2$, write $\psi_{a,b,c}(r + sx + ty) = \zeta_3^{ar+bs+ct}$. The two nontrivial characters with $(b,c) = (0,0)$ have $I_\psi = (x,y)$, and each of the remaining 24 has $I_\psi = \mathbb{F}_3(cx - by)$. Thus all 26 nontrivial characters collapse the pairing through a proper quotient.

Scope. The radical formula is uniform in the residue characteristic. The square-zero-ring census records the obstruction at one designated seed only; it does not construct a quantum system for a degenerate pairing and does not state a general quotient-collapse theorem.

Provenance. ids: D12, D13, D14, FCR-RAD; proved in: theory/fcr-local.md §2; tested by: theory/checks/fcr_local_check.py (G2,G3); exhaustive 26-character seed census in G1; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Testing $\omega((a,b),-)$ against $(0,r)$ and $(r,0)$ shows that a radical vector must have $aR, bR \subseteq \ker\psi$, hence $a, b \in I_\psi$. The reverse inclusion follows because I_ψ is an ideal. Bilinearity and strong alternation are immediate from the displayed determinant formula, and tests against $(0,1)$ and $(1,0)$ prove R-nondegeneracy. In the square-zero example every subspace of (x,y) is an ideal; restricting the character to that two-dimensional space gives exactly the displayed kernel line, or all of (x,y) when the restriction vanishes. □

9.3 Odd residue characteristic: algebra and representation

PROVED

Theorem 9.8 (Weyl commutation over a finite local ring)**.** *Let R be a finite commutative local ring, let $\psi \in X(R)$, and let $\beta \in \mathrm{Adm}(\omega)$. For every $v, v' \in V(R)$,*

$$W_\beta(v)W_\beta(v') = \psi(\omega(v,v'))W_\beta(v')W_\beta(v),$$

and

$$W_\beta(u)W_\beta(v)W_\beta(u)^{-1} = \psi(\omega(u,v))W_\beta(v).$$

Scope. The identities are uniform in the residue characteristic and do not require ψ to be generating. They do not by themselves imply simplicity of the Weyl algebra.

Provenance. ids: D14, D15, D16, FCR-COMM; proved in: theory/fcr-local.md §3; tested by: theory/checks/fcr_local_check.py (G2,G4); reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. The two products have the same basis term $W_\beta(v+v')$. Their scalar ratio is $\psi(\beta(v,v') - \beta(v',v)) = \psi(\omega(v,v'))$; conjugation gives the second identity. □

PROVED†

Theorem 9.9 (Odd-characteristic transport of polarizing cocycles)**.** *Let R be a finite commutative local ring with $2 \in R^\times$. Then $\mathrm{Adm}(\omega)$ is a torsor under $\mathrm{Sym}_R(V(R))$ and has $|R|^3$ elements. Its unique antisymmetric member is $\omega/2$. For every $\beta \in \mathrm{Adm}(\omega)$ and every symmetric s, the map*

$$\varphi_s(t,v) := \left(t + \frac{s(v,v)}{2}, v\right)$$

is an isomorphism $H_\beta(R) \to H_{\beta+s}(R)$ fixing the centre pointwise.

Scope. The condition $2 \in R^\times$ is essential to the displayed representative and transport. No symmetry group of the ring-side phase space is analyzed here.

Provenance. ids: D14, D15, D16, FCR-BETA-ODD; proved in: theory/fcr-local.md §4; tested by: theory/checks/fcr_local_check.py (G9); reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. The difference of two admissible forms is symmetric, and adding a symmetric form preserves admissibility. A symmetric form on R^2 is freely determined by three entries. Antisymmetry and admissibility give $2\beta = \omega$, hence the unique member $\omega/2$. Finally, for $f(v) = s(v,v)/2$, symmetry gives $f(v+w) = f(v) + f(w) + s(v,w)$, which is exactly the coboundary identity needed for φ_s to respect the two group laws. □

PROVED†

Theorem 9.10 (Central simplicity and finite Stone–von Neumann theorem)**.** *Let R be a finite commutative local ring, let $\psi \in \mathrm{Gen}(R)$, let $\beta \in \mathrm{Adm}(\omega)$, and suppose $2 \in R^\times$. Then $A_{\psi,\beta}(V(R))$ is simple, has centre $\mathbb{C}1$, has complex dimension $|R|^2$, and is noncanonically isomorphic to $M_{|R|}(\mathbb{C})$. The group $H_\beta(R)$ has exactly one irreducible unitary representation with central character ψ, up to unitary equivalence; its dimension is $|R|$, unitary intertwiners are unique up to $U(1)$, and every complex algebra automorphism of the Weyl algebra is inner.*

Scope. The condition $2 \in R^\times$ is used to transport an arbitrary cocycle to the fixed matrix model. The matrix isomorphism depends on choices of realization and basis and is not part of the algebra datum. No frame-preserving automorphism group is computed.

Provenance. ids: D12–D16, FCR-ALG, FCR-SVN; proved in: theory/fcr-local.md §5–§6; tested by: theory/checks/fcr_local_check.py (G4,G5,G6), including the G6 arena mutation; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Averaging conjugation by all Weyl basis elements kills every nonzero basis coefficient because the pairing is nondegenerate. It follows that the centre is scalar and that every nonzero two-sided ideal contains 1. The reference operators $Z(-b)X(a)$ give a nonzero representation on $\ell^2(R)$; simplicity makes it injective and the equal dimensions make it onto $\mathrm{End}_{\mathbb{C}}(\ell^2(R))$. The preceding transport handles arbitrary β.

Modules with central character ψ are precisely modules over this full matrix algebra. Its column module is the unique simple module and has dimension $|R|$. Schur's argument makes intertwiners scalar; polar rescaling gives the $U(1)$ statement. Twisting the column module by an algebra automorphism shows that the automorphism is conjugation by an invertible operator. □

PROVED †

Theorem 9.11 (Lagrangians and Schrödinger models over a local ring)**.** *Let R be a finite commutative local ring, let $\psi \in \mathrm{Gen}(R)$, let $\beta \in \mathrm{Adm}(\omega)$, and suppose $2 \in R^\times$. Every Lagrangian $L \subseteq V(R)$ has $|L| = |R|$. The coordinate axes are Lagrangian, and for every ideal $I \subseteq R$ so is*

$$I \oplus \mathrm{Ann}(I).$$

If R is not a field, then

$$L_* := \mathrm{soc}(R) \oplus \mathfrak{m}$$

is a nonfree Lagrangian. Every algebra character $\chi : A_L \to \mathbb{C}$ produces a simple induced module $M_{L,\chi}$ of dimension $|R|$, and all these modules are isomorphic.

Scope. This is a catalogue containing a general ideal family, not a classification of all Lagrangians over every local ring. In particular, it does not claim that every Lagrangian is free or belongs to the displayed ideal family.

Provenance. ids: D12–D16, FCR-POL; proved in: theory/fcr-local.md §7; tested by: theory/checks/fcr_local_check.py (G7), full submodule census at the two order-nine seeds; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Character orthogonality in the two orders gives $|L|\,|L^{\perp_\psi}| = |R|^2$, so a self-perpendicular module has order $|R|$. Annihilator duality gives $\mathrm{Ann}(\mathrm{Ann}(I)) = I$, hence the ideal family. For a nonfield local ring, $\mathrm{Ann}(\mathfrak{m}) = \mathrm{soc}(R)$ and $\mathrm{Ann}(\mathrm{soc}(R)) = \mathfrak{m}$, so L_* belongs to that family. Its quotient by $\mathfrak{m}L_*$ needs at least two generators, whereas a free module of order $|R|$ would have rank one.

The restriction of β to a Lagrangian is symmetric, and $\chi_0(W(l)) = \psi(-\beta(l,l)/2)$ is an algebra character. The Weyl basis makes the algebra right-free over A_L on representatives for $V(R)/L$, so the induced module has dimension $|R|$; uniqueness of the simple module then proves that all labels give isomorphic realizations. □

PROVED †

Proposition 9.12 (The two presentation data and bare-algebra ambiguity)**.** *Let R be a finite commutative local ring with $2 \in R^\times$ and $\mathrm{Gen}(R) \neq \varnothing$. Beyond R, the presentation uses exactly a generating character $\psi \in \mathrm{Gen}(R)$ and a cocycle $\beta \in \mathrm{Adm}(\omega)$. Distinct generating characters give inequivalent representations of the same group $H_\beta(R)$ because their central characters differ. All resulting bare algebras are isomorphic, but the torsor of bare-algebra isomorphisms has no point fixed by every target automorphism.*

Scope. The last assertion concerns all bare-algebra isomorphisms. It neither computes nor identifies a frame-preserving isomorphism torsor and makes no claim about a ring-side Weil or metaplectic symmetry.

Provenance. ids: D12–D16, FCR-CHOICE; proved in: theory/fcr-local.md §8; tested by: theory/checks/fcr_local_check.py (G8,G9); reviewed in: theory/verdicts/fcr-local-r1.md.

9.4 Clause-level return to an odd finite field

PROVED †

Theorem 9.13 (Clause-level compatibility with the odd-field system)**.** *Let $R = \kappa$ be a finite field of odd characteristic and put $q := |\kappa|$. Specialization recovers on the nose:*

(i) the nontrivial-character set and its $\kappa^\times$-torsor of order $q-1$;

(ii) bilinearity, strong alternation, and nondegeneracy of ω, together with the zero radical of $\psi \circ \omega$;

(iii) the Weyl commutator and conjugation identities;

(iv) simplicity, scalar centre, dimension q^2, and a noncanonical $M_q(\mathbb{C})$ realization;

(v) the unique irreducible unitary representation of dimension q, its $U(1)$-unique intertwiners, and innerness of algebra automorphisms;

(vi) exactly $q+1$ Lagrangian lines, q algebra characters per line, $q(q+1)$ model labels, and one module-isomorphism class;

(vii) the torsor of admissible cocycles of order q^3, its unique antisymmetric member $\omega/2$, and the centre-fixing transports;

(viii) the two named presentation data, the distinct-central-character obstruction, and isomorphism of the resulting bare algebras.

The specialized definitions give the same $V(\kappa)$, form, admissibility equation, Weyl and group laws, and fixed operator $W_{\beta_0} = Z(-b)X(a)$ with the same sign.

Scope. This is deliberately a clause-level comparison. It does not compare the field theory's frame-preserving isomorphism torsor of order $q^2|SL_2(\kappa)|$; the ring-side frame-preserving analysis is future work. It also does not compare the field theorem's cocycle or automorphism-group clauses, the $SL_2(\kappa)$-invariance of $\omega/2$, or the particular polarization-dependent matrix construction used in the field proof.

Provenance. ids: D12–D16, FCR-GEN, FCR-RAD, FCR-COMM, FCR-BETA-ODD, FCR-ALG, FCR-SVN, FCR-POL, FCR-CHOICE, FCR-REG; proved in: theory/fcr-local.md §9; tested by: theory/checks/fcr_local_check.py (G10), exact counts, form and radical tables, fixed matrices, rank, commutant, lines, half-census, and group profile; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. For a field, $\mathfrak{m} = 0$, $\mathrm{soc}(R) = R$, and the only ideals are 0 and R, so every nontrivial character is generating. Substitution in the preceding results proves every listed clause. A self-perpendicular subspace of the nondegenerate two-dimensional space is a line; nonzero vectors modulo $\kappa^\times$ give $(q^2-1)/(q-1) = q+1$ lines, and each line has q additive characters. No excluded symmetry or frame clause follows from these calculations. □

10 The five smallest local rings, worked in full

The three smallest orders of finite commutative local rings are 2, 3, and 4, and order 4 is a three-way tie. This section exercises every definition of Section 9 on the resulting five rings

$$\mathbb{F}_2, \qquad \mathbb{F}_3, \qquad \mathbb{F}_4, \qquad \mathbb{Z}/4, \qquad \mathbb{F}_2[\varepsilon]/(\varepsilon^2),$$

computing their Weyl–Heisenberg groups and complete irreducible representation catalogues. Every number in this section is verified by the exact-arithmetic census checker `theory/checks/small_rings_cat` written and run by an independent verifier lane that recomputed the orchestrator's catalogue blind; the checker runs green and carries eight mutation modes, each of which fails at its registered gate. Statements here are census facts about these five rings; no generality beyond them is claimed except where a claim identifier is cited.

10.1 The data, ring by ring

All five rings are Frobenius: the socle is one-dimensional over the residue field, so generating characters exist and form a single free orbit of the unit group (the existence criterion and torsor structure of Section 9).

R	$\mathfrak{m}$	κ	$\operatorname{soc}(R)$	$\lvert R^\times\rvert = \lvert\operatorname{Gen}(R)\rvert$	reference ψ
$\mathbb{F}_2$	0	$\mathbb{F}_2$	R	1	$(-1)^x$
$\mathbb{F}_3$	0	$\mathbb{F}_3$	R	2	ζ_3^x
$\mathbb{F}_4$	0	$\mathbb{F}_4$	R	3	$(-1)^{\operatorname{Tr}(x)}$
$\mathbb{Z}/4$	(2)	$\mathbb{F}_2$	(2)	2	i^x
$\mathbb{F}_2[\varepsilon]$	(ε)	$\mathbb{F}_2$	(ε)	2	$(-1)^{a+b}$ for $x = a + b\varepsilon$

Provenance. ids: FCR-GEN; proved in: theory/fcr-local.md §1; tested by: small_rings_catalogue_check.py (ring-data gates); reviewed in: theory/verdicts/smallest-rings-verification.md.

10.2 The Heisenberg groups are unitriangular matrix groups

Proposition 10.1 (Matrix form of the reference-cocycle Heisenberg group)**.** *For every finite commutative unital ring R, the map*

$$(t, (a, b)) \longmapsto \begin{pmatrix} 1 & a & t \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

is an isomorphism from the Heisenberg group of the reference cocycle $\beta_0((a, b), (a', b')) = ab'$ onto the unitriangular group $\mathrm{UT}_3(R)$.

Proof. Multiplying the two displayed matrices gives upper-right entry $t+t'+ab'$, which is exactly the group law of the Heisenberg group of β_0; the map is a bijection by construction. □

Scope. The identification is specific to the reference cocycle: at residue characteristic 2 other admissible cocycles can give non-isomorphic groups (the quaternion type over $\mathbb{F}_2$ below), and those are not unitriangular in this way.

Provenance. ids: D16; proved in: this section; tested by: small_rings_catalogue_check.py (cocycle gate); reviewed in: theory/verdicts/smallest-rings-verification.md.

The five groups, exactly:

R	$\lvert H\rvert$	centre	exponent	order profile	identification
$\mathbb{F}_2$	8	$\mathbb{Z}/2$	4	1:1, 2:5, 4:2	dihedral D_4
$\mathbb{F}_3$	27	$\mathbb{Z}/3$	3	1:1, 3:26	extraspecial 3^{1+2}_+
$\mathbb{F}_4$	64	$(\mathbb{Z}/2)^2$	4	1:1, 2:27, 4:36	characteristic-two Heisenberg
$\mathbb{Z}/4$	64	$\mathbb{Z}/4$	8	1:1, 2:7, 4:40, 8:16	$\mathrm{UT}_3(\mathbb{Z}/4)$
$\mathbb{F}_2[\varepsilon]$	64	$(\mathbb{Z}/2)^2$	4	1:1, 2:31, 4:32	$\mathrm{UT}_3(\mathbb{F}_2[\varepsilon])$

The three order-64 groups are pairwise non-isomorphic: $\mathbb{Z}/4$ is separated by its cyclic centre and its sixteen elements of order 8, while $\mathbb{F}_4$ and $\mathbb{F}_2[\varepsilon]$ share centre and exponent but differ in involution count. The order-8 elements over $\mathbb{Z}/4$ come from the square law $(t, v)^2 = (2t + \beta_0(v, v),\ 2v)$: the fourth power is $(2\beta_0(v, v), 0)$, nonzero exactly when ab is odd. Over $\mathbb{F}_2[\varepsilon]$ the identity $2v = 0$ kills this mechanism: the jet direction deepens the ring without deepening the clock. Mixed versus equal characteristic is thus visible in the bare group.

Remark. At $\mathbb{F}_3$ the group is independent of the cocycle choice (the odd-characteristic transport of Section 9). At $\mathbb{F}_2$ the eight admissible cocycles split six-to-two into a dihedral type and a quaternion type, separated by the Arf invariant of $Q_\beta(v) = \beta(v, v)$; the reference cocycle is of dihedral type. For $\mathbb{Z}/4$ and $\mathbb{F}_2[\varepsilon]$ the classification of the 64-element cocycle torsor is open and is the target of the running residue-characteristic-two increment.

Provenance. ids: WH-BETA-d, FCR-BETA-ODD; proved in: theory/wh-kappa-choice.md §6; theory/fcr-local.md §4; tested by: beta_census.py; small_rings_catalogue_check.py; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

10.3 The complete representation catalogues

The commutator subgroup is the full centre, so the abelianization is the phase module $R \oplus R$, and irreducible representations are stratified by the character of the centre, $\psi_u = \psi(u\,\cdot)$ for $u \in R$.

Proposition 10.2 (Catalogue census for the five smallest local rings)**.** *For each of the five rings above and each $u \in R$, the number of irreducible unitary representations of the Heisenberg group with central character ψ_u is $\lvert \operatorname{Ann}(u)\rvert^2$, and each has dimension $\lvert R\rvert/\lvert \operatorname{Ann}(u)\rvert$. In particular the total count of irreducibles equals the number of conjugacy classes,*

$$\#\{\textit{classes}\} = \lvert R\rvert^2 + \sum_{u\neq 0} \lvert \operatorname{Ann}(u)\rvert^2 \qquad (5,\ 11,\ 19,\ 22,\ 22\ \textit{respectively}),$$

and the sum of squared dimensions equals $\lvert R\rvert^3$.

Scope. A census statement about these five rings only, established by exhaustive computation; the same stratification pattern for a general finite commutative local ring is expected but not claimed here — its general proof belongs to the planned non-Frobenius/collapse increment.

Provenance. ids: (census); proved in: small_rings_catalogue_check.py (class-count and catalogue gates); tested by: same checker, mutation modes red-class-count, red-catalogue; reviewed in: theory/verdicts/smallest-rings-verification.md.

The three strata carry distinct physics:

- *Bottom, $u = 0$:* the $\lvert R\rvert^2$ characters of the abelianization — the classical shadow.
- *Top, $u \in R^\times$:* one Schrödinger representation per generating character, of dimension $\lvert R\rvert$, in the fixed model $W(a, b) = Z(-b)X(a)$ on $\ell^2(R)$: the qubit Pauli pair for $\mathbb{F}_2$; the qutrit clock–shift for $\mathbb{F}_3$; two $\mathbb{F}_4$-labelled qubits (three inequivalent copies, one per generating character); the ququart clock–shift $Z = \mathrm{diag}(1, i, -1, -i)$, $ZX = iXZ$ for $\mathbb{Z}/4$ (two copies); and the jet-paired four-dimensional representation for $\mathbb{F}_2[\varepsilon]$ (two copies). Irreducibility is pinned by exact trace-orthogonality of the $\lvert R\rvert^2$ Weyl operators.

- *Middle,* $u \in \mathfrak{m} \setminus 0$ (only the two thickened rings): four two-dimensional representations each, with central character supported on the socle. These are qubit representations of the collapsed quotient $R/\operatorname{Ann}(u) \cong \mathbb{F}_2$ twisted by characters of the radical of the degenerate phase pairing. The verifier's decomposition of the four-dimensional model with this central character found exactly two inequivalent two-dimensional blocks, each of multiplicity one — the model contains two of the four classes.

A ring is thickened exactly when its catalogue has a middle stratum: the representation theory detects the nilpotents. Two honest cautions from the blind verification: $\mathbb{F}_4$ has *three* top-stratum representations, not two; and as bare dimension–multiplicity tables the $\mathbb{Z}/4$ and $\mathbb{F}_2[\varepsilon]$ catalogues coincide ($16 \times \dim 1$, $4 \times \dim 2$, $2 \times \dim 4$) — the catalogue separates the two rings only through its central-character labelling and the underlying groups, not through the numbers alone.

10.4 Lagrangian labels and models

R	Lagrangians	free	non-free	model labels
$\mathbb{F}_2$	3	3	0	6
$\mathbb{F}_3$	4	4	0	12
$\mathbb{F}_4$	5	5	0	20
$\mathbb{Z}/4$	7	6	1	28
$\mathbb{F}_2[\varepsilon]$	7	6	1	28

The fields show the $q+1$ lines of the projective line; the thickened rings show the six free lines of $\mathbb{P}^1(R)$ plus the non-free witness $\operatorname{soc}(R) \oplus \mathfrak{m}$ of Section 9. All labels with a fixed generating central character present one representation, but the non-free label is a genuinely different presentation: position and momentum are each sampled at depth $\mathfrak{m}$, a mixed frame with no field analogue.

Provenance. ids: FCR-POL; proved in: theory/fcr-local.md §7; tested by: small_rings_catalogue_check.py (Lagrangian gate, mutation mode red-drop-nonfree); reviewed in: theory/verdicts/fcr-local-adjudication.md.

11 Sidequest: quantum mechanics over the field with one element

Opened 6 September 2026. This is a separate, literature-grounded exploration of arithmetic quantum mechanics over $\mathbb{F}_1$, also written $\mathbb{F}_{\mathrm{un}}$. Its north star is a precisely formulated quantum theory, with particular attention to the analogue of a qubit and to composition. The different established analogies are retained as separate candidates.

North-star clarification, 7 September 2026. The primary object is the family of subsystems and their assembly rules. An acceptable endpoint must admit a quantum operational realization by C*-algebras, normalized positive functionals or density operators, CP dynamics and Born probabilities. The proposed strategy extracts a compositional category that remembers the prime before formulating its field-with-one-element counterpart. Fibonacci is a test of the kind of composition sought, not an asserted endpoint. A strong monoidal fibre functor to ordinary Hilbert spaces is not a universal requirement: composite observable algebras may contain additional fusion observables. Compatible preparations and extensions of channels to composites are part of the target. Classical or combinatorial state sets alone do not satisfy this operational criterion.

The accompanying research map is `docs/sidequests/f1-qm.md`; primary source bodies, versions and retrieval hashes are indexed in `refs/LEDGER.md`. The finite-ring campaign is not superseded.

There is substantial relevant literature, but the constructions preserve different structures. A field may specify the arithmetic base, the state coefficients, a counting parameter, or a deformation parameter. Confusing these roles would make the comparison misleading. In the present campaign, the arithmetic base is a finite field or ring and the amplitudes remain complex. The direct "quantum $\mathbb{F}_{\mathrm{un}}$" papers instead change the coefficient object of the states themselves.

11.1 What could count as a qubit?

Two distinguishable states, a two-dimensional Hilbert space, a simple object of categorical dimension two, and two simple objects of a category are different notions. The following table makes the distinction explicit.

Approach	Qubit analogue	Composition
Pointed $\mathbb{F}_1$ modules	A basepoint and two basis points. Complex linearization of normal partial maps gives $(\mathbb{C}^2, M_2(\mathbb{C}))$.	Smash product has four nonzero basis points and realizes as the Hilbert tensor product. Native basis states have no superpositions.
Cyclotomic modules	Two free μ_N-orbits. For configuration C_2 and phase level two, clock and shift give the Heisenberg qubit.	Balanced smash of modules; central product of Heisenberg groups; ordinary tensor after complex realization.
Absolute Cartesian frames	Rank two has $N+2$ projective rays. At bare $\mathbb{F}_1$ these are $10, 01, 11$.	The rank-four frame has fifteen rays at $N=1$: nine product rays and six non-product rays. The operation on pairs is the Kronecker rule.
Thin symplectic geometry	Two coordinate polarizations, suggestive of position and momentum. These are not two Hilbert-state vectors.	Orthogonal sums of phase geometry; Hilbert tensor requires a realization.

Approach	Qubit analogue	Composition
Blueprint Heisenberg geometry	A chosen $\mathbb{F}_2$ fibre gives the reference order-eight group and its two-dimensional central-character module.	Ring realization and phase data precede the physical tensor product.
Bands and crowds	A relational Heisenberg object, with 27 signed or eight Krasner points. Neither number is a count of qubit states.	Geometric products exist; physical composition needs a specified phase and representation construction.
Hall/groupoid oscillator	No preferred qubit in one oscillator. Two colours give a two-dimensional one-particle sector after realization.	Full Fock spaces tensor on disjoint colour sets. Two identical bosons in two modes have dimension three; two distinguishable qubits have dimension four.
Representation categories	The two-dimensional simple object of $\mathrm{Rep}(D_8)$ or $\mathrm{Rep}(Q_8)$.	Internal fusion splits its square into four charges. Independent copies instead use an external tensor and a matching-central-character quotient.
Quantum torus	$VU = -UV$, $U^2 = V^2 = 1$ gives $M_2(\mathbb{C})$.	Two selected central fibres give $M_4(\mathbb{C})$.
Arithmetic descent and Bost–Connes	A specified finite fibre or a chosen two-level encoding; no canonical qubit is supplied by the global arithmetic data.	Frobenius and transfer correspondences; tensor products of statistical systems after realization.
Tropical and motivic routes	Support or incidence information, without a specified positive two-state quantum system.	The relevant semiring, geometric or categorical tensor must be named.

11.2 The developed geometric and physical analogies

Counting and thin geometry. The projective count $1 + q + \cdots + q^{r-1}$ specializes to r at $q = 1$; Gaussian coefficients specialize to subset counts. Tits's thin geometry therefore retains coordinate points, subsets and permutation symmetries. For symplectic rank r the reflection group is $W(C_r) = (C_2)^r \rtimes S_r$, acting on symplectic pairs. A coordinate Lagrangian selects one direction from each pair, giving 2^r choices. This is the incidence content of the familiar specialization of $\prod_{i=1}^{r}(q^i + 1)$. The geometric analogies are developed in Deitmar's *Schemes over* $\mathbb{F}_1$, section 5, and Lorscheid's Tits–Weyl models. Here a *Weyl group* is a reflection group, not the group of Weyl displacement operators. Moreover, $q^3 \mapsto 1$ for the order of $\mathrm{UT}_3(\mathbb{F}_q)$ is a polynomial specialization, not a limit of groups or a proof that its $\mathbb{F}_1$ geometry is empty.

Direct quantum $\mathbb{F}_{\mathrm{un}}$. Chang, Lewis, Minic and Takeuchi construct a $q = 1$ coordinate-state specialization of finite-field amplitude models: superposition disappears. This is different from this campaign's complex Hilbert space $\ell^2(\mathbb{F}_q)$. Thas's *Absolute Quantum Theory* uses larger Cartesian frames and develops monomial dynamics and quantum-information analogies. Its Hermitian expression is partial: a sum is allowed only when at most one term is nonzero. Orthogonality is disjointness of supports. The involution determines the unitary subgroup; inversion of phases makes all phase-permutation matrices unitary in this sense. Its cloning discussion distinguishes simple projective rays, which can be copied, from all frame states. These statements do not supply an ordinary positive-definite Hilbert inner product on the frame.

An existing geometric Heisenberg model. Lorscheid's Proposition 5.5 constructs Tits–Weyl models for standard parabolic unipotent radicals of GL_n. For the upper-triangular Borel of GL_3, this is the reference Heisenberg group

$$h(a,b,t) = \begin{pmatrix} 1 & a & t \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad h(a,b,t)h(c,d,u) = h(a+c, b+d, t+u+ad).$$

Its underlying blue scheme is affine three-space. Its Weyl extension has one point, while ring base extension recovers the full unipotent group. Thus a positive $\mathbb{F}_1$ model already exists; selecting a central character and constructing its representation are further steps. The whole polarizing-cocycle torsor, especially its characteristic-two types, is not supplied by that reference model. Likewise, a Tits lift of a reflection is not automatically a metaplectic or Weil lift.

Cyclotomic conventions. For the raw phase monoid $\{0\} \cup \mu_N$, integral realization gives $\mathbb{Z}[\mu_N] \simeq \mathbb{Z}[t]/(t^N - 1)$. A cyclotomic blueprint may instead impose additive relations giving $\mathbb{Z}[\zeta_N]$. A faithful complex phase character is another named realization datum. These conventions are compared in Lorscheid's *A blueprinted view on $\mathbb{F}_1$-geometry*, section 1.1.4. The group μ_{q-1} reflects multiplicative field data; finite-field Weyl phases come from additive characters and lie in μ_p. These two phase counts must not be identified.

11.3 Pointed modules, ordinary qubits and absolute frames

Definition 11.1 (Normal pointed maps and their two products)**.** For $r \geq 0$ let $V_r = \{*, 1, \dots, r\}$. A normal pointed map preserves $*$ and has at most one preimage of each non-basepoint; it is a partial injection of the nonzero sets. Wedge identifies basepoints, while smash collapses all pairs containing a basepoint in the Cartesian product. Put $L(S) = \mathbb{C}[S \setminus \{*\}]$, with its basis orthonormal, and let $B(S)$ be the complex span of realized normal endomorphisms in $\mathrm{End}_{\mathbb{C}} L(S)$.

Scope. These objects have no native addition of vectors. The image algebra $B(S)$ is explicitly a complex realization.

Provenance. ids: D1010; proved in: definitions.md; tested by: f1_check.py (M13); reviewed in: supporting comparison.

Proposition 11.2 (The pointed-set qubit after operator realization)**.** SKETCH *For every positive-rank finite pointed set S, $B(S) = \mathrm{End}_{\mathbb{C}} L(S)$. For finite pointed sets S, T,*

$$L(S \wedge T) \simeq L(S) \otimes L(T).$$

For positive ranks this identifies the image algebras with their tensor product. In particular, V_2 realizes to $(\mathbb{C}^2, M_2(\mathbb{C}))$, and $V_2 \wedge V_2 = V_4$ realizes to two qubits.

Scope. The realization retains a distinguished basis. The concrete image algebra is not the unreduced partial-injection monoid algebra. Algebras are transported along bijections; arbitrary partial maps do not give unital maps between full endomorphism algebras.

Provenance. ids: F1-NORMAL; proved in: f1-comparisons.md, section 3; tested by: f1_check.py (M13); reviewed in: supporting SKETCH.

Proof. The partial map taking j to i and everything else to the basepoint realizes as E_{ij}. These matrix units span the full algebra, and their partial inverses realize adjoints. Smash products of these maps give $E_{ij} \otimes E_{k\ell}$. The basis identification proves the tensor claim. □

At rank two the partial-injection monoid has seven elements, including zero. Its contracted monoid algebra has dimension six, whereas its image has dimension four: linear realization imposes relations such as $I = E_{00} + E_{11}$. Keeping only the two bijections gives the commutative span of I and the swap. Thus the class of allowed morphisms matters. The pointed-module, normal-map and base-change conventions come from Szczesny, pages 5–8.

Definition 11.3 (Cyclotomic projective Cartesian frames)**.** Fix an abstract cyclic group μ_N of order $N \geq 1$ and put $F_N = \{0\} \sqcup \mu_N$, with absorbing zero. For $r \geq 1$, define

$$P_N(r) = \left(F_N^r \setminus \{(0, \dots, 0)\}\right)/\mu_N$$

by simultaneous scalar multiplication. The product-state map $P_N(r) \times P_N(s) \to P_N(rs)$ sends $([x], [y])$ to $[(x_i y_j)_{i,j}]$. Rays outside its image are called nonproduct rays in this factorization sense.

Scope. A Cartesian frame is different from a free pointed module. No probability rule or total Hermitian form is stipulated.

Provenance. ids: D1009; proved in: definitions.md; tested by: f1_check.py (M11); reviewed in: supporting comparison.

Proposition 11.4 (Frame qubits and their nonproduct rays)**.** **SKETCH** *For every $N \geq 1$,*

$$|P_N(r)| = \frac{(N+1)^r - 1}{N}.$$

The rank-two product-state map is injective, so the rank-four frame has $(N+2)^2$ product rays and $N(N+1)(N+2)$ nonproduct rays.

Scope. Nonproduct means failure of tensor factorization. It does not by itself establish a Born rule, interference or Bell correlations.

Provenance. ids: F1-FRAME; proved in: f1-comparisons.md, section 2; tested by: f1_check.py (M11); reviewed in: supporting SKETCH.

Proof. Scalar multiplication acts freely on any nonzero tuple. For a nonzero product array, ratios within a nonzero row and column recover both factors up to scalars. The ray counts then follow by subtraction. $\square$

For $N = 1$ the one-system rays are $10, 01, 11$; the composite has fifteen rays, nine product and six nonproduct. The diagonal array $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ is a nonproduct because product supports are rectangular. This larger frame is the reason a nonadditive theory can retain a factorization notion of entanglement while the coordinate-only model does not. In the strict pointed-map category, only simple vectors are morphisms from the unit; the additional frame states require an explicitly enlarged state notion.

11.4 A finite cyclotomic Weyl–Heisenberg formulation

Definition 11.5 (Cyclotomic modules and named complex realization)**.** Fix $N \geq 1$, an abstract cyclic group μ_N of order N, and a named faithful character $\iota : \mu_N \to \mathbb{C}^\times$. Let $\mathrm{Free}_*^{\mu_N}$ be finite pointed sets with a μ_N action fixing the basepoint and free on its complement; maps are pointed equivariant maps. Define

$$\mathbb{C}_\iota(S) = \mathbb{C}[S]/(e_0,\ e_{us} - \iota(u)e_s)$$

and use the induced linear maps on morphisms.

Scope. The phase embedding is part of the realization datum. The objects are free phase sets, not Cartesian frames.

Provenance. ids: D1001; proved in: definitions.md; tested by: f1_check.py; reviewed in: f1-r1.md.

Definition 11.6 (Hyperbolic phase data and the pointed Schrödinger action)**.** For a finite abelian group A with exponent dividing N, put

$$A^\vee = \mathrm{Hom}(A, \mu_N), \quad V_A = A \times A^\vee, \quad c_A((a,\chi),(b,\eta)) = \eta(a)^{-1},$$
$$\kappa_A((a,\chi),(b,\eta)) = \chi(b)\eta(a)^{-1}.$$

Define $H_N(A) = \mu_N \times V_A$ by $(u,v)(u',w) = (uu'c_A(v,w), v+w)$ and adjoin an absorbing zero to get $H_N(A)^0$. Put $S_A = \{0\} \sqcup (\mu_N \times A)$ and stipulate

$$(u,a,\chi)[z,x] = [uz\chi(x+a), x+a], \qquad 0_H s = 0_S.$$

On its realization let $X(a)e_x = e_{x+a}$, $Z(\chi)e_x = \iota(\chi(x))e_x$, and $W(a,\chi) = Z(\chi)X(a)$.

Scope. The abelian configuration group is extra data. No field structure, division by two or identification $A \simeq A^\vee$ is assumed.

Provenance. ids: D1002,D1003; proved in: definitions.md; tested by: f1_check.py (M1–M3); reviewed in: f1-r1.md.

Theorem 11.7 (The finite cyclotomic Schrödinger system)**.** **PROVED** *Fix $N \geq 1$, cyclic μ_N of order N, a faithful $\iota : \mu_N \to \mathbb{C}^\times$, and a finite abelian A of exponent dividing N. For the associated phase system, $|A^\vee| = |A|$, characters extend from subgroups and separate points, and κ_A is perfect. The displayed multiplication makes $H_N(A)$ a group with centre exactly μ_N, acting faithfully on S_A. After the named complex realization,*

$$W(a,\chi)W(b,\eta) = \iota(\eta(a)^{-1})W(a+b,\chi\eta), \qquad \mathrm{Tr}(W(v)^*W(w)) = |A|\delta_{v,w}.$$

These operators span $\mathrm{End}_{\mathbb{C}}(\mathbb{C}[A])$*. The irreducible unitary representation of $H_N(A)$ with central character ι is unique up to equivalence, has dimension $|A|$, and has intertwiners unique up to $U(1)$.*

Scope. This is finite Heisenberg theory for abelian configurations, with an explicit pointed module. A universal representation theorem for all proposed $\mathbb{F}_1$ geometries is not asserted.

Provenance. ids: F1-DUAL,F1-WEYL,F1-REAL; proved in: f1-cyclotomic.md, sections 0–2; tested by: f1_check.py (M1–M3); reviewed in: f1-adjudication.md.

Proof. Character extension follows by adjoining a cyclic generator and choosing a root in μ_N; the exponent hypothesis ensures the required root exists. Restriction and induction on the group order give $|A^\vee| = |A|$. A nontrivial character sums to zero by translating its sum. Character evaluation verifies the cocycle identity and action. Testing the two coordinate subgroups proves perfection. The trace vanishes unless both translation and character are trivial; the resulting $|A|^2$ orthogonal operators form a basis of the full matrix algebra. Matrix units then prove uniqueness of its simple module and scalar uniqueness of intertwiners. □

Definition 11.8 (Fixed-level isomorphisms and balanced composition)**.** Let $\mathrm{FinAb}_N^{\simeq}$ have the above configurations and their group isomorphisms. For $f : A \to B$, put $f_*\chi = \chi \circ f^{-1}$, $S(f)[u,x] = [u, f(x)]$ and $H(f)(u,a,\chi) = (u, f(a), f_*\chi)$. Define

$$S \wedge_{\mu_N} T = (S \wedge T)/([us,t] \sim [s,ut]), \qquad H \boxdot K = (H \times K)/\{(u,u^{-1}) : u \in \mu_N\}.$$

In the last expression the two groups have specified central copies of μ_N.

Scope. Functoriality is asserted on isomorphisms at a common phase level. Arbitrary configuration homomorphisms are not silently included.

Provenance. ids: D1004; proved in: definitions.md; tested by: f1_check.py (M6); reviewed in: f1-r1.md.

Theorem 11.9 (Composition of independent phase systems)**.** PROVED *Fix $N \geq 1$ and a faithful phase realization ι. The module, group and Hilbert assignments on* $\mathrm{FinAb}_N^{\simeq}$ *are strong symmetric monoidal, with comparisons*

$$S_A \wedge_{\mu_N} S_B \simeq S_{A\times B}, \qquad H_N(A) \boxdot H_N(B) \simeq H_N(A \times B),$$
$$\mathbb{C}[A] \otimes \mathbb{C}[B] \simeq \mathbb{C}[A \times B].$$

At $N = 1$ the exponent condition forces $A = 0$ and a one-dimensional phase-system realization, while free pointed modules of arbitrary rank still exist.

Scope. The group tensor is a central quotient, not a Cartesian product retaining two independent centres. The level-one statement is restricted to perfect phase systems, not all $\mathbb{F}_1$ quantum routes.

Provenance. ids: F1-FUNCT,F1-ONE; proved in: f1-cyclotomic.md, sections 3,5; tested by: f1_check.py (M1,M6); reviewed in: f1-adjudication.md.

Proof. The phase-set comparison sends $[[u, a], [v, b]]$ to $[uv, (a, b)]$. The corresponding group map multiplies central phases and takes the external character. Its kernel is the antidiagonal centre. Every associativity diagram multiplies the same phases and associates the same tuple, proving coherence. The Hilbert comparison is the basis map. □

Definition 11.10 (Central pushout from a finite local ring)**.** Let R be finite commutative unital local with $1 \neq 0$, and let $\psi : (R, +) \to \mu_N$ be named, with exponent of $(R, +)$ dividing N. Assume $\iota\psi$ is generating, meaning its kernel contains no nonzero ideal. Let $H_{\beta_0}(R)$ have triples (t, a, b) and product $(t + u + ad, a + c, b + d)$. Put

$$\chi_b(x) = \psi(-bx), \qquad P_\psi = \big(\mu_N \times H_{\beta_0}(R)\big)/\{(\psi(t)^{-1}, (t, 0, 0)) : t \in R\}.$$

Scope. A generating character need not be faithful on the additive group. The raw Heisenberg group and its phase image are distinguished.

Provenance. ids: D1005,D12,D13,D16; proved in: definitions.md; tested by: f1_check.py (M4,M5); reviewed in: f1-r1.md.

Theorem 11.11 (Exact recovery of the arithmetic reference convention)**.** PROVED *Let R be a finite commutative unital local ring with $1 \neq 0$, exponent of $(R, +)$ dividing $N \geq 1$, and a named $\psi : (R, +) \to \mu_N$ such that the faithful complex realization $\iota\psi$ is generating. Then $b \mapsto \chi_b$ identifies $(R, +)$ with its phase dual and $P_\psi \simeq H_N((R, +))$ by $[u; (t, a, b)] \mapsto (u\psi(t), a, \chi_b)$. The raw group map has central kernel* $\ker\psi$*. Its realized operators are*

$$W(a, \chi_b)e_y = (\iota\psi)(-b(y + a))e_{y+a} = Z(-b)X(a)e_y,$$

with product cocycle $(\iota\psi)(ab')$, uniformly including characteristic two.

Scope. Only the reference cocycle is recovered here. This does not identify all raw polarizing-cocycle groups with one phase group.

Provenance. ids: F1-RING; proved in: f1-cyclotomic.md, section 4; tested by: f1_check.py (M4,M5); reviewed in: f1-adjudication.md.

For example, over $\mathbb{F}_4$ a nontrivial trace character has a two-element kernel. Its raw Heisenberg group has order 64 and its phase image order 32. Over $\mathbb{F}_2$, the order-eight reference group acts on the usual qubit. This supplies a checked bridge to the main campaign.

11.5 Fourier transform and a precisely scoped additive extension

Definition 11.12 (Quadratic phases, strict normalizer and Fourier kernel)**.** A quadratic phase is $\theta : A \to \mu_N$, $\theta(0) = 1$, for which $B_\theta(a, x) = \theta(a + x)\theta(a)^{-1}\theta(x)^{-1}$ is a bicharacter. For $t \in A$, $f \in \mathrm{Aut}(A)$ put $T_{t,f,\theta}[u, x] = [u\theta(x), f(x) + t]$. The strict normalizer is the normalizer of the μ_N-level Weyl group in $\mathrm{Aut}_{\mathrm{Free}_*^{\mu_N}}(S_A)$. Define

$$F_A e_x = \sum_{\rho \in A^\vee} \iota(\rho(x))e_\rho, \quad \mathrm{ev}_a(\rho) = \rho(a), \quad J_A(a, \chi) = (\chi^{-1}, \mathrm{ev}_a), \quad r_A(a, \chi) = \chi(a).$$

Scope. A self-Fourier endomorphism additionally needs a pairing identifying A with its dual. No such pairing is suppressed.

Provenance. ids: D1006; proved in: definitions.md; tested by: f1_check.py (M7); reviewed in: f1-r1.md.

Theorem 11.13 (What remains monomial and what Fourier requires)**.** PROVED *Fix $N \geq 1$, a faithful ι, and finite abelian A of exponent dividing N. Its strict normalizer consists exactly of the $T_{t,f,\theta}$ and central phases. Its phase-space maps are*

$$(a, \chi) \longmapsto \big(f(a), (\chi B_\theta(a, -)) \circ f^{-1}\big).$$

For $|A| > 1$, F_A has full column support and cannot be the realization of a strict pointed map. Nevertheless,

$$F_A^* F_A = |A|I, \qquad F_A W(v) = \iota(r_A(v))W(J_A v)F_A.$$

Scope. Only bicharacters admitting a μ_N-valued quadratic refinement occur in the strict subgroup. Enlarging phases alone does not make Fourier monomial. No canonical Weil splitting is asserted.

Provenance. ids: F1-MON; proved in: f1-cyclotomic.md, section 6; tested by: f1_check.py (M7, Fourier only); reviewed in: f1-adjudication.md.

Definition 11.14 (Framed cyclotomic kernels and lifted phase isomorphisms)**.** Put $K_N = \mathbb{Q}[\mu_N]/\ker\iota$. The category $\mathrm{Mat}(K_N)$ has finite sets I as objects and $J \times I$ matrices over K_N as morphisms; tensor is Cartesian product of sets and Kronecker product of matrices. The associated modules are explicitly framed by $[1, i]$ in $S_I = \{0\} \sqcup (\mu_N \times I)$. Let $\mathrm{PMat}(K_N)$ be invertible matrices modulo $K_N^\times$, realized by entrywise evaluation.

The groupoid $\mathrm{LiftHyp}_N$ has configurations A as objects. An arrow is (g, r) with $g : V_A \to V_B$ a group isomorphism preserving κ and $r : V_A \to \mu_N$ satisfying

$$r(v)r(w)c_B(gv, gw) = c_A(v, w)r(v + w).$$

It denotes the centre-fixing map $(u, v) \mapsto (ur(v), g(v))$; composition composes these maps. Tensor is $A \times B$ on objects and $(g \times h, (v, w) \mapsto r(v)s(w))$ on arrows, with the canonical reordering of phase factors. The unit is $A = 0$ and the symmetry is the swap.

Scope. This target is an additive cyclotomic envelope. It is not claimed to be a category of $\mathbb{F}_1$ schemes; arbitrary free torsors are not given coordinate matrices without a frame.

Provenance. ids: D1007; proved in: definitions.md; tested by: no general checker; reviewed in: f1-adjudication.md, O1–O3 repair.

Proposition 11.15 (Projective realization of lifted phase symmetries)**.** SKETCH *For $N \geq 1$ and a named faithful ι, there is a projective strong symmetric monoidal functor*

$$Q_N : \mathrm{LiftHyp}_N \longrightarrow \mathrm{PMat}(K_N)$$

defined by the Egorov intertwining equations. Its complex realization is projectively unitary; affine-quadratic maps and Fourier have the values above.

Scope. The lift is input data. Geometric descent of the additive envelope and a lift of every symplectic transformation remain separate questions.

Provenance. ids: F1-CORR; proved in: f1-cyclotomic.md, section 7; tested by: not claimed as computationally proved; reviewed in: supporting SKETCH after scope repair.

Proof. The intertwining equations have coefficients in K_N and a one-dimensional solution space after embedding in $\mathbb{C}$. Gaussian elimination has the same rank over either field, so there is a nonzero solution over K_N. Irreducibility makes it invertible; uniqueness up to scalar gives projective composition and tensor coherence. Unitary normalization is performed only after complex realization. □

11.6 Relational Heisenberg geometry over bands

Definition 11.16 (Bands and the upper-unitriangular crowd)**.** A band is a commutative pointed multiplicative monoid B with a null ideal N_B of formal sums, in which every a has a unique $-a$ with $a + (-a) \in N_B$; morphisms preserve null sums. The regular partial field $\mathbb{F}_1^{\pm} = \{0, 1, -1\}$ uses sums vanishing in $\mathbb{Z}$. The Krasner hyperfield $\mathbb{K} = \{0, 1\}$ declares a sum null exactly when it has zero or at least two nonzero terms. Define $H_{\text{crowd}}(B) = B^3$, represented by $h(a, b, t)$ as above, with identity $(0, 0, 0)$. Its colaw consists of triples for which

$$\sum_i a_i, \qquad \sum_i b_i, \qquad t_1 + t_2 + t_3 + a_1 b_2 + a_1 b_3 + a_2 b_3$$

and both cyclic variants of the last sum lie in N_B. The inverse set of h consists of k with $(h, k, 1)$ in the colaw; the product set $h * k$ consists of c for which some d has both (h, k, d) and $(c, d, 1)$ in the colaw.

Scope. Nullity is part of the band, not an invented binary addition. The signed band and the raw pointed phase monoid have different structure.

Provenance. ids: D1008; proved in: definitions.md; tested by: f1_check.py (M10); reviewed in: supporting comparison.

Proposition 11.17 (A band-level Heisenberg crowd and its ring fibres)**.** SKETCH *The functor $B \mapsto H_{\text{crowd}}(B)$ on bands, with the displayed colaw, is an affine algebraic crowd. For every commutative ring R, its fibre is exactly $\mathrm{UT}_3(R)$. The signed fibre has 27 points, 319 colaw triples and 23 points admitting an inverse inside that fibre. The Krasner fibre has eight points and 152 colaw triples, with multivalued inverses.*

Scope. These finite fibres are not ordinary groups. Even a crowd product with the identity may be empty when its required inverse mediator is absent. No Hilbert representation is inferred from the point counts.

Provenance. ids: F1-CROWD; proved in: f1-comparisons.md, section 1; tested by: f1_check.py (M10); reviewed in: supporting SKETCH.

This specializes the algebraic SL_3 crowd of Lorscheid–Thas (2023), section 5.4. Restricting to an identity-containing subset preserves the crowd axioms; the displayed coordinate relations are affine. Over rings they say $h_1 h_2 h_3 = I$. The signed example retains $h(1, 0, 0) * h(0, 1, 0) = \{h(1, 1, 1)\}$, and hence the central cross-term, although $h(1, 0, 0)^2$ is absent. This provides a constructive second Heisenberg formulation, with a representation theory still to specify.

11.7 Tensor categories and the qubit fusion object

Definition 11.18 (The categorical phase system)**.** Let $\mathcal{C}_N(A) = \mathrm{Rep}_{\mathbb{C}}(H_N(A))$, with all linear intertwiners and internal tensor using the diagonal group action. Let $\mathcal{C}_N(A)_\lambda$ be the full subcategory on which central u acts as $\lambda(u)I$. The unitary version chooses invariant Hermitian inner products and has the same intertwining spaces. Write ω_A for the forgetful tensor functor. For groups H, K with specified central μ_N, define $\mathrm{Rep}(H) \boxtimes_{\mathrm{match}} \mathrm{Rep}(K)$ as the full subcategory of $\mathrm{Rep}(H \times K)$ on which all (u, u^{-1}) act trivially.

Scope. The matching condition is explicit. It is not an unspecified relative tensor product, and it is different from internal fusion.

Provenance. ids: D1011; proved in: definitions.md; tested by: f1_check.py (examples); reviewed in: supporting comparison.

Proposition 11.19 (Central sectors explain why the category matters)**.** SKETCH *For $N \geq 1$ and finite abelian A of exponent dividing N, $\mathcal{C}_N(A)$ is a symmetric fusion category with a faithful tensor functor ω_A, and*

$$\mathcal{C}_N(A) = \bigoplus_{\lambda \in \widehat{\mu_N}} \mathcal{C}_N(A)_\lambda, \qquad \mathcal{C}_N(A)_\lambda \otimes \mathcal{C}_N(A)_\nu \subseteq \mathcal{C}_N(A)_{\lambda\nu}.$$

Moreover, for finite groups H, K with specified central embeddings of μ_N,

$$\mathrm{Rep}(H \boxdot K) \simeq \mathrm{Rep}(H) \boxtimes_{\mathrm{match}} \mathrm{Rep}(K).$$

On simple external tensors, the matching condition is equality of the two central characters. In particular a fixed nontrivial sector is not a tensor category by itself.

Scope. The unit of matching composition on this family is $\mathrm{Rep}(\mu_N)$, with one simple per grade. It is not the unit for the unrestricted Deligne product.

Provenance. ids: F1-CAT; proved in: f1-comparisons.md, section 5; tested by: f1_check.py (M6,M12; examples); reviewed in: supporting SKETCH.

Proof. Average an inner product over the finite group to obtain semisimplicity. Central character eigenspaces give the grading, and diagonal tensor multiplies eigencharacters. A representation factors through the central quotient precisely when its antidiagonal centre acts trivially; inflation and descent are inverse tensor functors. On an external simple the condition is $\lambda(u)\nu(u)^{-1} = 1$. □

Definition 11.20 (Dihedral and quaternion qubit data)**.** For $e \in \{0, 1\}$ let $G_e = \mathbb{F}_2^3$ with

$$(t, a, b)(u, c, d) = (t + u + ad + e(ac + bd), a + c, b + d).$$

Put

$$\sigma_e(t, a, b) = (-1)^t i^{e(a+b)} Z^b X^a, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

using representatives $a, b \in \{0, 1\}$ in the exponent of i. Define $\nu_2(\sigma_e) = |G_e|^{-1} \sum_g \mathrm{Tr}(\sigma_e(g^2))$.

Scope. The group law distinguishes the two cocycle choices. The quaternion matrices require fourth-root phases in this monomial realization.

Provenance. ids: D1012; proved in: definitions.md; tested by: f1_check.py (M12); reviewed in: supporting comparison.

Proposition 11.21 (Two qubit objects with the same fusion rules)**.** SKETCH $G_0 = D_8$ *of order eight and* $G_1 = Q_8$*. Each has four one-dimensional characters and the single two-dimensional irreducible* σ_e*. Writing the four characters as* $1, \alpha, \beta, \alpha\beta$*,*

$$\sigma_e \otimes \sigma_e \simeq 1 \oplus \alpha \oplus \beta \oplus \alpha\beta, \qquad \nu_2(\sigma_0) = 1, \quad \nu_2(\sigma_1) = -1.$$

Their sourced tensor-category identifications are

$$\mathrm{Rep}(D_8) \simeq \mathrm{TY}(\mathbb{F}_2^2, \chi_{\mathrm{alt}}, 1/2), \qquad \mathrm{Rep}(Q_8) \simeq \mathrm{TY}(\mathbb{F}_2^2, \chi_{\mathrm{alt}}, -1/2).$$

Scope. The displayed tensor square uses one diagonal symmetry group. It does not replace the external tensor describing independent qubits.

Provenance. ids: F1-FUSION; proved in: f1-comparisons.md, section 6; Shimizu, p. 20; tested by: f1_check.py (M12); reviewed in: supporting SKETCH.

The two-dimensional character is $2, -2$ on the two central elements and zero elsewhere. Squaring it gives the sum of the four quotient characters. Six elements square to 1 in D_8 and only two do in Q_8, giving the indicator signs. The categorical identifications and their distinction are in Shimizu's *Frobenius–Schur indicators in Tambara–Yamagami categories*, Definition 3.1, Table 1 and page 20.

For two independent reference qubits the relevant group instead is $H_2(C_2) \boxdot H_2(C_2) = H_2(C_2 \times C_2)$, of order 32. Its nontrivial-central-character irreducible has dimension four. Both this representation and internal fusion use the underlying Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$, but their symmetry actions differ. This is a concrete reason to retain the tensor category, its sectors and its fibre functor as part of the quantum object.

The fibre functor is essential to this particular formulation:

$$\mathrm{End}_{\mathrm{Rep}(H)}(\sigma) = \mathbb{C}, \qquad \mathrm{Hom}_{\mathrm{Rep}(H)}(1, \sigma) = 0$$

for the nontrivial central character. The equivariant category alone does not give every state vector or every qubit observable. Those are recovered as vectors of $\omega(\sigma) = \mathbb{C}^2$ and elements of $\mathrm{End}_{\mathbb{C}}(\omega(\sigma)) = M_2(\mathbb{C})$. An alternative categorical state notion would have to be specified.

11.8 The independent Hall and categorical-oscillator route

Definition 11.22 (Pointed-set Hall algebra and Fock realization)**.** On the normal pointed-set category above, the rational Hall vector space has basis u_r indexed by rank. Its product counts pointed subsets T with the prescribed isomorphism classes of T and S/T. Put $\mathcal{F}_{\mathrm{alg}} = \mathbb{C}[x]$ with $\langle x^m, x^n \rangle = \delta_{mn} n!$, define $a^\dagger f = xf$, $af = df/dx$, and let $\mathcal{F}_{\mathrm{bos}}$ be the Hilbert completion.

Scope. Hall coefficients and the Fock space are additive realizations of the pointed-set data. Creation is unbounded.

Provenance. ids: D1010; proved in: definitions.md; tested by: f1_check.py (M8,M9); reviewed in: supporting comparison.

Proposition 11.23 (A bosonic Heisenberg algebra from finite sets)**.** SKETCH *The rational Hall algebra of finite normal pointed sets is* $\mathbb{Q}[x]$ *under*

$$u_m u_n = \binom{m+n}{m} u_{m+n}, \qquad u_n \longmapsto x^n/n!.$$

On $\mathcal{F}_{\mathrm{alg}}$*, creation and annihilation are adjoints and* $[a, a^\dagger] = I$*. In the normalized basis* $x^n/\sqrt{n!}$ *their coefficients are* $\sqrt{n+1}$ *and* $\sqrt{n}$*.*

Scope. The polynomial Hall algebra alone is not the Weyl algebra; annihilation is additional operator structure. No finite-dimensional characteristic-zero CCR representation is asserted.

Provenance. ids: F1-HALL; proved in: f1-comparisons.md, section 4; tested by: f1_check.py (M8,M9); reviewed in: supporting SKETCH.

The subset count proves the Hall formula. The operator calculation is $d(xf)/dx - x\,df/dx = f$. The source construction is Szczesny's Hall algebra, section 3. Baez–Dolan give the factorial groupoid weights and the positive categorical relation

$$AA^\dagger \simeq A^\dagger A \oplus \mathrm{id}.$$

Morton–Vicary realize the categorified Heisenberg algebra using spans of finite-set groupoids: add/remove histories split according to whether the removed point was just added. Their two-linearization retains the tower of $\mathrm{Rep}(S_n)$, with the stated finiteness/completion caveats. The connection to $\mathbb{F}_1$ is the equivalence between finite sets and the core of finite pointed modules. It is not a claim that these authors construct one universal $\mathbb{F}_1$ scheme theory.

This supplies a categorical north star different from the finite Weyl system. Degree two of the representation-category tower has two simples in $\mathrm{Rep}(S_2)$; that is not automatically a qubit. With two colours, the one-particle complex space is a qubit and the two-boson sector is $\mathrm{Sym}^2(\mathbb{C}^2)$, of dimension three. A one-mode cutoff to $|0\rangle, |1\rangle$ instead has projected commutator $\mathrm{diag}(1, -1)$, and is not closed under creation. The Hall operation uses wedge and particle-number addition; the finite configuration tensor uses smash and rank multiplication.

11.9 Toric, analytic and arithmetic-statistical formulations

Definition 11.24 (A quantum torus and its selected central fibre)**.** For a named $\xi \in \mathbb{C}^\times$ put

$$\mathcal{T}_\xi = \mathbb{C}\langle U^{\pm 1}, V^{\pm 1}\rangle/(VU - \xi UV).$$

If ξ is a primitive Nth root, put $\mathcal{T}_\xi(1,1) = \mathcal{T}_\xi/(U^N - 1, V^N - 1)$.

Scope. The parameter ξ, its order N and a finite-field cardinality are different data. Central values are explicitly selected.

Provenance. ids: D1013; proved in: definitions.md; tested by: cyclic Weyl checks; reviewed in: supporting comparison.

Proposition 11.25 (The qubit as a quantum-torus fibre)**.** SKETCH *For ξ primitive of order N, $\mathcal{T}_\xi(1,1) \simeq M_N(\mathbb{C})$ via $Ue_j = e_{j+1}$ and $Ve_j = \xi^j e_j$ on $\mathbb{Z}/N$. For $N = 2$ this is a qubit algebra; two independent fibres tensor to $M_4(\mathbb{C})$.*

Scope. The universal quantum torus has not been identified with a finite matrix algebra. At $\xi = 1$ it is commutative.

Provenance. ids: F1-TORUS; proved in: f1-comparisons.md, section 7; tested by: f1_check.py (cyclic cases); reviewed in: supporting SKETCH.

The N^2 clock/shift words are trace-orthogonal and span, which proves the selected-fibre statement. This is the twisted-lattice picture of Neeb's rational quantum tori. Monoidal tori retain character lattices; a chosen skew pairing is additional symplectic data. Manin's cyclotomic analytic geometry supplies Habiro completions and evaluation or Taylor maps at roots of unity. It suggests an arithmetic interpolation problem for quantum fibres, but does not itself give a common Hilbert space or compatible Weil intertwiners. A deformation limit $\xi \to 1$ is different from all the preceding $q \to 1$ operations.

Frobenius descent. Borger's lambda geometry treats lambda structure as descent data below $\operatorname{Spec}\mathbb{Z}$. For flat schemes it can be expressed as commuting Frobenius lifts. Monoid algebras have the lift $m \mapsto m^p$. Its naive coordinatewise application to the integral Heisenberg group fails already on the additive subgroup: $(1+1)^p \neq 1^p + 1^p$ in $\mathbb{Z}$. This rejects that particular group-compatible lift, not all arithmetic descent. A quantum extension should specify whether arithmetic maps act by homomorphisms, noninvertible operators or correspondences. No qubit or Born rule is selected by lambda structure alone.

A published quantum-statistical realization. Connes–Consani–Marcolli's *Fun with* $\mathbb{F}_1$, Proposition 6.1 and Theorem 6.2, constructs an $\mathbb{F}_1$ model of the Bost–Connes endomotive, with the rational system recovered by scalar extension. It uses a cyclotomic tower and Frobenius/transfer correspondences. Its complex realization has the time evolution

$$\sigma_t(\mu_n) = n^{it}\mu_n, \qquad \sigma_t(e(r)) = e(r).$$

In the standard representation on $\ell^2(\mathbb{N}_{>0})$, $He_n = (\log n)e_n$ and $\operatorname{Tr}(e^{-\beta H}) = \zeta(\beta)$ for $\beta > 1$; see Bost–Connes (1995). This is genuine quantum statistical dynamics, not a finite Stone–von Neumann representation. Choosing two energy levels provides an encoding but not a sector invariant under its full observable algebra.

Lieber–Manin–Marcolli, published in 2022, develop lifts to Grothendieck rings, assembler categories, spectra and Nori motives. This is an independent literature-based reason to explore categorical targets, without asserting that these categories are the fusion categories of the finite Heisenberg construction.

Tropical and motivic boundaries. The band/Krasner framework retains combinatorial flags beyond a single coordinate apartment. Boolean semirings, Krasner hyperfields, signed partial fields and pointed monoids have different additive information. Connes–Consani's Arithmetic Site uses a tropical semiring on a multiplicative-semigroup topos to realize arithmetic correspondences. It does not prescribe a local Weyl algebra or positive quantum states. Relative and homotopical geometries broaden the candidate bases and targets, but no general Heisenberg representation theorem for those bases is established in this sidequest. Finally, Bejleri–Marcolli's *Quantum field theory over* $\mathbb{F}_1$ concerns torifications and varieties attached to Feynman integrals; its introduction explicitly distinguishes that problem from physical Lagrangians and Feynman rules over $\mathbb{F}_1$.

An additional tensor analogy: fermionic occupancy. Lin–Wang–Wu–Yang (2013) categorify the one-mode fermion algebra with $c^2 = (c^\dagger)^2 = 0$ and $cc^\dagger + c^\dagger c = 1$. Occupancy subsets suggest a combinatorial exterior-power route; this is our proposed $\mathbb{F}_1$ bridge, not their claim. Hadzihasanovic–de Felice–Ng (2018), section 2, give a precise monoidal category of local fermionic modes and maps of definite parity. Its basic object has underlying space $\mathbb{C}^{1|1}$; tensor adds parity and the physical fermionic swap negates $|11\rangle$. The two occupancy labels therefore do not supply unrestricted qubit operations. Parity-preserving one-mode observables are diagonal, while a fixed-parity two-mode sector can encode a qubit. Signed phases are a candidate carrier for exchange signs over $\mathbb{F}_1$; the cited complex categorical construction does not prove that descent. This is a further composition law to compare with the bosonic and distinguishable-system routes.

11.10 Precisely formulated outputs and open comparisons

The finite arithmetic candidate is the configuration-to-phase-system functor, with a named phase realization and a checked tensor law. The categorical candidate retains

$$(\mathcal{C}_N(A),\ \mathcal{C}_N(A)_\iota,\ \omega_A)$$

together with internal fusion and matching external composition. The pointed normal-map candidate already realizes an ordinary qubit over the bare combinatorial base. The Heisenberg crowd is a geometric object with correct ring fibres, while the Hall/groupoid route gives an independent categorical oscillator. Bost–Connes supplies the arithmetic quantum-statistical test case. These are several precise formulations, not evidence that all the analogies must coincide.

The next comparison problems are concrete: relate a chosen Tits reflection lift to its projective Fourier operator; extend the geometric Heisenberg model to nonreference cocycles; identify a genuine geometric home for the additive phase kernels; compare finite cyclotomic fibres with arithmetic transfer maps; and distinguish tensor products of distinguishable systems from bosonic particle-number sectors. A theory on arbitrary $\mathbb{F}_1$ schemes still needs its input category, morphisms and realization functor specified. It is a research target, not a theorem or a well-formed universal conjecture supplied by the present literature.

Seven finite-kernel claims are PROVED; the extensions remain SKETCH. Thirteen exact example gates and their mutations check the finite computations. The cited literature and our deductions are distinguished throughout.

12 Operational sidequest: quantum mechanics in subsystem composition

7 September 2026. We now have a concrete positive candidate for the subsystem-first programme. Its endpoint assigns

$$\mathcal{A}(S) = \mathbb{C}[\mathrm{Sym}(S)]$$

to a finite set of constituents. The algebras have normalized positive states, CP dynamics, Born probabilities and proper assembly inclusions. One constituent has only scalar observables; three constituents have an $M_2(\mathbb{C})$ sector. Two disjoint three-constituent clusters can carry a Bell state. The construction is a quantum family, not a proposed single-particle wavefunction over a one-element coefficient field.

The arithmetic precursor is the positive type-A Hecke family. At finite field cardinality q, it is the invariant transition algebra of complete flags in a chosen configuration Lagrangian. Two overlapping subsystem embeddings determine $q \geq 1$ through the intrinsic Born overlap $q/(q+1)^2$, even though their abstract algebra types are independent of q. The endpoint at one is an actual specialization of algebraic presentations.

The limitation is explicit: this is a *flag-context sector extracted from polarized Weyl quantum mechanics*. It is not yet a specialization of the entire Weyl algebra, its phase data and every change of polarization. The research report is `docs/sidequests/f1-operational.md`.

12.1 A positive arithmetic family

Definition 12.1 (Hecke systems and their reference trace)**.** Fix a real $q > 0$. Put $H_0(q) = H_1(q) = \mathbb{C}$. For $n \geq 2$, define the unital complex algebra $H_n(q)$ by generators $T_1, \ldots, T_{n-1}$, involution $T_i^* = T_i$, and relations

$$(T_i - q)(T_i + 1) = 0, \qquad T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \qquad T_iT_j = T_jT_i \quad (|i-j| > 1).$$

For $w \in S_n$, let T_w be a reduced-word product and let $\ell(w)$ be its Coxeter length. Define the coefficient functional by $\tau_n(T_w) = \delta_{w,1}$.

Scope. Basis independence, positivity and the C*-norm are proved below. The real parameter becomes a field cardinality only at the specified arithmetic fibres.

Provenance. ids: D1101; proved in: definitions.md; tested by: f1_operational_check.py (H1,H2); reviewed in: f1-operational-adjudication.md.

Theorem 12.2 (Positive Hecke quantum algebras)**.** PROVED *For every $q > 0$ and $n \geq 0$, the T_w form a basis of $H_n(q)$, $T_w^* = T_{w^{-1}}$, and*

$$\tau_n(T_u^* T_v) = \delta_{u,v} q^{\ell(u)}.$$

Consequently τ_n is a faithful normalized positive trace, and left multiplication on $L^2(H_n(q), \tau_n)$ gives a faithful finite C-algebra representation.*

Scope. This covers all positive real parameters, including one. It does not assert positivity at arbitrary complex roots of unity.

Provenance. ids: F1-HCK-POS; proved in: f1-operational/hecke.md, section 1; tested by: H1,H2; reviewed in: f1-operational-adjudication.md.

Proof. On a basis b_w, the standard generator raises Coxeter length with coefficient one, and on a descent acts by $qb_{s_iw} + (q-1)b_w$. The two-string matrix is $\begin{pmatrix} 0 & q \\ 1 & q-1 \end{pmatrix}$. The rank-two strings verify the defining relations. Reduced words span and are independent by their action on b_1. With $\langle b_u, b_v\rangle = \delta_{u,v} q^{\ell(u)}$, every generator is self-adjoint. This proves the Gram identity and $\tau_n(x^*x) = \sum_w |x_w|^2 q^{\ell(w)}$. The basis identity also gives traciality. The faithful regular *-representation has closed finite-dimensional image. □

Definition 12.3 (Finite-field flag context algebra)**.** Fix a named finite field $k = \mathbb{F}_Q$ and an n-dimensional k-space L. A complete flag is $0 = U_0 < U_1 < \cdots < U_n = L$ with $\dim_k U_i = i$. Put

$$\mathrm{Cxt}(L) = \mathrm{End}_{GL(L)}\, \ell^2(\mathrm{Fl}(L)).$$

Let $(A_w f)(F) = \sum_{\mathrm{pos}(F,F')=w} f(F')$, labelling relative positions by reduced gallery words in adjacency-product order. Use the normalized operator trace $\mathrm{tr}_{\mathrm{Fl}}(a) = |\,\mathrm{Fl}(L)|^{-1}\,\mathrm{Tr}(a)$. If L is a named configuration Lagrangian of a Weyl system, call this its complete-flag context algebra.

Scope. The context register is a separate Hilbert space from the Weyl configuration register. The Lagrangian is named data.

Provenance. ids: D1104; proved in: definitions.md; tested by: H9; reviewed in: f1-operational-adjudication.md.

Theorem 12.4 (Arithmetic realization by invariant flag transitions)**.** PROVED *For every finite field $\mathbb{F}_Q$ and finite-dimensional L of dimension n, the adjacency map $T_w \mapsto A_w$ is a trace-preserving *-isomorphism*

$$H_n(Q) \simeq \mathrm{Cxt}(L).$$

Scope. The operator convention is direct; a right-convolution convention would require the usual opposite/inversion adjustment.

Provenance. ids: F1-HCK-FLAG; proved in: f1-operational/hecke.md, section 2; Iwahori 1964; tested by: H9, fields of orders 2 and 3; reviewed in: f1-operational-adjudication.md.

This is the finite-field construction in Iwahori (1964), Lemma 3.1 and Theorems 3.2 and 4.1. Pair-orbits of flags are indexed by S_n and their kernels span the commutant. An adjacent panel contains $Q+1$ flags, so its nonidentity adjacency has square $QI + (Q-1)A_{s_i}$. Transposition inverts relative position, and only the identity kernel has nonzero diagonal. These facts fix the trace.

Definition 12.5 (Ordered assembly and coarse graining)**.** For real $q > 0$ and $m, n \geq 0$, embed $S_m \times S_n$ as consecutive block permutations in S_{m+n}. Define

$$\iota_{m,n}(T_u \otimes T_v) = T_{u\times v}, \qquad E_{m,n}(T_w) = \begin{cases} T_w, & w \in S_m \times S_n, \\ 0, & \text{otherwise,} \end{cases}$$

identifying the retained basis with $H_m(q) \otimes H_n(q)$.

Scope. These are ordered maps. No symmetric block swap is stipulated at $q \neq 1$.

Provenance. ids: D1102; proved in: definitions.md; tested by: H3; reviewed in: f1-operational-adjudication.md.

Theorem 12.6 (Coherent positive assembly)**.** PROVED *For $q > 0$ and $m, n \geq 0$, $\iota_{m,n}$ is an injective unital trace-preserving *-homomorphism. The map $E_{m,n}$ is the unique faithful trace-preserving UCP conditional expectation onto its image. Both families satisfy ordered three-block coherence.*

Scope. Complete positivity includes all matrix amplifications. The finite checker samples this statement; the proof has no rank bound.

Provenance. ids: F1-HCK-TOWER; proved in: f1-operational/hecke.md, section 3; tested by: H3,H13, stated finite samples; reviewed in: f1-operational-adjudication.md.

Proof. The two sets of block generators commute and their basis images are distinct. The coefficient trace restricts to the product trace. Orthogonality of the basis makes E the trace-orthogonal projection. The trace-pairing characterization gives its bimodule property and positivity; repeating the projection argument in every matrix amplification gives complete positivity. The two three-block routes retain precisely $S_l \times S_m \times S_n$. The classical source for the expectation is Umegaki (1954). □

Definition 12.7 (Operational states and processes of a Hecke level)**.** Fix real $q > 0$ and $n \geq 0$. A state is a positive functional $\omega : H_n(q) \to \mathbb{C}$ with $\omega(1) = 1$, represented by $h \geq 0$, $\tau_n(h) = 1$, through $\omega(a) = \tau_n(ha)$. An effect satisfies $0 \leq e \leq 1$ and has probability $\tau_n(he)$. Deterministic Heisenberg channels are UCP; outcome maps are subunital CP and an instrument has a unital sum. The ordered product density is $\iota_{m,n}(h \otimes k)$.

A retained local Kraus list satisfies $\sum_i K_i^* K_i = 1$ and gives

$$\Phi_K(a) = \sum_i K_i^* a K_i, \qquad \mathcal{T}_K(h) = \sum_i K_i h K_i^*.$$

In a larger block, embed the operators before applying these formulas.

Scope. Equality of isolated CP maps is not assumed to survive assembly. Not every CP map is asserted to have this local inner form.

Provenance. ids: D1106; proved in: definitions.md; tested by: H3,H5,H8; reviewed in: f1-operational-adjudication.md.

Theorem 12.8 (Quantum operational semantics of the positive family)**.** PROVED *For $q > 0$, the Hecke states have unique positive normalized densities, effects give Born probabilities, and the stated restriction and preparation maps are CP. Ordered product densities are normalized and associative. Retained local Kraus processes extend coherently through ordered block assembly.*

Scope. This is a finite C*-quantum sector. A tensor law on arbitrary isolated CP shadows has not been asserted.

Provenance. ids: F1-HCK-OPS; proved in: f1-operational/hecke.md, section 4; tested by: H3,H5,H8; reviewed in: f1-operational-adjudication.md.

The density/functional identification is blockwise matrix duality for a faithful trace. The equality

$$\tau_{m+n}\big(\iota(h \otimes k)a\big) = (\omega_h \otimes \omega_k)(E_{m,n}(a))$$

identifies product preparation with precomposition by the conditional expectation. Positivity and normalization give the Born interval $[0, 1]$.

12.2 The parameter lives in the subsystem diagram

Definition 12.9 (Unlabelled overlap and marked trace)**.** Let a finite C*-algebra have a unique noncommutative simple summand $M_2(\mathbb{C})$ and two specified unital subalgebras $B_1, B_2 \simeq \mathbb{C}^2$. When the two pairs of minimal projections compress to rank-one projections in that block, define

$$a(\mathcal{A}; B_1, B_2) = \min_{P \in B_1,\, Q \in B_2} \mathrm{Tr}_2(PQ),$$

where the minimum runs over those compressed minimal projections and Tr_2 is ordinary matrix trace. Separately, in $H_2(q)$ define $e_{\mathrm{triv}} = (T_1 + 1)/(q + 1)$ and $r_2 = \tau_2(e_{\mathrm{triv}})$.

Scope. The first invariant does not label the projections or use a chosen reference state. The second invariant explicitly marks the trivial projection.

Provenance. ids: D1103,D1105; proved in: definitions.md; tested by: H4; reviewed in: f1-operational-adjudication.md.

Theorem 12.10 (Arithmetic memory in overlapping subsystems). **PROVED** *For every $q > 0$,*

$$H_1(q) = \mathbb{C}, \quad H_2(q) \simeq \mathbb{C}^2, \quad H_3(q) \simeq \mathbb{C}^2 \oplus M_2(\mathbb{C}).$$

For the left and right H_2 inclusions in H_3, the unlabelled overlap is

$$a(q) = \frac{q}{(q+1)^2}, \qquad q = \frac{1 - 2a + \sqrt{1-4a}}{2a} \quad (q \geq 1).$$

It determines exactly $\{q, q^{-1}\}$, and uniquely determines q on $q \geq 1$. The marked trace instead satisfies $r_2 = 1/(q+1)$ and determines q without that range restriction.

Scope. The algebra types alone do not determine the parameter. The two overlapping embeddings, or the separately marked trace, do.

Provenance. ids: F1-HCK-LOW,F1-HCK-Q; proved in: f1-operational/hecke.md, sections 5–6; tested by: H4,H5; reviewed in: f1-operational-adjudication.md.

In the matrix block one may calculate with

$$P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_2 = \begin{pmatrix} a & \sqrt{a(1-a)} \\ \sqrt{a(1-a)} & 1-a \end{pmatrix}.$$

The four unlabelled overlaps are $a, a, 1-a, 1-a$, and $a \leq 1/4$. This matrix calculation proves a basis-independent statement about the abstract C*-diagram. The reciprocal *-isomorphism is $T_i \mapsto -qS_i$ into $H_n(q^{-1})$.

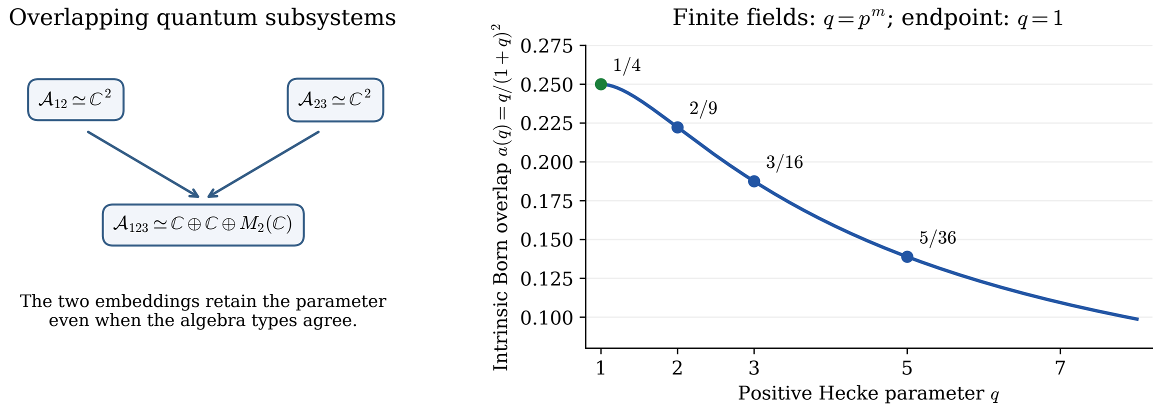

Figure 1: The abstract algebra types stay fixed, while the relative subsystem embeddings retain the positive parameter. The markers are exact rational probabilities; the intervening curve illustrates the proved formula.

The coefficient trace on $H_3(q)$ is

$$\tau_3(a_+, a_-, a_{\text{std}}) = \frac{a_+ + q^3 a_-}{(q+1)(q^2+q+1)} + \frac{q}{q^2+q+1} \mathrm{Tr}_2(a_{\text{std}}).$$

Thus a standard-block pure state with ordinary density P_2 has full density $h = P_2/\gamma$, $\gamma = q/(q^2+q+1)$. Let $u_1 = 2e_1 - 1$, $e_1 = (T_1+1)/(q+1)$. Its isolated action on the left commutative H_2 is identity, but the same retained unitary in H_3 changes the return probability for effect P_2 from 1 to

$$(1 - 2a(q))^2.$$

At one the full density is $3P_2$ and the return probability is $1/4$. Preparing it by measuring P_2 in the reference state has success probability $1/3$. This is a complete finite quantum experiment.

12.3 A functor on normal maps over the field with one element

Definition 12.11 (The symmetric-group quantum net)**.** For a finite set S, put $G(S) = \mathrm{Sym}(S)$ and $\mathcal{A}(S) = \mathbb{C}[G(S)]$, with

$$\Big(\sum_g a_g g\Big)^* = \sum_g \overline{a_g} g^{-1}, \qquad \tau_S\Big(\sum_g a_g g\Big) = a_1.$$

An injection $f : S \to T$ extends permutations by the identity off $f(S)$, giving i_f. Its reverse coefficient map E_f deletes terms outside that subgroup. Disjoint union gives the block inclusion $\mu_{S,T}$. States use $h \geq 0$, $\tau_S(h) = 1$, and $\omega_h(a) = \tau_S(ha)$. Local Kraus lists satisfy $\sum_i K_i^* K_i = 1$; their action in T retains the embedded list $i_f(K_i)$. Lists are identified under scalar unitary mixing after padding by zero operators.

Scope. The coefficient trace is normalized. It is different from the unnormalized categorical trace used later for a general fusion category.

Provenance. ids: D1125; proved in: definitions.md; tested by: H5,H7,H8; reviewed in: f1-operational-adjudication.md.

Theorem 12.12 (A collective quantum endpoint)**.** PROVED *The symmetric-group assignment on finite injections, with its coefficient expectations, states and retained local Kraus processes, is a finite quantum net. It has coherent proper disjoint assembly and CP restriction/preparation. Its three-constituent standard sector has the full density $3P_2$ and the Born witness $1 \to 1/4$.*

Scope. The net supplies a sufficient coherent class of local processes, not a canonical extension of every isolated CP map.

Provenance. ids: F1-OP-SYM; proved in: f1-operational/cp.md, CPOP-5; tested by: H5,H7,H8; reviewed in: f1-operational-adjudication.md.

Definition 12.13 (Partial-injection realization)**.** A normal map of finite pointed $\mathbb{F}_1$ modules preserves the basepoint and has at most one preimage of each non-basepoint. On nonzero sets it is a partial injection $f : S \rightharpoonup T$, a bijection $D \to E$ between subsets. Define

$$\mathcal{R}(f) = i_{E\subset T} \circ \mathcal{A}(f : D \simeq E) \circ E_{D\subset S}.$$

The target has finite tracial C*-algebras, unital trace-preserving CP maps and trace adjoints as daggers. Wedge of pointed modules becomes disjoint union of the nonzero sets.

Scope. This is the normal-map convention; arbitrary pointed maps are not included.

Provenance. ids: D1010,D1127; proved in: definitions.md; Szczesny 1204.5395; tested by: H12; reviewed in: f1-operational-adjudication.md.

Theorem 12.14 (Operational realization of normal F1 maps)**.** PROVED *On finite sets and partial injections $f : D \subseteq S \to E \subseteq T$, put $\mathcal{A}(S) = \mathbb{C}[\mathrm{Sym}(S)]$ with coefficient trace, and $\mathcal{R}(f) = i_E \mathcal{A}(f : D \simeq E) E_D$, using subgroup inclusion and coefficient expectation. This functor is dagger and lax symmetric monoidal, with structure maps*

$$\mu_{S,T} : \mathcal{A}(S) \otimes \mathcal{A}(T) \hookrightarrow \mathcal{A}(S \sqcup T).$$

An empty partial map has image $a \mapsto \tau_S(a)1_T$.

Scope. The laxators need not be isomorphisms. The source zero map is a reference reset, not zero CP; this is a nonadditive functor.

Provenance. ids: F1-OP-PMAP; proved in: f1-operational/contexts.md, PMAP-1; tested by: H12; reviewed in: f1-operational-adjudication.md.

Proof. A group-basis permutation survives precisely when its support is in D; then it is relabelled and extended by identity. Two successive survival conditions say that the support lies in the domain of the composed partial map. This proves functoriality. Inclusion and coefficient expectation are trace adjoints, proving the dagger statement. The same support calculation proves naturality under disjoint union; associativity and symmetry follow from the canonical set bijections. □

The source wedge operation is therefore collective assembly. Classical alternatives can separately use C*-algebra direct sums, with their central probabilities. They are not the same operation, and neither should be silently identified with unrestricted coherent Hilbert direct sum.

12.4 An exact process quotient and a sharp context bound

Definition 12.15 (Kraus Gram and contextual equivalence)**.** Fix a nonempty finite S, $n = |S|$, $H = \mathrm{Sym}(S)$, and normalized local lists. Write $K_i = \sum_h a_i(h)h$ and put

$$J_K(h,k) = \sum_i a_i(h)\overline{a_i(k)}.$$

Two lists are contextually equivalent when their Heisenberg maps $\Phi_{K,f}(a) = \sum_i i_f(K_i)^* a\, i_f(K_i)$ agree after every finite injection into an ambient set. Stable scalar-unitary equivalence pads lists by zeros and applies a unitary matrix to the Kraus index. An admissible Gram is $J \geq 0$ with

$$\sum_h J(hx,h) = \delta_{x,1} \qquad (x \in H).$$

Scope. The Gram is process data, not just the isolated CP action. It has trace one by the constraint at $x = 1$.

Provenance. ids: D1126; proved in: definitions.md; tested by: H8,H11; reviewed in: f1-operational-adjudication.md.

Theorem 12.16 (Finite contextual recovery of a quantum process)**.** **PROVED** *Fix a finite set S of size $n \geq 1$ and finite lists in $\mathbb{C}[\mathrm{Sym}(S)]$ normalized by $\sum_i K_i^* K_i = 1$, extended through permutation inclusions along finite injections $S \to T$. Kraus Gram equality, stable scalar-unitary equivalence and equality of all ambient CP actions are equivalent. One ambient set of size $2n-1$ suffices. For $n \geq 2$ this universal size bound is sharp, with explicit positive-state Born witnesses.*

Scope. The theorem concerns retained local Kraus processes of the symmetric-group net. It is not a claim about every structural partial map or every quantum channel in every fusion category.

Provenance. ids: F1-OP-KCF; proved in: f1-operational/contexts.md, KCF-1; tested by: H11 at q=1; reviewed in: f1-operational-adjudication.md.

Proof. In an ambient set of size $2n-1$, choose an involution g exchanging $n-1$ local points with the outside points and fixing the remaining one. Then $H \cap gHg^{-1} = 1$, so all words $h^{-1}gk$ are distinct. The expansion

$$\Phi_K(g) = \sum_{h,k} J_K(k,h)h^{-1}gk$$

recovers every Gram coefficient. Conversely equal Grams give equal maps in all contexts. Equality of finite Grams gives exactly stable unitary mixing by extending an isometry between coefficient-vector spans.

For sharpness let p_+, p_- be the local trivial and sign central projections and $p_0 = 1-p_+-p_-$. The normalized lists $(p_+ + p_-, p_0)$ and $(p_+ - p_-, p_0)$ differ only in cross terms between the trivial and sign sectors. In every smaller possible ambient set, each support intersection has at least two points. Its transposition forces those cross terms to vanish. At size $2n-1$ they do not vanish. For $e = (1+g)/2$, set $\Delta = \Phi_+(e) - \Phi_-(e)$. Both images are effects and have equal trace, so $h = 1 + \Delta$ is a positive normalized density, with Born difference $\tau(\Delta^2) > 0$. □

Thus $n-1$ ancillary constituents can be necessary. At $n = 2$ all inner channels are the isolated identity on $\mathbb{C}^2$, while their Grams form the disk

$$J = \begin{pmatrix} r & ib \\ -ib & 1-r \end{pmatrix}, \qquad b^2 \le r(1-r).$$

A three-constituent context distinguishes them. The corresponding positive-Hecke context-bound extension remains **SKETCH**: its minimal-double-coset argument is written down and exact examples at $q = 2$, $n = 2, 3$ pass, but only the symmetric-group theorem was admitted.

Provenance. ids: F1-OP-KCF-Q; proved in: f1-operational/contexts.md, separated extension; tested by: H11 at q=2; reviewed in: supporting SKETCH.

12.5 The partial-flag category is explicitly constructible

Definition 12.17 (Partial-flag corners and their specialization)**.** Fix real $q > 0$. A composition $\alpha = (\alpha_1, \ldots, \alpha_r)$ of n has positive parts summing to n; include the empty composition of zero. Let W_α preserve its consecutive blocks and put

$$P_\alpha(q) = \sum_{w \in W_\alpha} q^{\ell(w)}, \qquad e_\alpha = P_\alpha(q)^{-1} \sum_{w \in W_\alpha} T_w.$$

Define $\Gamma_q(\alpha, \beta) = e_\beta H_n(q) e_\alpha$ for equal totals, and zero Hom spaces for different totals. Composition is multiplication, identity is e_α, adjoint is $*$, and tensor concatenates compositions through ι. Let $x = (1)$. Finite formal sums and splitting self-adjoint idempotents give the additive completion; only its nonzero objects are used as normalized quantum systems. The generic coefficient ring is $\mathbb{Z}[t, t^{-1}, \{P_\alpha(t)^{-1}\}_\alpha]$, evaluated at $t = q$.

For $L = \mathbb{F}_Q^n$, partial flags have successive quotient dimensions α. If r_α forgets steps, define

$$J_\alpha e_E = P_\alpha(Q)^{-1/2} \sum_{F : r_\alpha(F) = E} e_F.$$

The normalized corner trace is $\tau_\alpha = P_\alpha(q)\tau_n$. A corner Kraus family $v_j \in e_\beta H_n e_\alpha$ is normalized when $\sum_j v_j^* v_j = e_\alpha$ and acts by $a \mapsto \sum_j v_j^* a v_j$.

Scope. The localization is explicit. There is no rigidity or cross-degree preparation in the amplitude category itself.

Provenance. ids: D1141; proved in: definitions.md; tested by: H9,H13; reviewed in: supporting SKETCH.

Proposition 12.18 (A positive monoidal category of flag contexts)**.** **SKETCH** *For $q > 0$, Γ_q is a graded monoidal C^*-category. At a finite field cardinality Q, its corner morphisms are exactly the $GL(L)$ intertwiners between partial-flag Hilbert spaces, through $a \mapsto J_\beta^* a J_\alpha$. Refinement is represented by the corresponding typed projection. At one,*

$$\mathrm{End}_{\Gamma_1}(x^{\otimes n}) = \mathbb{C}[S_n],$$

the C-linear finite-bijection skeleton, with partial-flag retracts added.

Scope. An operational envelope supplies normalized states and additional preparations. The corner trace rescaling must be retained.

Provenance. ids: F1-OP-FLAGCAT; proved in: f1-operational/flag-category.md, sections 1–3; tested by: H9,H13; reviewed in: supporting SKETCH.

The calculation is elementary: $T_s \sum_{W_\alpha} T_w = q \sum_{W_\alpha} T_w$ implies $e_\alpha^2 = e_\alpha = e_\alpha^*$. Block length additivity gives $\iota(e_\alpha \otimes e_\beta) = e_{\alpha\|\beta}$. Refinement gives $e_\beta e_\alpha = e_\beta$, and $J_\alpha J_\alpha^* = e_\alpha$. A density transported by a corner Kraus family uses the factor P_α / P_β between the two normalized traces.

12.6 A concrete coupling back to Weyl quantum mechanics

Definition 12.19 (Weyl constraint flags and controlled tests)**.** Fix $k = \mathbb{F}_Q$ of characteristic p, an n-dimensional named configuration Lagrangian L, and nontrivial additive $\psi : k \to U(1)$. On $\mathbb{C}[L]$ use $X(u)e_y = e_{y+u}$ and $Z(\chi)e_y = \chi(y)e_y$, with additive characters $\chi : L \to U(1)$; the k-linear dual is identified through ψ. For a k-subspace U, let U^{ann} be the characters trivial on it and define

$$P_U^X = |U|^{-1} \sum_{u \in U} X(u), \qquad P_U^Z = |U^{\mathrm{ann}}|^{-1} \sum_{\chi \in U^{\mathrm{ann}}} Z(\chi).$$

On $K_{\mathrm{ctx}}(L) = \mathbb{C}[\mathrm{Fl}(L)]$, declare the flag basis orthonormal and put, on its tensor with the Weyl register,

$$C_i = \sum_F |F\rangle\langle F| \otimes P^Z_{U_i(F)}.$$

The Levi action is $R(g)e_y = e_{gy}$, together with permutation of flags.

Scope. The field, Lagrangian, origin and phase datum are retained. The context and Weyl Hilbert spaces are different registers.

Provenance. ids: D1142; proved in: definitions.md; tested by: character identities in the proof; reviewed in: supporting SKETCH.

Proposition 12.20 (Flags as quantum constraint chains)**.** SKETCH *Fix a finite field $k = \mathbb{F}_Q$, an n-dimensional configuration Lagrangian L and a nontrivial additive character $\psi : k \to U(1)$. Its complete flags inject $GL(L)$-equivariantly into each of the two nested Weyl projector chains, with commutation within each chain. The C_i are commuting orthogonal projections on the joint context/Weyl register, invariant under the diagonal Levi action.*

Scope. This is an extraction and coupling of sectors. It is not an equivalence with full Weyl quantum mechanics or a proof of independence from the polarization.

Provenance. ids: F1-HCK-CONTEXT,F1-OP-COUPLE; proved in: f1-operational/flag-category.md, section 4; tested by: explicit character derivation; reviewed in: supporting SKETCH.

Character orthogonality gives $P_U^Z = \sum_{y \in U} |y\rangle\langle y|$; P_U^X averages on cosets. The two projector-chain ranks are Q^i and Q^{n-i}. These formulas make the coupling explicit, while the dimensions Q^n and $[n]_Q!$ prevent an unmentioned identification of the two registers.

Definition 12.21 (Coordinate-apartment comparison)**.** For a finite field $\mathbb{F}_Q$, $L = \mathbb{F}_Q^n$ and an ordered basis of L, embed the coordinate complete flags by the isometry $J_{\mathrm{ap}} : \mathbb{C}[S_n] \to \mathbb{C}[\mathrm{Fl}(L)]$, using the gallery/regular-action convention. Put $\Omega_Q(a) = J_{\mathrm{ap}}^* a J_{\mathrm{ap}}$ on the context commutant.

Scope. This is an operational comparison at a fixed field, not the definition of substituting one into parameter-dependent coefficients.

Provenance. ids: D1144; proved in: definitions.md; tested by: H9 apartment probes; reviewed in: supporting SKETCH.

Proposition 12.22 (A direct CP map from arithmetic contexts to the endpoint). **SKETCH** *For every prime power Q, apartment compression is basis-independent after permutation labelling and gives a trace-preserving UCP map $H_n(Q) \to \mathbb{C}[S_n]$, $T_w \mapsto w$. It commutes with ordered block inclusions and coefficient expectations.*

Scope. For $Q > 1$ it is not multiplicative and is not automatically natural for every partial-flag corner compression.

Provenance. ids: F1-OP-APART; proved in: f1-operational/flag-category.md, section 7; tested by: H9 at Q=2,3; reviewed in: supporting SKETCH.

Each relative position has one neighbour within the coordinate apartment. Changing its basis conjugates by $GL(L)$, which fixes the commutant. These facts prove the range and independence; isometric compression proves complete positivity. The limits are visible in $\Omega_Q(T_s^2) = (Q-1)s + Q1 \neq 1 = \Omega_Q(T_s)^2$. A coarse flag compression gives scalar Q before the apartment map but scalar one after it. Compatible refinement-process comparison is therefore a concrete remaining problem.

12.7 The broader categorical operational framework

Definition 12.23 (Nonzero categorical subsystem data). Let $\mathcal{C}$ be a unitary fusion category with its positive spherical trace and a tensor-closed replete family of nonzero objects containing the unit. Define

$$A_X = \mathrm{End}_{\mathcal{C}}(X), \quad d_X = \mathrm{Tr}_X(1_X) > 0, \quad \tau_X = \mathrm{Tr}_X / d_X,$$

and $j_{X,Y}(a \otimes b) = a \otimes b \in A_{X \otimes Y}$. Retain associators and the trace-preserving assembly expectations, including their iterated comparison maps. A state has density $\rho \geq 0$, $\mathrm{Tr}_X(\rho) = 1$, representing $\omega(a) = \mathrm{Tr}_X(\rho a)$; effects lie between zero and one. Assembly restriction is the expectation and product-subalgebra preparation embeds its density by j.

Scope. Zero objects are excluded from normalized state systems. This definition uses unnormalized categorical trace, unlike the coefficient trace convention of the group net.

Provenance. ids: D1121,D1122; proved in: definitions.md; tested by: finite expectation examples H3; reviewed in: f1-operational-adjudication.md.

Theorem 12.24 (Positive assembly without a strong Hilbert fibre). **PROVED** *Let $\mathcal{C}$ be a unitary fusion category with its positive spherical trace and a tensor-closed replete family of nonzero objects containing the unit. Then $A_X = \mathrm{End}_{\mathcal{C}}(X)$ are finite C*-algebras with faithful positive traces. Assembly is injective and trace-multiplicative and has a unique UCP trace-preserving expectation. States, restriction, preparation and Born probabilities are well defined and coherent under nested assembly.*

Scope. No strong monoidal Hilbert-space fibre functor is required. The inclusion may be proper.

Provenance. ids: F1-OP-ALG; proved in: f1-operational/cp.md, CPOP-1–2; tested by: H3, examples only; reviewed in: f1-operational-adjudication.md.

In simple-sector coordinates, $A_X = \bigoplus_a M_{m_a}(\mathbb{C})$ and $\mathrm{Tr}_X = \sum_a d_a \mathrm{Tr}_{m_a}$, with $d_a > 0$. This proves faithfulness and identifies the quantum-trace weights. The expectation is orthogonal projection for the trace inner product; matrix amplification proves its complete positivity. The relevant internal operator-algebra analogue is developed by Jones–Penneys; their internal algebra objects are not being identified with this net without a comparison.

Definition 12.25 (Retained categorical Kraus and dilation processes)**.** Let $\mathcal{C}$ be a unitary fusion category with positive spherical trace, and fix a tensor-closed replete family of nonzero objects containing the unit. For family objects X, Y, a Kraus process is a finite list $K_i : X \to Y$ with $\sum_i K_i^* K_i = 1_X$, modulo stable scalar-unitary mixing. It acts by $\Phi_K(a) = \sum_i K_i^* a K_i$ and $\mathcal{T}_K(\rho) = \sum_i K_i \rho K_i^*$; its retained right-context operators are $K_i \otimes 1_Z$. Composition uses $L_j K_i$, and tensor uses $K_i \otimes L_j$, with the category associators.

A charge-carrying dilation is a named nonzero environment E and isometry $v : X \to Y \otimes E$, with $\Phi_v(a) = v^*(a \otimes 1_E)v$. Sequential composition retains its ordered environment. Parallel composition requires named unitary interchange, supplied for example by a unitary braiding or a specified environment half-braiding.

Scope. Not every isolated CP map is asserted to be represented by these local lists, and arbitrary environments are not silently interchangeable in a merely monoidal category.

Provenance. ids: D1123,D1124; proved in: definitions.md; tested by: H8; finite examples; reviewed in: f1-operational-adjudication.md.

Theorem 12.26 (Coherent quantum processes from retained operators)**.** PROVED *In a unitary fusion category with positive spherical trace, fix a tensor-closed replete family of nonzero objects containing the unit. On that family, normalized Kraus lists compose and tensor modulo stable scalar-unitary mixing. Their shadows are UCP and their density maps preserve categorical trace in every retained context. The stated dilations compose sequentially and in parallel when the named interchange data are provided.*

Scope. This does not quotient by equality of isolated shadows. That coarser identification fails in the examples above.

Provenance. ids: F1-OP-PROC; proved in: f1-operational/cp.md, CPOP-3; tested by: H8; reviewed in: f1-operational-adjudication.md.

The essential identity is $\sum_{i,j} K_i^* L_j^* L_j K_i = 1_X$; cyclicity gives trace preservation. Scalar unitary mixing cancels the Kraus indices in every context. These are the data that an isolated CP map can fail to retain.

For comparison, use the standard unitary Fibonacci data

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \qquad R = \operatorname{diag}(e^{-4\pi i/5}, e^{3\pi i/5}).$$

On the two-anyon algebra $\mathbb{C}^2$, Ad_R is identity. In the charge-τ matrix block of three anyons it is nontrivial: the $|+\rangle$ return probability becomes $(5-\sqrt{5})/8$. The braid FRF has $|0\rangle$ return probability $(3-\sqrt{5})/2$. These sourced comparisons remain SKETCH and do not identify Fibonacci with the arithmetic endpoint.

Provenance. ids: F1-OP-FIB; proved in: f1-operational/cp.md, CPOP-4; Ahmadi–Kissinger 2211.03855; tested by: independent exact critic recomputation; reviewed in: supporting SKETCH.

12.8 Entanglement and alternative branches

Definition 12.27 (A Bell density on two collective sectors)**.** In $\mathbb{C}[S_3]$, let $s = (12)$, $t = (23)$, $z = (2 - (123) - (132))/3$, and put $Z_* = zs$, $X_{\text{num}} = z(s+2t)$, $Y_{\text{anti}} = z(st - ts)$. For the block inclusion into $\mathbb{C}[S_6]$, define

$$P_{\text{Bell}} = \frac{1}{4}\mu\left(z \otimes z + Z_* \otimes Z_* + \frac{X_{\text{num}} \otimes X_{\text{num}} + Y_{\text{anti}} \otimes Y_{\text{anti}}}{3}\right), \qquad h_{\text{Bell}} = 9P_{\text{Bell}}.$$

On the standard sector the normalized observables are Z_* and $X_* = X_{\text{num}}/\sqrt{3}$.

Scope. The coefficient trace on the full six-constituent algebra fixes the density factor nine.

Provenance. ids: D1143; proved in: definitions.md; tested by: H14; reviewed in: supporting SKETCH.

Proposition 12.28 (A Bell state of two disjoint collective qubits)**.** SKETCH *The stated P_{Bell} is a positive projection, $\tau_6(h_{\mathrm{Bell}}) = 1$, and both subgroup marginals are $(3/2)z$, the maximally mixed standard-sector qubit states. Its XX and ZZ correlations are one and its crossed correlations zero, giving the ordinary CHSH value $2\sqrt{2}$ with the corresponding qubit measurements.*

Scope. This uses two disjoint size-three regions and their local matrix blocks; it does not assert that all composite observables factor locally.

Provenance. ids: F1-OP-BELL; proved in: f1-operational/flag-category.md, section 6; tested by: H14; reviewed in: supporting SKETCH.

In the standard block $X_{\mathrm{num}} = \sqrt{3}X$ and $Y_{\mathrm{anti}} = i\sqrt{3}Y$, so the formula is the usual Bell projector. Each standard block contributes trace weight $1/3$; their product gives $1/9$. The conditional expectation extends the Bell functional to the full composite while preserving these local correlations.

The type-C flag route remains distinct: at one its signed permutation groups have a rank-two D_8 qubit sector, but $B_m \times B_n$ is not parabolic in B_{m+n}. A separate assembly theorem is needed. The rank-two Temperley–Lieb route also needs extra input: its coefficient is $\delta^{-2} = q/(q+1)^2$, with $\delta = \sqrt{q} + 1/\sqrt{q}$. At one it gives the $SU(2)$ spin tensor-power centralizer tower after the quotient and a Jones/Markov trace. The faithful flag trace cannot descend through the nonzero quotient: at level two the flag weights are $(1/2, 1/2)$, whereas spin singlet/triplet weights are $(1/4, 3/4)$. Full rigidity additionally requires duality conventions. The Fibonacci quotient is a different root-of-unity construction with its corresponding positive star/trace structure.

The substantive outcome is a quantum subsystem family with intrinsic arithmetic memory, an explicit positive endpoint and a sharp description of its process data. The remaining central task is comparison with the full Weyl system and all its refinement/polarization processes. Those missing comparisons are stated as such; they are not hidden inside the word "canonical."

13 A continuous Hecke limit and its complete category

The positive Hecke parameter gives a genuine continuous path from arithmetic flag algebras to the symmetric-group fibre at $q = 1$. The construction below keeps one parameter throughout: tensor products of section algebras are balanced over $C(I)$, whereas tensor products inside one fibre are ordinary spatial tensor products. Completing the parabolic objects then produces all finite self-adjoint matrix corners, including off-diagonal maps between summands that happen to have the same total degree.

13.1 Continuous Hecke sections and parabolic corners

Definition 13.1 (Positive Hecke algebra and normalized regular frame)**.** Fix $q > 0$. Let $H_0(q) = H_1(q) = \mathbb{C}$, and for $n \geq 2$ let $H_n(q)$ be generated by self-adjoint $T_1, \ldots, T_{n-1}$ subject to

$$(T_i - q)(T_i + 1) = 0, \qquad T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \qquad T_iT_j = T_jT_i \quad (|i-j| > 1).$$

For $w \in S_n$, write T_w for a reduced-word product and $\ell(w)$ for its Coxeter length. The coefficient trace is $\tau_{n,q}(T_w) = \delta_{w,1}$. Put

$$b_w(q) = q^{-\ell(w)/2}T_w(q).$$

The vectors $b_w(q)$ are identified with the standard basis δ_w of $K_n = \ell^2(S_n)$, and $B_w(q)$ denotes left multiplication by $b_w(q)$ on this fixed Hilbert space.

Scope. The embedding in the fixed regular Hilbert space is named data. It does not make the Hecke generators independent of q, and no positivity is asserted for a non-real or non-positive parameter.

Provenance. ids: D1101, D1201; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: theory/checks/f1_limit_check.py (L1); theory/checks/f1_operational_check.py (H1,H2); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 13.2 (Continuous positive Hecke section algebra)**.** For a nonempty compact interval $I \subset (0, \infty)$, define

$$H_n^{\mathrm{cts}}(I) = \left\{ q \longmapsto \sum_{w \in S_n} f_w(q)B_w(q) : f_w \in C(I) \right\} \subset C\big(I, \mathrm{End}(K_n)\big).$$

Operations are pointwise and the norm is the supremum operator norm. Its $C(I)$-valued trace is

$$\tau_n^{\mathrm{cts}}(x)(q) = \tau_{n,q}(x(q)).$$

Scope. This is a continuous section algebra over a compact positive interval. It is not a generic algebra over a Laurent-polynomial coefficient ring.

Provenance. ids: D1202; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: finite-fibre probes L1,H1,H2; closedness is proved analytically; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 13.3 (Continuous regular Hecke algebras)**.** PROVED *For every nonempty compact $I \subset (0, \infty)$ and every $n \geq 0$, $H_n^{\mathrm{cts}}(I)$ is a unital C^*-subalgebra of $C(I, \mathrm{End}(K_n))$. Evaluation at $q_0 \in I$ is a surjective *-homomorphism onto the faithful regular copy of $H_n(q_0)$, and τ_n^{cts} is continuous, tracial, positive and faithful.*

Scope. The assertion is uniform in the rank and on every compact positive interval. The finite checks test fibres; they do not prove norm-closedness or continuity.

Provenance. ids: F1-LIM-CTS; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: theory/checks/f1_limit_check.py (L1); theory/checks/f1_operational_check.py (H1,H2); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. On a length pair $w, s_i w$ with $\ell(s_i w) = \ell(w) + 1$, left multiplication by T_i has matrix

$$\begin{pmatrix} 0 & \sqrt{q} \\ \sqrt{q} & q-1 \end{pmatrix}$$

in the normalized basis. Hence all $B_w(q)$ vary continuously. The Hecke relations show that their $C(I)$-span is a unital *-algebra. If a sequence of sections converges uniformly, applying it to δ_1 recovers every coefficient function uniformly; the limit therefore remains in the displayed span. This proves closedness. Constant coefficient functions prove surjectivity of every evaluation. Finally,

$$\tau_{n,q}(x^* x) = \sum_w |c_w|^2 q^{\ell(w)} \quad \text{for } x = \sum_w c_w T_w,$$

so fibre traces, and consequently the section trace, are faithful and positive. □

Definition 13.4 (Continuous parabolic corner category)**.** For a composition $\alpha = (\alpha_1, \dots, \alpha_r)$ of n, let $W_\alpha \leq S_n$ preserve its consecutive blocks and set

$$P_\alpha(q) = \sum_{w \in W_\alpha} q^{\ell(w)}, \qquad e_\alpha(q) = P_\alpha(q)^{-1} \sum_{w \in W_\alpha} T_w(q).$$

The objects of Γ_I^{cts} are compositions, including the empty composition of zero, and

$$\mathrm{Hom}(\alpha, \beta) = \begin{cases} e_\beta H_n^{\mathrm{cts}}(I) e_\alpha, & |\alpha| = |\beta| = n, \\ 0, & |\alpha| \neq |\beta|. \end{cases}$$

Composition, identity and dagger are multiplication, e_α and matrix adjoint. Ordered tensor concatenates compositions. On section algebras its domain is the balanced product

$$H_m^{\mathrm{cts}}(I) \otimes_{C(I)} H_n^{\mathrm{cts}}(I),$$

the section algebra of the pointwise spatial fibre products, and the map is the contiguous-block inclusion. On the endomorphism corner the normalized trace is

$$\tau_{\alpha,q} = P_\alpha(q) \tau_{n,q}.$$

The germ category at one identifies two morphism sections when they agree on some smaller interval $[1-\varepsilon, 1+\varepsilon]$. A germ is positive when it has a positive representative on one such interval.

Scope. The ordinary tensor product over $\mathbb{C}$ is not the section tensor: for nonconstant f, the nonzero element $f \otimes 1 - 1 \otimes f$ is killed by restriction to the diagonal. Only evaluation at one is canonical on a germ, and no C*-norm is assigned to germ Hom spaces.

Provenance. ids: D1141, D1203, D1204; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: theory/checks/f1_operational_check.py (H3,H13); theory/checks/f1_limit_check.py (L4,L14); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 13.5 (Continuous corners and their endpoint germ)**.** PROVED *For every compact positive interval I, the preceding corners form an ordered monoidal C*-category. Their normalized traces are faithful and multiplicative under ordered tensor; the corner inclusions and their trace-preserving expectations are continuous and obey the three-block tower laws. Fibre evaluation is full. Germs at one form an ordered monoidal *-category with a full canonical endpoint evaluation to the $q = 1$ corner category.*

Scope. No block exchange is supplied for $q \neq 1$. A nonzero germ may vanish at the endpoint, which is why endpoint norm is not used as a C*-norm.

Provenance. ids: F1-LIM-CORNER, F1-LIM-GERM; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: H3,H13; L4,L14 on finite fibres and representatives; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Writing $x_\alpha = \sum_{w \in W_\alpha} T_w$, the Hecke multiplication rule gives $T_i x_\alpha = q x_\alpha$ for every simple reflection in the parabolic. Thus $x_\alpha^2 = P_\alpha x_\alpha$ and e_α is a self-adjoint projection. Block length additivity sends $B_u \otimes B_v$ to the distinct section $B_{u \times v}$, proving injectivity of the balanced tensor map. Moreover $P_{\alpha\|\beta} = P_\alpha P_\beta$ and $\iota(e_\alpha \otimes e_\beta) = e_{\alpha\|\beta}$. The ambient coefficient expectation is bimodular, so it restricts to each corner, fixes the included product algebra and preserves the normalized trace. All operations commute with restriction and evaluation. To lift a fibre corner morphism, lift it with constant normalized-basis coefficients and compress by the source and target projection sections. This proves fullness, both on an interval and after passing to endpoint germs. □

Definition 13.6 (Arithmetic and endpoint evaluations)**.** An arithmetic fibre evaluates the continuous construction at the positive real number $q = Q = p^r$ and may then apply the complete- or partial-flag realization over $\mathbb{F}_Q$. Endpoint specialization evaluates at $q = 1$. Coordinate-apartment compression at a fixed arithmetic fibre is a separate UCP map and is not an evaluation map.

Scope. No sequence of prime powers tends to one. The construction retains the chosen flag context and does not specialize the full Weyl system or all polarizations.

Provenance. ids: D1216; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: theory/checks/f1_operational_check.py (H9,H13); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proposition 13.7 (Arithmetic partial-flag realization)**.** PROVED *For every prime power Q and every composition α of n, the fibre corner $e_\alpha H_n(Q) e_\alpha$ is the full $GL_n(\mathbb{F}_Q)$-commutant on partial flags of type α, and $P_\alpha(Q)\tau_{n,Q}$ is normalized physical operator trace. These identifications preserve corner morphisms, composition and dagger.*

Scope. This is an evaluation followed by a finite-field realization, rather than a limit over finite fields. Apartment compression remains nonmultiplicative when $Q > 1$.

Provenance. ids: F1-LIM-ARITH; proved in: theory/sidequests/f1-limit/continuous-corners.md; tested by: theory/checks/f1_operational_check.py (H9,H13); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. A partial flag of type α has $P_\alpha(Q)$ complete refinements. The normalized fibre-sum isometry J_α has range projection e_α. Consequently $F \mapsto J_\beta F J_\alpha^*$ identifies every partial-flag intertwiner with $e_\beta H_n(Q) e_\alpha$, with inverse $x \mapsto J_\beta^* x J_\alpha$. Ordinary trace on the range of J_α and the flag-count ratio give the asserted normalized trace. □

13.2 All finite self-adjoint summands

Definition 13.8 (Finite graded self-adjoint completion)**.** For a nonempty compact positive interval I, put $A_n(I) = H_n^{\mathrm{cts}}(I)$. An object X is a finite-support family $(r_n, p_{X,n})_{n \geq 0}$, where $r_n \geq 0$ and $p_{X,n} \in M_{r_n}(A_n(I))$ is a self-adjoint projection. For $Y = (s_n, p_{Y,n})$, define

$$\mathrm{Hom}(X, Y) = \bigoplus_n p_{Y,n} M_{s_n, r_n}(A_n(I)) p_{X,n}.$$

Composition and dagger are degreewise matrix multiplication and adjoint, the identity is $(p_{X,n})_n$, and the norm is the maximum of the finitely many supremum C*-norms. Direct sum uses block-diagonal projections within each degree, so its endomorphism algebra retains the off-diagonal rectangular corners between equal-degree summands. Different degrees have zero Homs. The fibre version at q is denoted $\mathcal{U}_q^\dagger$.

Scope. This is the finite additive dagger Karoubi completion. It does not assume duals, rigidity, a spherical trace or finitely many simple objects.

Provenance. ids: D1261; proved in: theory/sidequests/f1-limit/karoubi-trace.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Remark. An additive object is a coherent multiplicity resource, not a classical alternative. For example, the direct sum of two tensor units has endomorphism algebra $M_2(\mathbb{C})$, including off-diagonal coherences. A two-outcome classical wire instead has the commutative algebra $\mathbb{C}^2$. Adding finite sums therefore does not change the original generator tower $\mathrm{End}(x^{\otimes n}) = H_n(q)$, and it does not give the bare one-constituent object qubit observables.

Definition 13.9 (Balanced ordered tensor and finite trace)**.** For objects X, Y, the degree-k multiplicity of $X \otimes Y$ is

$$t_k = \sum_{m+n=k} r_m s_n.$$

Its projection is block diagonal, indexed by (m, n) and lexicographic matrix indices, with block

$$\iota_{m,n}^{(r_m,s_n)}(p_{X,m} \otimes p_{Y,n}).$$

Morphism tensor is the analogous block-diagonal amplification. The full endomorphism corner in $M_{t_k}(A_k(I))$ includes off-diagonal blocks between different pairs with the same sum. The unit is the degree-zero object $(1, 1)$, and associators are the permutation matrices identifying the two triple indexings. Section tensor is balanced over $C(I)$.

For $a \in \mathrm{End}(X)$ put

$$\theta_X(a) = \sum_n (\mathrm{Tr}_{r_n} \otimes \tau_n)(a_n), \qquad w_X = \theta_X(1_X), \qquad \tau_X = \theta_X / w_X.$$

Here matrix trace is unnormalized. Operational systems are nonzero objects; a density satisfies $\rho \geq 0$ and $\tau_X(\rho) = 1$.

Scope. Associators reorder bookkeeping indices but never exchange ordered Hecke blocks. The trace is a finite graded matrix/coefficient trace and is not claimed to arise from categorical duality.

Provenance. ids: D1262, D1263; proved in: theory/sidequests/f1-limit/karoubi-trace.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1,K2); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 13.10 (Full continuous completion and its trace)**.** PROVED *For every nonempty compact positive interval, the finite projection objects above form an additive ordered monoidal C*-category with every self-adjoint idempotent split. The additive self-adjoint completion of the parabolic corner category is unitarily strong-monoidally equivalent to it. The trace θ is continuous, positive, faithful and cyclic across rectangular Homs; on a connected interval every nonzero object has $w_X(q) > 0$, and*

$$w_{X\otimes Y} = w_X w_Y, \qquad \tau_{X\otimes Y}\big(j(a \otimes b)\big) = \tau_X(a)\tau_Y(b).$$

At a fibre this is the self-adjoint model of the full finite Hecke composition category.

Scope. The dagger statement concerns self-adjoint projection presentations. An algebraic idempotent is merely isomorphic, not canonically unitarily equal, to its support projection.

Provenance. ids: F1-LIM-KAR, F1-LIM-KTRACE; proved in: theory/sidequests/f1-limit/karoubi-trace.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1,K2); uniform assertions by proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Each Hom is a closed rectangular corner of a finite direct sum of matrix C*-algebras. Block diagonal sums give biproducts, and a self-adjoint idempotent is itself a new projection object. The normalized block basis proves injectivity of the balanced tensor, while permutation matrices on triple indices give the associator and pentagon. The complete parabolic composition has projection 1 in every degree, so its sums and retracts generate all objects.

For a rectangular a, direct expansion gives

$$(\mathrm{Tr}\otimes\tau_n)(a^*a)=\sum_{i,j}\tau_n(a_{ij}^*a_{ij}),$$

which proves faithfulness. Cyclicity follows entrywise. A continuous projection has constant ordinary rank on a connected interval, so a nonzero object cannot disappear at one fibre and w_X stays positive. Finally the matrix trace and coefficient trace both multiply on each (m,n) block; summing over all pairs proves both displayed tensor identities. ☐

Definition 13.11 (Traced assembly of completed objects)**.** For nonzero X,Y, let $B_X=\mathrm{End}(X)$ and let

$$j_{X,Y}:B_X\otimes B_Y\longrightarrow B_{X\otimes Y}$$

be morphism tensor; on intervals its domain is $C(I)$-balanced. Define $E_{X,Y}:B_{X\otimes Y}\to B_X\otimes B_Y$ by

$$(\tau_X\otimes\tau_Y)(b^*E_{X,Y}(a))=\tau_{X\otimes Y}(j_{X,Y}(b)^*a)$$

for all b. Assembly is the process from the separated word $[X][Y]$ to the collective atom $[X\otimes Y]$ represented in Heisenberg orientation by $E_{X,Y}$; split has the reverse type and is represented by $j_{X,Y}$.

Scope. The separated algebra and the collective endomorphism algebra are different. The trace is the finite graded trace just defined, not a pivotal trace.

Provenance. ids: D1264; proved in: theory/sidequests/f1-limit/karoubi-trace.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1,K5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 13.12 (Continuous traced expectations)**.** PROVED *For nonzero completed objects on a connected compact positive interval, $j_{X,Y}$ is an injective unital trace-preserving *-map and $E_{X,Y}$ is the unique normalized-trace-preserving UCP retraction. These expectations act continuously on sections, satisfy $Ej=\mathrm{id}$ and the ordered three-object tower laws, while jE is generally a proper conditional expectation. On parabolic objects they recover the earlier corner maps.*

Scope. No rigidity or fusion hypothesis is used. Properness means that collective same-degree off-diagonal observables can be erased by jE.

Provenance. ids: F1-LIM-KEXPECT; proved in: theory/sidequests/f1-limit/karoubi-trace.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1,K5); complete positivity and continuity by proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. In one fibre, orthogonal projection onto $j(B_X \otimes B_Y)$ for the faithful trace inner product fixes the subalgebra and is bimodular. If the image of a positive element had a negative spectral projection in the subalgebra, the trace-pairing identity would be simultaneously negative and nonnegative. Applying the same argument to matrix amplifications proves complete positivity. A local continuous basis turns the trace equations into $G(q)c(q) = t(q)$ with $G(q) > 0$; hence $c(q) = G(q)^{-1}t(q)$ is continuous. Uniqueness proves agreement on overlaps and the three-object tower law. $\square$

14 Typed operations, classical histories and collective context

The amplitude category has morphisms only within a fixed total degree. An operational theory needs more: preparation, discard, measurement and channels must be arrows between observable algebras. We build those arrows in Heisenberg orientation, retain their Kraus data, and keep two kinds of composition separate. Juxtaposing registers produces a spatial tensor algebra; assembling them produces one larger collective corner whose observable algebra usually contains more operators.

14.1 Registers and primitive processes

Definition 14.1 (Quantum atoms, classical wires and register words)**.** For every nonempty composition α, let $[\alpha]$ be a quantum atom with observable algebra

$$A_\alpha(q) = e_\alpha(q) H_{|\alpha|}(q) e_\alpha(q), \qquad \tau_{\alpha,q} = P_\alpha(q)\tau_{|\alpha|,q}.$$

A register word is a finite ordered word of quantum atoms and classical wires. The algebra and reference trace of the word are the ordered spatial tensor products of the factor algebras and traces; the empty word has algebra $\mathbb{C}$. For a nonempty finite set O, the wire $\underline{O}$ has algebra $\mathbb{C}^O$ and uniform trace

$$\tau_O(f) = |O|^{-1} \sum_{o\in O} f(o).$$

The separated word $[\alpha][\beta]$ and collective atom $[\alpha \parallel \beta]$ are different objects.

Scope. The empty outcome set is excluded from deterministic instruments. A classical wire has a commutative algebra; a coherent additive direct sum in the completed amplitude category instead has matrix off-diagonal observables.

Provenance. ids: D1205; proved in: theory/sidequests/f1-limit/operational-category.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_wiring_check.py (W1–W3,W5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 14.2 (Finite Heisenberg UCP category)**.** Objects are finite-dimensional unital C*-algebras. An arrow $A \to B$ is a unital completely positive map $\Phi : B \to A$. If $A \to B \to C$ are represented by $\Phi : B \to A$ and $\Psi : C \to B$, their composite is $\Phi \circ \Psi : C \to A$; tensor is the spatial tensor of maps. For faithful reference traces, the density dual is the unique map satisfying

$$\tau_B\big(\Phi_*(\rho)a\big) = \tau_A\big(\rho\Phi(a)\big).$$

Scope. This target records operational UCP maps and their density duals. It does not assert that every such map possesses a preferred Hecke-corner Kraus label.

Provenance. ids: D1206; proved in: theory/sidequests/f1-limit/operational-category.md; tested by: finite UCP examples in theory/checks/f1_operational_check.py; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 14.3 (Preparations, measurements and retained instruments)**.** For a register word X, a preparation is a density $h \geq 0$, $\tau_X(h) = 1$, with Heisenberg functional $a \mapsto \tau_X(ha)$. Discard is $\lambda \mapsto \lambda 1_X$. A POVM with outcome set O is a family $e_o \geq 0$, $\sum_o e_o = 1_X$, and realizes

$$\mathbb{C}^O \longrightarrow A_X, \qquad f \longmapsto \sum_o f(o)e_o.$$

An effect $0 \leq e \leq 1$ is the two-outcome POVM $(e, 1 - e)$.

For equal-total compositions α, β, a corner instrument is a finite family

$$K_{o,i} \in e_\beta H_n(q) e_\alpha, \qquad \sum_{o,i} K_{o,i}^* K_{o,i} = e_\alpha,$$

with type

$$K : [\alpha] \longrightarrow [\beta]\underline{O}$$

and Heisenberg realization

$$\Phi_K((a_o)_o) = \sum_{o,i} K_{o,i}^* a_o K_{o,i}.$$

At one fibre, two labels are equal when the normalized-basis coefficient Gram matrices agree separately in each outcome. Equivalently, after zero padding they differ by a scalar-unitary mixing of the Kraus index within each outcome.

Scope. The quantum output of an instrument is retained. A POVM alone specifies effects but no postmeasurement state. Kraus indices may be mixed within an outcome, never across different classical outcomes.

Provenance. ids: D1207; proved in: theory/sidequests/f1-limit/operational-category.md; tested by: theory/checks/f1_operational_check.py (H8); theory/checks/f1_limit_wiring_check.py (W3,W4); reviewed in: theory/verdicts/f1-limit-adjudication.md.

The branch density for outcome o is not the raw matrix sum when source and target corner traces have different normalizations. It is

$$T_o(\rho) = \frac{P_\alpha(q)}{P_\beta(q)} \sum_i K_{o,i} \rho K_{o,i}^*.$$

Relative to $\tau_\beta \otimes \tau_O$, the o-block of the full recorded density is $|O| T_o(\rho)$. Cyclicity proves that its trace is precisely the branch Born weight and that the sum of all branch masses is one.

Proposition 14.4 (UCP realization of primitive operations)**.** PROVED *For every $q > 0$, every preparation, discard, finite POVM and normalized corner instrument just defined is a typed UCP process. Its Schrödinger dual is positive and trace preserving, and the branch density has the displayed corner and classical cardinality factors.*

Scope. This is a sufficient retained process class. It makes no completeness claim for arbitrary CP maps between two corner algebras.

Provenance. ids: F1-LIM-OP; proved in: theory/sidequests/f1-limit/operational-category.md; tested by: theory/checks/f1_operational_check.py (H8); theory/checks/f1_limit_wiring_check.py (W3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Preparations are positive normalized functionals and discards are unital *-maps. At matrix level a positive element of $M_r(\mathbb{C}^O)$ is a family of positive scalar matrices, so tensoring each with e_o and summing proves complete positivity of the POVM map. Each instrument summand has the form $a \mapsto K^* a K$; completeness makes their sum unital. The density formula follows from cyclicity:

$$\tau_\beta(T_o(\rho) a) = \tau_\alpha\left(\rho \sum_i K_{o,i}^* a K_{o,i}\right).$$

□

14.2 Assembly is a retraction, not an identification

Definition 14.5 (Assembly and split)**.** For parabolic atoms, let

$$\mathrm{asm}_{\alpha,\beta} : [\alpha][\beta] \longrightarrow [\alpha \parallel \beta]$$

be represented in Heisenberg orientation by the trace-preserving conditional expectation

$$E_{\alpha,\beta} : A_{\alpha\|\beta} \longrightarrow A_\alpha \otimes A_\beta.$$

Let $\mathrm{spl}_{\alpha,\beta}$ have the reverse type and be represented by the corner inclusion $j_{\alpha,\beta} : A_\alpha \otimes A_\beta \to A_{\alpha\|\beta}$. The defining relations include the three-block tower relations and

$$\mathrm{spl} \circ \mathrm{asm} = \mathrm{id}_{[\alpha][\beta]}\,.$$

The reverse composite is the collective coarse graining jE.

Scope. The map jE is generally proper and is not set equal to the collective identity. No assembly retraction is promoted to an isomorphism.

Provenance. ids: D1208; proved in: theory/sidequests/f1-limit/operational-category.md; tested by: theory/checks/f1_limit_check.py (L14); theory/checks/f1_operational_check.py (H3,H13); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 14.6 (Coherent collective assembly)**.** **PROVED** *Assembly and split are associative UCP processes, $Ej = \mathrm{id}$, and jE is the conditional expectation onto the separated product subalgebra. Preparing densities h, k on separated registers and then assembling gives the collective density $j(h \otimes k)$.*

Scope. The conclusion uses contiguous ordered blocks and the normalized corner traces. It supplies neither a block exchange nor equality of separated and collective observables.

Provenance. ids: F1-LIM-ASM; proved in: theory/sidequests/f1-limit/operational-category.md; tested by: L14; H3,H13; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The Hecke block inclusion is a trace-preserving *-monomorphism and the coefficient expectation fixes its range, giving $Ej = \mathrm{id}$. The two three-block inclusions and expectations agree on the standard basis, so the processes are associative. Trace duality gives $\tau(j(h \otimes k)a) = (\tau_\alpha \otimes \tau_\beta)((h \otimes k)E(a))$, proving the density statement. □

Definition 14.7 (Retained collective context and parallel lists)**.** For $K : [\alpha] \to [\beta]\underline{O}$ and a right context γ, retain

$$K \triangleleft \gamma : [\alpha \parallel \gamma] \longrightarrow [\beta \parallel \gamma]\underline{O}, \qquad (K \triangleleft \gamma)_{o,i} = \iota(K_{o,i} \otimes e_\gamma).$$

For a second list $L : [\alpha'] \to [\beta']\underline{P}$, the direct ordered collective parallel instrument has type

$$K \boxtimes_{\mathrm{c}} L : [\alpha \parallel \alpha'] \longrightarrow [\beta \parallel \beta']\underline{O \times P}$$

and list $(\iota(K_{o,i} \otimes L_{p,j}))_{(o,p),(i,j)}$.

Scope. These lists act on full collective corner algebras. They are not inferred from the isolated CP shadows and preserve the displayed left-to-right order.

Provenance. ids: D1209; proved in: theory/sidequests/f1-limit/operational-category.md; tested by: theory/checks/f1_operational_check.py (H5,H8); reviewed in: theory/verdicts/f1-limit-adjudication.md.

14.3 Classical history wires

Definition 14.8 (Classical products, relabeling and routing)**.** Fix a singleton $*$. A bijection $r : O \to P$ gives $\mathrm{rel}_r : \underline{O} \to \underline{P}$, represented by $r^* : C^P \to C^O$, $r^* f = f \circ r$. Classical product and unit are the *-isomorphisms

$$\mathrm{mul}_{O,P} : \underline{O}\,\underline{P} \longrightarrow \underline{O \times P}, \qquad \mathrm{unit}_{\mathrm{c}} : \underline{*} \longrightarrow \mathbf{1},$$

induced by $\delta_{(o,p)} \mapsto \delta_o \otimes \delta_p$ and $\mathbb{C}^{\{*\}} \cong \mathbb{C}$; their inverses are included. Cartesian associators and singleton unitors are the corresponding relabelings.

For every object word X, the routing isomorphism

$$\mathrm{route}_{O,X} : \underline{O}, X \longrightarrow X\,\underline{O}$$

is represented by the spatial flip $A_X \otimes C^O \to C^O \otimes A_X$. It is natural, involutive after exchanging source and target, and satisfies the usual nesting and unit relations.

For instruments $K : [\alpha] \to [\beta]\underline{O}$ and $L : [\beta] \to [\gamma]\underline{P}$, define

$$\begin{aligned}&\mathrm{Seq}(L, K) = (\mathrm{id} \otimes \mathrm{mul}_{O,P})(\mathrm{id} \otimes \mathrm{route}_{P,\underline{O}})(L \otimes \mathrm{id}_{\underline{O}})K,\\&\mathrm{Seq}(L, K) : [\alpha] \to [\gamma]\underline{O \times P}.\end{aligned}$$

Its branch list is $(L_{p,j}K_{o,i})_{(o,p),(j,i)}$. For $K' : [\alpha'] \to [\beta']\underline{P'}$, define

$$\begin{aligned}&\mathrm{Par}(K, K') = (\mathrm{id} \otimes \mathrm{mul}_{O,P'})(\mathrm{id}_{[\beta]} \otimes \mathrm{route}_{O,[\beta']} \otimes \mathrm{id}_{\underline{P'}})(K \otimes K'),\\&\mathrm{Par}(K, K') : [\alpha][\alpha'] \to [\beta][\beta']\underline{O \times P'}.\end{aligned}$$

Sequential associativity uses $((o, p), r) \leftrightarrow (o,(p, r))$; sequential–parallel interchange uses

$$((o, p),(o', p')) \longleftrightarrow ((o, o'),(p, p')).$$

One-outcome instruments use the singleton unitor. The single Kraus list (e_α), followed by that unitor, is the identity process on $[\alpha]$; its Heisenberg map is $a \mapsto e_\alpha a e_\alpha = a$.

Scope. Only a classical wire moves through a quantum output. These maps do not provide a quantum block commutor at generic q.

Provenance. ids: D1218; proved in: theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_wiring_check.py (W1–W5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 14.9 (Presented operational categories and evaluation)**.** At a fibre, take the strict ordered monoidal category generated by the register objects, preparations, discards, POVMs, retained instruments, assembly/split, context/parallel lists and classical wiring above. Equality is the congruence generated by typed planar isomorphism, category and monoidal axioms, literal state and POVM equality, outcome-wise coefficient-Gram equality, the routed history relations, context identities and assembly tower relations. It is not equality of one isolated CP shadow.

On an interval, labels are continuous and two Kraus labels are equal when their coefficient Grams agree at every parameter. At the endpoint, circuit germs are equal when the presentation relation, and in particular Gram equality, holds on a common smaller interval. No continuous choice of a mixing unitary is required. Realization sends objects to their algebras and generators to the stated UCP maps. Evaluation substitutes the parameter in every label; only endpoint evaluation is canonical for an arbitrary germ.

Scope. The realization may be nonfaithful. Local lifting of one circuit will not imply that every extra equation true only at the endpoint also lifts.

Provenance. ids: D1210, D1211; proved in: theory/sidequests/f1-limit/operational-category.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_wiring_check.py (W1–W5); H8,L14; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 14.10 (Sound typed operational category). PROVED *For every positive fibre, compact positive interval and endpoint germ, the preceding generators and relations define an ordered monoidal category with a monoidal UCP realization. Routed sequential and parallel instruments are normalized, history associativity and interchange are typed by the displayed bijections, and pointwise or eventual Gram equality is a congruence.*

Scope. No classical retyping is hidden in planar isotopy. No quantum exchange is asserted away from one, and no arbitrary CP-map completeness is claimed.

Provenance. ids: F1-LIM-OP; proved in: theory/sidequests/f1-limit/operational-category.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_wiring_check.py (W1–W5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Every wiring box is a trace-preserving *-isomorphism represented by a finite permutation matrix. Sequential completeness follows from

$$\sum_{o,p,i,j} K_{o,i}^* L_{p,j}^* L_{p,j} K_{o,i} = 1,$$

and parallel completeness factors into the two original sums. The two associative histories have the same branch operator and differ only by their Cartesian bracketing. Interchange follows from multiplicativity of the ordered block inclusion and the stated history bijection. At each fibre, equal coefficient Grams are equivalent to outcome-wise scalar-unitary mixing; tensoring those finite unitaries proves congruence pointwise, without choosing them continuously. ☐

14.4 A finite collective context detects retained Kraus data

Definition 14.11 (Local contextual recovery map). For the complete n-constituent object, expand a retained list in the normalized basis and put

$$J_K(k,h) = \sum_i a_i(k)\overline{a_i(h)}.$$

Embed the local regular operator explicitly into rank $2n-1$ by

$$\widetilde{B}_h(q) = \iota_{n,n-1}(B_h(q) \otimes 1).$$

Choose an involution $d \in S_{2n-1}$ fixing one local point and pairing the other $n-1$ local points with the complement. The recovery map is

$$J \longmapsto \sum_{h,k} J(k,h)\widetilde{B}_h(q)^* B_d(q)\widetilde{B}_k(q).$$

The endpoint minor uses the distinct rows indexed by $h^{-1}dk$.

Scope. The neighborhood on which the minor stays invertible may depend on n. No sharp all-positive-parameter context bound is included.

Provenance. ids: D1215; proved in: theory/sidequests/f1-limit/lifting.md; tested by: theory/checks/f1_limit_check.py (L6); theory/checks/f1_operational_check.py (H11 at the endpoint); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 14.12 (Local contextual recovery of the Kraus Gram). PROVED *For every fixed $n \geq 1$, some interval about one has the following property: one retained context of total rank $2n-1$ recovers the full normalized-basis Kraus Gram. On that interval, equality in all retained contexts is equivalent to pointwise Gram equality, and at each fibre to stable scalar-unitary mixing.*

Scope. The scalar unitary supplied by the finite Gram lemma is fibrewise. A continuous family of such unitaries is neither obtained nor required for equality in the repaired core category.

Provenance. ids: F1-LIM-CTX; proved in: theory/sidequests/f1-limit/lifting.md; tested by: L6; H11 at q=1; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. At one the intersection of the local subgroup with its d-conjugate is trivial, so the $(n!)^2$ permutations $h^{-1}dk$ are distinct. The selected coefficient minor is therefore a permutation matrix. Its entries vary continuously, so its determinant remains nonzero near one. The displayed map is injective there. Equality of its values forces equality of Grams; conversely every retained channel is linear in the same Gram coefficients. □

14.5 Operations on the full additive completion

Definition 14.13 (Controlled operations on completed objects). Let $X = (r_n, p_{X,n})$ be a nonzero finite-support family of self-adjoint matrix projections and put

$$B_X = \bigoplus_n p_{X,n} M_{r_n}(H_n^{\mathrm{cts}}) p_{X,n},$$
$$\theta_X(a) = \sum_n (\mathrm{Tr}_{r_n} \otimes \tau_n)(a_n), \qquad w_X = \theta_X(1_X), \quad \tau_X = \theta_X / w_X.$$

Quantum atoms are such objects; separated words use spatial tensor algebras and product traces, while classical wires retain the uniform trace above. Include every normalized preparation, discard and finite POVM. An instrument $X \to Y\underline{O}$ is a finite family

$$K_{o,i} \in \mathrm{Hom}(X, Y), \qquad \sum_{o,i} K_{o,i}^* K_{o,i} = 1_X,$$

realized by $(a_o)_o \mapsto \sum_{o,i} K_{o,i}^* a_o K_{o,i}$. Retain $K_{o,i} \otimes 1_Z$, ordered parallel lists, traced assembly/split and finite typed circuits.

The presentation uses the explicit sound classical wiring above. At a fibre, Kraus equality is coefficient-Gram equality, equivalently stable scalar-unitary mixing within each outcome. On an interval it is pointwise coefficient-Gram equality, and for germs it is such equality on a common smaller interval. Scalar-unitary cancellation proves congruence in each fibre; no continuous choice of mixing unitary is required. Any finite classical control is a typed blockwise family of the admitted labels and contributes no additional arbitrary CP map. The resulting fibre, interval and germ categories retain the realization and evaluation conventions above; their full circuit realizations are not stipulated to be faithful.

Scope. The construction assumes neither rigidity nor fusion and includes no arbitrary CP-map data. Fibrewise stable mixing does not assert the existence of a continuous environmental unitary.

Provenance. ids: D1266; proved in: theory/sidequests/f1-limit/karoubi-lifting.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1–K6, primitive examples); theory/checks/f1_limit_wiring_check.py (W1–W5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 14.14 (Operational completion of all finite projection objects). **PROVED** *With the displayed classical-wiring presentation and its UCP relations, every nonzero object in the full finite completion carries normalized states, effects, POVMs, retained instruments, traced assembly and finite protocols. These generate an ordered UCP circuit category with continuous and germ evaluations. Restriction to parabolic atoms recovers the preceding operational formulas.*

Scope. This statement uses the wiring and equality convention just stated. It asserts neither all-CP completeness, rigidity, fusion, faithful circuit realization nor preservation of every additional endpoint equation.

Provenance. ids: F1-LIM-KOP; proved in: theory/sidequests/f1-limit/karoubi-lifting.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_karoubi_check.py (K1–K6, primitive examples); W1–W5; categorical typing by proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Preparations, discards, POVMs and Kraus lists give UCP maps exactly as in the parabolic case. The completed trace supplies the correct density dual, and the unique traced expectations supply assembly and split. Classical wiring types every history relation and is parameter independent. The free typed category may therefore be quotiented by the proved congruence, and its generator realization descends to a monoidal UCP functor. The restriction to rank-one parabolic projections identifies all formulas term by term. ☐

15 Local lifting, Born limits and a three-constituent experiment

The continuous family becomes operational only if concrete endpoint data can be continued to a neighborhood of one. The continuation need not be global or canonical. It is enough to lift the finitely many objects and labels in a chosen circuit, normalize them by positive functional calculus, and intersect their finitely many neighborhoods. This distinction is decisive: lifting an individual circuit does not lift every equation that happens to hold only in the symmetric endpoint fibre.

15.1 Local lifts of objects and labels

Definition 15.1 (Raw local corner lift)**.** Fix $q_0 > 0$ and $a_0 \in e_\beta(q_0)H_n(q_0)e_\alpha(q_0)$. Expand

$$a_0 = \sum_{w \in S_n} c_w b_w(q_0).$$

On a compact positive interval containing q_0, its raw lift is

$$a_{\mathrm{raw}}(q) = e_\beta(q) \left(\sum_w c_w b_w(q) \right) e_\alpha(q).$$

For register words, raw lifts are finite sums in the $C(I)$-balanced tensor product of such sections.

Scope. This is local existence data and is not a preferred trivialization of the continuous family. Compression is part of the formula and keeps the source and target corners typed.

Provenance. ids: D1213; proved in: theory/sidequests/f1-limit/lifting.md; tested by: analytic coefficient-lift proof; no finite gate claims generality; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 15.2 (Normalized local operational lifts)**.** For raw Kraus lifts $\widehat{K}_r(q)$, set

$$S(q) = \sum_r \widehat{K}_r(q)^* \widehat{K}_r(q), \qquad \widetilde{K}_r(q) = \widehat{K}_r(q) S(q)^{-1/2}$$

on a neighborhood where S is invertible in the source corner. To lift a state, lift its positive square root c_0, put $H = c^*c$, and divide by the positive scalar $\tau(H)$. To lift a POVM, lift the square roots of its effects, put $A_o = c_o^* c_o$, $S = \sum_o A_o$, and define

$$e_o = S^{-1/2} A_o S^{-1/2}.$$

These constructions may be applied simultaneously to any finite family by shrinking to one common interval.

Scope. The inverse square root is taken in the moving source corner, whose unit is its projection. Zero Kraus branches and rank-deficient state densities are allowed. No global normalization is claimed.

Provenance. ids: D1214; proved in: theory/sidequests/f1-limit/lifting.md; tested by: theory/checks/f1_limit_check.py (L2–L4, finite normalizations); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 15.3 (Local lifting of parabolic operational circuits)**.** PROVED *Fix a finite endpoint circuit built from normalized states, effects, finite POVMs, normalized corner instruments, retained context, preparation, discard, assembly, split and classical wiring. There is an endpoint neighborhood on which all its labels have continuous lifts of the same types, and the lifted circuit evaluates to the original circuit at one. Pointwise coefficient-Gram equality is preserved on intervals and eventual Gram equality on germs.*

Scope. The lift is neither unique nor global. It lifts one chosen circuit and the relations of the presented interval category; it does not preserve every additional endpoint equation or provide a section of the evaluation functor.

Provenance. ids: F1-LIM-LIFT; proved in: theory/sidequests/f1-limit/lifting.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: L2–L4; W1–W5 for finite typed wiring; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Raw coefficient lifts exist for every corner element. At the endpoint, $S(q_0)$ is the source identity, so invertibility persists locally and the binomial functional-calculus series for $S^{-1/2}$ converges uniformly. The displayed formulas then give exact normalization throughout a smaller interval. There are only finitely many labels and contextual-recovery neighborhoods, so their intersection is still an endpoint neighborhood. Assembly, split and classical wiring are already global continuous labels. Reusing one lift for every repeated label gives the required circuit. Congruence of Gram equality is proved fibre by fibre and needs no continuous environmental unitary. $\square$

Definition 15.4 (Germs of full projection objects)**.** An object germ is represented on some interval $[1-\varepsilon, 1+\varepsilon]$ by a finite-support family of fixed matrix multiplicities and continuous self-adjoint projection matrices. Two such presentations are equal as germs when they agree literally after restriction to a smaller interval; isomorphic presentations are related by explicit morphisms instead. Morphisms are germs of the corresponding rectangular corner sections. Operations and the permutation-matrix associators descend under restriction. Evaluation at one is canonical, evaluation away from one requires a representative, and no C*-norm is placed on germ Hom spaces.

Scope. A local lift is one representative on one neighborhood. Essential surjectivity is asserted only for endpoint germs, not for evaluation from an arbitrarily fixed large interval.

Provenance. ids: D1265; proved in: theory/sidequests/f1-limit/karoubi-lifting.md; tested by: theory/checks/f1_limit_karoubi_check.py (K3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 15.5 (Full endpoint projection lifting)**.** **PROVED** *Every finite family of self-adjoint projection objects in a positive fibre has simultaneous local continuous lifts. Evaluation is full on already chosen section objects, and endpoint germ evaluation is full and essentially surjective onto the finite self-adjoint completion. A unitary between two chosen endpoint object lifts has a local unitary lift.*

Scope. The construction uses positive spectral cutoff and is not algebraic base change over the generic coefficient ring. It gives no global canonical choice of object lifts.

Provenance. ids: F1-LIM-KLIFT; proved in: theory/sidequests/f1-limit/karoubi-lifting.md; tested by: theory/checks/f1_limit_karoubi_check.py (K3); spectral-gap and fullness arguments by proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Lift the entries of an endpoint projection p_0 with constant normalized Hecke coefficients and take the self-adjoint part a. Put $h = 2a - 1$. After shrinking, h^2 remains within 1/2 of the identity, so

$$j = h(h^2)^{-1/2}, \qquad p = \frac{1+j}{2}$$

is a continuous self-adjoint projection with $p(q_0) = p_0$. Finitely many such cutoffs share a common interval. A fibre morphism lifts entrywise and is compressed by the lifted source and target projections, proving fullness. For an endpoint unitary, normalize a raw rectangular lift a to $a(a^*a)^{-1/2}$. Its final projection is continuous and equals the target projection at the endpoint; after shrinking their difference has norm less than one, forcing equality because every nonzero projection has norm one. $\square$

Theorem 15.6 (Normalized processes on lifted completed objects)**.** PROVED *Fix finitely many nonzero completed objects with chosen local lifts. Every specified endpoint state, effect, finite POVM and normalized retained instrument family lifts locally. For $K_{o,i} : X \to Y$, its unnormalized branch density is*

$$T_o(\rho) = \frac{w_Y}{w_X} \sum_i K_{o,i} \rho K_{o,i}^*,$$

and the o-block of the recorded quantum–classical density is $|O| T_o(\rho)$. On parabolic corners the dimension ratio is P_α / P_β, and tensoring a right context leaves it unchanged.

Scope. For continuous labels, equality means pointwise coefficient-Gram equality; for germs it means eventual pointwise Gram equality. Fibrewise stable mixing does not provide, and the construction does not require, a continuous family of environmental unitaries.

Provenance. ids: F1-LIM-KPROC; proved in: theory/sidequests/f1-limit/karoubi-lifting.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_karoubi_check.py (K2,K4,K6); W1–W5; finite-family lifting by proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. States, POVMs and Kraus families are normalized by the same square-root formulas as above, now in finite matrix corners. Cross-object cyclicity of the unnormalized trace gives

$$\tau_Y(T_o(\rho)a) = \tau_X\left(\rho \sum_i K_{o,i}^* a K_{o,i}\right),$$

which fixes the factor w_Y / w_X. The uniform classical trace contributes $|O|$. Tensor multiplicativity of d cancels the dimension of a retained right context. For a parabolic projection, $w_\alpha = 1/P_\alpha$, yielding the stated specialization. □

15.2 Finite protocols and positive-limit conditioning

Definition 15.7 (Finite typed protocol and postselection)**.** A finite protocol is a rooted finite instrument tree. Its root carries a continuous normalized density on a finite register word. Every internal vertex carries a typed finite-outcome circuit and may depend on the preceding finite outcome history; following an edge selects one named outcome while retaining the declared quantum output. Each leaf carries an effect. Branch weights are obtained by composing the outcome CP maps and applying the final Born pairing; event weights are finite sums of branch weights. A conditioned probability $N(q)/D(q)$ is considered near one only when $D(1) > 0$.

Scope. Only finite trees and specified instruments are included. A POVM does not determine its postmeasurement state, and no conditioned limit is asserted when the limiting denominator vanishes.

Provenance. ids: D1212; proved in: theory/sidequests/f1-limit/protocols.md; tested by: theory/checks/f1_limit_wiring_check.py (W1,W2,W5); theory/checks/f1_limit_check.py (L11); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 15.8 (Continuity of finite Born experiments)**.** PROVED *Every finite continuous typed instrument tree has continuous nonnegative branch and event Born weights, and evaluation commutes with running the protocol. If a conditioning event has $D(1) > 0$, then on a smaller endpoint neighborhood*

$$\lim_{q \to 1} \frac{N(q)}{D(q)} = \frac{N(1)}{D(1)}.$$

The same statement holds for a finite germ protocol after choosing one common representative interval.

Scope. The theorem is analytic and is not inferred from finite samples. If $D(1) = 0$, even continuous normalized instruments can have conditioned probabilities with different subsequential limits.

Provenance. ids: F1-LIM-BORN; proved in: theory/sidequests/f1-limit/protocols.md; tested by: L11 gives the zero-denominator falsifier; W1,W2,W5 test finite routing; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. All generator matrices, trace pairings and wiring permutations vary continuously. Composition and tensor use finite matrix products and Kronecker products. Induction on tree depth therefore gives continuous positive unnormalized branch densities. Their trace pairings with leaf effects are continuous and nonnegative. Instrument unitality conserves the sum of child masses, and evaluation commutes with every finite operation. If $D(1) > 0$, continuity makes $D(q) > D(1)/2$ nearby, where ordinary quotient continuity gives the result. □

The positivity condition cannot be deleted. On $1 \le q \le 3/2$, let $t = q - 1$ and choose a unitary $U(q)$ that oscillates between the identity and $2e_1(q) - 1$. The continuous Kraus pair

$$K_s(q) = tU(q), \qquad K_f(q) = \sqrt{1 - t^2}\,1,$$

extended by $(0, 1)$ at one, is normalized. Its success probability is $t^2 \to 0$, while a conditioned return experiment has subsequential limits 1 and $1/4$.

15.3 A Lüders experiment on three constituents

Definition 15.9 (Three-constituent retained-context protocol)**.** In $H_3(q)$ put

$$e_1 = \frac{T_1 + 1}{q + 1}, \qquad e_2 = \frac{T_2 + 1}{q + 1}, \qquad \alpha = (1, 2).$$

Prepare the normalized corner density e_2 and refine to the complete three-part object with the isometry $v = e_2$, producing the full density $(q+1)e_2$. If z_{std} is the central support of the unique $M_2(\mathbb{C})$ summand, put $P_2 = z_{\mathrm{std}}e_2$. Apply the Lüders instrument

$$K_s = P_2, \qquad K_f = 1 - P_2, \qquad [x^3] \longrightarrow [x^3]\underline{\{s, f\}},$$

postselect the successful branch while retaining its quantum output, apply the right-context extension of $u_1 = 2e_1 - 1$ from the first two constituents, and finally measure the POVM $(P_2, 1 - P_2)$.

Scope. The successful Kraus operator is part of the experiment. Replacing it by u_1P_2 leaves the first POVM unchanged but changes the later return probability, so the effects alone do not define this protocol.

Provenance. ids: D1217; proved in: theory/sidequests/f1-limit/protocols.md; tested by: theory/checks/f1_limit_wiring_check.py (W4); theory/checks/f1_limit_check.py (L5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 15.10 (Three-constituent Lüders witness)**.** **PROVED** *For every $q > 0$, the preceding protocol has first success probability, conditional return probability and joint probability*

$$\begin{aligned} s(q) &= \frac{q(q+1)}{1 + q + q^2}, \\ r(q) &= \left(1 - \frac{2q}{(q+1)^2}\right)^2, \\ p(q) &= s(q)r(q). \end{aligned}$$

The successful density is

$$h_s(q) = \frac{1+q+q^2}{q} P_2(q).$$

At $q = 1$ these become

$$s(1) = \frac{2}{3}, \qquad h_s(1) = 3P_2(1), \qquad r(1) = \frac{1}{4}, \qquad p(1) = \frac{1}{6}.$$

Scope. The unitary u_1 has the identity conjugation channel on the isolated two-constituent commutative algebra, but its retained action on the full three-constituent algebra is not erased. The theorem uses the coefficient trace and the specified Lüders backaction.

Provenance. ids: F1-LIM-H3; proved in: theory/sidequests/f1-limit/protocols.md; tested by: theory/checks/f1_limit_wiring_check.py (W4); theory/checks/f1_limit_check.py (L5); theory/checks/f1_operational_check.py (H5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The parabolic trace normalization sends the corner state to $h = (q+1)e_2$. Write $d = 1 + q + q^2$ and $\gamma = \tau_3(P_2) = q/d$. The first probability is

$$\tau_3(hP_2) = (q+1)\gamma = \frac{q(q+1)}{d}.$$

Because the successful Kraus operator is P_2, $P_2hP_2 = (q+1)P_2$; division by $s(q)$ gives $h_s = P_2/\gamma$. In the standard two-dimensional block, $u_1 = 2P_1 - 1$ and the two rank-one projections have overlap $\mathrm{Tr}_2(P_1P_2) = q/(q+1)^2$. Therefore the return amplitude is $1 - 2q/(q+1)^2$ and its squared modulus is $r(q)$. Multiplying branch probabilities and substituting $q = 1$ proves the remaining formulas. $\square$

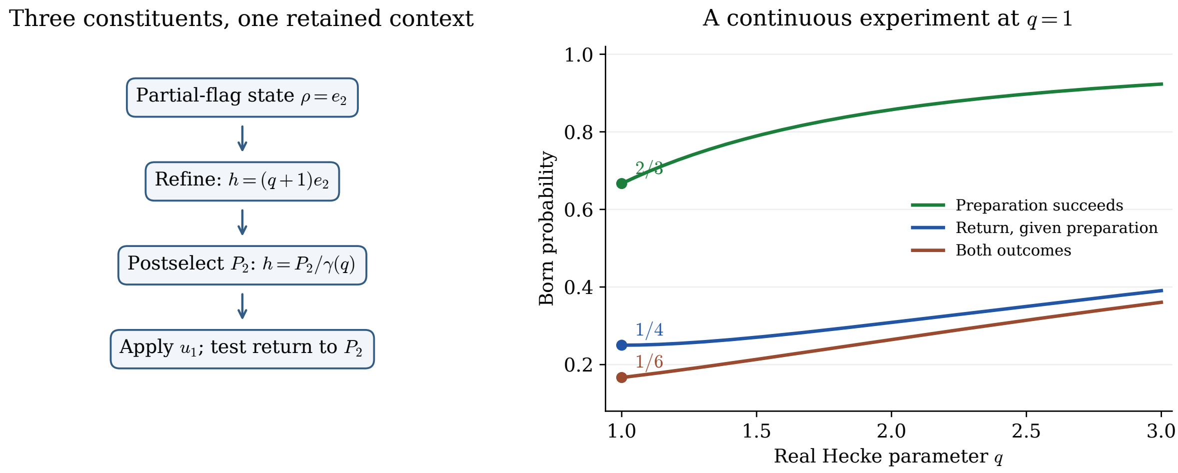

Figure 2: The three-constituent Lüders protocol. Refinement is followed by the specified backaction $K_s = P_2$, postselection succeeds with probability $2/3$ at the endpoint, retained conjugation returns with conditional probability $1/4$, and the joint endpoint probability is $1/6$.

16 Universal Hecke composition and physical exchange

The positive Hecke tower contains more than its familiar partial-flag corners. Closing the tower under finite sums and self-adjoint retracts gives a finite graded composition category, while passing to finite-dimensional right modules gives the same category algebraically. The tensor product is induction along the ordered block inclusion. This point matters: the plain vector-space tensor product of modules over two different Hecke algebras is not yet a module over the next Hecke algebra.

There are also two different notions of exchange. The usual Hecke braid is an algebraic isomorphism, but it is not unitary for the coefficient-trace C^*-structure away from $q = 1$. Polar decomposition of the longest Hecke elements instead produces canonical unitary reversals. Their block commutors obey cactus, rather than braid, coherence.

16.1 The finite completion of a multiplicative sequence

Definition 16.1 (Multiplicative sequence, skeleton, and variance)**.** A multiplicative sequence over $\mathbb{C}$ is a family of unital algebras $A_* = (A_n)_{n\geq 0}$, with $A_0 = \mathbb{C}$, and unital algebra maps

$$\mu_{m,n} : A_m \otimes A_n \longrightarrow A_{m+n}$$

whose two composites from $A_l \otimes A_m \otimes A_n$ to A_{l+m+n} agree. The unit maps are the scalar identifications $\mu_{0,n}(\lambda \otimes a) = \lambda a$ and $\mu_{n,0}(a \otimes \lambda) = \lambda a$.

Its skeleton $\mathbb{C}[A_*]$ has objects $[n]$, zero morphisms between unequal degrees, $\mathrm{End}([n]) = A_n$, composition given by algebra multiplication, and tensor induced by μ. A linear presheaf is a functor $\mathbb{C}[A_*]^{\mathrm{op}} \to \mathrm{Vect}_{\mathbb{C}}$. At degree n it is a right A_n-module:

$$m \cdot a = F(a)m.$$

The representable $\mathrm{Hom}(-, [n])$ is the right regular module, because precomposition sends x to xa.

Scope. The direction of the module action is part of the definition. No opposite algebra is inserted later, and no morphisms between different degrees are created at the skeletal stage.

Provenance. ids: D1221; proved in: theory/sidequests/f1-limit/universal.md; tested by: structural proof; no dedicated finite gate; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.2 (Finite Day–Schur–Weyl completion)**.** Assume every A_n is finite-dimensional over $\mathbb{C}$. Define

$$\mathrm{SW}_{\mathrm{fin}}(A_*) = \bigoplus_{n\geq 0} \mathrm{mod}_{\mathrm{fd}}\text{-}A_n,$$

where objects have finite degree support. For a right A_m-module M and a right A_n-module N, set

$$M \star N = (M \otimes_{\mathbb{C}} N) \otimes_{A_m \otimes A_n} A_{m+n}.$$

Here A_{m+n} is a left $A_m \otimes A_n$-module through $\mu_{m,n}$ and a right module by multiplication. The unit is $\mathbb{C} = A_0$, and the associator is the canonical balanced-tensor-product isomorphism.

Scope. Finite-dimensionality of every level algebra is required for this finite subcategory to be closed under $\star$. The tensor is induction, not the unextended tensor product $M \otimes_{\mathbb{C}} N$.

Provenance. ids: D1222; proved in: theory/sidequests/f1-limit/universal.md; tested by: structural balanced-tensor proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.3 (Finite projective composition category)**.** The additive Karoubi completion $\mathrm{Proj}(A_*)$ has degree-n objects (r, e), where $r \geq 0$ and $e \in M_r(A_n)$ is an idempotent. It represents the right module eA_n^r. For two objects in degree n,

$$\mathrm{Hom}((r, e), (s, f)) = fM_{s,r}(A_n)e,$$

with matrix multiplication as composition; unequal degrees have zero Hom. Tensor uses the matrix amplification of $\mu_{m,n}$ and the idempotent $\mu_{m,n}^{(r,s)}(e \otimes f)$.

If every A_n is a finite-dimensional C^*-algebra and every $\mu_{m,n}$ is a star-homomorphism, the positive version uses self-adjoint projections and matrix adjoint as its dagger. Its comparison with bare right modules is algebraic: the bare module category carries no chosen Hilbert form or dagger.

Scope. The positive construction requires star-preserving structural maps. An algebraic module equivalence does not manufacture a dagger on its target.

Provenance. ids: D1223; proved in: theory/sidequests/f1-limit/universal.md; tested by: finite matrix-corner derivation; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.4 (Positive Hecke sequence and its universal completion)**.** For real $q > 0$, put $H_0(q) = H_1(q) = \mathbb{C}$. For $n \geq 2$, let $H_n(q)$ be generated by self-adjoint $T_1, \ldots, T_{n-1}$ with

$$(T_i - q)(T_i + 1) = 0, \quad T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \quad T_iT_j = T_jT_i \quad (|i - j| > 1).$$

For $w \in S_n$, T_w is the reduced-word basis element and $\tau_n(T_w) = \delta_{w,1}$. Ordered block permutations define

$$\iota_{m,n}(T_u \otimes T_v) = T_{u \times v}.$$

These maps are injective unital trace-preserving star-homomorphisms and make $H_*(q)$ a positive multiplicative sequence.

Let

$$R = \mathbb{Z}[t, t^{-1}, \{P_\alpha(t)^{-1}\}_\alpha]$$

be the localized coefficient ring used by the generic based Hecke algebras. The generic category is the finite additive self-adjoint-idempotent envelope

$$\mathcal{U}_R = \mathrm{Kar}_*\big(\mathrm{Add}(\mathbb{C}[H_*(R)])\big).$$

Evaluation is defined for its displayed objects and arrows. The full positive fibre is

$$\mathcal{U}_q = \mathrm{Proj}_*(H_*(q)).$$

For $x = [1]$, $\mathrm{End}_{\mathcal{U}_q}(x^{\otimes n}) = H_n(q)$. A nonzero projection corner uses the coefficient trace divided by the trace of its corner unit. Normalized states and effects live on these finite C^*-endomorphism algebras.

Scope. The generic envelope contains self-adjoint projectors presented over R. It need not evaluate essentially surjectively onto projections constructed separately by functional calculus in one positive fibre. This is an amplitude category; its operational completion is supplied separately and does not require a fusion hypothesis.

Provenance. ids: D1101,D1102,D1224,F1-HCK-POS,F1-HCK-TOWER; proved in: theory/sidequests/f1-limit/universal.md; tested by: theory/checks/f1_operational_check.py (H13, finite corner samples); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.5 (Partial-flag completion)**.** For a composition $\alpha = (\alpha_1, \dots, \alpha_r)$ of n, let $W_\alpha \le S_n$ preserve its consecutive blocks and define

$$P_\alpha(q) = \sum_{w \in W_\alpha} q^{\ell(w)}, \qquad e_\alpha(q) = P_\alpha(q)^{-1} \sum_{w \in W_\alpha} T_w.$$

The partial-flag category has

$$\mathrm{Hom}(\alpha, \beta) = e_\beta H_n(q) e_\alpha$$

for equal totals, zero Homs otherwise, identity e_α, adjoint inherited from $H_n(q)$, and concatenation tensor. Its projection-model functor sends

$$\alpha \longmapsto e_\alpha H_n(q), \qquad z \longmapsto L_z : e_\alpha H_n(q) \to e_\beta H_n(q).$$

The extension under finite sums and self-adjoint retracts is the universal positive Hecke category above.

Scope. The normalization by P_α is essential. The completion statement is proved below from the complete-flag object, not assumed from an older sketch about partial flags.

Provenance. ids: D1141,D1225; proved in: theory/sidequests/f1-limit/universal.md; tested by: theory/checks/f1_operational_check.py (H13, levels 3 and 4 at the declared parameters); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.6 (Davydov–Molev parameter and finite module category)**.** Put $v = \sqrt{q} > 0$. If t_i^{DM} satisfies

$$(t_i^{\mathrm{DM}} - v)(t_i^{\mathrm{DM}} + v^{-1}) = 0,$$

then the project conventions are

$$q_{\mathrm{project}} = v^2, \qquad T_i^{\mathrm{project}} = v, t_i^{\mathrm{DM}}.$$

Let $\mathrm{DM}_{\mathrm{fin}}(v)$ be the finite-support, finite-dimensional right-module subcategory of the associated Hecke Schur–Weyl category. Its tensor is the induced module just defined. The projection realization $(r, e) \mapsto eH_n(q)^r$ is the proposed strong monoidal comparison

$$\mathcal{U}_q \simeq \mathrm{DM}_{\mathrm{fin}}(\sqrt{q}).$$

Scope. This comparison is algebraic. It respects the multiplicative sequence and induction tensor, but it asserts no dagger on the bare-module target.

Provenance. ids: D1226,D1227; proved in: theory/sidequests/f1-limit/universal.md; tested by: parameter identity and structured proof; no dedicated exact gate; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 16.7 (Universal finite Hecke composition)**.** PROVED *For every $q > 0$, the category $\mathcal{U}_q$ is algebraically strong-monoidally equivalent to the finite-support category of finite-dimensional right modules over the Hecke multiplicative sequence, with tensor*

$$(M \otimes N) \otimes_{H_m(q) \otimes H_n(q)} H_{m+n}(q).$$

It is also the additive self-adjoint Karoubi completion of the partial-flag corner category. Generic evaluation applies to the self-adjoint projectors presented over R.

Scope. The bare-module equivalence is algebraic. Generic evaluation is not claimed essentially surjective on all fibrewise functional-calculus projections, and no cross-degree amplitude morphisms are added.

Provenance. ids: F1-LIM-UNIV,D1221,D1222,D1223,D1224,D1225; proved in: theory/sidequests/f1-limit/universal.md; tested by: theory/checks/f1_operational_check.py (H13, finite parabolic corners only); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Contravariance gives $(m \cdot a) \cdot b = m \cdot (ab)$, fixing right-module variance. Both bracketings of three induced modules reduce canonically to

$$(L \otimes M \otimes N) \otimes_{A_l \otimes A_m \otimes A_n} A_{l+m+n},$$

so associativity and the pentagon follow from the common balanced universal map. The representables multiply because $((A_m \otimes A_n) \otimes_{A_m \otimes A_n} A_{m+n}) \cong A_{m+n}$.

Finite sums of representables give free right modules; splitting an idempotent e gives eA_n^r, and $\mathrm{Hom}_{A_n}(eA_n^r, fA_n^s) = fM_{s,r}(A_n)e$. Since a finite-dimensional C^*-algebra is semisimple, all of its finite-dimensional modules are projective. This proves the algebraic equivalence. Star preservation of the block maps sends tensor projections to projections.

For partial flags, write $x_\alpha = \sum_{w \in W_\alpha} T_w$. Pairing terms along a simple parabolic reflection gives $T_s x_\alpha = q x_\alpha$ and its right-handed star version. Hence

$$x_\alpha^2 = P_\alpha x_\alpha, \qquad x_\alpha^* = x_\alpha, \qquad \tau_n(e_\alpha) = P_\alpha^{-1}.$$

Block length additivity gives $P_{\alpha\|\beta} = P_\alpha P_\beta$ and $\iota(e_\alpha \otimes e_\beta) = e_{\alpha\|\beta}$. Finally the complete composition $(1, \ldots, 1)$ has $e = 1$ in each degree, so its finite sums and retracts generate every object of $\mathcal{U}_q$. □

Theorem 16.8 (Algebraic Hecke Schur–Weyl comparison). **PROVED** *With $q_{\text{project}} = v^2$ and $T_i^{\text{project}} = v t_i^{\mathrm{DM}}$, the category $\mathcal{U}_q$ is algebraically strong-monoidally equivalent to* $\mathrm{DM}_{\text{fin}}(v)$.

Scope. Only finite degree support and finite-dimensional modules are included. The comparison does not turn the algebraic Hecke braiding into a physical unitary or endow the bare-module category with a dagger.

Provenance. ids: F1-LIM-DM,D1226,D1227; proved in: theory/sidequests/f1-limit/universal.md; tested by: exact parameter calculation; categorical equivalence by proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Substitution $t_i^{\mathrm{DM}} = v^{-1} T_i$ turns its quadratic relation into $(T_i - v^2)(T_i + 1) = 0$. The braid and distant-commutation relations are homogeneous, and the same scaling respects every ordered block inclusion. Thus the full multiplicative sequences agree. The preceding theorem then identifies their finite right-module categories with their induction tensor. □

16.2 Algebraic braids and unitary cactus exchange

Definition 16.9 (Adjacent spectral sign). In $H_n(q)$ define

$$u_i = \frac{2T_i + 1 - q}{q + 1} = 2\frac{T_i + 1}{q + 1} - 1.$$

This is $+1$ on the q-eigenspace of T_i and -1 on its (-1)-eigenspace. Its adjacent braid defect is

$$u_i u_{i+1} u_i - u_{i+1} u_i u_{i+1} = -\frac{(q-1)^2}{(q+1)^2}(u_i - u_{i+1}).$$

Scope. The definition is fibrewise for $q > 0$. It produces local unitary exchanges, not a braiding away from one.

Provenance. ids: D1228; proved in: theory/sidequests/f1-limit/exchange.md; tested by: theory/checks/f1_limit_check.py (L7, six declared rational parameters); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.10 (Polar longest-element reversal)**.** Let $w_0^{(n)}$ be the longest permutation and $D_n = T_{w_0^{(n)}}$. In the positive fibre put

$$|D_n| = (D_n^2)^{1/2}, \qquad J_n = D_n|D_n|^{-1}, \qquad J_0 = J_1 = 1.$$

The square root is the unique positive C^*-functional-calculus root. For an interval I of consecutive tensor positions, J_I is the shifted copy of $J_{|I|}$.

Scope. This definition uses positivity and functional calculus at real $q > 0$; it is not an elementwise construction over the localized generic coefficient ring.

Provenance. ids: D1229; proved in: theory/sidequests/f1-limit/exchange.md; tested by: theory/checks/f1_limit_check.py (L13, ranks at most 4 and q=1,4,9); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.11 (Unitary block commutor)**.** For $m, n \geq 0$, define on tensor powers

$$\sigma_{m,n} = J_{m+n}\, \iota_{m,n}(J_m \otimes J_n).$$

On projection-completed objects, restrict this operator to the source and target corners; on finite sums also apply the canonical matrix-index flip. Its coherence consists of naturality, symmetry $\sigma_{n,m}\sigma_{m,n} = 1$, the unit laws, and the cactus square formed by the interval reversals.

Scope. The commutor is ordered and coboundary. Symmetry here means involutivity of the commutor, not the braid relation for three adjacent factors.

Provenance. ids: D1230; proved in: theory/sidequests/f1-limit/exchange.md; tested by: theory/checks/f1_limit_check.py (L13, finite cactus samples); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 16.12 (Analytic domain of physical exchange)**.** The block commutor is extra canonical structure on every positive real C^*-fibre and varies continuously for $q > 0$. It is the polar unitary of the positive minimal Hecke braid exchanging the two blocks. At $q = 1$ it is the ordinary unitary block permutation. Its functional-calculus coefficients need not belong to the localized generic ring.

Scope. No algebraic generic commutor is stipulated, and no braid coherence is claimed for $q \neq 1$.

Provenance. ids: D1231; proved in: theory/sidequests/f1-limit/exchange.md; tested by: theory/checks/f1_limit_check.py (L13, declared fibres only); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 16.13 (Spectral unitaries do not braid away from one)**.** PROVED *For $q > 0$, the standard algebraic Hecke exchange is invertible but unitary for the coefficient-trace C^*-structure only at $q = 1$. The adjacent spectral signs are self-adjoint unitaries with the exact defect displayed above, and therefore do not braid when $q \neq 1$.*

Scope. This separates algebraic Yang–Baxter exchange from physical unitary exchange. It does not obstruct a coboundary commutor.

Provenance. ids: F1-LIM-EXCH,D1226,D1228; proved in: theory/sidequests/f1-limit/exchange.md; tested by: theory/checks/f1_limit_check.py (L7, six rational parameters); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The algebraic generator $v^{-1}T_i$ has eigenvalues v and $-v^{-1}$, so both have modulus one exactly at $v = 1$. Meanwhile $e_i = (T_i + 1)/(q + 1)$ is a self-adjoint projection and $u_i = 2e_i - 1$. Writing $u_1 = ax + b$, $u_2 = ay + b$, with $a = 2/(q + 1)$, $b = (1 - q)/(q + 1)$ and $x = T_1, y = T_2$, braid cancellation gives

$$u_1u_2u_1 - u_2u_1u_2 = a^2b(x^2 - y^2) + ab^2(x - y).$$

Since $x^2 - y^2 = (q - 1)(x - y)$, this is the stated multiple of $u_1 - u_2$. The standard basis makes $T_1 - T_2$ nonzero. □

Theorem 16.14 (Canonical unitary cactus exchange). **PROVED** *For every $q > 0$, the polar reversals J_n define a continuous natural unitary coboundary commutor on all of $\mathcal{U}_q$. It is symmetric, satisfies the cactus relations, restricts to every partial-flag and arbitrary projection corner, is the polar unitary of the block braid, and becomes ordinary symmetric exchange at $q = 1$.*

Scope. The theorem is analytic on positive fibres and positive intervals. It makes no claim over the generic localized ring and no claim of braiding away from one.

Provenance. ids: F1-LIM-COB,D1229,D1230,D1231; proved in: theory/sidequests/f1-limit/exchange.md; tested by: theory/checks/f1_limit_check.py (L13, n at most 4 and q=1,4,9); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The longest element is self-adjoint and invertible. The relation $w_0s_i = s_{n-i}w_0$ gives $D_nT_i = T_{n-i}D_n$, hence D_n^2 is central. Its positive modulus is therefore central, so J_n is a canonical self-adjoint unitary satisfying $J_nT_iJ_n = T_{n-i}$. Shifted copies on disjoint intervals commute, and unitary conjugation carries a subinterval reversal to its reflected subinterval reversal. Thus

$$J_{[p,r]}J_{[k,l]} = J_{[p+r-l,p+r-k]}J_{[p,r]},$$

which is the interval presentation of the cactus relations.

The product $J_{m+n}(J_m \otimes J_n)$ reverses internally and then totally, so the internal reversals cancel and the two ordered block algebras swap. This proves naturality; the same interval identities prove symmetry and the cactus square. Conjugation carries any tensor projection to its swapped projection, so compression extends the commutor to all finite sums and retracts.

If $B_{m,n} = D_{m+n}(D_m \otimes D_n)^{-1}$ is the positive minimal block braid, then

$$\sigma_{m,n}^* B_{m,n} = |D_{m+n}|\,|D_m \otimes D_n|^{-1} > 0.$$

Uniqueness of polar decomposition identifies $\sigma_{m,n}$ as its polar unitary. Functional calculus varies continuously on positive invertible matrices. At one, every T_i is a transposition, so the construction is the ordinary block permutation. □

17 Schur–Weyl quotients and the boundary of fusion

At the symmetric endpoint, the Hecke tower becomes the symmetric-group tower. Its action on tensor powers of $\mathbb{C}^d$ gives a concrete monoidal quotient, but it changes the physical trace: a permutation is weighted by its number of cycles rather than only by whether it is the identity. The difference disappears at fixed tensor degree as d grows, and the associated conditional expectations converge with an explicit finite-d bound.

Rigidity and fusion require further data. The nonnegative grading of the universal amplitude category rules out evaluation and coevaluation maps inside that category. A rigid comparison must name a target, a monoidal functor, dual objects, cups and caps, dagger and trace compatibility. A Temperley–Lieb or Fibonacci branch adds quotient ideals and a different trace; neither is an automatic consequence of the positive-real endpoint.

17.1 The tensor-power quotient

Definition 17.1 (The symmetric endpoint amplitude category)**.** For real $q > 0$, let $H_n(q)$ be the positive type-A Hecke algebra with self-adjoint generators T_i, relations

$$(T_i - q)(T_i + 1) = 0, \qquad T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \qquad T_i T_j = T_j T_i \quad (|i-j| > 1),$$

and coefficient trace $\tau_n(T_w) = \delta_{w,1}$. Ordered block inclusion maps $T_u \otimes T_v$ to $T_{u\times v}$. The finite positive composition category $\mathcal{U}_q$ consists of finite sums and self-adjoint projection retracts of the graded regular objects; unequal degrees have zero Homs. At $q = 1$, $H_n(1) = \mathbb{C}[S_n]$ and the generator $x = [1]$ satisfies $\mathrm{End}(x^{\otimes n}) = \mathbb{C}[S_n]$.

Scope. This is an amplitude category with finite C^*-endomorphism algebras. Its nonnegative grading is retained, and no internal dual for x is assumed.

Provenance. ids: D1101,D1102,D1224; proved in: theory/sidequests/f1-limit/universal.md; tested by: theory/checks/f1_operational_check.py (H13, finite corner samples); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 17.2 (Classical Schur–Weyl quotient)**.** Fix $d \geq 1$, put $V_d = \mathbb{C}^d$, and let

$$P_\sigma(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

Define

$$\rho_{d,n} : \mathbb{C}[S_n] \longrightarrow \mathrm{End}_{U(d)}(V_d^{\otimes n}), \qquad \rho_{d,n}(\sigma) = P_\sigma.$$

Its kernel is the sum of Wedderburn blocks indexed by partitions of n with more than d rows. Equivalently, for $n \geq d + 1$, it is the ideal generated by the antisymmetrizer on the first $d + 1$ positions. The map is faithful exactly when $d \geq n$.

Scope. The target is the full $U(d)$-commutant on one tensor power. For $d < n$ the map is a quotient, and no faithful embedding is claimed.

Provenance. ids: D1232; proved in: theory/sidequests/f1-limit/schur-weyl.md; tested by: theory/checks/f1_limit_check.py (L8, n and d at most 4); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 17.3 (Quotient assembly)**.** Under the standard associator $V_d^{\otimes(m+n)} \cong V_d^{\otimes m} \otimes V_d^{\otimes n}$, define

$$\nu_{m,n}\big(\rho_{d,m}(a) \otimes \rho_{d,n}(b)\big) = \rho_{d,m+n}\big(\iota_{m,n}(a \otimes b)\big).$$

These are associative unital algebra maps. On right modules, the associated Schur–Weyl functor is

$$M \longmapsto M \otimes_{\mathbb{C}[S_n]} V_d^{\otimes n},$$

where the tensor power has its left permutation action.

Scope. Well-definedness uses compatibility of the kernels with block inclusion. The source tensor on modules is balanced induction, not an objectwise S_{m+n}-action on $M \otimes N$.

Provenance. ids: D1233; proved in: theory/sidequests/f1-limit/schur-weyl.md; tested by: theory/checks/f1_limit_check.py (L8, finite rank/trace samples; assembly by proof); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 17.4 (Coherent finite-rank Schur–Weyl quotient)**.** PROVED *For every $d \geq 1$ and $n \geq 0$, $\rho_{d,n}$ is a surjective star-homomorphism onto the tensor-power $U(d)$-commutant. Its kernel is exactly the sum of partition blocks with more than d rows, equivalently the antisymmetrizer-generated ideal, and it is faithful exactly for $d \geq n$. The maps $\nu_{m,n}$ form a multiplicative sequence and induce a strong monoidal functor from finite right modules with induction tensor.*

Scope. The theorem is at the symmetric endpoint. It permits a quotient kernel and does not identify induction with the bare vector-space tensor product of modules over different symmetric groups.

Provenance. ids: F1-LIM-SW,D1232,D1233; proved in: theory/sidequests/f1-limit/schur-weyl.md; tested by: theory/checks/f1_limit_check.py (L8, n,d at most 4; general kernel by proof); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The permutation convention gives $P_\sigma P_\pi = P_{\sigma\pi}$ and $P_\sigma^* = P_{\sigma^{-1}}$. Position permutations commute with the diagonal $U(d)$ action. Conversely, the complex span of the Lie algebra of $U(d)$ is $\mathfrak{gl}_d(\mathbb{C})$, so the unitary and classical general-linear commutants agree; classical Schur–Weyl duality makes the map surjective.

If $d \geq n$, apply a putative linear relation among the P_σ to $e_1 \otimes \cdots \otimes e_n$; the resulting $n!$ words are distinct. If $d < n$, the $(d+1)$-strand antisymmetrizer acts through $\bigwedge^{d+1} V_d = 0$. In the decomposition

$$V_d^{\otimes n} \cong \bigoplus_{\substack{\lambda \vdash n \\ \mathrm{rows}(\lambda) \leq d}} L_d(\lambda) \otimes S^\lambda,$$

the block $\mathrm{End}(S^\lambda)$ survives exactly for partitions with at most d rows. The branching rule says precisely those excluded blocks contain the sign representation of the embedded S_{d+1}, proving the ideal description.

On elementary tensors, block permutation gives

$$\rho_{d,m+n}(\iota(\sigma \otimes \pi)) = \rho_{d,m}(\sigma) \otimes \rho_{d,n}(\pi).$$

This proves kernel compatibility and associativity. Cancelling consecutive extension and restriction of scalars identifies the image of $(M \star N)$ with $(M \otimes_{\mathbb{C}[S_m]} V_d^{\otimes m}) \otimes (N \otimes_{\mathbb{C}[S_n]} V_d^{\otimes n})$, giving the strong monoidal structure. □

17.2 Which trace does the quotient see?

Definition 17.5 (Tensor trace and physical subgroup expectation)**.** Define the normalized tensor trace

$$\mathrm{tr}_{d,n}(a) = d^{-n}\,\mathrm{Tr}(\rho_{d,n}(a)).$$

For a permutation σ, this is

$$\mathrm{tr}_{d,n}(\sigma) = d^{\mathrm{cyc}(\sigma)-n}.$$

When $d \geq n$ and $K = S_m \times S_{n-m}$, the physical trace-preserving expectation $E_{d,K} : \mathbb{C}[S_n] \to \mathbb{C}[K]$ is characterized by

$$\mathrm{tr}_{d,n}(k^* E_{d,K}(x)) = \mathrm{tr}_{d,n}(k^* x), \qquad k \in K.$$

In the group basis its coefficients solve $G_d c = b$, where

$$G_d(k,h) = d^{\mathrm{cyc}(k^{-1}h)-n}, \qquad b_k = d^{\mathrm{cyc}(k^{-1}\sigma)-n}.$$

Scope. Faithfulness on the full group algebra, and therefore this Gram inversion there, requires $d \geq n$. On the quotient image the tensor trace remains faithful for all d.

Provenance. ids: D1234; proved in: theory/sidequests/f1-limit/schur-weyl.md; tested by: theory/checks/f1_limit_check.py (L8,L9, declared finite ranks); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 17.6 (Fixed-degree stable trace comparison)**.** Let τ_n be coefficient trace and let $E_{\infty,K}$ delete group-basis coefficients outside K. For fixed n and $x = \sum_\sigma a_\sigma \sigma$,

$$|\mathrm{tr}_{d,n}(x) - \tau_n(x)| \leq \frac{1}{d} \sum_{\sigma \neq 1} |a_\sigma|.$$

Put $N = |K|$. If both $d \geq n$ and $d > N - 1$, then for $\sigma \notin K$ every coefficient of $E_{d,K}(\sigma)$ has modulus at most

$$\frac{1}{d - (N-1)},$$

whereas $E_{d,K}(\sigma) = \sigma$ for $\sigma \in K$. Thus the physical expectation converges coefficientwise to coefficient deletion at fixed degree.

Scope. Both inequalities on d are required: one for faithfulness and one for the Neumann-series bound. No equality between the two traces or expectations is claimed at fixed finite d.

Provenance. ids: D1235; proved in: theory/sidequests/f1-limit/schur-weyl.md; tested by: theory/checks/f1_limit_check.py (L8,L9; general bound analytic); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 17.7 (Stable physical trace and expectation)**.** PROVED *At fixed n, the normalized tensor trace converges coefficientwise to τ_n. If $K = S_m \times S_{n-m}$, $d \geq n$, and $d > |K| - 1$, then $E_{d,K}$ converges coefficientwise to $E_{\infty,K}$ with the bound above. For fixed $h \geq 0$, $\tau_n(h) = 1$, and $0 \leq e \leq 1$, put*

$$\Delta_d(y) = \frac{1}{d} \sum_{\sigma \neq 1} |y_\sigma|.$$

Whenever $\Delta_d(h) < 1$, the normalized quotient density has Born probability $\mathrm{tr}_{d,n}(he)/\mathrm{tr}_{d,n}(h)$ and

$$\left| \frac{\mathrm{tr}_{d,n}(he)}{\mathrm{tr}_{d,n}(h)} - \tau_n(he) \right| \leq \frac{\Delta_d(he) + \Delta_d(h)}{1 - \Delta_d(h)}.$$

Scope. All convergence is at fixed tensor degree and for fixed coefficient data. There is no uniform claim when the degree, coefficient norms, subgroup, or experiment varies with d.

Provenance. ids: F1-LIM-SWTRACE,D1234,D1235; proved in: theory/sidequests/f1-limit/schur-weyl.md; tested by: theory/checks/f1_limit_check.py (L8 for n,d at most 4; L9 finite S2-in-S3 sentinel); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. A tensor-basis word is fixed by P_σ exactly when it is constant on each cycle, so there are $d^{\mathrm{cyc}(\sigma)}$ fixed words. A nonidentity permutation has at most $n-1$ cycles, proving the trace bound. Moreover

$$\mathrm{tr}_{d,n}(x^* x) = d^{-n} \operatorname{Tr}(\rho_{d,n}(x)^* \rho_{d,n}(x)),$$

so its radical is exactly the Schur–Weyl kernel.

The conditional expectation is orthogonal projection for this trace. Its Gram matrix is $G_d = I + R$, with $\|R\|_\infty \leq (N-1)/d$. Under the two displayed hypotheses,

$$\|G_d^{-1}\|_\infty \leq \frac{1}{1 - (N-1)/d}.$$

For $\sigma \notin K$, the right-hand Gram vector has norm at most $1/d$, which proves the coefficient bound. The finite-rank difference is real: for $n = 3$, $K = \langle(12)\rangle$, one obtains $E_{d,K}((23)) = d^{-1}1$, while coefficient deletion gives zero.

Finally the trace estimate applied to h makes its physical normalization at least $1 - \Delta_d(h)$. Apply the same estimate to he and divide by this lower bound to obtain the stated Born inequality. □

17.3 Rigid and fusion targets require additional data

Definition 17.8 (Rigid target datum)**.** A rigid comparison consists of an additive idempotent-complete monoidal category $\mathcal{R}$, a named strong monoidal functor $i : \mathcal{U}_q \to \mathcal{R}$, and a named left and right dual $y^\vee$ of $y = i(x)$. It includes

$$\mathrm{coev}_R : \mathbf{1} \to y \otimes y^\vee, \quad \mathrm{ev}_R : y^\vee \otimes y \to \mathbf{1},$$

$$\mathrm{coev}_L : \mathbf{1} \to y^\vee \otimes y, \quad \mathrm{ev}_L : y \otimes y^\vee \to \mathbf{1},$$

satisfying the four zigzags with the stated associators. The target is generated from $y, y^\vee$ by tensor, finite sums and retracts. The functor may have a kernel. A positive dagger version additionally requires a C^*-category, a star-preserving i, compatible dagger/pivotal data, and a positive spherical structure. Duals of sums and retracts are formed in the additive Karoubi closure.

Scope. Full faithfulness is additional and is never inferred from the existence of cups and caps. The target closure is part of the datum.

Provenance. ids: D1236; proved in: theory/sidequests/f1-limit/fusion.md; tested by: theory/checks/f1_limit_check.py (L8, finite quotient witness only); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 17.9 (Fusion quotient datum)**.** A fusion quotient consists of a monoidal ideal $\mathcal{I}$ in a chosen rigid extension, a compatible dagger on the quotient, a positive pivotal trace, and the instruction to quotient and then take additive Karoubi completion. Any trace radical or negligible ideal is named explicitly. The result is called fusion only after proving semisimplicity, simple unit, finite-dimensional Hom spaces, duals, and finitely many simple isomorphism classes.

Scope. A family of degreewise algebra ideals is insufficient unless it is closed under tensor, composition, dagger, and the chosen duality operations.

Provenance. ids: D1237; proved in: theory/sidequests/f1-limit/fusion.md; tested by: structural proof; no finite sample proves fusion; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 17.10 (Temperley–Lieb branch)**.** For $q > 0$, set

$$f_i = \frac{q - T_i}{q + 1}, \qquad \delta = \sqrt{q} + \frac{1}{\sqrt{q}}.$$

A Temperley–Lieb branch imposes the tensor-ideal relations

$$f_i f_{i+1} f_i = \delta^{-2} f_i, \qquad f_{i+1} f_i f_{i+1} = \delta^{-2} f_{i+1},$$

equivalently killing the excluded three-strand Hecke summand. It also supplies diagrammatic cups and caps, dagger, and the Jones–Markov trace. The faithful coefficient trace is not declared to descend through this nonzero ideal.

Scope. This is a named quotient branch with new rigid and trace data. It is not a property of the full positive flag/Hecke category.

Provenance. ids: D1238; proved in: theory/sidequests/f1-limit/fusion.md; tested by: theory/checks/f1_operational_check.py (H10, finite trace obstruction); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 17.11 (Fibonacci root branch)**.** The unitary Fibonacci branch is the even part of the level-three $SU(2)$/Jones fusion quotient. Choose

$$q_H = e^{2\pi i/5}, \qquad v = e^{\pi i/5}, \qquad v^2 = q_H, \qquad q_{\mathrm{ILZ}} = -v,$$

and distinguish the loop generator from the projection by

$$U_i^{\mathrm{ILZ}} = \delta f_i, \qquad \delta = -(q_{\mathrm{ILZ}} + q_{\mathrm{ILZ}}^{-1}) = v + v^{-1} = 2\cos(\pi/5).$$

Quotient by the Jones trace radical, take additive Karoubi completion, and retain the even labels $\{0, 2\}$, with

$$2 \otimes 2 \cong 0 \oplus 2.$$

Scope. The square parameter has order five. This root-of-unity construction is not the $q = 1$ specialization of the positive-real branch, where $\delta = 2$ and the spin tower is untruncated.

Provenance. ids: D1239; proved in: theory/sidequests/f1-limit/fusion.md; tested by: primary-source parameter derivation; no cyclotomic checker; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 17.12 (A positive rigid target at the symmetric endpoint)**.** PROVED *The category $\mathcal{U}_q$ is not rigid: its degree-one generator has no left or right dual. A positive rigid comparison requires the complete datum above. At $q = 1$, for every $d \geq 1$, the full replete C^*-tensor subcategory of* $\mathrm{Rep}(U(d))$ *generated by V_d, V_d^*, finite sums and subobjects is a concrete positive rigid target of the Schur–Weyl quotient image. The named functor $i_d : \mathcal{U}_1 \to \mathcal{R}_d$ may have a kernel.*

Scope. The comparison is not asserted fully faithful in degrees $n > d$. The target has infinitely many simple objects as degrees vary and is not thereby a fusion category.

Provenance. ids: F1-LIM-RIGID,D1236,D1237; proved in: theory/sidequests/f1-limit/fusion.md; tested by: theory/checks/f1_limit_check.py (L8, finite Schur–Weyl kernel samples); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Every summand of $x \otimes y$ has degree at least one for any object y of the nonnegatively graded category, while the unit has degree zero. Thus an evaluation $x \otimes y \to \mathbf{1}$ must vanish and cannot satisfy a zigzag.

At the endpoint, send x to V_d, symmetric-group arrows to their tensor permutations, and a projection retract to the range of its image projection. This is a named strong monoidal star-functor, with precisely the Schur–Weyl kernel. The usual pairing and copairing

$$\mathrm{ev}(\varphi \otimes v) = \varphi(v), \qquad \mathrm{coev}(1) = \sum_i e_i \otimes e_i^*$$

satisfy the zigzags by basis contraction. The Hilbert dagger and $V_d \cong V_d^{\vee\vee}$ give positive dimension d and ordinary matrix trace. Closing under sums and retracts supplies duals for the whole generated target. ∎

Theorem 17.13 (Temperley–Lieb and Fibonacci are separate quotient branches)**.** PROVED *The Temperley–Lieb branch requires its named tensor ideal, cups, caps, dagger, and Jones–Markov trace. The faithful coefficient trace cannot descend through the nonzero quotient ideal. At $q = 1$ and $d = 2$, coefficient-trace weights $(1/2, 1/2)$ on the symmetric and alternating projections become physical weights $(3/4, 1/4)$. The Fibonacci branch further requires the displayed order-five parameter, Jones trace radical quotient, additive Karoubi completion, and even subcategory.*

Scope. Neither the $d = 2$ spin tower nor Fibonacci is identified with the full positive-real endpoint. Fibonacci is not obtained from any real $q > 0$ in the displayed loop normalization.

Provenance. ids: F1-LIM-TL,D1237,D1238,D1239; proved in: theory/sidequests/f1-limit/fusion.md; tested by: theory/checks/f1_operational_check.py (H10, finite TL/trace sample; no root-of-unity gate); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The element f_i is the spectral projection for the (-1)-eigenvalue of T_i. The quotient relation gives

$$f_i f_{i+1} f_i = \frac{q}{(q+1)^2} f_i = \delta^{-2} f_i.$$

Yet

$$\tau_n(f_i) = \frac{q}{q+1}, \qquad \mathrm{tr}_J(f_i) = \delta^{-2} = \frac{q}{(q+1)^2}.$$

If the faithful coefficient trace descended through a nonzero quotient kernel, a nonzero z in that kernel would satisfy both $\tau_n(z^*z) > 0$ and $\tau_n(z^*z) = 0$, a contradiction. At $q = 1$, the two projections $(1 \pm (12))/2$ have coefficient weights $1/2, 1/2$, whereas their ranks on $(\mathbb{C}^2)^{\otimes 2}$ are 3 and 1.

On the positive-real branch, $\delta = \sqrt{q} + q^{-1/2} \geq 2$. The Fibonacci value $2\cos(\pi/5) = (1+\sqrt{5})/2$ is smaller than two, so it requires the complex root and sign convention in the definition. At that root, the raw Temperley–Lieb algebra is semisimplified by its Jones trace radical; the even surviving labels give the stated Fibonacci fusion rule. The three traces—coefficient, finite tensor, and Jones—therefore belong to three different comparison maps and cannot be silently identified. □

18 The affine-vector commutant and its Fourier-polarized form

Adding a vector to a pair of flags produces a finite arithmetic observable algebra that remembers more than the ordinary flag Hecke algebra. Its extra generator is a controlled translation projector. Fourier transformation turns these translation constraints into controlled phase constraints, but the flag register must simultaneously be dualized and reversed. We fix all indices and signs below, including the square of the Fourier bridge.

18.1 Affine invariant kernels

Definition 18.1 (Arithmetic affine-vector context algebra)**.** Let $k = \mathbb{F}_Q$, let L be a finite-dimensional k-vector space, $G = GL(L)$, and $X = \mathrm{Fl}(L)$ the set of complete flags. The affine group

$$\mathrm{Aff}(L) = G \ltimes L, \qquad (g, a)(h, b) = (gh, a + gb),$$

acts on $Y_L = X \times L$ by

$$(g, a)(F, x) = (gF, a + gx).$$

Define

$$R_X(L) = \mathrm{End}_{\mathrm{Aff}(L)} \mathbb{C}[Y_L]$$

with Hilbert adjoint and normalized operator trace $\tau_Q = \mathrm{Tr}/(|X||L|)$. The invariant-kernel convention is

$$K_f((F, x), (F', x')) = f(F, F', x' - x).$$

Scope. The field cardinality is denoted Q. The symbol q below is reserved for a real deformation parameter. The affine group symbol does not reuse the notation for a Weyl subalgebra.

Provenance. ids: D1241; proved in: theory/sidequests/f1-limit/mirabolic.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B1); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 18.2 (Affine convolution as a physical commutant)**.** PROVED *For every finite field $\mathbb{F}_Q$ and finite-dimensional L, convolution on $GL(L)$-orbits of triples (F, F', v) is *- and trace-preservingly the commutant $R_X(L)$. In particular its normalized operator trace is faithful and positive.*

Scope. This is an arithmetic finite-Hilbert-space theorem. It does not infer positivity of a generic deformation from a polynomial presentation.

Provenance. ids: F1-MIR-AFF; proved in: theory/sidequests/f1-limit/mirabolic.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B1, n=2 and Q=2,3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. If $B \leq GL(L)$ stabilizes a reference complete flag, then $\mathrm{Aff}(L)/(B, 0) \cong X \times L$ by $(g, a)B \mapsto (gB, a)$. An operator commutes with the affine action exactly when its matrix kernel is constant on diagonal affine orbits. Translation by $-x$ reduces a pair to $((F, 0), (F', x' - x))$, leaving diagonal $GL(L)$-invariance. Matrix multiplication becomes

$$(f * g)(F, F', v) = \sum_{H \in X} \sum_{u \in L} f(F, H, u) g(H, F', v - u),$$

which is the stated convolution. Taking adjoints gives $f^*(F, F', v) = \overline{f(F', F, -v)}$. Normalized matrix trace restricts to a faithful positive trace on every finite-dimensional *-subalgebra. $\square$

18.2 The controlled translation generator

For $u \in L$ and an additive character χ of L, write $X(u)e_x = e_{x+u}$ and $Z(\chi)e_x = \chi(x)e_x$. For a subspace $U \le L$, put

$$P_U^X = |U|^{-1}\sum_{u\in U} X(u), \qquad P_U^Z = |U^{\mathrm{ann}}|^{-1}\sum_{\chi\in U^{\mathrm{ann}}} Z(\chi).$$

Definition 18.3 (Controlled translation and phase constraints)**.** For a complete flag $F = (U_i(F))_{i=0}^n$, with $U_0 = 0$ and $U_n = L$, define on $\mathbb{C}[\mathrm{Fl}(L)] \otimes \mathbb{C}[L]$

$$C_i^X = \sum_F |F\rangle\langle F| \otimes P^X_{U_i(F)}, \qquad C_i^Z = \sum_F |F\rangle\langle F| \otimes P^Z_{U_i(F)}, \qquad 0 \le i \le n.$$

Thus $C_0^Z = I \otimes |0\rangle\langle 0|$ and $C_n^Z = I$. Let $\mathrm{Ctx}_X(L)$ be the *-algebra generated by all relative-position flag operators $A_w \otimes I$ and all C_i^X. For $n \ge 1$, its marked projection and unnormalized vector generator are

$$e = C_1^X, \qquad T_0 = Qe - I.$$

The same definitions apply on the dual vector space.

Scope. The endpoints $i = 0, n$ are included. In particular the index-zero phase constraint is non-scalar in rank one. A named additive character is used when linear duals are identified with additive characters.

Provenance. ids: D1142, D1242; proved in: theory/sidequests/f1-limit/mirabolic.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B2); theory/checks/f1_limit_bridge_check.py (B3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 18.4 (Generation by flags and one controlled line)**.** PROVED *For L of dimension $n \ge 1$ over $\mathbb{F}_Q$, the embedded flag Hecke algebra acts as $A_w \otimes I$, the marked element $(T_0 + I)/Q$ is exactly C_1^X, and these operators generate the full affine commutant $R_X(L)$. Moreover*

$$e^2 = e = e^*, \qquad \tau_Q(e) = Q^{-1}, \qquad T_0^2 = (Q-2)T_0 + (Q-1)I.$$

Scope. The additional constraints C_i^X are useful tests but are redundant as algebra generators. The theorem fixes the orbit basis, adjoint, trace and Hecke inclusion, rather than asserting only an abstract algebra isomorphism.

Provenance. ids: F1-MIR-GEN; proved in: theory/sidequests/f1-limit/mirabolic.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B2, n=2 and Q=2,3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The zero-vector orbit kernels act as $A_w \otimes I$. The kernel of $T_0 + I$ is one precisely when the two flags agree and the vector difference lies in the controlled line $U_1(F)$. Dividing by Q averages translations along that line, hence gives C_1^X. A finite-group average is a self-adjoint projection; its rank in each flag block is $|L|/Q$, which gives the trace. The quadratic formula follows from $T_0 = Qe - I$. Rosso's generation theorem then gives the reverse inclusion from these concrete operators to the full invariant-kernel algebra. □

18.3 Fourier duality with exact indices

Definition 18.5 (Fourier dual flag reversal)**.** Fix a named nontrivial additive character $\psi : k \to U(1)$. Define

$$\mathcal{F}_{L,\psi} e_x = |L|^{-1/2} \sum_{\lambda\in L^\vee} \psi\big(-\lambda(x)\big) e_\lambda.$$

If $F = (U_i)_{i=0}^n$, its reversed dual flag is

$$D_L(F)_j = U_{n-j}^{\perp}, \qquad 0 \le j \le n.$$

Let D_L also denote the induced flag-basis unitary and put $W_{L,\psi} = D_L \otimes \mathcal{F}_{L,\psi}$. Define

$$R_Z(L^\vee) = W_{L,\psi} R_X(L) W_{L,\psi}^*.$$

The phase-polarized groupoid has finite-dimensional k-spaces as objects, linear isomorphisms as arrows, and retains the pair $(R_X(L), R_Z(L^\vee))$, their Hecke inclusions, all controlled constraints and the named $W_{L,\psi}$. A same-register realization additionally names a linear self-duality $\sigma : L \to L^\vee$; morphisms act on flags, vectors and duals by their induced permutation unitaries.

Scope. The negative Fourier kernel, the phase ψ, dual-flag reversal and any same-register self-duality are named choices. Forgetting them retains only the corresponding conjugacy class of algebra comparisons.

Provenance. ids: D1243, D1244; proved in: theory/sidequests/f1-limit/mirabolic.md; tested by: theory/checks/f1_limit_bridge_check.py (B3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 18.6 (Fourier exchange of controlled constraints). **PROVED** *For every finite field $k = \mathbb{F}_Q$, every n-dimensional L, every named nontrivial $\psi : k \to U(1)$ and every $0 \le i \le n$,*

$$W_{L,\psi} C_i^X W_{L,\psi}^* = C_{n-i}^Z,$$
$$W_{L,\psi}(A_w \otimes I) W_{L,\psi}^* = A_{w_0 w w_0} \otimes I.$$

The operators on the right generate exactly $R_Z(L^\vee)$, including $C_0^Z = I \otimes |0\rangle\langle 0|$. Under canonical double duality,

$$W_{L^\vee,\psi} W_{L,\psi} e_{(F,x)} = e_{(F,-x)}.$$

The construction is natural for linear isomorphisms.

Scope. Only the vector Fourier factor is tensor-preserving under direct sums. The full flag–Fourier bridge is not monoidal on complete-flag registers and does not identify either commutant with the full Weyl matrix algebra.

Provenance. ids: F1-MIR-FOURIER; proved in: theory/sidequests/f1-limit/mirabolic.md; tested by: theory/checks/f1_limit_bridge_check.py (B3, Fourier constraints, reversal, square and ranks); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Character orthogonality makes $\mathcal{F}_{L,\psi}$ unitary and gives

$$\mathcal{F}_{L,\psi} P_U^X \mathcal{F}_{L,\psi}^* = P_{U^\perp}^Z.$$

Annihilators reverse a full flag, so the index i becomes $n - i$. The intersection-rank matrix of a flag pair transforms by $r'(i,j) = i + j - n + r(n-i, n-j)$, which is the rank matrix of $w_0 w w_0$. This proves both conjugation formulas and the generation statement. Applying the negative Fourier kernel twice yields

$$|L|^{-1} \sum_{y \in L} \sum_{\lambda \in L^\vee} \psi\big(-\lambda(x+y)\big) e_y = e_{-x};$$

dual reversal squares to the original flag. In characteristic two this vector negation is the identity. $\square$

Definition 18.7 (Vector-polarization bridge package)**.** In rank n, a bridge package retains the affine commutant and orbit data, the controlled translation and phase families for every $0 \le i \le n$, the based Hecke inclusion, conjugate-linear orbit star, reference trace, algebraic and trace-supported endpoints, the named phase and Fourier unitary, and the regular one-sided operational boundary. When composition is needed, it additionally names an arithmetic symplectic or affine decomposable-shuffle correspondence with its ternary shuffle coherence.

Only $\mathcal{F}_{L\oplus M} = \mathcal{F}_L \otimes \mathcal{F}_M$ is a literal tensor identity. On the decomposable flag–vector subspace the full bridge uses

$$W_{L\oplus M} J_{L,M} = J_{L^\vee, M^\vee}\big((W_L \otimes W_M) \otimes \mathrm{Rev}\big),$$

where Rev reverses the shuffle word. An isomorphism of packages preserves all named UCP maps, the regular label class and quotient boundary functor, and the exact square $W_{L^\vee,\psi} W_{L,\psi}(F, x) = (F, -x)$.

Scope. An abstract algebra isomorphism is not an isomorphism of bridge packages. The decomposable flag subspace is generally proper; even adjoining shuffles does not tensor-factor the full complete-flag Hilbert space.

Provenance. ids: D1259; proved in: theory/sidequests/f1-limit/mirabolic.md; theory/sidequests/f1-limit/type-c.md; tested by: B1–B7 in the two bridge checkers, at their stated finite scopes; reviewed in: theory/verdicts/f1-limit-adjudication.md.

19 Positive mirabolic fibres and their regular boundary

The affine commutant at a finite field has a physical matrix trace. To move away from arithmetic parameters, one needs an algebraic reason that the same coefficient trace stays positive. Orbital valencies supply that reason for every real $q > 1$. At $q = 1$ the form becomes degenerate: the full algebraic specialization remains, while its reference GNS representation retains only the ordinary symmetric-group Hecke algebra. The operational boundary is therefore built from one-sided sections positive at every nearby real parameter, rather than from values sampled only at prime powers.

19.1 Orbit labels, valencies and the positive Gram form

Definition 19.1 (Mirabolic orbit poset and based star-trace)**.** For $w \in S_n$, define a partial order on $\{1, \ldots, n\}$ by

$$i \prec_w j \quad \Longleftrightarrow \quad i < j \;\text{ and }\; w^{-1}(i) < w^{-1}(j).$$

For an antichain A, set

$$\downarrow_w A = \{i : i = a \text{ or } i \prec_w a \text{ for some } a \in A\}.$$

The orbit label (w, A) denotes the affine orbital whose flag relative position is w and whose vector has maximal-support antichain A; its adjacency basis element is $T_{w,A}$. The empty antichain is the zero-vector Hecke basis.

On the polynomial orbit-basis algebra define the conjugate-linear star by

$$T^*_{w,A} = T_{w^{-1},w^{-1}(A)}$$

and the coefficient trace by

$$\tau_q(T_{w,A}) = \begin{cases} 1, & w = 1, A = \varnothing, \\ 0, & \text{otherwise.} \end{cases}$$

The proposed positive domain is real $q > 1$; evaluation at one is retained as a positive-semidefinite boundary form.

Scope. Vector negation is part of the orbital adjoint but does not change support antichains. Positivity is not stipulated and is proved below. No positive trace is claimed for $0 < q < 1$.

Provenance. ids: D1245, D1246; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: theory/checks/f1_limit_check.py (L12); B1,B4 in theory/checks/f1_limit_mirabolic_check.py; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 19.2 (Antichain classification and orbital valency)**.** PROVED *Let $L = \mathbb{F}_Q^n$, fix a reference complete flag, and let $w \in S_n$. The residual stabilizer orbits on L are indexed by antichains A in the preceding poset. The full affine orbital (w, A) has exact valency*

$$k_{w,A}(Q) = Q^{\ell(w)+|\downarrow_w A|-|A|}(Q-1)^{|A|}.$$

Scope. This holds for every finite field, including characteristic two and extension fields. It is an orbit-counting theorem, not a positivity statement at a non-arithmetic parameter.

Provenance. ids: F1-MIR-VAL; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: theory/checks/f1_limit_check.py (L12, n at most 3 and Q=2,3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The intersection of the two Borels for relative position w contains the diagonal torus and precisely the shears $I + cE_{ij}$ with $i \prec_w j$. Allowed shears change only coordinates below an existing support coordinate, so the maximal nonzero support is an invariant antichain. Conversely, diagonal scaling normalizes the coordinates on A, and shears from each $a \in A$ create arbitrary coordinates in its strict downset. The residual orbit therefore has $(Q-1)^{|A|}Q^{|\downarrow_w A|-|A|}$ elements. The Bruhat cell of w has $Q^{\ell(w)}$ flags, giving the formula. □

Theorem 19.3 (Faithful positive mirabolic trace for all real parameters above one)**.** PROVED
For every $n \geq 1$ and every real $q > 1$, the orbit-basis trace has diagonal Gram form

$$\tau_q(T^*_{w,A}T_{v,C}) = \delta_{(w,A),(v,C)}q^{\ell(w)+|\downarrow_w A|-|A|}(q-1)^{|A|}.$$

It is a faithful normalized trace, and its left regular representation makes the fibre a finite C-algebra.*

Scope. The all-real conclusion follows from a polynomial identity and the displayed sign, not from numerical sampling at arithmetic values. For $0 < q < 1$, odd $|A|$ gives a negative Gram entry.

Provenance. ids: F1-MIR-POS; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: L12 checks finite orbit counts; the uniform positivity statement is proved analytically; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. At every prime power, distinct orbital adjacency matrices have disjoint support, and their normalized Hilbert–Schmidt squared norms equal their valencies. Both the product coefficient and the valency are polynomials in q; equality at infinitely many prime powers makes them identical. For $q > 1$ every displayed diagonal entry is strictly positive. Polynomial extension of matrix-trace cyclicity gives traciality. Left multiplication is faithful because $L_x 1 = x$, and its adjoint is L_{x^*}. □

19.2 Hecke retraction and the trace-supported endpoint

Definition 19.4 (Hecke inclusion and vector expectation)**.** Let

$$i_q : H_n(q) \longrightarrow R_n(q), \qquad T_w \longmapsto T_{w,\varnothing},$$

and define coefficient deletion by

$$E_q(T_{w,A}) = \begin{cases} T_w, & A = \varnothing, \\ 0, & A \neq \varnothing. \end{cases}$$

Scope. These are fixed-rank maps. They do not provide a generic cross-rank map $R_m(q) \otimes R_n(q) \to R_{m+n}(q)$; in particular they leave room for separately constructed right module actions.

Provenance. ids: D1247; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B4); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 19.5 (Vector coarse graining as a conditional expectation)**.** PROVED *For every real $q > 1$, i_q is a trace-preserving unital *-monomorphism and E_q is its trace-preserving UCP conditional expectation, with $E_q i_q = \mathrm{id}$. Hence Hecke densities, effects and retained Kraus lists lift unchanged to the mirabolic fibre.*

Scope. The reverse operation $i_q E_q$ is generally a proper vector-sector coarse graining. The statement concerns fixed-rank operational data only.

Provenance. ids: F1-MIR-EXPECT; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B4); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The Gram formula makes E_q the trace-orthogonal projection onto the unital Hecke *-subalgebra. Traciality gives bimodularity. Testing a positive input against elements of the subalgebra proves positivity, and the identical argument in matrix amplifications proves complete positivity. The basis formulas give trace preservation and $E_q i_q = \mathrm{id}$. □

Definition 19.6 (Algebraic and trace-supported endpoints)**.** The algebraic endpoint is the full based specialization $R_n(1)$ with the specialized trace τ_1. Its trace-supported endpoint is

$$R_n(1)/\mathcal{N}_{\tau_1}, \qquad \mathcal{N}_{\tau_1} = \{x : \tau_1(x^*x) = 0\}.$$

The two endpoint algebras are not identified by definition.

Scope. No C*-norm is placed on the degenerate algebraic endpoint. Only its GNS quotient carries the faithful reference representation.

Provenance. ids: D1248; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B4); theory/checks/f1_limit_check.py (L10); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 19.7 (The mirabolic reference quotient at one)**.** PROVED *At $q = 1$, the coefficient trace is positive semidefinite and*

$$\mathcal{N}_{\tau_1} = \mathrm{span}\{T_{w,A} : A \neq \varnothing\}.$$

*This nullspace is a two-sided *-ideal, and*

$$R_n(1)/\mathcal{N}_{\tau_1} \cong H_n(1) = \mathbb{C}[S_n]$$

*as a traced *-algebra. The map E_1 is exactly the quotient map under the empty-orbit basis identification.*

Scope. The full algebraic specialization still contains the vector ideal. The claim concerns the GNS quotient selected by the reference trace.

Provenance. ids: F1-MIR-END; proved in: theory/sidequests/f1-limit/mirabolic-trace.md; tested by: B4, exact rank-two interpolation and null Gram; L10 rank one; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Setting $q = 1$ in the diagonal Gram formula leaves norm one exactly for $A = \varnothing$ and norm zero otherwise. The nullspace of a positive tracial form is a two-sided *-ideal by Cauchy–Schwarz. The surviving basis classes multiply by the Hecke rule at one, giving the group algebra and its coefficient trace. Since E_1 deletes precisely the null basis, it is the quotient. □

19.3 The actual one-sided operational category

Definition 19.8 (Regular one-sided labels)**.** Fix $J_\varepsilon = [1, 1+\varepsilon]$. A regular mirabolic section is

$$h(q) = \sum_{w,A} h_{w,A}(q) T_{w,A}(q), \qquad h_{w,A} \in C(J_\varepsilon),$$

with pointwise polynomial multiplication, the orbit star and continuous coefficient trace. Hecke factors use their continuous section algebra on the same interval; separated register words use continuous coefficients in the product bases and the $C(J_\varepsilon)$-balanced common-parameter tensor.

A regular density additionally satisfies $h(q) \geq 0$ and $\tau_q(h(q)) = 1$ for every real $1 < q \leq 1+\varepsilon$. A regular effect satisfies $0 \leq e(q) \leq 1$ there; a regular POVM consists of finitely many such positive sections summing to one. Regular instrument coefficients are continuous and obey Kraus completeness at every real $q > 1$ in the interval.

Scope. There is no C*-norm on the boundary section algebra. Positivity only at prime powers is insufficient. Coefficient-pole densities are excluded and require separate singular boundary data.

Provenance. ids: D1249; proved in: theory/sidequests/f1-limit/mirabolic-boundary.md; tested by: uniform theorem by proof; L10 and B4 give boundary examples; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 19.9 (Regular separated operational category)**.** For each real $q > 1$, quantum atoms are $[H_n]$ and $[R_n]$, $n \geq 1$, with their faithful coefficient traces; nonempty finite classical wires have uniform trace. The empty word has algebra $\mathbb{C}$, and ordered word tensor is the actual spatial tensor product. The interval version uses the regular common-parameter sections just defined.

Raw morphisms are finite typed planar circuits generated by all regular preparations, discards and finite POVMs; every retained same-register instrument

$$K : X \to X\underline{O}, \qquad \sum_{o,i} K_{o,i}(q)^* K_{o,i}(q) = 1;$$

the explicit classical product, singleton, bijective relabeling and classical-wire routing maps; and

$$\begin{aligned} \mathrm{up}_n : [H_n] \to [R_n] \qquad & (E_q \text{ in Heisenberg form}), \\ \mathrm{down}_n : [R_n] \to [H_n] \qquad & (i_q \text{ in Heisenberg form}). \end{aligned}$$

Instrument lists are retained under separated context tensor and factorwise inclusion. With the uniform classical trace, the o-block of the output density is $|O| \sum_i K_{o,i} h K_{o,i}^*$.

Composition is typed gluing and ordered tensor is horizontal juxtaposition. Equality is the congruence generated by category, monoidal and explicit classical-wiring relations, literal state/POVM equality, pointwise equality of coefficient Kraus Grams within each outcome, the routed list identities, and $\mathrm{down}_n \, \mathrm{up}_n = \mathrm{id}_{[H_n]}$. The reverse composite is $i_q E_q$ and is not an identity. Germs identify circuits when these relations hold on a common smaller one-sided interval. There is no cross-rank mirabolic assembly generator.

Scope. The classical wiring is the explicit parameter-independent UCP presentation already proved for the core category; no unresolved wiring hypothesis remains. Equality is not an isolated-CP quotient and requires no continuous family of Kraus mixing matrices. In particular $[R_m][R_n]$ is not identified with $[R_{m+n}]$.

Provenance. ids: D1257, D1218; proved in: theory/sidequests/f1-limit/mirabolic-boundary.md; theory/sidequests/f1-limit/classical-wiring.md; tested by: theory/checks/f1_limit_wiring_check.py (W1–W5); B4; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 19.10 (Regular quotient boundary functor)**.** Let $\Pi_n : R_n(1) \to H_n(1)$ be the GNS quotient E_1. For a register word X, let Q_X tensor Π_n on every mirabolic factor and the identity on Hecke and classical factors; write $\overline{X}$ for the word obtained by replacing every mirabolic atom by its Hecke atom. The endpoint category has these Hecke/classical words, the same wiring, all normalized preparations, POVMs and retained same-register instruments, the same label relations, and its UCP realization.

The boundary functor evaluates continuous coefficients at one, applies Q_X to every density, effect and Kraus entry, and sends every up/down generator to the identity. Evaluation at $q > 1$ requires a chosen representative whose interval contains q.

Scope. Arbitrary UCP families with merely regular superoperator coefficients are not source morphisms. The regular density and retained-Kraus restrictions are part of the functor's domain.

Provenance. ids: D1258; proved in: theory/sidequests/f1-limit/mirabolic-boundary.md; tested by: structured descent proof; B4 and W1–W5 test primitive data; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 19.11 (Regular operational mirabolic boundary). **PROVED** *The regular one-sided circuits and their germs form actual ordered monoidal categories with UCP realizations. Coefficient evaluation followed by the factorwise GNS quotient defines a monoidal endpoint functor to the separated Hecke operational category. It preserves all specified generators and typed relations. Finite protocol Born weights converge, and conditioned weights converge whenever the limiting conditioning probability is positive. Every individual finite endpoint Hecke circuit has a local regular lift.*

Scope. All source densities, effects and instruments are positive or normalized for every nearby real $q > 1$. Arithmetic-only positivity and arbitrary coefficient-regular UCP superoperators are excluded. No generic $R_m(q) \otimes R_n(q)$ assembly is asserted.

Provenance. ids: F1-MIR-REG; proved in: theory/sidequests/f1-limit/mirabolic-boundary.md; tested by: B4 and W1–W5 support primitive formulas; functoriality and positivity by structured proof; reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Polynomial structure constants and the parameter-independent star make regular sections a *-algebra. For a regular positive density h and any endpoint Hecke element x, lift x through the Hecke inclusion. Positivity of $\tau_q(x(q)^*h(q)x(q))$ for every real $q > 1$ passes to one by coefficient continuity. Faithfulness of the endpoint quotient trace then forces $Q_X(h(1)) \geq 0$; normalization also passes to the limit. The same argument handles effects and POVMs, while Kraus completeness passes algebraically and multiplicativity of Q_X makes every branch map descend.

The explicit wiring commutes with factorwise quotients. Pointwise Gram equality is a congruence fibre by fibre and descends under the linear quotient map. Hence the circuit quotient and endpoint functor are well defined. A finite branch density has continuous coefficients by induction over its circuit, so trace pairings and finite event sums converge. Positive-limit conditioning is ordinary quotient continuity. Local endpoint lifts use square-root normalization in the continuous Hecke subalgebra followed by i_q, on one common neighborhood for the finitely many labels. ☐

The real-neighborhood hypothesis is sharp. In $H_2(q)$, if e_+, e_- are the two spectral projections, then

$$h(q) = (q - frac32)e_+ + \frac{5}{2q}e_-$$

is normalized and positive at every prime power $Q \geq 2$, but has a negative e_+ coefficient for $1 < q < 3/2$. It is not a regular density. Conversely, the rank-one singular state $q(q-1)^{-1}(1-e)$ is positive and normalized for every $q > 1$, but has a coefficient pole and does not annihilate the endpoint null ideal. It is a legitimate punctured-fibre state requiring additional boundary data, not an object of the regular category.

20 Decomposable-shuffle correspondences in type C and the affine bridge

Orthogonal direct sum of symplectic spaces does not produce a standard parabolic inclusion of type-C Hecke algebras. It does, however, select a canonical subspace of decomposable isotropic flags. Component flags and a shuffle word form an orthonormal basis of that subspace. Compression to it and scaled averaging back give trace-adjoint UCP maps, while ternary shuffle bijections make the correspondence associative. The same construction works for affine flag–vector spaces at prime-power fibres.

20.1 The decomposable type-C arena

Definition 20.1 (Arithmetic type-C context algebra)**.** Let $k = \mathbb{F}_Q$ and let V be a $2n$-dimensional symplectic k-space. Write $X_C(V)$ for its complete isotropic flags

$$0 = U_0 < U_1 < \cdots < U_n, \qquad \dim U_i = i,$$

where U_n is Lagrangian. With the flag basis of $K_V = \mathbb{C}[X_C(V)]$ orthonormal, define

$$A_C(V) = \mathrm{End}_{Sp(V)}(K_V)$$

with Hilbert adjoint and normalized operator trace.

Scope. At a prime power this is the finite type-C_n flag Hecke commutant. No generic positive-real type-C Hecke category is stipulated here.

Provenance. ids: D1250; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 20.2 (Decomposable isotropic flags and shuffle isometry)**.** Let V, W have symplectic ranks m, n. A complete isotropic flag in $V \perp W$ is decomposable when every step is $U_a \oplus Z_b$ for steps of component flags. Let $\mathrm{Sh}(m, n)$ be the two-letter words with m letters V and n letters W. If (a_r, b_r) counts the two letters in the first r positions of σ, then

$$(U, Z, \sigma) \longmapsto (U_{a_r} \oplus Z_{b_r})_{r=0}^{m+n}$$

is the decomposable flag. Linear extension gives the basis isometry

$$J_{V,W} : K_V \otimes K_W \otimes \ell^2(\mathrm{Sh}(m, n)) \longrightarrow K_{\mathrm{dec}}.$$

Let $p_{V,W} = J_{V,W} J_{V,W}^*$.

Scope. The decomposable subspace is usually proper inside the full composite flag space. The shuffle factor is retained as a matrix-coherent register, rather than discarded as a classical label.

Provenance. ids: D1251; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B5,B6); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Definition 20.3 (Local-symmetry product-shuffle corner)**.** Put

$$H_{V,W} = Sp(V) \times Sp(W), \qquad \mathcal{D}_{V,W} = \mathrm{End}_{H_{V,W}}(K_{V \perp W}).$$

The product-shuffle arena is the corner $p_{V,W} \mathcal{D}_{V,W} p_{V,W}$, whose unit is $p_{V,W}$ and whose trace is ordinary operator trace normalized by $\mathrm{rank}(p_{V,W})$.

Scope. This local-symmetry commutant is larger than the full-symplectic commutant. The corner type and its non-scalar unit are part of the construction.

Provenance. ids: D1252; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 20.4 (Decomposable flags are component flags plus a shuffle). PROVED *For symplectic spaces V, W over any finite field, the preceding flag map is a bijection, and*

$$p_{V,W}\mathcal{D}_{V,W}p_{V,W} \cong A_C(V) \otimes A_C(W) \otimes M_{\mathrm{Sh}(m,n)}(\mathbb{C})$$

*as a traced *-algebra.*

Scope. This is an arithmetic finite-dimensional commutant identity. It is not an algebra isomorphism between the full composite context and the product corner, and it includes all shuffle coherences.

Provenance. ids: F1-TC-DEC; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B5, Sp4 over F2); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The dimensions of intersections with V and W recover the shuffle word and the nonrepeated intersections recover both component flags. Thus the displayed map is bijective. The subgroup $Sp(V) \times Sp(W)$ acts on the two component flag factors and trivially on the shuffle factor. Its commutant on the decomposable space is therefore the tensor product of the two component commutants and the full matrix algebra on shuffles. □

20.2 Restriction and preparation

Definition 20.5 (Trace-adjoint type-C channels). Let $G = Sp(V \perp W)$ and let

$$\mathcal{E}_G(x) = |G|^{-1} \sum_{g \in G} U_g x U_g^*, \qquad r_{V,W} = \frac{\mathrm{rank}(p_{V,W})}{\dim K_{V \perp W}}.$$

For $a \in A_C(V \perp W)$ and $x \in p_{V,W}\mathcal{D}_{V,W}p_{V,W}$, extended by zero on the orthogonal complement, define

$$\Phi_{V,W}(a) = p_{V,W}\, a\, p_{V,W}, \qquad \Psi_{V,W}(x) = r_{V,W}^{-1}\mathcal{E}_G(x).$$

Scope. The scale $r_{V,W}^{-1}$ is required for unitality and trace preservation. Compression is not asserted to be multiplicative on a generic full symplectic transition.

Provenance. ids: D1253; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B5); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 20.6 (Bidirectional positive type-C correspondence). PROVED *For every prime power Q and finite symplectic V, W over $\mathbb{F}_Q$, $\Phi_{V,W}$ and $\Psi_{V,W}$ are unital trace-preserving completely positive maps, and they are adjoint for the normalized traces:*

$$\mathrm{tr}_p(\Phi_{V,W}(a)x) = \mathrm{tr}_G(a\Psi_{V,W}(x)).$$

Scope. These are restriction and preparation channels between two different observable algebras. Mutual trace adjointness does not make them inverse *-isomorphisms.

Provenance. ids: F1-TC-CP; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B5, exact averaging and trace pairing); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Compression is CP and sends the full unit to the corner unit. Transitivity of $Sp(V \perp W)$ on complete isotropic flags gives $\mathcal{E}_G(p_{V,W}) = r_{V,W} I$, so scaled averaging is also UCP. For a commuting with G,

$$\operatorname{Tr}(pa) = \operatorname{Tr}(\mathcal{E}_G(p)a) = r \operatorname{Tr}(a),$$

which proves trace preservation. The same averaging identity applied to ax gives the displayed adjointness. □

Definition 20.7 (Ternary shuffle coherence)**.** For ranks ℓ, m, n, let $\operatorname{Sh}(\ell, m, n)$ be the three-letter words with those multiplicities. The two coherence maps are the canonical bijections

$$\operatorname{Sh}(\ell, m) \times \operatorname{Sh}(\ell + m, n) \longrightarrow \operatorname{Sh}(\ell, m, n),$$
$$\operatorname{Sh}(m, n) \times \operatorname{Sh}(\ell, m + n) \longrightarrow \operatorname{Sh}(\ell, m, n),$$

obtained by expanding each combined letter with the next letter of the inner word. All iterated flag isometries, compression channels and trace-adjoint preparations are compared using these bijections and the ordinary Hilbert associator.

Scope. The coherence maps preserve the order of the three summands. They are specified bijections, not silent equality of differently nested shuffle sets.

Provenance. ids: D1254; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B6); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 20.8 (Associativity of the type-C correspondence)**.** PROVED *For three finite symplectic spaces over one finite field, the two iterations of decomposable restriction agree after the canonical ternary shuffle bijections. Their trace-adjoint preparation maps agree in the reverse direction.*

Scope. The theorem is arithmetic and finite-dimensional. It retains the shuffle register and supplies no symmetric exchange beyond the named reorderings.

Provenance. ids: F1-TC-COH; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B6, three rank triples); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Both nested isometries send a triple of component flags and a three-letter word to the same flag whose current step is the direct sum of the three current component steps. Hence both iterated projections are the same diagonal projection. Compression maps coincide. Their normalized-trace adjoints coincide as well; equivalently, the nested rank fractions multiply to the rank fraction of this common projection. □

20.3 The thin signed-permutation fibre

Definition 20.9 (Thin type-C block comparison)**.** At $q = 1$, let B_n be the signed permutation group. Embed $B_m \times B_n$ in B_{m+n} by preserving the two coordinate blocks. Fix the convention that every signed permutation factors as a block signed permutation followed by one unsigned (m, n) shuffle. The induced group-algebra *-inclusion is the thin block assembly.

Scope. This definition concerns the group-algebra endpoint only. It includes no generic type-C Hecke lift of the block map.

Provenance. ids: D1255; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B6, signed factorizations); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 20.10 (Unsigned shuffles are the thin block cosets)**.** PROVED *For $m, n \geq 0$, every element of B_{m+n} has the fixed unique factorization above, so*

$$\ell^2(B_{m+n}) \cong \ell^2(B_m \times B_n) \otimes \ell^2(\mathrm{Sh}(m,n)).$$

*For $m, n > 0$ the block subgroup is proper and nonparabolic, although its group algebra embeds by a unital *-map.*

Scope. The theorem leaves open generic bimodules or right Hecke actions. It states only that no standard generic $H(B_m, q) \otimes H(B_n, q)$ parabolic assembly has been supplied.

Provenance. ids: F1-TC-ONE; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_bridge_check.py (B6, four signed block factorizations); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. Move the unsigned shuffle of the two absolute-value blocks to the chosen side; what remains preserves both coordinate blocks and carries all signs. This is unique. The subgroup index is $\binom{m+n}{m}$. It has full Coxeter rank $m+n$, whereas a proper parabolic subgroup has smaller rank, proving nonparabolicity. □

20.4 Arithmetic affine decomposable composition

Definition 20.11 (Affine decomposable product-shuffle arena)**.** For finite vector spaces L, M over one finite field, let $p^{\mathrm{aff}}_{L,M}$ project onto basis vectors consisting of a decomposable complete flag in $L \oplus M$ and an arbitrary vector $(x, y) \in L \oplus M$. The local symmetry is

$$\mathrm{Aff}(L) \times \mathrm{Aff}(M).$$

The affine product-shuffle corner, restriction and preparation maps are the local-symmetry commutant corner, compression by $p^{\mathrm{aff}}_{L,M}$, and inverse-rank-fraction scaled averaging over $\mathrm{Aff}(L \oplus M)$, exactly as in the symplectic formulas above.

Scope. This is arithmetic data at a prime-power fibre. It is not a generic-real mirabolic assembly map and is not a generator of the regular separated boundary category.

Provenance. ids: D1256; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B7); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Theorem 20.12 (Arithmetic affine product-shuffle correspondence)**.** PROVED *Let L, M be finite-dimensional over $\mathbb{F}_Q$. Decomposable flags and all vector labels identify the affine local-symmetry corner with*

$$R_X(L) \otimes R_X(M) \otimes M_{\mathrm{Sh}(\dim L, \dim M)}(\mathbb{C}).$$

Compression from the full affine commutant and scaled full-affine averaging back are mutually trace-adjoint trace-preserving UCP maps, coherent for three summands under the ternary shuffle bijections.

Scope. The hypotheses require a prime-power field. The result does not infer positivity or composition for all real $q > 1$, and it does not identify the full flag register with the product register.

Provenance. ids: F1-MIR-DEC; proved in: theory/sidequests/f1-limit/type-c.md; tested by: theory/checks/f1_limit_mirabolic_check.py (B7, Q=2,3); reviewed in: theory/verdicts/f1-limit-adjudication.md.

Proof. The decomposable flag–vector basis is the tensor product of the two flag–vector bases and the shuffle basis, equivariantly for $\mathrm{Aff}(L) \times \mathrm{Aff}(M)$. Taking commutants gives the displayed corner. The full affine group is transitive on flag–vector basis elements, so averaging the decomposable projection gives its rank fraction times the identity. The compression, scaled-average and ternary coherence proofs are therefore identical to the symplectic arguments. □

21 Frobenius, subfield codes and higher multiplication gates

A finite extension field supplies both a quantum register and arithmetic operations on it. The trace pairing identifies its position and momentum labels over the prime field. Relative trace then gives a logical momentum coordinate for subfield codes and a Fourier-dual transfer between registers. Multiplication in separate registers supplies reversible gates at every higher Clifford level. These statements hold in characteristic two as well as at odd primes, with the choices of phase and computational basis explicit.

21.1 Trace geometry and arithmetic Frobenius

Definition 21.1 (Trace-framed arithmetic register)**.** Fix a prime p and a named primitive complex p-th root ζ_p. For a finite field $E/\mathbb{F}_p$ of cardinality p^r, the trace-framed arithmetic register retains the field E, its Hilbert space $\mathcal{H}_E = \ell^2(E)$ with orthonormal basis $|x\rangle$ indexed by $x \in E$, and the trace character

$$\mathrm{Tr}_{E/\mathbb{F}_p}(x) = \sum_{j=0}^{r-1} x^{p^j}, \qquad \psi_E(x) = \zeta_p^{\mathrm{Tr}_{E/\mathbb{F}_p}(x)}.$$

Its prime-field phase space is $S_E = E \oplus E$ with form

$$\Omega_E((a,b),(a',b')) = \mathrm{Tr}_{E/\mathbb{F}_p}(ab' - a'b).$$

The reference operators are

$$X_E(a)|x\rangle = |x+a\rangle, \qquad Z_E(b)|x\rangle = \psi_E(bx)|x\rangle, \qquad W_E(a,b) = Z_E(-b)X_E(a).$$

An ordered list of registers has the tensor-product Hilbert space and direct-sum prime-field phase space. The empty list has Hilbert space $\mathbb{C}$. The field structures and ordered factors are retained data.

Scope. The root of unity and reference operator convention are named choices. No restriction beyond the stated finite-field hypotheses is imposed.

Provenance. ids: D3, D8, D1301; proved in: theory/sidequests/frobenius-hierarchy/foundation.md; tested by: theory/checks/frobenius_hierarchy_check.py (A1–A2); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Definition 21.2 (Arithmetic Frobenius operator)**.** For a trace-framed arithmetic register with named p, ζ_p, put

$$\sigma_E(x) = x^p, \qquad U_E|x\rangle = |\sigma_E(x)\rangle.$$

For a list, Frobenius means the tensor product of these operators.

Scope. This is the Frobenius of the named field structure. It has the same field as source and target.

Provenance. ids: D1302; proved in: theory/sidequests/frobenius-hierarchy/foundation.md; tested by: theory/checks/frobenius_hierarchy_check.py (A1); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Definition 21.3 (Named embedding and transported relative trace)**.** For trace-framed arithmetic registers over the same named p, ζ_p, a named field embedding $i : K \to E$, $|K| = p^s$, and $r = ds$, define

$$T_i(x) = i^{-1}\left(\sum_{j=0}^{d-1} x^{p^{sj}}\right), \qquad \kappa_i = |\ker T_i|.$$

Here i^{-1} is restricted to $i(K)$. For $j : L \to K$ the composite trace is denoted $T_{ij} : E \to L$.

Scope. The trace formula taking values in $i(K)$ and its surjectivity are results below, including when p divides the extension degree.

Provenance. ids: D1303; proved in: theory/sidequests/frobenius-hierarchy/foundation.md; tested by: theory/checks/frobenius_hierarchy_check.py (A3); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Theorem 21.4 (Relative trace and prime-field symplectic geometry). **PROVED** *For every named embedding $i : K \to E$ of finite fields over $\mathbb{F}_p$, the transported relative trace is well defined, surjective and K-linear, and*

$$\kappa_i = \frac{|E|}{|K|}, \qquad \mathrm{Tr}_{E/\mathbb{F}_p}(i(a)x) = \mathrm{Tr}_{K/\mathbb{F}_p}(aT_i(x)).$$

For every tower $L \xrightarrow{j} K \xrightarrow{i} E$, $T_{ij} = T_j T_i$. The absolute trace pairing on E is nondegenerate, and Ω_E is alternating and nondegenerate over $\mathbb{F}_p$.

Scope. There is no division by the extension degree. All prime characteristics are included; no restriction beyond the stated hypotheses is imposed.

Provenance. ids: FRB-TRACE; proved in: theory/sidequests/frobenius-hierarchy/foundation.md, section 1; tested by: theory/checks/frobenius_hierarchy_check.py (A1–A3); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. Put $q = |K|$ and $S(x) = \sum_{j=0}^{d-1} x^{q^j}$. Since $x^{q^d} = x$, one has $S(x)^q = S(x)$; the q roots of $t^q - t$ in E are precisely $i(K)$. The polynomial S has degree $q^{d-1} < q^d$ and nonzero leading coefficient, so it cannot vanish on every point of E. It is $i(K)$-linear and therefore surjects onto $i(K)$. Equal-size additive fibres give the kernel cardinality. In a tower, expanding the two trace sums lists every Frobenius exponent in the composite trace exactly once. This proves transitivity and the pairing identity. If $c \neq 0$, choose u with absolute trace one and put $x = c^{-1}u$; then $\mathrm{Tr}_{E/\mathbb{F}_p}(cx) = 1$. Thus the trace pairing is nondegenerate. Pairing a vector (a, b) with $(0, x)$ and then $(x, 0)$ proves nondegeneracy of Ω_E; $\Omega_E(v, v) = 0$ directly in every characteristic. ☐

Theorem 21.5 (Exact Frobenius covariance and atomic factorization). **PROVED** *For every trace-framed register $E/\mathbb{F}_p$, Frobenius is a field automorphism, U_E is unitary, both have order $r = [E : \mathbb{F}_p]$, and*

$$\Omega_E(\sigma_E v, \sigma_E w) = \Omega_E(v, w), \qquad U_E W_E(a, b) U_E^* = W_E(a^p, b^p).$$

For a named $\mathbb{F}_p$-basis $(e_j)_{j=1}^r$ of E, there is a unique trace-dual basis $(e^j)_{j=1}^r$. If $a = \sum_j a_j e_j$ and $b = \sum_j b_j e^j$, the coordinate unitary $|\sum_j x_j e_j\rangle \mapsto \bigotimes_j |x_j\rangle$ sends $W_E(a, b)$ to $\bigotimes_j W_{\mathbb{F}_p}(a_j, b_j)$.

Scope. Position and momentum use dual bases. No self-dual basis is required, and no theorem about a lift of every symplectic map is asserted.

Provenance. ids: FRB-FROB; proved in: theory/sidequests/frobenius-hierarchy/foundation.md, section 2; tested by: theory/checks/frobenius_hierarchy_check.py (A1); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. The absolute trace is invariant under the cyclic shift of Frobenius powers. Consequently $U_E X_E(a) U_E^* = X_E(a^p)$ and $U_E Z_E(b) U_E^* = Z_E(b^p)$, as is checked on the computational basis. Multiplying in the reference order gives the exact Weyl formula. The same trace invariance proves preservation of Ω_E. Frobenius has order r: its r-th power is the identity, while an identity $x^{p^k} = x$ for all $x \in E$ with $0 < k < r$ would contradict the root bound for the polynomial $t^{p^k} - t$. Nondegeneracy of the trace matrix supplies its inverse and hence the dual basis. Then $\mathrm{Tr}_{E/\mathbb{F}_p}(b(x + a)) = \sum_j b_j(x_j + a_j)$, which factors the reference Weyl operator into the displayed tensor. ☐

21.2 Subfield support and Fourier-dual transfers

Definition 21.6 (Inclusion and normalized trace-fibre transfers)**.** For a named embedding $i : K \to E$ of trace-framed registers, define linear maps $J_i, V_i : \mathcal{H}_K \to \mathcal{H}_E$ by

$$J_i|a\rangle = |i(a)\rangle, \qquad V_i|a\rangle = \kappa_i^{-1/2} \sum_{T_i(x)=a} |x\rangle,$$

using the positive real square root. Their reverse arrows are their Hilbert adjoints J_i^*, V_i^*.

Scope. The source, target and positive normalization are fixed data; independent phases on the transfers are not introduced.

Provenance. ids: D1304; proved in: theory/sidequests/frobenius-hierarchy/transfers-example.md; tested by: theory/checks/frobenius_hierarchy_check.py (A3); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Definition 21.7 (Subfield support code and logical phase quotient)**.** For a named embedding $i : K \to E$ of trace-framed registers, put

$$\mathcal{C}_i = \operatorname{span}_{\mathbb{C}}\{|i(a)\rangle : a \in K\}, \qquad P_i = J_i J_i^*, \qquad N_i = \{0\} \oplus \ker T_i, \qquad G_i = \{Z_E(b) : b \in \ker T_i\}.$$

The proposed logical phase space is $N_i^{\perp}/N_i$, where perpendicularity uses Ω_E. The logical coordinate map is

$$[(i(a), b)] \longmapsto (a, T_i(b)) \quad \text{whenever } N_i^{\perp} = i(K) \oplus E.$$

Its well-definedness and symplectic property are claims, not stipulations.

Scope. The code is defined by computational-basis support in the embedded field. It is not defined as the invariant-vector space of Frobenius.

Provenance. ids: D1305; proved in: theory/sidequests/frobenius-hierarchy/foundation.md; tested by: theory/checks/frobenius_hierarchy_check.py (A2); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Theorem 21.8 (Subfield stabilizers and logical Weyl system)**.** PROVED *For every named embedding $i : K \to E$ of trace-framed finite fields, $\mathcal{C}_i$ is exactly the common $+1$ eigenspace of G_i, and*

$$\dim \mathcal{C}_i = |K|, \qquad P_i = \frac{1}{\kappa_i} \sum_{b \in \ker T_i} Z_E(b), \qquad N_i^{\perp} = i(K) \oplus E.$$

The logical coordinate map is a symplectic isomorphism $N_i^{\perp}/N_i \cong S_K$. Exactly the Weyl operators whose position labels belong to $i(K)$ preserve the code, and

$$J_i^* W_E(i(a), b) J_i = W_K(a, T_i(b)).$$

Scope. Logical momentum is a trace quotient. The statement does not identify it with inclusion of momentum labels from K.

Provenance. ids: FRB-CODE; proved in: theory/sidequests/frobenius-hierarchy/foundation.md, section 3; tested by: theory/checks/frobenius_hierarchy_check.py (A2); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. The trace-pairing identity and nondegeneracy over K show that the annihilator of $i(K)$ is $\ker T_i$. Taking annihilators again gives $(\ker T_i)^{\perp} = i(K)$. A basis vector is fixed by every phase stabilizer precisely when its label belongs to this latter annihilator. Character summation gives the displayed projector. The same annihilator calculation yields $N_i^{\perp} = i(K) \oplus E$; the displayed coordinate map is surjective with kernel N_i and preserves the form by trace transitivity. Finally, the Weyl operator sends a support coset to its translate, and its phase on $|i(x)\rangle$ is $\psi_K(-T_i(b)(x + a))$. This proves the restriction formula. □

For an extension of degree d, relative trace gives $T_i(i(a)) = da$. Consequently $\psi_E(i(a)) = \psi_K(a)^d$. In particular, the included momentum labels can all act trivially on the code when $p \mid d$. The trace quotient remains valid because the full relative trace is surjective in this case as well.

Definition 21.9 (Negative-kernel arithmetic Fourier transform)**.** For a trace-framed register define

$$F_E|x\rangle = |E|^{-1/2} \sum_{y \in E} \psi_E(-xy)|y\rangle,$$

with positive real square root. A subscript on a tensor factor means that this operator acts there and the identity acts on every other factor.

Scope. The negative sign and positive normalization fix the Fourier convention. No restriction beyond the finite-field hypotheses is imposed.

Provenance. ids: D1306; proved in: theory/sidequests/frobenius-hierarchy/transfers-example.md; tested by: theory/checks/frobenius_hierarchy_check.py (A3); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Theorem 21.10 (Coherent inclusion and Fourier-dual trace transfer)**.** PROVED *For every named embedding $i : K \to E$ of trace-framed finite fields, J_i, V_i are isometries, F_E is unitary, and*

$$F_E J_i = V_i F_K, \qquad U_E J_i = J_i U_K, \qquad U_E V_i = V_i U_K.$$

For every tower $L \xrightarrow{j} K \xrightarrow{i} E$,

$$J_i J_j = J_{ij}, \qquad V_i V_j = V_{ij}.$$

Identity embeddings give identity transfers. Moreover,

$$F_E X_E(a) F_E^* = Z_E(-a), \qquad F_E Z_E(b) F_E^* = X_E(b).$$

Scope. These are exact identities of maps with the displayed domains and positive fibre normalizations. Their adjoints reverse the domains.

Provenance. ids: FRB-TRANSFER; proved in: theory/sidequests/frobenius-hierarchy/transfers-example.md, section 1; tested by: theory/checks/frobenius_hierarchy_check.py (A3); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. Different trace fibres are disjoint and each has κ_i elements, which proves the isometry assertion. Character orthogonality proves Fourier unitarity. The coefficient of $|x\rangle$ on the two sides of $F_E J_i = V_i F_K$ is, respectively,

$$|E|^{-1/2} \psi_E(-i(a)x), \qquad (|K|\kappa_i)^{-1/2} \psi_K(-aT_i(x)).$$

They agree by the trace-pairing identity and fibre cardinality. In a tower, each output point occurs once, through its unique intermediate trace label, and fibre cardinalities multiply. Finally, $T_i(x^p) = T_i(x)^p$, so Frobenius permutes corresponding fibres with their normalization unchanged. The two Fourier covariance formulas follow by translating the kernel summation index. $\square$

21.3 Multiplication at exact higher Clifford levels

Definition 21.11 (Prime-field Pauli group and Clifford hierarchy)**.** For an ordered list $A = (E_1, \dots, E_n)$ of trace-framed arithmetic registers, let

$$\mathcal{P}_A = \left\{ \lambda \bigotimes_j W_{E_j}(a_j, b_j) : \lambda \in U(1) \right\}.$$

Set $\mathcal{C}_1(A) = \mathcal{P}_A$ and, recursively for $k \geq 1$,

$$\mathcal{C}_{k+1}(A) = \{U \text{ unitary} : UPU^* \in \mathcal{C}_k(A) \text{ for every } P \in \mathcal{P}_A\}.$$

The exact level of U is the least positive k with $U \in \mathcal{C}_k(A)$, if such a k exists. All scalar phases are included, in particular at $p = 2$. These sets are not stipulated to be groups beyond the second level.

Scope. The Pauli frame is the prime-field frame on all constituent atomic factors. The finite group with phases restricted to μ_p is a different datum.

Provenance. ids: D1307; proved in: theory/sidequests/frobenius-hierarchy/hierarchy.md; tested by: theory/checks/frobenius_hierarchy_check.py (A5); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Definition 21.12 (Reversible multiplication gate)**.** For a trace-framed register and integer $d \geq 1$, define the linear operator on $\mathcal{H}_E^{\otimes(d+1)}$ by

$$M_E^{(d)} |x_1, \dots, x_d, z\rangle = \left| x_1, \dots, x_d, z + \prod_{j=1}^{d} x_j \right\rangle.$$

Each argument occupies a distinct register.

Scope. Repeated powers of a variable within a single register are outside this definition; the number of separate control registers is d.

Provenance. ids: D1308; proved in: theory/sidequests/frobenius-hierarchy/hierarchy.md; tested by: theory/checks/frobenius_hierarchy_check.py (A4–A5); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Definition 21.13 (Phase polynomials and additive differences)**.** For an ordered list of trace-framed registers with configuration space $A = \prod_j E_j$ and a function $f : A \to \mathbb{F}_p$, put

$$D_f |t\rangle = \zeta_p^{f(t)} |t\rangle, \qquad \delta_h f(t) = f(t) - f(t-h) \quad (h \in A).$$

A polynomial of total degree at most m means a polynomial expression in the coordinates of named $\mathbb{F}_p$-bases, representing f, with that degree bound. The condition concerns existence of such an expression; no unique polynomial representative is stipulated.

Scope. The differences are operations on functions. A polynomial expression can have a larger degree than another expression defining the same function.

Provenance. ids: D1309; proved in: theory/sidequests/frobenius-hierarchy/hierarchy.md; tested by: theory/checks/frobenius_hierarchy_check.py (A5); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Theorem 21.14 (Exact hierarchy level of reversible multiplication). **PROVED** *For every finite field $E/\mathbb{F}_p$ with named primitive phase and every integer $d \geq 1$, the reversible multiplication gate is unitary and*

$$M_E^{(d)} \in \mathcal{C}_{d+1}(E, \ldots, E) \setminus \mathcal{C}_d(E, \ldots, E).$$

Thus its exact prime-field Clifford level is $d+1$.

Scope. Every characteristic and every positive d is included. The theorem uses distinct registers and does not infer exact level from ordinary polynomial degree alone.

Provenance. ids: FRB-HIERARCHY; proved in: theory/sidequests/frobenius-hierarchy/hierarchy.md, sections 1–3; tested by: theory/checks/frobenius_hierarchy_check.py (A5); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. Subtracting the product gives the inverse permutation. Fourier transform on the target gives

$$F_{\text{target}} M_E^{(d)} F_{\text{target}}^* = D_f, \qquad f(x_1, \ldots, x_d, y) = -\operatorname{Tr}_{E/\mathbb{F}_p}\left(y \prod_j x_j\right).$$

Fourier is Clifford, and Clifford conjugation preserves each hierarchy level. Multiplication on either side by a Pauli also preserves each level; both facts follow inductively from the defining recursion. The exact commutator identity is

$$D_f X(h) D_f^* X(h)^* = D_{\delta_h f}.$$

A difference lowers the degree of a coordinate polynomial by at least one. Affine phases are Paulis by trace duality, so induction gives $D_f \in \mathcal{C}_{d+1}$, since f is multilinear in $d+1$ separate arguments. Conversely, a diagonal phase in $\mathcal{C}_k$ has every k-fold additive difference constant: absorb the Pauli in the displayed identity and induct down to a diagonal Pauli.

Now translate each of the d distinct control registers by 1_E. The resulting d-fold difference is $-\operatorname{Tr}_{E/\mathbb{F}_p}(y)$, which is nonconstant because absolute trace is surjective. Thus $D_f \notin \mathcal{C}_d$. No factorial occurs in these separate-register differences, including when $p \leq d$. Fourier conjugation transfers both conclusions to $M_E^{(d)}$. □

Theorem 21.15 (Arithmetic covariance and encoded multiplication). **PROVED** *For every named embedding $i : K \to E$ of trace-framed finite fields and every $d \geq 1$, multiplication commutes with simultaneous Frobenius and satisfies*

$$U_E^{\otimes(d+1)} M_E^{(d)} = M_E^{(d)} U_E^{\otimes(d+1)}, \qquad M_E^{(d)} J_i^{\otimes(d+1)} = J_i^{\otimes(d+1)} M_K^{(d)}.$$

The latter identity also holds for inverse gates. Multiplication preserves $\mathcal{C}_i^{\otimes(d+1)}$; its encoded logical action is $M_K^{(d)}$ and has exact level $d+1$ for the logical prime-field Pauli frame supplied by the trace quotient.

Scope. The intertwining uses the support encoding J_i. No corresponding multiplicative intertwining is asserted for trace-fibre encoding V_i.

Provenance. ids: FRB-NATURAL; proved in: theory/sidequests/frobenius-hierarchy/hierarchy.md, section 4; tested by: theory/checks/frobenius_hierarchy_check.py (A4); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. Frobenius and field embeddings preserve sums and products, so the two orders of each composition agree on every computational basis point. Compressing by the tensor encoding's adjoint gives the logical gate. The trace quotient identifies its Pauli frame with that over K, to which the exact-level theorem applies. □

Proposition 21.16 (Coherent multiplication return probability). **PROVED** *For a trace-framed finite field E, put $Q = |E|$ and*

$$|+_E\rangle = Q^{-1/2}\sum_x |x\rangle, \qquad D = F_{\text{target}} M_E^{(2)} F_{\text{target}}^*.$$

The return amplitude of D on $|+_E\rangle^{\otimes 3}$ is $(2Q-1)/Q^2$, and its return probability is $(2Q-1)^2/Q^4$.

Scope. The input, unitary and return effect are fixed as stated. The rational formula alone does not supply a specialization functor or reference trace.

Provenance. ids: FRB-NATURAL; proved in: theory/sidequests/frobenius-hierarchy/hierarchy.md, section 5; tested by: theory/checks/frobenius_hierarchy_check.py (A5-interference); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. The amplitude is

$$\frac{1}{Q^3}\sum_{x,y,t\in E}\psi_E(-txy) = \frac{2Q-1}{Q^2},$$

because summing over t gives Q exactly when $xy = 0$ and zero otherwise. There are $2Q-1$ pairs with $xy = 0$. The return probability is $(2Q-1)^2/Q^4$, giving $9/16$ at $Q = 2$, whereas identity evolution gives one. The rational function takes the value one at $Q = 1$; that calculation does not supply a specialization functor or its reference trace. □

21.4 The binary degree-four tower

Definition 21.17 (Binary degree-four field tower). Put

$$K = \mathbb{F}_2[a]/(a^2+a+1), \qquad E = K[b]/(b^2+b+a).$$

The named embeddings are the constant embeddings $\mathbb{F}_2 \to K \to E$. Write an element of E as $(u,v) = u + vb$, with $u, v \in K$, and use $\zeta_2 = -1$. Irreducibility and the resulting field orders are claims.

Scope. The polynomial models and embeddings fix concrete coordinates for the example. No basis-independent coordinate decomposition is asserted.

Provenance. ids: D1310; proved in: theory/sidequests/frobenius-hierarchy/transfers-example.md; tested by: theory/checks/frobenius_hierarchy_check.py (A6); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Theorem 21.18 (Frobenius, code and transfer in the binary quartic field). **PROVED** *For the displayed binary tower, K and E are fields of cardinalities four and sixteen. For $u, v \in K$,*

$$\sigma_E(u,v) = (u^2+av^2, v^2), \qquad \sigma_E^2(u,v) = (u+v, v), \qquad \sigma_E^4 = \mathrm{id}, \qquad \mathrm{ord}(\sigma_E) = 4.$$

For $i : K \to E$, the relative and absolute traces are

$$T_i(u,v) = v, \qquad \mathrm{Tr}_{E/\mathbb{F}_2}(u,v) = v + v^2.$$

The subfield code has basis $|u,0\rangle$, dimension four, and transfers

$$J_i|u\rangle = |u,0\rangle, \qquad V_i|v\rangle = \frac{1}{2}\sum_{u\in K}|u,v\rangle.$$

The full trace transfer from $\mathbb{F}_2$ has coefficient $1/\sqrt{8}$ on each absolute-trace fibre. The fixed-vector space of U_E^2 has dimension ten, whereas the support code has dimension four.

Scope. The two different invariant spaces are computed in these named coordinates; this does not identify Hilbert-space fixed vectors with arithmetic fixed field elements in general.

Provenance. ids: FRB-EXAMPLE; proved in: theory/sidequests/frobenius-hierarchy/transfers-example.md, section 3; tested by: theory/checks/frobenius_hierarchy_check.py (A6); reviewed in: theory/verdicts/frobenius-hierarchy-algebra-r1.md.

Proof. The first quadratic has no root in $\mathbb{F}_2$. In K, the values of x^2+x are $0,0,1,1$ at $0,1,a,a+1$, so the second quadratic has no root either. Using $b^2 = b + a$ gives the first Frobenius formula. Squaring again and using $a^2 + a = 1$ gives the second, which is nonidentity and squares to the identity. The relative trace $x + x^4$ is therefore $(v, 0)$, giving the trace and fibre formulas. Logical position k may be represented by $(k, 0)$ and logical momentum m by $(0, m)$; the momentums $(m, 0)$ are phase stabilizers. Finally U_E^2 has four singleton basis orbits and six orbits of length two. A fixed vector has constant coefficients on each orbit, so its fixed space has dimension ten. □

The tower also supports the two-control multiplication gate over K inside that over E: both physical and logical gates have exact level three, and the displayed Frobenius commutes with them on their respective registers. The field multiplication, Frobenius permutation, support code and trace fibres remain explicitly available to the process construction.

22 Arithmetic circuits with retained quantum reductions

A field extension supplies two distinct quantum embeddings: inclusion of its basis labels, and a normalized sum over each relative-trace fibre. Their adjoints are successful reductions to the smaller field. To compose these reductions as physical operations, we retain their failure outcomes and the quantum system on which each outcome lives. Independent tensor composition then preserves partial success on either system.

Definition 22.1 (Named finite arithmetic registers)**.** Fix a prime p and a named primitive root $\zeta_p \in \mathbb{C}$. For a finite field $E/\mathbb{F}_p$, put

$$H_E = \ell^2(E), \qquad \psi_E(x) = \zeta_p^{\mathrm{Tr}_{E/\mathbb{F}_p}(x)}, \qquad W_E(a,b) = Z_E(-b)X_E(a),$$

where $X_E(a)|x\rangle = |x+a\rangle$ and $Z_E(b)|x\rangle = \psi_E(bx)|x\rangle$. The attached prime-field symplectic space is $E \oplus E$ with form

$$\Omega_E((a,b),(a',b')) = \mathrm{Tr}_{E/\mathbb{F}_p}(ab' - a'b).$$

Choose a finite nonempty collection $\mathcal{F}$ of explicit finite fields and a finite category $\mathcal{I}$ of named injective field maps between them, containing identities and composites. Equality of field arrows means equality of their functions on the named finite sets. The polynomial presentations, element names and embeddings are retained choices. Use the coefficient field $R_p = \mathbb{Q}(\zeta_p, \sqrt{p})$, with its given complex embedding, positive square root and complex conjugation.

For $i : K \hookrightarrow E$, let $T_i : E \to K$ be relative trace followed by i^{-1}, and set

$$J_i|a\rangle = |i(a)\rangle, \qquad V_i|a\rangle = \frac{1}{\sqrt{|\ker T_i|}} \sum_{T_i(x)=a} |x\rangle.$$

The support code is $C_i = \operatorname{ran} J_i$, its projector is $P_i = J_i J_i^*$, and its phase stabilizer labels form $N_i = \{0\} \oplus \ker T_i$. Its logical symplectic space is $N_i^\perp/N_i$, identified with $K \oplus K$ by $[(i(a), b)] \mapsto (a, T_i(b))$.

The quantum atoms are Q_E and the ambient-labeled code atoms C_i. A quantum word is a finite ordered sequence of atoms, with the empty word $\mathbf{1}$ allowed; tensor is concatenation. Attach the symplectic space $E \oplus E$ to Q_E, the quotient $N_i^\perp/N_i$ to C_i, and orthogonal direct sums to words. An isotropic context is a flag of prime-field isotropic subspaces in the attached space. Contexts remain geometric data, without quotienting by a symmetry group; no generator for every context is stipulated. A code atom retains its ambient field, embedding and isotropic reduction label. No identification of a tensor of field atoms with one extension-field atom is imposed.

Scope. Characteristic two is included. There R_p is real, so this is the stated arithmetic gate fragment rather than the full Clifford category. All field models and embeddings are named choices; they are not reconstructed from an unmarked matrix algebra. Contexts are not the sole objects.

Provenance. ids: D1301,D1303–D1305,D1321; proved in: definitions.md; theory/sidequests/frobenius-hierarchy/foundation.md; tested by: theory/checks/frobenius_hierarchy_check.py (A1–A3); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Definition 22.2 (Presented arithmetic amplitudes)**.** On these quantum words, take finite R_p-linear combinations of well-typed circuits generated by identities, swaps, composition, tensor, dagger, and the arrows

arrow	source and target
$w_E(a,b), f_E, u_E$	$Q_E \longrightarrow Q_E$
$m_{E,d}$	$Q_E^{\otimes(d+1)} \longrightarrow Q_E^{\otimes(d+1)}, \quad d \geq 1$
j_i, v_i	$Q_K \longrightarrow Q_E$
e_E	$\mathbf{1} \longrightarrow Q_E$
t_i	$Q_K \longrightarrow C_i$
c_i	$C_i \longrightarrow Q_E.$

Dagger reverses types and conjugates coefficients. Coefficient scalars in R_p act centrally; general endomorphisms of the empty word can also contain closed circuit diagrams and are not asserted to be coefficient scalars. Equality is the least typed congruence generated by the linear dagger strict symmetric monoidal axioms and the following equations and their daggers. Write $r = [E : \mathbb{F}_p]$, and use $i : K \to L$, $j : L \to E$ for towers.

1. $w_E(0,0) = 1$, each $w_E(a,b)$ is unitary, and
$$w_E(a,b)w_E(a',b') = \psi_E(ab')w_E(a+a',b+b').$$
2. $f_E, u_E, m_{E,d}$ are unitary, $f_E^4 = 1$, $u_E^r = 1$, $m_{E,d}^p = 1$, and $f_E u_E = u_E f_E$.
3. Frobenius satisfies
$$u_E w_E(a,b) = w_E(a^p, b^p)u_E, \qquad u_E^{\otimes(d+1)} m_{E,d} = m_{E,d} u_E^{\otimes(d+1)}.$$
4. $j_i^\dagger j_i = v_i^\dagger v_i = 1$, $j_{\mathrm{id}} = v_{\mathrm{id}} = 1$, $j_j j_i = j_{ji}$ and $v_j v_i = v_{ji}$.
5. For $i : K \to E$,
$$u_E j_i = j_i u_K, \qquad u_E v_i = v_i u_K, \qquad f_E j_i = v_i f_K.$$
6. For $a \in K$ and $b \in E$,
$$w_E(i(a),b)j_i = j_i w_K(a, T_i(b)), \qquad m_{E,d} j_i^{\otimes(d+1)} = j_i^{\otimes(d+1)} m_{K,d}.$$
7. $e_E^\dagger e_E = 1$, $t_i^\dagger t_i = t_i t_i^\dagger = 1$, and $c_i t_i = j_i$.

8. With $b_{E,a} = w_E(a,0)e_E$,

$$b^\dagger_{E,a} b_{E,a'} = \delta_{a,a'}, \qquad \sum_{a\in E} b_{E,a} b^\dagger_{E,a} = 1.$$

The trace for the composite embedding is the transported composite trace. There are no further arithmetic equations by default. Scalars are retained; there are no infinite sums, projective identifications, matrix-equality quotients or completely-positive-map equality quotients.

Scope. The presentation specifies marked arithmetic circuits. It asserts neither completeness of its equations nor faithfulness of a realization. Higher Clifford levels label multiplication generators; a fixed higher level is not declared to be closed under composition.

Provenance. ids: D1322,D1323; proved in: theory/sidequests/frobenius-hierarchy/amplitude-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A1–A4,A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Definition 22.3 (Hilbert interpretation of the arithmetic presentation)**.** Interpret Q_E as H_E, C_i as its actual support-code subspace with orthonormal basis $|i(a)\rangle$, and words as ordered Hilbert tensors, with $H(\mathbf{1}) = \mathbb{C}$. Interpret w, j, v as W, J, V above, and use

$$F_E|x\rangle = |E|^{-1/2} \sum_y \psi_E(-xy)|y\rangle, \quad U_E|x\rangle = |x^p\rangle,$$

$$M_E^{(d)}|x_1, \ldots, x_d, z\rangle = |x_1, \ldots, x_d, z + x_1 \cdots x_d\rangle$$

for f, u, m. Interpret e_E as $|0\rangle$, t_i as the unitary $|a\rangle \mapsto |i(a)\rangle$ with codomain the code, and c_i as its inclusion into H_E. Swaps are ordinary tensor swaps, dagger is adjoint, and coefficients use the named complex embedding. Denote this candidate interpretation by H.

Scope. The negative Fourier kernel and fibre normalization are part of the data. No basis or character compatibility beyond the named trace construction is silently imposed.

Provenance. ids: D1302,D1304,D1306,D1308,D1324; proved in: theory/sidequests/frobenius-hierarchy/amplitude-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A1–A4); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Theorem 22.4 (A small arithmetic amplitude category)**.** PROVED *For every finite named field diagram and coefficient datum above, the presentation is a small linear dagger symmetric monoidal category, and H is a strong monoidal dagger functor. If the prime field belongs to the diagram and $B = (\beta_1, \ldots, \beta_r)$ is a named prime-field basis of E, the coordinate circuit*

$$a_B = \sum_{x=\sum_l x_l\beta_l \in E} \left(\bigotimes_l b_{\mathbb{F}_p, x_l} \right) b^\dagger_{E,x} : Q_E \longrightarrow Q_{\mathbb{F}_p}^{\otimes r}$$

is a source unitary. Its interpretation factors the field Weyl operators into prime-field Weyls when positions use B and momenta its trace-dual basis.

Scope. The coordinate identification depends on its named basis. The theorem asserts no reconstruction of the field from the unmarked linear envelope, no faithfulness, and no full Clifford or all-process completeness.

Provenance. ids: FRP-CAT,D1327; proved in: theory/sidequests/frobenius-hierarchy/amplitude-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A1–A4,A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Every relation is checked against the independent formulas. For example, trace adjunction gives $F_E J_i = V_i F_K$ and the logical Weyl relation with $T_i(b)$ as momentum. The basis-ket equations prove the coordinate circuit unitary by cancelling off-diagonal terms in $a_B^\dagger a_B$ and $a_B a_B^\dagger$. These identities supply arithmetic content to a source whose equality was fixed before interpretation.

Definition 22.5 (Source-certified processes and their equality)**.** Fix a countable naming universe containing the field-element labels and closed under finite ordered tuples. All external tag sets and hidden index sets are finite subsets of this universe, with products formed by ordered tuples in that universe. This naming convention is presentation data. A process object is a finite nonempty tagged family $X = (X_a)_{a \in A}$ of quantum words. An arrow $k : X \to Y$ consists of finite hidden index sets $J_{b,a}$, possibly empty, and source amplitudes $k_{b,a,j} : X_a \to Y_b$. For each a there must be finitely many source arrows $h_{a,l} : X_a \to Z_{a,l}$, to arbitrary quantum words, and a finite source derivation of

$$1_{X_a} - \sum_{b,j} k_{b,a,j}^\dagger k_{b,a,j} = \sum_l h_{a,l}^\dagger h_{a,l}.$$

The certificate is evidence, not additional arrow data. An arrow is normalized when its displayed squared-amplitude sum equals 1_{X_a}. Equality is a bijection of each hidden index set carrying amplitudes to equal source arrows. It does not allow general Kraus mixing, zero deletion, external-tag relabeling or equality merely of realized CP maps. Distinct external tags remain distinct even when their quantum words agree.

The identity has amplitude 1 on each diagonal block and empty lists elsewhere. For $k : X \to Y$ and $l : Y \to Z$, composition has hidden triples (b, j, t) on block (c, a) and amplitude $l_{c,b,t} k_{b,a,j}$. Tensor uses Cartesian products of external and hidden index sets and amplitudes $k \otimes l$, in the stated factor order. The tensor unit is the singleton empty word. Structural maps use canonical tag bijections and word swaps. A deterministic tag map $g : A \to B$ has identity amplitude in block $(g(a), a)$ and requires the same quantum word on each routed block; this includes forgetting a classical tag, but cannot change its quantum type. A coefficient-scalar process is a singleton coefficient amplitude on the empty word with the same certificate; general endomorphisms of that object may have closed-diagram amplitudes or several hidden indices.

Scope. Composition sums over an intermediate classical tag. A history to be retained must be part of the final external tag. No arbitrary complex-state or arbitrary-CP-map generators are stipulated.

Provenance. ids: D1325; proved in: theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Definition 22.6 (Ordinary-trace process interpretation)**.** Assign the block algebra

$$B_X = \bigoplus_a \mathrm{End}_{\mathbb{C}} H(X_a), \qquad \mathrm{Tr}_X(\rho) = \sum_a \mathrm{Tr}(\rho_a).$$

A density has $\mathrm{Tr}_X(\rho) = 1$. The target arrows are CP maps that do not increase this trace. Tensor is ordinary finite-dimensional tensor, distributed over tag blocks. The candidate process functor is

$$R(k)(\rho)_b = \sum_{a,j} H(k_{b,a,j}) \rho_a H(k_{b,a,j})^*.$$

The Born weight of tag b is the ordinary trace of its block. Conditioning divides by this weight only when positive and is not a linear process arrow. Retaining a history means placing it in an external tag; ordinary composition sums over its intermediate tag, unless that value was also routed to output.

Scope. These are ordinary trace-one Hilbert densities, distinct from uniform classical-trace and Hecke coefficient-trace densities. The density convention is not changed by using an arithmetic source.

Provenance. ids: D1326; proved in: theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Theorem 22.7 (Positive independent composition)**.** PROVED *The source-certified processes form a small symmetric monoidal category. Its normalized arrows form a symmetric monoidal subcategory. The map R is a strong monoidal functor to trace-nonincreasing CP maps and takes normalized arrows to trace-preserving maps. It is not faithful.*

Scope. No comparison with a collective Hecke assembly map or a changing-parameter endpoint is asserted. External outcomes and source amplitudes remain more detailed than their CP interpretation.

Provenance. ids: FRP-CP; proved in: theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

The source certificate proves positivity without deciding equality by realization. If $S = \sum k^\dagger k$ and $T = \sum l^\dagger l$, the tensor deficit splits as

$$1 \otimes 1 - S \otimes T = (1 - S) \otimes 1 + S \otimes (1 - T),$$

a sum of squares built from the two certificates. Sequential composition has the corresponding deficit $1 - S + \sum k^\dagger(1 - T)k$. After interpretation these sums give nonnegative trace losses at every matrix amplification. The singleton amplitude processes $\{1\}$ and $\{-1\}$ are distinct at source but induce the same scalar CP map, proving the stated failure of faithfulness directly.

Definition 22.8 (Retained decoding, preparation and register circuits)**.** For a source isometry $s : X \to Y$, put $p_s = ss^\dagger$ and $q_s = 1 - p_s$. The retained decoder has source (Y) and target $(\mathrm{success} : X, \mathrm{failure} : Y)$, with amplitudes $s^\dagger$ and q_s. Its success-only arrow $d_s : (Y) \to (X)$ has amplitude $s^\dagger$ and residual certificate q_s; encoding has amplitude s. Apply this to j_i, v_i, c_i, their composites and independent tensors, without stipulating a reset completion. For $a : Q_K^{\otimes n} \to Q_K^{\otimes n}$, its code lift is $a^{[i]} = t_i^{\otimes n} a(t_i^\dagger)^{\otimes n}$, in particular for Frobenius, Weyl and multiplication gates. The Fourier comparison after decoding applies f_K on success and f_E on failure.

Zero and basis preparations have the single amplitudes e_E and $b_{E,a}$. Quantum discard has one external output and hidden amplitudes $b_{E,a}^\dagger$, indexed by $a \in E$. Code discard first applies $t_i^\dagger$; word discard tensors the atom discards; tagged-family discard sums all input blocks into the singleton empty-word output. A normalized source-generated state followed by a retained decoder is a preparation/test protocol. For a named basis B with the prime field in the diagram, the coordinate circuit is the displayed a_B above, retaining B as its label. No basis-independent identification is imposed. The realized span of basis matrix units contains all R_p-valued matrices; the field structure is in the marked source data and relations, not reconstructed from that span.

Scope. Preparations, effects and discards are those supplied by these source circuits and certificates. A retained failure branch keeps its stated quantum output type. Conditioning is used only at positive success weight.

Provenance. ids: D1327; proved in: theory/sidequests/frobenius-hierarchy/amplitude-category.md; theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Theorem 22.9 (Coherent successful descent with complete outcome records). PROVED *For the inclusion, trace-fibre and code isometries, retained decoding is normalized, and successful decoding is trace-nonincreasing with probability* $\mathrm{Tr}(H(p_s)\rho)$*. Successful decodes compose along field towers and tensor independently. Full decoders keep their specified histories. Fourier conjugation intertwines both success and failure branches, and code-lifted Frobenius and multiplication intertwine their physical gates through the encoding.*

Scope. The complete two-stage tower instrument is not identified with the two-outcome decoder of the composite. The tensor of complete instruments has four outcomes, including partial successes. No fixed higher hierarchy level is declared a category.

Provenance. ids: FRP-DESCENT; proved in: theory/sidequests/frobenius-hierarchy/amplitude-category.md; theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A3,A4,A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

For isometries $s : X \to Y$ and $t : Y \to Z$, the three retained tower histories have amplitudes q_t, $q_s t^\dagger$, $s^\dagger t^\dagger$, with output words Z, Y, X. Their effects sum to $q_t + tq_s t^\dagger + tp_s t^\dagger = 1$. The all-success amplitude is $(ts)^\dagger$. For independent isometries, the four effects are $p_s \otimes p_t$, $p_s \otimes q_t$, $q_s \otimes p_t$, $q_s \otimes q_t$. These equations explain why no reset convention is needed for composition.

A support code and its Fourier-dual trace code

Take $\mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$. The inclusion code is spanned by $|0\rangle, |1\rangle$, while the relative trace fibres are $\{0, 1\}$ and $\{\alpha, \alpha + 1\}$. Thus

$$J|0\rangle = |0\rangle, \quad J|1\rangle = |1\rangle, \qquad V|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad V|1\rangle = \frac{|\alpha\rangle + |\alpha + 1\rangle}{\sqrt{2}}.$$

On $(|0\rangle + |\alpha\rangle)/\sqrt{2}$, inclusion decoding succeeds with probability $1/2$, returning the conditional state $|0\rangle$. The failure also has probability $1/2$, retaining the state $|\alpha\rangle$ in the larger register. The maximally mixed input $I_4/4$ likewise succeeds with probability $1/2$. The Fourier relation identifies the whole instrument:

$$V^* F_4 = F_2 J^*, \qquad (1 - VV^*)F_4 = F_4(1 - JJ^*).$$

Provenance. ids: FRB-TRANSFER,FRP-DESCENT; proved in: theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A3,A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

Two independent inclusion decoders have output dimensions $4, 8, 8, 16$ in the order success–success, success–failure, failure–success, failure–failure. On the entangled state

$$(|0, 0\rangle + |\alpha, \alpha\rangle)/\sqrt{2},$$

the first and last outcomes each have probability $1/2$, and both partial successes have probability zero. Joint success is computed by $P_J \otimes P_J$; multiplying the marginal success probabilities would incorrectly give $1/4$. In the tower $\mathbb{F}_2 \subset \mathbb{F}_4 \subset \mathbb{F}_{16}$, the three history outputs have dimensions $16, 4, 2$. Only the all-success branch is the direct composite reduction to the prime field.

Provenance. ids: FRP-DESCENT; proved in: theory/sidequests/frobenius-hierarchy/process-category.md; tested by: theory/checks/frobenius_hierarchy_check.py (A7); reviewed in: theory/verdicts/frobenius-hierarchy-category-r1.md.

The next comparison at an incidence endpoint

The proved arithmetic formulas do not yet define a specialization at one. A concrete next proposal separates an incidence parameter q from a fixed cyclotomic phase level N. For extension-field isotropic contexts the arithmetic point is $q = p^r, N = p$; prime-field contexts after restriction of scalars have different rank and incidence parameter p. The proposed comparison must carry the actual multiplication tables, inclusion and trace matrices, code constraints and Frobenius processes to a coupled positive context algebra. The type-C decomposable corner and shuffle correspondence provide a composition test; their nonparabolic block does not supply a generic Hecke tensor inclusion.

The reference trace matters. With phase level fixed, a normal-basis cyclic Frobenius has phase trace N^{1-r}. Substituting $p = 1$ into the different physical arithmetic expression p^{1-r} would force trace one, and a unitary of trace one for a faithful positive normalized trace τ, with $\tau(1) = 1$, equals the identity. A viable next construction must name its positive trace and comparison functor, preserve an observed arithmetic process, and lift the relation diagrams as well as their individual gates. The specific next candidate and six witness tests are recorded in the accompanying endpoint proposal; none is an admitted deformation theorem.

Provenance. ids: D1250–D1255,D1001–D1007; proved in: docs/research-plans/frobenius-coupled-limit.md; theory/sidequests/f1-limit/type-c.md; tested by: next-stage witness specification; reviewed in: proposal only.

23 A conditional Frobenius boundary with quantum composition

There is a concrete way for arithmetic Frobenius to survive at one: retain the quantum systems carried by its moving orbits, together with their conditional reference states. The smallest example is a qubit on which Frobenius exchanges two distinguishable states. Its weight in the original register vanishes, but its normalized internal physics does not. The construction below includes independent composition and standard subfield inclusion. It is a selected observable family; extending it to the full arithmetic multiplication and Fourier processes remains open. The statements below have passed the capped review. The "orbit checker" cited below is `theory/checks/frobenius_boundary_check.py`.

23.1 The arithmetic comparison and its positive continuation

For a prime p and an integer $r \geq 1$, let $E = \mathbb{F}_{p^r}$, $H_E = \ell^2(E)$ with orthonormal basis $|x\rangle$, and $U_E|x\rangle = |x^p\rangle$. The ordinary physical reference trace is $\mathrm{Tr}\,/p^r$. We continue the prime parameter by a real variable $t \geq 1$, with r fixed. This defines an algebra family; it does not posit finite fields of noninteger cardinality.

Provenance. ids: D1301, D1302; proved in: the preceding arithmetic sections; tested by: the arithmetic Frobenius checker; reviewed in: the arithmetic adjudication.

Definition 23.1 (Marked Frobenius orbit algebra)**.** Fix $r \geq 1$ and a real variable $t \geq 1$, distinct from a field characteristic. Let $\mu(n)$ be zero when a prime square divides n and otherwise (-1) to the number of its prime factors. Let $\varphi(d)$ count the integers $1 \leq a \leq d$ coprime to d. Set

$$c_d(t) = \sum_{e|d} \mu(d/e)t^e, \qquad \mathcal{O}_r = \bigoplus_{d|r} M_d(\mathbb{C}), \qquad S_d|j\rangle = |j+1 \bmod d\rangle, \qquad u_r = \bigoplus_{d|r} S_d.$$

At $t = p$, choose $E = \mathbb{F}_{p^r}$ and an origin b_O in each Frobenius orbit O. Label that orbit by $\sigma_E^j(b_O)$, $0 \leq j < |O|$. The representation $R_{p,r}$ acts with the same matrix a_d on every marked orbit of length d. For a tower of standard subfields in one finite field, use the same origin wherever an orbit occurs.

Scope. The comparison includes the origin choices. Compatible origins for all named embeddings or field automorphisms are not stipulated.

Provenance. ids: D1331; proved in: definitions.md; tested by: orbit checker B1; reviewed in: frobenius-boundary-adjudication.md.

Proposition 23.2 (Arithmetic orbit counts and matrix comparison). PROVED *For every prime p, every $r \geq 1$, and each $d \mid r$, exactly $c_d(p)$ field labels have Frobenius period d. The marked representation is faithful, unital and preserves adjoints, and*

$$R_{p,r}(u_r) = U_E, \qquad \frac{\operatorname{Tr}(R_{p,r}(a))}{p^r} = \sum_{d|r} \frac{c_d(p)}{p^r} \operatorname{tr}_d(a_d), \qquad \operatorname{tr}_d = \frac{\operatorname{Tr}}{d}.$$

Changing the origins gives an orbitwise unitary conjugation commuting with U_E.

Scope. Marked rank-one observables can change with the origins. This representation retains common matrices on equal-length orbits, not the full field matrix algebra and not the commutant of Frobenius.

Provenance. ids: FRL-ORBIT; proved in: orbit-boundary.md, Sections 1–2; tested by: orbit checker B1–B2; reviewed in: frobenius-boundary-adjudication.md.

For $d \mid r$, the polynomial $X^{p^d} - X$ divides $X^{p^r} - X$ and has derivative -1, so it has exactly p^d roots in E. Decomposing these fixed labels by their least period and applying divisor inversion gives $c_d(p)$. Each matrix a_d occurs $c_d(p)/d$ times, giving the trace formula. This is the usual necklace polynomial multiplied by d; see Trevor Hyde, *Cyclotomic factors of necklace polynomials*, introduction, p. 1. The matrix and trace continuation is the construction here. It uses actual coordinate tests: summing $|b_O\rangle\langle b_O|$ over length-d orbit origins, then multiplying on the left and right by U_E^i and U_E^{-j}, produces the represented matrix unit $|i\rangle\langle j|_d$. Frobenius and the marked origin tests therefore generate the selected matrix algebra.

Definition 23.3 (Conditional orbit boundary and reference state). For $r \geq 2$ let e_r be identity on the blocks $d > 1$ and zero on the $d = 1$ block, and put

$$\mathcal{O}_r^+ = e_r \mathcal{O}_r e_r, \qquad u_r^+ = e_r u_r e_r, \qquad \theta_{r,t}(a) = \sum_{d|r} \frac{c_d(t)}{t^r} \operatorname{tr}_d(a_d), \qquad w_r(t) = 1 - t^{1-r}.$$

The identity of $\mathcal{O}_r^+$ is e_r. For $t > 1$ define $\omega_{r,t}(a) = \theta_{r,t}(a)/w_r(t)$ on this corner. At one define

$$\omega_{r,1}(a) = \sum_{\substack{d|r \\ d>1}} \frac{\varphi(d)}{r-1} \operatorname{tr}_d(a_d).$$

Write $\lambda_{r,d}(t) = c_d(t)/(t^r - t)$ for $t > 1$ and $\lambda_{r,d}(1) = \varphi(d)/(r-1)$. The ordinary block-trace density is $\sigma_{r,t} = \bigoplus_{d|r,d>1} \lambda_{r,d}(t) I_d/d$. At $t = p$ the physical conditioning projection is $P_r^{\text{mov}} = \sum_{x^p \neq x} |x\rangle\langle x|$.

Scope. This removes fixed basis labels. It does not remove every invariant vector: a moving orbit still contains an invariant uniform superposition.

Provenance. ids: D1332; proved in: definitions.md; tested by: orbit checker B2–B3; reviewed in: frobenius-boundary-adjudication.md.

Proposition 23.4 (A faithful positive conditional endpoint). PROVED *For every $r \geq 2$, $\theta_{r,t}$ is a faithful normalized positive trace for $t > 1$, and $\omega_{r,t}$ extends continuously to the displayed faithful normalized positive trace on $\mathcal{O}_r^+$ at $t = 1$. For every integer k, with $\gcd(r, 0) = r$,*

$$\omega_{r,t}((u_r^+)^k) = \frac{t^{\gcd(r,k)} - t}{t^r - t} \quad (t > 1), \qquad \omega_{r,1}((u_r^+)^k) = \frac{\gcd(r,k) - 1}{r - 1}.$$

For any block $d > 1$, the state and effect $|0\rangle\langle 0|$ have Born probability one before Frobenius and zero afterwards.

Scope. This is a positive family of selected marked orbit observables. It does not specialize all arithmetic amplitudes. The unconditioned trace at one has a different quotient, retaining only the scalar block.

Provenance. ids: FRL-POS; proved in: orbit-boundary.md, Sections 2–4; tested by: orbit checker B2–B3; reviewed in: frobenius-boundary-adjudication.md.

Positivity holds throughout the real interval, not just at arithmetic points. Put $t = e^h$ with $h > 0$. For $d > 1$,

$$c_d(e^h) = \sum_{k \geq 1} \frac{h^k}{k!} d^k \prod_{\substack{\ell \mid d \\ \ell \text{ prime}}} (1 - \ell^{-k}).$$

This convergent series has positive terms. Its first coefficient is $\varphi(d)$, so $c_d'(1) = \varphi(d)$. Since $\sum_{d|r,d>1} c_d(t) = t^r - t$, division and cancellation at one give the stated weights, which are positive and sum to one. A cyclic shift has trace zero unless its power fixes every index, yielding the moment formula. Thus $C([1,T], \mathcal{O}_r^+)$, for any finite $T > 1$, is an explicit continuous section algebra with these reference states.

For $r = 2$, the entire conditional family is $M_2(\mathbb{C})$ with trace tr_2 and Frobenius S_2. At $p = 2$ its physical orbit is (α, α^2) in $\mathbb{F}_4$, where $\alpha^2 = \alpha + 1$. The ambient probability of this qubit sector is $(t-1)/t$; the normalized qubit remains after division by that vanishing weight. For $r = 4$ the conditional algebra is $M_2 \oplus M_4$:

$$(\lambda_{4,2}(t), \lambda_{4,4}(t)) = \left(\frac{1}{t^2 + t + 1}, \frac{t(t+1)}{t^2 + t + 1} \right).$$

Its weights change from $(1/7, 6/7)$ at $t = 2$ to $(1/3, 2/3)$ at one. Frobenius remains measurable on both blocks.

23.2 Composition and retained subfield information

Definition 23.5 (Orbit processes and coherent independent composition)**.** Objects are finite ordered words $R = (r_1, \ldots, r_n)$, $r_j \geq 2$, and finite nonempty tagged families of such words. A word has algebra $\mathcal{O}_R^+ = \bigotimes_j \mathcal{O}_{r_j}^+$, product reference $\omega_{R,t} = \bigotimes_j \omega_{r_j,t}$, and ambient weight $W_R(t) = \prod_j w_{r_j}(t)$. The empty word has algebra $\mathbb{C}$; tagged families have direct sum algebras. Processes are complex-linear CPTP maps in the Schrödinger orientation, with ordinary sum-of-blocks trace. Equality is equality of linear maps; composition is composition and independent tensor pairs tags. Frobenius is conjugation by $\bigotimes_j u_{r_j}^+$.

For $2 \leq r \mid s$, let $e_{r,s}$ be identity on the blocks $d \mid r$, $d > 1$, in $\mathcal{O}_s^+$ and zero elsewhere. The encoding $j_{r,s}$ inserts zero blocks. The retained decoder sends ρ to

$$\left((\rho_d)_{d|r,d>1},\ (1 - e_{r,s})\rho(1 - e_{r,s})\right) \quad \text{in } \mathcal{O}_r^+ \oplus \mathcal{O}_s^+,$$

with success and failure tags, respectively. Denote its reference success probability by $h_{r,s}(t) = (t^r - t)/(t^s - t)$ for $t > 1$, and by $h_{r,s}(1) = (r-1)/(s-1)$ at one.

For $d, e \geq 1$, set $g = \gcd(d, e)$ and $l = \operatorname{lcm}(d, e)$. The reblocking is

$$B_{d,e} : \mathbb{C}^g \otimes \mathbb{C}^l \longrightarrow \mathbb{C}^d \otimes \mathbb{C}^e, \qquad |a, k\rangle \longmapsto |k \bmod d, a + k \bmod e\rangle, \quad 0 \leq a < g, \quad 0 \leq k < l.$$

Its multiplicity factor is quantum. Iterated reblockings are compared through the same ordered Cartesian basis. The word length records order of ambient vanishing and is not a Clifford level.

Scope. The CP envelope is not asserted to be the image of all arithmetic source processes. Arithmetic comparison covers the marked observables and the specified gates and block instruments.

Provenance. ids: D1333; proved in: definitions.md; tested by: orbit checker B4–B5; reviewed in: frobenius-boundary-adjudication.md.

Proposition 23.6 (Positive composition and conditional descent). **PROVED** *The word and tagged algebras above form symmetric monoidal CPTP categories with continuous product reference states for every $t \geq 1$. Continuous CPTP matrix families evaluate functorially at one. Standard subfield inclusions with coherent orbit origins give the stated block encodings and retained decoders, commute with Frobenius, and compose on success along towers. Their reference success probability is $h_{r,s}(t)$. The reblocking is unitary and satisfies*

$$B_{d,e}^*(S_d \otimes S_e)B_{d,e} = I_g \otimes S_l, \qquad M_d \otimes M_e \simeq M_g \otimes M_l.$$

Iterated reblockings have coherent unitary connectors through the common Cartesian basis.

Scope. Independent tensor retains coherence between simultaneous Frobenius orbits. It is not identified with a field compositum or with a classical sum over those orbits. Arbitrary named embeddings require their own orbit-coordinate transport. Full decoder histories keep their tags. The prime-field register has no moving sector and is not a conditional atom.

Provenance. ids: FRL-COMP; proved in: orbit-composition.md, Sections 1–3; tested by: orbit checker B4–B5; reviewed in: frobenius-boundary-adjudication.md.

The cycle-pair formula follows by solving two congruences for k, with a the residue of the difference of the two output labels modulo g. Its set-level counterpart is the gcd/lcm product in the necklace algebra (Yoshida, Section 2.2, p. 127). For two degree-two cycles, the quantum algebra is M_4, and Frobenius becomes $I_2 \otimes S_2$ after reblocking. Replacing it by $M_2 \oplus M_2$ dephases the two orbits $\{00, 11\}$ and $\{01, 10\}$. On the state $(|00\rangle + |01\rangle)/\sqrt{2}$, that dephasing reduces the return probability from one to $1/2$.

Inclusion is equally concrete: a label belongs to $\mathbb{F}_{p^r}$ inside $\mathbb{F}_{p^s}$ exactly when its period divides r. The matching divisor blocks give the decoder. At the conditional boundary, the degree-two inclusion into degree four succeeds on the reference state with probability $1/3$. Two independent decoders retain all four probabilities $1/9, 2/9, 2/9, 4/9$.

23.3 Higher interactions and the remaining mixed construction

Definition 23.7 (Multiplication cuts and Fourier-transported sectors). Let E be a finite field of size Q, and let $d \geq 1$ separate controls and a separate target carry the gate

$$M_E^{(d)}|x_1, \ldots, x_d, z\rangle = |x_1, \ldots, x_d, z + \textstyle\prod_j x_j\rangle.$$

Put $P_{E,d}^{\mathrm{nz}} = (I - |0\rangle\langle 0|)^{\otimes d} \otimes I$, $\tau_{E,d} = \operatorname{Tr}/Q^{d+1}$, and $v_{E,d} = P_{E,d}^{\mathrm{nz}} M_E^{(d)} P_{E,d}^{\mathrm{nz}}$. The corner trace is $\tau_{E,d}^{\mathrm{nz}}(a) = \tau_{E,d}(a)/\tau_{E,d}(P_{E,d}^{\mathrm{nz}})$. On a specified word, P_A^{nz} tests nonzero labels precisely on the register subset A. For $E = \mathbb{F}_{p^r}$ with its named primitive root ζ_p, let $F_E|x\rangle = Q^{-1/2} \sum_y \zeta_p^{-\operatorname{Tr}_{E/\mathbb{F}_p}(xy)}|y\rangle$ and put $P_r^{\mathrm{dual}} = F_E P_r^{\mathrm{mov}} F_E^*$.

Scope. Nonzero controls, moving Frobenius labels and their Fourier-transported cuts are distinct tests.

Provenance. ids: D1306, D1308, D1334; proved in: definitions.md; tested by: orbit checker B6; reviewed in: frobenius-boundary-adjudication.md.

Proposition 23.8 (Active weights and interaction order). PROVED *For every finite field E, every $d \geq 1$, and the traces just specified, write $w = ((Q-1)/Q)^d$. Then*

$$\tau_{E,d}(P^{\mathrm{nz}}_{E,d}) = w, \quad \tau_{E,d}(M^{(d)}_E) = 1 - w, \quad \tau_{E,d}((M^{(d)}_E - I)^*(M^{(d)}_E - I)) = 2w.$$

The corner gate is unitary, with trace zero and conditional squared difference from its corner identity equal to two. Frobenius and field inclusions preserve these nonzero-control cuts. On one word, $P^{\mathrm{nz}}_A P^{\mathrm{nz}}_B = P^{\mathrm{nz}}_{A\cup B}$, with reference weight $((Q-1)/Q)^{|A\cup B|}$. Independently conditioned moving-orbit words obey

$$W_R(t) = \left(\prod_{j=1}^{n}(r_j - 1)\right)(t-1)^n + O((t-1)^{n+1}).$$

Scope. These are trace and counting identities, not operator-norm convergence. The multiplication family's order d matches its interaction arity and exact Clifford level $d+1$; no general equivalence is asserted.

Provenance. ids: FRL-ACTIVE; proved in: orbit-composition.md, Sections 4–5; tested by: orbit checker B6; reviewed in: frobenius-boundary-adjudication.md.

There are $(Q-1)^d$ active control tuples. Each gives a nonzero translation of the target, with no fixed labels; all other tuples give the identity. This proves the traces by counting permutation fixed points. Overlapping controls are counted once, which explains why their vanishing orders need not add. Independent moving factors do multiply, and must retain that order information even after their individual normalizations.

The extension to mixed arithmetic processes needs more structure. In three $\mathbb{F}_4$ registers, multiplication sends $(\alpha, \alpha, \alpha^2)$ to $(\alpha, \alpha, 0)$, leaving the tensor of moving-label sectors. Also, for the normalized trace-fibre inclusion $V|a\rangle = |\ker \mathrm{Tr}_{E/K}|^{-1/2} \sum_{\mathrm{Tr}_{E/K}(x)=a} |x\rangle$, the map $\mathbb{F}_2 \subset \mathbb{F}_4$ has $V|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$; its nonzero cut does not intertwine with V. Finally the Fourier matrix has constant squared entry modulus $1/Q$, giving

$$\frac{\mathrm{Tr}(P^{\mathrm{mov}}_r P^{\mathrm{dual}}_r)}{p^r} = (1 - p^{1-r})^2.$$

This is an overlap of possibly noncommuting projections, not an intersection dimension. It supplies a definite second-order mixed datum. The next construction must retain the two charts, their measured transitions and the arithmetic multiplication and transfer relations in one positive section algebra. The present Frobenius orbit family supplies a concrete positive starting point for that task.

Provenance. ids: D1304, FRL-ACTIVE; proved in: orbit-composition.md, Section 5; tested by: orbit checker B6; reviewed in: supporting scope calculations; no additional promotion.

24 Galois embedding registers over an arbitrary field

A finite separable extension carries a finite set of embeddings into a separable closure, even when the ground field is infinite. This set gives a finite quantum register with a natural Galois action. Its dimension is the extension degree. Over a finite field it realizes a single primitive Frobenius orbit. The construction below includes every finite separable field tower and the finite étale algebras required by independent tensor. The three results have passed the capped independent review.

Definition 24.1 (Finite étale embedding datum)**.** Fix a field K and a named separable closure $\Omega = K^{\mathrm{sep}}$. Put $G_K = \mathrm{Aut}_K(\Omega)$ with its Krull topology. Let Et_K^+ have as objects nonzero finite étale commutative unital K-algebras and as arrows unital K-algebra homomorphisms $f : A \to B$ for which B is finite locally free over A of one constant positive rank r_f. Equivalently, B is isomorphic to A^{r_f} as an A-module; no such module isomorphism is chosen. The tensor is $\otimes_K$ and its unit is K. Set

$$X_A = \mathrm{Hom}_{K\text{-alg}}(A, \Omega), \qquad n_A = \dim_K A, \qquad H_A^{\mathrm{emb}} = \ell^2(X_A),$$

$$\mathcal{B}_A^{\mathrm{emb}} = \mathrm{End}_{\mathbb{C}}(H_A^{\mathrm{emb}}), \qquad U_A^{\mathrm{emb}}(g)|\tau\rangle = |g \circ \tau\rangle, \qquad \rho_A^{\mathrm{mix}} = I/n_A.$$

The basis of $\ell^2(X_A)$ is orthonormal for counting measure. The observable algebra is the full matrix algebra, including coherent matrix entries between distinct G_K-orbits. The reference trace is ordinary matrix trace, and the named reference density is ρ_A^{mix}.

Scope. Constant rank is a condition on every component of A. It includes every finite separable field extension, with no restriction on characteristic or on whether K is perfect. The superscript distinguishes this register from the register with all field elements as basis labels.

Provenance. ids: D1401; proved in: theory/sidequests/arithmetic-limits/embedding-functor.md, section 1; tested by: theory/checks/arithmetic_limits_check.py, G1–G3; reviewed in: galois-embedding-adjudication.md.

Definition 24.2 (Embedding-fibre transfers and comparison maps)**.** For a finite étale embedding datum and $f : A \to B$, define

$$R_f : X_B \longrightarrow X_A, \quad R_f(\tau) = \tau \circ f, \qquad s_f|\sigma\rangle = \frac{1}{\sqrt{r_f}} \sum_{R_f(\tau)=\sigma} |\tau\rangle.$$

The square root is the positive real square root. For objects A, C, let

$$\mu_{A,C} : H_A^{\mathrm{emb}} \otimes H_C^{\mathrm{emb}} \longrightarrow H_{A \otimes_K C}^{\mathrm{emb}}$$

send $|\sigma\rangle \otimes |\tau\rangle$ to the basis vector of the homomorphism $a \otimes c \mapsto \sigma(a)\tau(c)$. The unit comparison sends $1 \in \mathbb{C}$ to the unique K-embedding of K into Ω. For a named K-isomorphism of separable closures $\eta : \Omega \to \Omega'$, define

$$W_{\eta,A}|\tau\rangle = |\eta \circ \tau\rangle, \qquad c_\eta(g) = \eta g \eta^{-1}.$$

Scope. No basis of A over K is chosen. A separable closure and any comparison between two such closures are named data. A single normalization for nonuniform embedding fibres is outside this definition.

Provenance. ids: D1402; proved in: theory/sidequests/arithmetic-limits/embedding-functor.md, sections 1–2; tested by: theory/checks/arithmetic_limits_check.py, G1–G3; reviewed in: galois-embedding-adjudication.md.

Theorem 24.3 (Coherent quantization of finite étale towers)**.** PROVED *For every field K and named separable closure Ω, the category Et_K^+ is symmetric monoidal. Each restriction map R_f is surjective with every fibre of cardinality r_f. The assignments $A \mapsto (H_A^{\mathrm{emb}}, U_A^{\mathrm{emb}})$ and $f \mapsto s_f$, with the displayed tensor and unit comparisons, give a faithful strong symmetric* *monoidal functor to finite continuous unitary G_K-representations and equivariant isometries. In particular,*

$$s_f^* s_f = I, \qquad s_{h\circ f} = s_h s_f, \qquad U_B^{\mathrm{emb}}(g) s_f = s_f U_A^{\mathrm{emb}}(g),$$
$$s_{f\otimes k}\mu_{A,C} = \mu_{B,D}(s_f \otimes s_k) \quad (f : A \to B,\ k : C \to D).$$

Every named change η of separable closure gives a natural monoidal unitary comparison, transports the Galois action through c_η, and satisfies $W_{\theta\eta,A} = W_{\theta,A} W_{\eta,A}$.

Scope. The result includes all finite separable field towers over every K and all allowed étale towers. Faithfulness concerns the specified algebra maps; it does not assert fullness among equivariant isometries. A different closure comparison η' changes W_η by $U^{\mathrm{emb}}(\eta'\eta^{-1})$ on the target.

Provenance. ids: GAL-FUNCTOR; proved in: theory/sidequests/arithmetic-limits/embedding-functor.md, sections 1–2; tested by: theory/checks/arithmetic_limits_check.py, G1–G3; reviewed in: galois-embedding-adjudication.md.

Proof. Splitting over Ω identifies $A \otimes_K \Omega$ with $\prod_{\sigma\in X_A} \Omega$. The coordinate idempotent at σ cuts $B \otimes_K \Omega$ down to one coordinate for each extension of σ to B. Constant rank makes this dimension r_f. Consequently the normalized columns of s_f have disjoint support and unit norm. A label in the top field of a tower has one uniquely determined intermediate restriction, so the composite sum has coefficient $(r_f r_h)^{-1/2}$ exactly once on every surviving label. This proves tower composition.

An embedding of $A \otimes_K C$ is uniquely the product of one embedding of each factor. All tensor comparisons therefore act on the same ordered tuples. Reassociation and swap commute on these tuples, and fibre sizes multiply. Galois postcomposition and closure changes biject every restriction fibre and preserve its normalization. Finally, the support of s_f recovers R_f, and the embedding-set equivalence recovers f. The source input is Milne, *Fields and Galois Theory*, version 5.10, Propositions 8.6, 8.9 and 8.20, Corollary 8.7 and Theorems 6.10 and 8.21; the same equivalence is Stacks Lemma 58.2.2. □

Definition 24.4 (Retained embedding decoding)**.** For a finite étale embedding datum and its embedding-fibre transfers, put $P_f = s_f s_f^*$ and $Q_f = I - P_f$, and use the retained-isometry decoder schema in the full complex matrix algebras. Its encoding, success map and retained decoder are

$$\mathcal{E}_f(\rho) = s_f \rho s_f^*, \qquad d_f(\rho) = s_f^* \rho s_f,$$
$$\mathcal{D}_f(\rho) = \big(d_f(\rho), Q_f \rho Q_f\big) : \mathcal{B}_B^{\mathrm{emb}} \longrightarrow \mathcal{B}_A^{\mathrm{emb}} \oplus \mathcal{B}_B^{\mathrm{emb}}.$$

The output tags are success:A and failure:B; trace on a tagged algebra is the sum of ordinary matrix traces. A completely positive map is a linear map whose identity-matrix amplifications preserve positive matrices; it is trace preserving if it preserves this specified trace. Independent decoders pair every output tag. A stopped tower decoder applies the next decoder only on the preceding success output and retains every failure at the object where it first occurs. Full instruments are compared with these histories, without identifying them with the binary decoder of a composite isometry.

Scope. Failure retains its quantum output. This is an instantiation of the retained decoder for an isometry, with ordinary trace on each tag.

Provenance. ids: D1403,D1327; proved in: theory/sidequests/arithmetic-limits/embedding-functor.md, section 3; tested by: theory/checks/arithmetic_limits_check.py, G1–G3; reviewed in: galois-embedding-adjudication.md.

Theorem 24.5 (Positive decoding with all independent outcomes). PROVED *For every arrow f of a finite étale embedding datum, $\mathcal{E}_f$ and $\mathcal{D}_f$ are G_K-equivariant completely positive trace-preserving maps, and d_f is completely positive and trace nonincreasing. They satisfy*

$$\mathcal{D}_f(\mathcal{E}_f(\rho)) = (\rho, 0), \qquad \operatorname{Tr} d_f(\rho_B^{\mathrm{mix}}) = \frac{1}{r_f}, \qquad d_f d_h = d_{h \circ f}.$$

Conditioned reference success gives ρ_A^{mix}. Stopped tower instruments and independent products of decoders are completely positive and trace preserving with every indicated history or paired output. For two independent maximally mixed inputs their four probabilities are

$$(r_f^{-1}, 1 - r_f^{-1}) \otimes (r_k^{-1}, 1 - r_k^{-1}).$$

Closure changes and Cartesian tensor comparisons transport all these branches.

Scope. Complete positivity holds on entangled inputs as well. Only the stated independent reference inputs force product probabilities. Full instruments retain their histories; no equality with the binary decoder of a composite or tensor isometry is asserted.

Provenance. ids: GAL-DESCENT; proved in: theory/sidequests/arithmetic-limits/embedding-functor.md, section 3; tested by: theory/checks/arithmetic_limits_check.py, G1–G3; reviewed in: galois-embedding-adjudication.md.

Proof. The decoder has amplitudes s_f^* and Q_f, whose squared sum is $P_f + Q_f^2 = I$. Their matrix amplifications are positive by the same conjugation formula, and the squared-sum identity gives trace preservation. Two independent decoders have all four tensor products of these amplitudes, with squared sum $(P_f + Q_f) \otimes (P_k + Q_k) = I$. In a tower $A \xrightarrow{f} B \xrightarrow{h} C$, the stopped history has amplitudes $s_f^* s_h^*$, $Q_f s_h^*$ and Q_h, with output registers A, B, C respectively. Their squared sum is $s_h(P_f + Q_f)s_h^* + Q_h = I$. Replacing any success by one more complete decoder preserves this identity, proving arbitrary finite towers.

The difference between histories and a binary decoder is measurable. Write $R = s_h Q_f s_h^*$, so $Q_{h \circ f} = R + Q_h$ with orthogonal summands. Re-encoding and merging the two failure histories gives $R\rho R + Q_h \rho Q_h$; the binary decoder gives $(R + Q_h)\rho(R + Q_h)$. When both summands are nonzero, a superposition between their ranges distinguishes these outputs by its off-diagonal terms. Retaining the actual history is therefore part of the process specification. ☐

Definition 24.6 (Embedding stabilizer and normal closure). For a finite étale embedding datum with A a field and a named $\tau_0 \in X_A$, put

$$H_{A,\tau_0} = \{g \in G_K : g \circ \tau_0 = \tau_0\}, \qquad C_A = \bigcap_{g \in G_K} g H_{A,\tau_0} g^{-1},$$

$$N_A = K\left(\bigcup_{\tau \in X_A} \tau(A)\right) \subset \Omega.$$

Thus N_A is the compositum of all embedded conjugates. Its definition does not use τ_0. The notation C_A anticipates the proved independence of its defining expression from the choice of τ_0.

Scope. The extension A/K is finite separable and need not be normal. The stabilizer itself depends on the chosen embedding.

Provenance. ids: D1404; proved in: theory/sidequests/arithmetic-limits/galois-image-orbits.md, section 1; tested by: theory/checks/arithmetic_limits_check.py, G1; reviewed in: galois-embedding-adjudication.md.

Definition 24.7 (Primitive finite-field orbit comparison)**.** For a finite étale embedding datum with $K = \mathbb{F}_p$, let $\mathrm{Frob}_p : \Omega \to \Omega$ be $x \mapsto x^p$. Let A be a degree-d field extension of $\mathbb{F}_p$, choose $a \in A$ with $A = \mathbb{F}_p(a)$, and choose $\tau_0 \in X_A$. Put $b = \tau_0(a)$ and $O_b = \{b, b^p, \ldots, b^{p^{d-1}}\}$. For a positive multiple r of d, let $E = \mathbb{F}_{p^r}$ be the unique such subfield of Ω and define

$$T_a : H_A^{\mathrm{emb}} \longrightarrow \ell^2(O_b) \subset \ell^2(E), \qquad T_a|\tau\rangle = |\tau(a)\rangle.$$

The orbit coordinate unitary $L_{\tau_0} : \mathbb{C}^d \to H_A^{\mathrm{emb}}$ sends $|j\rangle$ to $|\,\mathrm{Frob}_p^j \circ \tau_0\rangle$ for $0 \le j < d$. In these coordinates write $S_d|j\rangle = |j + 1 \bmod d\rangle$. The word primitive here means degree exactly d over $\mathbb{F}_p$, and does not require a generator of A's multiplicative group.

Scope. The comparison names a primitive element and cyclic origin. It compares one orbit subspace; the full field-label register has dimension p^r.

Provenance. ids: D1405; proved in: theory/sidequests/arithmetic-limits/galois-image-orbits.md, section 3; tested by: theory/checks/arithmetic_limits_check.py, G3; reviewed in: galois-embedding-adjudication.md.

Theorem 24.8 (Normal-closure symmetry and primitive Frobenius atoms)**.** PROVED *For every finite separable field extension A/K and named separable closure, N_A/K is the finite Galois normal closure and*

$$\ker U_A^{\mathrm{emb}} = C_A, \qquad \operatorname{Im} U_A^{\mathrm{emb}} \simeq \mathrm{Gal}(N_A/K).$$

The isomorphism sends $U_A^{\mathrm{emb}}(g)$ to $g|_{N_A}$. It is independent of the choice of τ_0. Every finite separable tower $A \to B$ gives $N_A \subset N_B$ and the corresponding restriction quotient of these images. Over all finite Galois subfields E of Ω,

$$G_K \simeq \varprojlim_E \operatorname{Im} U_E^{\mathrm{emb}}$$

as profinite groups, with the stated restriction transitions. For $K = \mathbb{F}_p$, a degree-d extension and the primitive orbit comparison above satisfy

$$L_{\tau_0}^* U_A^{\mathrm{emb}}(\mathrm{Frob}_p) L_{\tau_0} = S_d.$$

The map T_a is a Frobenius-equivariant unitary onto the primitive length-d orbit subspace. Its full observable algebra and maximally mixed state identify with the cycle block $M_d(\mathbb{C})$ and I/d. For $d > 1$ this is a nonfixed-label orbit block.

Scope. Nonabelian images are included. The inverse limit retains the finite Galois actions and their restriction maps; bare matrix algebras alone do not reconstruct them. The finite-field comparison specifies the orbit block and normalized block trace, with no assertion about orbit multiplicities, boundary mixture weights, field-label code transfers, Fourier transforms or multiplication processes.

Provenance. ids: GAL-IMAGE,D1331,D1332; proved in: theory/sidequests/arithmetic-limits/galois-image-orbits.md, sections 1–3; tested by: theory/checks/arithmetic_limits_check.py, G1–G3; reviewed in: galois-embedding-adjudication.md.

Proof. Every embedding extends to a Galois automorphism of the separable closure, so $X_A \simeq G_K/H_{A,\tau_0}$. An element fixes every coset precisely when it lies in every conjugate stabilizer. Fixing those embeddings is equivalent to fixing their image compositum pointwise. The Galois correspondence therefore identifies this kernel with $\mathrm{Gal}(\Omega/N_A)$ and its quotient with $\mathrm{Gal}(N_A/K)$. This is the normal-core description of the normal closure in Milne, Remark 3.18. Stacks Lemmas 9.22.2–9.22.3 and Theorem 9.22.4 give the restriction quotients and the profinite inverse limit.

Over $\mathbb{F}_p$, the embeddings are $\tau_j = \mathrm{Frob}_p^j \tau_0$, $0 \le j < d$, by Milne, Propositions 4.19–4.23. Evaluation at a primitive element makes their images distinct, so it identifies their basis with one length-d orbit. Changing τ_0 shifts the cyclic coordinates; changing the primitive element gives the explicit comparison $T_{a'}T_a^*$ between the named orbit spaces. Both comparisons intertwine Frobenius. □

For standard subfields $\mathbb{F}_{p^d} \subset \mathbb{F}_{p^e}$ with $d \mid e$, the cyclic coordinates give

$$s_f|j\rangle = \sqrt{\frac{d}{e}} \sum_{k=0}^{e/d-1} |j+kd\rangle.$$

Thus this transfer pulls back a quotient of cycles. For example its dimensions for $\mathbb{F}_2 \subset \mathbb{F}_4$ are $1 \to 2$, whereas the field-label inclusion has dimensions $2 \to 4$. The independent tensor of degree-d and degree-e atoms has $\gcd(d, e)$ simultaneous-Frobenius orbits of length $\mathrm{lcm}(d, e)$, but retains the full $M_{de}(\mathbb{C})$, including coherence between different orbits.

Separability is essential to the stated formula. If a finite extension L/K is not separable, a K-embedding into K^{sep} would make every element of L separable, a contradiction. Its embedding set here is empty. Using an algebraic closure instead counts separable degree and still loses inseparable multiplicity. The positive construction above therefore supplies the Galois part of the arithmetic problem uniformly over all base fields, while additional arithmetic data remain necessary for a construction that also retains Fourier and multiplication diagrams.

25 Arithmetic Fourier charts and coherent tower refinement

The inclusion of a finite field and the uniform pullback along its trace give two quantum codes in the larger register. Their exact relative angle depends on whether the extension degree vanishes in the characteristic. All degrees admit a uniform Gram classification. A positive conditional matrix chart survives when its angle closes, and degrees invertible in the characteristic admit coherent refinement through towers of every finite length. The three positive results have passed the capped review; the full mixed arithmetic construction remains conjectural. The checker is `theory/checks/arithmetic_limits_check.py`.

25.1 The exact finite-field sectors

Definition 25.1 (Arithmetic register and transfer conventions)**.** Fix a prime p and a named primitive complex pth root ζ_p. For a finite field $E/\mathbb{F}_p$ of cardinality p^r, retain the field, the Hilbert space $H_E = \ell^2(E)$ with orthonormal basis $|x\rangle$, and the character

$$\psi_E(x) = \zeta_p^{\mathrm{Tr}_{E/\mathbb{F}_p}(x)}, \qquad \mathrm{Tr}_{E/\mathbb{F}_p}(x) = \sum_{j=0}^{r-1} x^{p^j}.$$

The prime-field phase space is $E \oplus E$ with alternating form $\Omega_E((a, b), (a', b')) = \mathrm{Tr}_{E/\mathbb{F}_p}(ab' - a'b)$. The reference operators are

$$X_E(a)|x\rangle = |x + a\rangle, \quad Z_E(b)|x\rangle = \psi_E(bx)|x\rangle, \quad W_E(a, b) = Z_E(-b)X_E(a).$$

An ordered register list has tensor-product Hilbert space and direct-sum phase space; the empty list has Hilbert space $\mathbb{C}$. Field structures and factor order remain part of the data. Frobenius and Fourier are

$$U_E|x\rangle = |x^p\rangle, \qquad F_E|x\rangle = |E|^{-1/2} \sum_{y\in E} \psi_E(-xy)|y\rangle.$$

Frobenius on a list is their tensor product. A subscript on a Fourier tensor factor means identity on the other factors. For a named embedding $i : K \to E$ over the same p, ζ_p, with $|K| = p^s$, $|E| = p^{ns}$, put

$$T_i(y) = i^{-1}\left(\sum_{j=0}^{n-1} y^{p^{sj}}\right), \qquad \kappa_i = |\ker T_i|,$$

where the inverse is restricted to $i(K)$. Composite traces use their composite named embeddings. The inclusion and trace-fibre transfers are

$$J_i|a\rangle = |i(a)\rangle, \qquad V_i|a\rangle = \kappa_i^{-1/2} \sum_{T_i(y)=a} |y\rangle.$$

All square roots are positive, and reverse arrows are Hilbert adjoints.

Scope. The negative Fourier kernel and the common named root of unity are fixed. These are finite-field registers; a separable embedding-set register over an arbitrary field is a separate Galois construction.

Provenance. ids: D1301–D1304, D1306; proved in: foundation.md; transfers-example.md; tested by: arithmetic Frobenius checker; reviewed in: frobenius-hierarchy-adjudication.md.

Definition 25.2 (Joint arithmetic code and Gram datum)**.** For a named embedding $i : K \to E$ of degree $n \geq 2$, put $q = |K| = p^s$, $Q = |E| = q^n$ and $\kappa_i = Q/q$. Write

$$\Gamma_i = J_i^* V_i, \quad P_i = J_i J_i^*, \quad \widetilde{P}_i = V_i V_i^*, \quad \mathcal{S}_i = \operatorname{ran} J_i + \operatorname{ran} V_i.$$

Let E_i project orthogonally onto $\mathcal{S}_i$, and use the physical trace $\tau_E = \operatorname{Tr}/Q$. The projection algebra is the unital star algebra generated by $P_i, \widetilde{P}_i$ in $\operatorname{End}(H_E)$; zero-dimensional summands are omitted. Put $|+_K\rangle = q^{-1/2} \sum_x |x\rangle$ and $R_K|x\rangle = |-x\rangle$. The joint Frobenius representation means U_E restricted to $\mathcal{S}_i$.

Scope. The joint range is a subspace of the larger register, with its inherited physical trace. Its normalized conditional trace will be specified separately.

Provenance. ids: D1421; proved in: definitions.md; tested by: arithmetic limits checker F1–F2; reviewed in: arithmetic-mixed-adjudication.md.

Definition 25.3 (Corrected trace frame and singular sectors)**.** If $p \nmid n$, define

$$D_n^K|x\rangle = |nx\rangle, \qquad L_i = V_i D_n^K, \qquad c_i = \kappa_i^{-1/2}, \qquad h_i = \sqrt{1 - c_i^2},$$
$$W_i = \frac{L_i - c_i J_i}{h_i}, \qquad B_i = (D_n^K)^{-1} F_K.$$

The map $\mathcal{T}_i : \mathbb{C}^2 \otimes H_K \to H_E$ has columns J_i, W_i. For any nonzero prime-field scalar a, scalar dilations on each field are $D_a|x\rangle = |ax\rangle$. If $p \mid n$, instead put

$$\alpha_i = J_i|+_K\rangle, \quad \beta_i = V_i|0\rangle, \quad d_i = \sqrt{q/\kappa_i}, \quad A_K = |+_K\rangle^\perp.$$

For $n > 2$ put $e_i = \sqrt{1 - d_i^2}$ and let the angle frame $\mathcal{T}_i^{\text{ang}} : \mathbb{C}^2 \to H_E$ have columns $\alpha_i, (\beta_i - d_i\alpha_i)/e_i$. For every $p \mid n$ the residual frame $\mathcal{T}_i^{\text{res}} : \mathbb{C}^2 \otimes A_K \to H_E$ has columns $J_i|_{A_K}, V_i F_K|_{A_K}$. If $p = 2, n = 2$, retain the common line $\mathbb{C}\alpha_i = \mathbb{C}\beta_i$ and only the residual frame.

Scope. The corrected degree dilation is defined only when the degree is invertible. The singular quadratic case has no angle frame requiring division by zero.

Provenance. ids: D1422, D1423; proved in: definitions.md; tested by: arithmetic limits checker F1–F2; reviewed in: arithmetic-mixed-adjudication.md.

Proposition 25.4 (Uniform Gram sectors, physical weights and symmetry)**.** PROVED *For every named finite-field embedding $i : K \to E$ of degree $n \geq 2$,*

$$\sqrt{\kappa_i}(\Gamma_i)_{x,a} = \delta_{a,nx}.$$

If $p \nmid n$, all q singular-value squares are $1/\kappa_i$. If $p \mid n$, there is one square $q/\kappa_i = q^{2-n}$ and $q-1$ zeros. The ranges intersect only when $p = 2, n = 2$, in the invariant line $\mathbb{C}\alpha_i$. The joint projection algebra and its physical block weights are

degree stratum	*joint algebra, with multiplicities*	*physical weights*
$p \nmid n$	$M_2 \otimes I_q$	$2q/Q$
$p \mid n,\ n > 2$	$M_2 \oplus \mathbb{C} \oplus \mathbb{C}\ (1, q-1, q-1)$	$2/Q,\ (q-1)/Q,\ (q-1)/Q$
$p = 2,\ n = 2$	$\mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}\ (1, q-1, q-1)$	$1/Q,\ (q-1)/Q,\ (q-1)/Q.$

Each matrix summand uses its normalized matrix trace. The complement is one scalar summand, of weight $1 - 2q/Q$ in the first two cases and $1 - (2q-1)/Q$ in the last. In the regular frame Fourier and Frobenius are

$$\mathcal{T}_i^* F_E \mathcal{T}_i = \begin{pmatrix} c_i & h_i \\ h_i & -c_i \end{pmatrix} \otimes B_i, \qquad \mathcal{T}_i^* U_E \mathcal{T}_i = I_2 \otimes U_K.$$

On the singular angle plane Fourier is the same matrix with parameters d_i, e_i, while Frobenius is identity. On the singular residual frame,

$$F_{\mathrm{res}} = \begin{pmatrix} 0 & R_K|_{A_K} \\ I_{A_K} & 0 \end{pmatrix}, \qquad U_{\mathrm{res}} = I_2 \otimes U_K|_{A_K}.$$

In quadratic characteristic two the common line is Fourier-fixed and $F_{\mathrm{res}} = X \otimes I_{A_K}$, where $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The joint Frobenius representation is two copies of U_K, with one invariant common line removed in that exceptional case. In every stratum $U_E^s = I$ on $\mathcal{S}_i$ when $q = p^s$.

Scope. These formulas hold for all finite fields and named embeddings of degree at least two. The projection algebra alone is commutative in the exceptional case; its residual matrix factor additionally uses compressed Fourier. The complement retains the actual ambient Fourier and Frobenius restrictions.

Provenance. ids: MIX-GRAM; proved in: gram-and-chart.md, Sections 1–4; tested by: arithmetic limits checker F1–F2; reviewed in: arithmetic-mixed-adjudication.md.

Derivation. The trace of $i(x)$ is nx, so evaluating a normalized trace-fibre column at $i(x)$ gives the Gram entry. If n is invertible it is a scaled permutation matrix. Otherwise it is $\sqrt{q/\kappa_i}|+_K\rangle\langle 0|$. A common range vector corresponds exactly to singular value one, giving the stated exception. Gram subtraction produces the displayed orthogonal frames. The negative Fourier kernel gives $F_E J_i = V_i F_K$, $F_E V_i = J_i F_K$ and $F_K^2 = R_K$; also $F_K D_n^K = (D_n^K)^{-1} F_K$. Substitution gives every Fourier block, including the residual reflection in odd characteristic. Frobenius intertwines both transfers and fixes the uniform and zero vectors. The block weights are their dimensions divided by Q. □

25.2 A positive conditional chart and its tangent

Definition 25.5 (Conditioned matrix chart)**.** For $0 \le c \le 1$ let $h = \sqrt{1-c^2}$, use $M_2(\mathbb{C})$ with faithful reference $\mathrm{tr}_2 = \mathrm{Tr}/2$, and set

$$p_c = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad q_c = \begin{pmatrix} c^2 & ch \\ ch & h^2 \end{pmatrix}, \quad f_c = \begin{pmatrix} c & h \\ h & -c \end{pmatrix}, \quad d_c = \begin{pmatrix} -h & c \\ c & h \end{pmatrix}.$$

For $c < 1$, $d_c = (q_c - p_c)/h$; at one retain its continuous displayed value $d_1 = X$, while $f_1 = Z = \mathrm{diag}(1, -1)$. Conditioning means dividing a represented chart's restricted physical trace by its support weight before varying c. With a retained logical space H, use $\mathrm{tr}_2 \otimes \mathrm{tr}_H$ on $M_2 \otimes \mathrm{End}(H)$, with chart matrices on the first factor.

Scope. This is a matrix continuation in an angle parameter. No field of noninteger cardinality is posited.

Provenance. ids: D1424; proved in: definitions.md; tested by: arithmetic limits checker R1–R2; reviewed in: arithmetic-mixed-adjudication.md.

Definition 25.6 (Continuous instruments and retained decoding)**.**

For finite-dimensional Hilbert spaces X, Y_b, a continuous instrument on a closed real parameter interval is a finite family of continuous matrices $K_b(t) : X \to Y_b$ satisfying $\sum_b K_b(t)^* K_b(t) = I_X$. It sends ρ to $\bigoplus_b K_b(t)\rho K_b(t)^*$, with ordinary matrix traces on the output blocks. For a continuous isometry $S(t) : X \to Y$, the retained decoder amplitudes are

$$S(t)^* : Y \to X, \qquad I_Y - S(t)S(t)^* : Y \to Y,$$

with distinct success and failure tags, respectively. Independent instruments use tensor-product Kraus amplitudes and retain all outcome tuples. Sequential instruments retain every history. A reference state supplies input probabilities and does not replace the ordinary trace in the channel normalization equation.

Scope. Failure retains the ambient state. Distinct sequential and direct measurement protocols are not identified by forgetting their histories.

Provenance. ids: D1426; proved in: definitions.md; tested by: arithmetic limits checker R2–R3; reviewed in: arithmetic-mixed-adjudication.md.

Proposition 25.7 (Faithful Fourier tangent after conditioning)**.** PROVED

For every regular-degree joint chart and every singular-degree angle plane with $n > 2$, physical conditioning gives the matrix chart just defined, with its stated logical factor when present. Throughout the closed angle interval,

$$d_c^2 = f_c^2 = I, \qquad f_c p_c f_c = q_c, \qquad f_c d_c = -d_c f_c.$$

At $c = 1$ the two projections coincide, while the retained tangent X and p_1 generate M_2 with faithful trace $\mathrm{Tr}/2$. Actual chart Fourier becomes Z and conjugates X to $-X$. With the regular logical factor retained, the full Fourier operator is $Z \otimes B_i$ at this endpoint. Continuous instruments, retained decoders and independent tensor remain completely positive and trace preserving at the endpoint. Quadratic characteristic two has instead a common invariant line and a residual factor $M_2 \otimes I_{A_K}$ generated by its compressed projections and Fourier $X \otimes I_{A_K}$, with normalized residual reference $\mathrm{tr}_2 \otimes \mathrm{tr}_{A_K}$.

Scope. The coalescing-plane continuation has the declared regular or singular nonquadratic scope. The exceptional residual factor is positive but is not obtained by dividing by a zero angle. Frobenius remains on the logical factor, and is identity on the singular angle plane.

Provenance. ids: MIX-CHART; proved in: gram-and-chart.md, Sections 5–6; tested by: arithmetic limits checker F2, R1–R2; reviewed in: arithmetic-mixed-adjudication.md.

Derivation. The four matrix identities follow by multiplication. The trace is faithful because $\mathrm{tr}_2(A^*A) = \frac{1}{2}\sum_{j,k} |A_{jk}|^2$. At one, preparing p_1 and testing p_1 gives probability one, whereas applying $d_1 = X$ first gives zero. Complete positivity follows because every amplification is a sum of positive quadratic forms $(I \otimes K_b)\rho(I \otimes K_b)^*$; the Kraus completeness equation gives trace preservation, including for tensor products. For the rank-one columns e_0 and $v = ce_0 + he_1$, $f_c e_0 = v$ gives the full Fourier decoder comparison on both success and failure. ☐

The order of operations matters. The formal unconditioned complement weight in the regular branch is $1 - 2/\kappa$, which is $-1/3$ at $\kappa = 3/2$. The positive continuation above first conditions the arithmetic chart and then varies its angle; it does not continue that whole physical trace as a positive reference state.

25.3 Refinement through arbitrary finite towers

Definition 25.8 (Path frame and refinement connector)**.** Let $K_0 \xrightarrow{i_1} \cdots \xrightarrow{i_\ell} K_\ell$, $\ell \geq 1$, be a named tower with degrees $n_r \geq 2$ invertible in characteristic p. Set $N = \prod_r n_r$. In outer-to-inner chart order $\ell, \dots, 1$, let

$$\mathcal{T}_{\mathrm{path}} = \mathcal{T}_{i_\ell}(I_2 \otimes \mathcal{T}_{i_{\ell-1}}) \cdots (I_{2^{\ell-1}} \otimes \mathcal{T}_{i_1}) : (\mathbb{C}^2)^{\otimes \ell} \otimes H_{K_0} \longrightarrow H_{K_\ell}.$$

For any $0 < c_r < 1$ put $h_r = \sqrt{1 - c_r^2}$, $v_r = c_r e_0 + h_r e_1$, $c_* = \prod_r c_r$, $h_* = \sqrt{1 - c_*^2}$. Define $C : \mathbb{C}^2 \to (\mathbb{C}^2)^{\otimes \ell}$ by

$$Ce_0 = |0 \cdots 0\rangle, \qquad Ce_1 = \frac{v_\ell \otimes \cdots \otimes v_1 - c_*|0 \cdots 0\rangle}{h_*}.$$

At arithmetic fibres use $c_r = \kappa_{i_r}^{-1/2}$. Consecutive blocks use the products of their constituent parameters in the same order.

Scope. Only the path frame requires a physical tower. The connector is also a well-defined matrix family at nonarithmetic parameter values.

Provenance. ids: D1425; proved in: definitions.md; tested by: arithmetic limits checker F3, R3; reviewed in: arithmetic-mixed-adjudication.md.

Definition 25.9 (Weighted endpoint connector)**.** For fixed positive weights $a_1, \dots, a_\ell$, put $A = \sum_r a_r$ and $c_r(t) = t^{-a_r/2}$ for $t > 1$. At $t = 1$ define

$$C_a(1)e_0 = |0 \cdots 0\rangle, \qquad C_a(1)e_1 = \sum_r \sqrt{\frac{a_r}{A}}\,|\,\mathrm{one}_r\rangle,$$

where $|\,\mathrm{one}_r\rangle$ has its sole entry one in edge r's factor. Consecutive blocks have summed weights. The direct chart uses tr_2; the whole path space uses tr_{2^ℓ}. Conditioning the latter on the connector range gives the former. Their unconditioned reference states are distinct.

Scope. Every weight is positive. Identity edges may be omitted; they require no local chart factor.

Provenance. ids: D1427; proved in: definitions.md; tested by: arithmetic limits checker R3; reviewed in: arithmetic-mixed-adjudication.md.

Proposition 25.10 (Coherent refinement and Fourier through every tower length)**.** PROVED
For every finite named tower of length $\ell \geq 1$ and degrees $n_r \geq 2$ invertible in the characteristic, the corrected trace maps compose exactly. The path frame and connector are isometries. For composite embedding k,

$$L_{i_\ell} \cdots L_{i_1} = L_k, \qquad \mathcal{T}_k = \mathcal{T}_{\text{path}}(C \otimes I_{H_{K_0}}).$$

All consecutive-block refinements give the same connector into the common path basis. The actual Fourier and Frobenius restrictions satisfy

$$\begin{aligned} F_{K_\ell}\mathcal{T}_{\text{path}} &= \mathcal{T}_{\text{path}}\left[(f_{c_\ell} \otimes \cdots \otimes f_{c_1}) \otimes (D_N^{K_0})^{-1}F_{K_0}\right], \\ U_{K_\ell}\mathcal{T}_{\text{path}} &= \mathcal{T}_{\text{path}}(I_{2^\ell} \otimes U_{K_0}), \\ (f_{c_\ell} \otimes \cdots \otimes f_{c_1})C &= Cf_{c_*}. \end{aligned}$$

For every positive weight vector the connector extends continuously to the displayed weighted endpoint, and refinement and Fourier identities continue there; in particular $Z^{\otimes \ell}C_a(1) = C_a(1)Z$. Retained decoder success amplitudes compose exactly. Full sequential history instruments and all independent tensor outcomes stay continuous, completely positive and trace preserving through the endpoint.

Scope. There is no bound on the finite tower length. Singular-degree refinements are not supplied by this formula. The direct connector occupies a two-dimensional subspace of the full independent path tensor. Equality of successful decoding does not identify a sequential history instrument with the binary decoder of the composite isometry.

Provenance. ids: MIX-TOWER; proved in: coherent-towers.md, Sections 1–6; tested by: arithmetic limits checker F3, R3; reviewed in: arithmetic-mixed-adjudication.md.

Derivation. Trace linearity gives $D_a^L V_i = V_i D_a^K$ for every nonzero prime-field scalar. Thus for consecutive degrees m,n, $V_j D_n^L V_i D_m^K = V_j V_i D_{mn}^K = L_{ji}$. The path frame sends the empty path to the direct inclusion and $v_\ell \otimes \cdots \otimes v_1$ to the direct corrected trace column; Gram subtraction gives C. Each block connector preserves these two vectors, so every grouping gives the same linear map. Fourier exchanges the two vectors in each factor and scalar dilations commute through the local frames, giving the precise logical scaling shown above. Writing $t = 1 + \epsilon$ gives $1 - t^{-a_r} = a_r\epsilon + O(\epsilon^2)$. The coefficient of a path with k excitations is of order $\epsilon^{(k-1)/2}$ in the second column. Only single excitations survive, with the stated square-root weights. Finally, an isometry S has retained Kraus completeness $SS^* + (I - SS^*)^2 = I$; all endpoint and tensor assertions follow continuously. □

For two edges j after i the nontrivial connector column is

$$\frac{c_j h_i|01\rangle + h_j c_i|10\rangle + h_j h_i|11\rangle}{\sqrt{1 - c_j^2 c_i^2}}.$$

For three weights $(1, 2, 4)$ the endpoint is $\sqrt{1/7}|\,\text{one}_1\rangle + \sqrt{2/7}|\,\text{one}_2\rangle + \sqrt{4/7}|\,\text{one}_3\rangle$. The encoded tangent has square CC^*, rather than identity on the entire path tensor. Higher excitation sectors remain available for independent processes, even though direct refinement does not fill them.

25.4 The remaining mixed arithmetic problem

Definition 25.11 (Finite mixed extension test diagram)**.** Take a finite diagram of named finite-field embeddings in one characteristic, closed under its named composites, with finite ordered register words. Its named amplitude operations are Frobenius, Fourier, inclusion and trace transfer and their adjoints, nonzero prime-field dilations, tensor identities, and the reversible multiplication gates

$$M_E^{(d)}|x_1,\dots,x_d,z\rangle = |x_1,\dots,x_d,z+\prod_j x_j\rangle$$

for each explicitly named positive finite arity $d \geq 1$, with their finite composites and adjoints. Retain their isometric encodings and decoding instruments, every sequential outcome history and every independent outcome tuple. Marked sector tests include the joint and singular sectors just defined, explicitly selected Frobenius orbit projections and active-control projections, and their actual Fourier transports. An orbit projection selects its named Frobenius basis orbits; an active-control projection requires each named control to be nonzero and leaves the target arbitrary. An equality is equality of the named arithmetic amplitude maps or tagged CP maps, not equality created by compressing intermediate gates separately.

Scope. Measurement placement is part of the source process. Retaining measurement histories does not identify a measured circuit with an unmeasured circuit.

Provenance. ids: D1428, D1308, D1327; proved in: definitions.md; tested by: arithmetic limits checker M1; reviewed in: arithmetic-mixed-adjudication.md.

Conjecture 25.12 (A positive mixed arithmetic process realization)**.** CONJECTURE *There is one positive filtered process realization natural in every finite mixed extension test diagram, including all characteristic-dividing tower sectors, and compatible with the separately specified general Galois embedding-set fragment through its stated orbit comparisons. It retains the named Fourier-transfer and multiplication relations, all instrument histories and independent tensor outcomes, and recovers the prescribed conditional chart, orbit and active-sector normalizations. Such a realization must specify its operator algebras, CP arrows, reference weights, continuation rule for mixed moments, and diagram comparison maps. It must give positive mixed moment Gram matrices at every finite matrix amplification and exhibit retained Frobenius and active multiplication state/effect distinctions in the common model.*

Scope. No such full realization is constructed here. The all-characteristic Gram classification and invertible-degree tower refinement supply constraints on it, while an all-characteristic tower connector and a compatible mixed-moment normalization remain open. Separate constant matrix envelopes or unrelated local continuations do not establish the conjecture.

Provenance. ids: MIX-ALL; proved in: coherent-towers.md, Section 7; tested by: arithmetic limits checker M1 is only a necessary finite-fibre diagnostic; reviewed in: arithmetic-mixed-adjudication.md.

The next explicit comparison uses both $\mathbb{F}_2 \to \mathbb{F}_4 \to \mathbb{F}_{16}$ and $\mathbb{F}_3 \to \mathbb{F}_9 \to \mathbb{F}_{81}$, adding $\mathbb{F}_3 \to \mathbb{F}_{27}$ for its singular odd-characteristic reflection. The arithmetic matrices already distinguish the full multiplication circuit from the circuit obtained by deleting every intermediate sector exit. A candidate must retain those exits and their subsequent transitions while preserving the probability of each specified measurement protocol. That is the next mixed requirement beyond the positive tower chart established here.

26 A conjectural refinement forced by Fourier symmetry

The relation $F_E^2|x\rangle = |-x\rangle$ means that retaining arithmetic Fourier also retains negation. Negation commutes with Frobenius, but in odd characteristic it does not fix 1 and is not a field automorphism. This is a closure of arithmetic operators, distinct from the normal closure of a field extension. It suggests a concrete test of how much additional information a filtered boundary must carry.

Definition 26.1 (Signed Frobenius orbit candidates)**.** Let p be odd and $r \geq 1$. The generators of $\Gamma_r^{\mathrm{sgn}} = C_r \times C_2$ act on $\mathbb{F}_{p^r}^\times$ by $x \mapsto x^p$ and $x \mapsto -x$. A point of exact Frobenius period d is internal when its negative lies in its Frobenius orbit, and external otherwise. Let t be a real variable. The integer Möbius function μ is zero on integers divisible by a prime square and otherwise (-1) to the number of prime factors. Write $\varphi(d) = |\{1 \leq a \leq d : \gcd(a,d) = 1\}|$ and $c_d(t) = \sum_{e|d} \mu(d/e)t^e$. For $d = 2^k m$ even, with $k \geq 1$ and m odd, define

$$A_d^{\mathrm{sgn}}(t) = \sum_{e|m} \mu(m/e)(t^{2^{k-1}e} - 1), \qquad B_d^{\mathrm{sgn}}(t) = \sum_{e|m} \mu(m/e)(t^{2^{k-1}e} - 1)^2.$$

For odd d set $A_d^{\mathrm{sgn}} = 0$, $B_1^{\mathrm{sgn}} = t - 1$, and $B_d^{\mathrm{sgn}} = c_d(t)$ for $d > 1$. Finally put

$$J_2(m) = m^2 \prod_{\substack{\ell | m \\ \ell \text{ prime}}} (1 - \ell^{-2}),$$

with empty product one.

Scope. These are candidate count polynomials. Characteristic two is excluded: there negation is identity and the group action changes.

Provenance. ids: D1331, D1441; proved in: definitions.md; tested by: theory/checks/arithmetic_signed_check.py; reviewed in: conjectural increment.

Conjecture 26.2 (First- and second-order signed orbit sectors)**.** CONJECTURE *For every odd prime p, every $r \geq 1$, and every $d \mid r$, $A_d^{\mathrm{sgn}}(p)$ and $B_d^{\mathrm{sgn}}(p)$ count the internal and external nonzero labels of exact Frobenius period d, respectively. Internal signed orbits occur only for even d and have size d; external signed orbits have size $2d$. Every nonzero candidate polynomial is positive for all $t > 1$. For even $d = 2^k m$,*

$$A_d^{\mathrm{sgn}}(t) = \varphi(d)(t-1) + O((t-1)^2),$$
$$B_d^{\mathrm{sgn}}(t) = 2^{2k-2} J_2(m)(t-1)^2 + O((t-1)^3).$$

For odd d, including $d = 1$, the first coefficient of $B_d^{\mathrm{sgn}}(t)$ is $\varphi(d)$.

Scope. The quantifiers range over all degrees and all odd primes. The finite census supports this conjecture and does not prove it. No full Fourier, multiplication or mixed-process limit follows from the proposed counts.

Provenance. ids: LIM-SIGNED; proved in: docs/research-plans/signed-orbit-filtration.md; tested by: theory/checks/arithmetic_signed_check.py, S1–S2; reviewed in: not submitted for theorem admission in this round.

For a degree-two extension, the formula predicts the following three types of nonzero labels.

Type	Point count	Signed orbit size	Vanishing order
Prime-field labels	$p-1$	2	1
Internal period-two labels	$p-1$	2	1
External period-two labels	$(p-1)^2$	4	2

The two first-order types suggest two matrix blocks M_2: on the first, Frobenius is identity and negation exchanges the points; on the second, both exchange them. The second-order type suggests M_4 with the regular $C_2 \times C_2$ action. This is a specific proposed mechanism by which closing the arithmetic symmetries retains additional orders of degeneration. It does not identify those orders with arbitrary Clifford levels.

The exact checker enumerates quotient fields of orders $3, 9, 27, 81, 5, 25$, obtains periods by iteration, and determines internality by orbit membership of the actual negative. It separately expands the sparse polynomials about one through degree 48. Both the incorrect-negation and incorrect-order mutations fail. The conjecture's general proof and its integration with Fourier remain future work.

27 A positive arithmetic counting boundary

A nontrivial arithmetic boundary can retain the actual field operations while changing the reference preparation. The construction here assigns positive mass to each absolute Frobenius period and matches the ordinary maximally mixed state at the actual characteristic. Its unnormalized density has positive Taylor coefficients; its normalized state is a Poisson mixture. Its boundary requires only a finite operator polynomial for every finite reference experiment. This polynomial preserves the leading probability and conditional state of every later arithmetic branch, including coherent mixed circuits. The characteristic and field tables remain named data. The six positive statements below have passed the capped review.

27.1 The reference and its positive coefficients

For a fixed actual prime p and named primitive root ζ_p, an arithmetic field $E = \mathbb{F}_{p^r}$ has register $H_E = \ell^2(E)$, orthonormal basis $|x\rangle$, and character $\psi_E(x) = \zeta_p^{\sum_{j=0}^{r-1} x^{p^j}}$. We use the actual matrices

$$U_E|x\rangle = |x^p\rangle, \qquad F_E|x\rangle = |E|^{-1/2} \sum_y \psi_E(-xy)|y\rangle,$$

and, for a named embedding $i : K \to E$, $|K| = p^s$, $r = ns$,

$$T_i(y) = i^{-1}\left(\sum_{j=0}^{n-1} y^{p^{sj}}\right), \quad \kappa_i = |E|/|K|, \quad J_i|a\rangle = |i(a)\rangle, \quad V_i|a\rangle = \kappa_i^{-1/2} \sum_{T_i(y)=a} |y\rangle.$$

The inverse embedding is restricted to its image and every square root is positive. Both transfers are isometries, compose in named towers, and obey $F_E J_i = V_i F_K$ and $F_E V_i = J_i F_K$ in every characteristic. Independent register lists use ordered tensor products.

Provenance. ids: D1301–D1304, D1306, FRB-TRACE, FRB-TRANSFER; proved in: foundation.md; transfers-example.md; tested by: arithmetic Frobenius checker; reviewed in: frobenius-hierarchy-adjudication.md.

Definition 27.1 (Period-uniform arithmetic reference)**.** Fix the actual p, ζ_p and a finite diagram of named arithmetic fields and embeddings. Let $h \geq 0$ and $t = e^h$; the prime p is retained data. For $E = \mathbb{F}_{p^r}$ set $P_{E,0} = |0\rangle\langle 0|$ and let $P^{\times}_{E,d}$ project onto the nonzero labels of exact absolute Frobenius period d, for $d \mid r$. Define

$$b_1(t) = t - 1, \qquad b_d(t) = \sum_{e|d} \mu(d/e)t^e \quad (d > 1),$$

where μ is zero on integers divisible by a prime square and otherwise is (-1) to the number of distinct prime factors. Put

$$D_E(h) = P_{E,0} + \sum_{d|r} \frac{b_d(e^h)}{b_d(p)} P^{\times}_{E,d}, \qquad \rho_E(h) = e^{-rh} D_E(h).$$

A period-uniform density lies in the real span of these orthogonal projections, without off-diagonal blocks. Its prescribed masses are e^{-rh} at zero and $e^{-rh} b_d(e^h)$ on the nonzero period blocks. For an ordered word $E_1, \ldots, E_m$ use tensor products D_{word} and ρ_{word}, and put $R = \sum_j r_j$. The empty word has $D = \rho = 1$ and $R = 0$. This reference is additional preparation data.

Scope. The actual maximally mixed preparation remains its original constant arithmetic map. The new reference varies a state, not a field cardinality or the matrices of the arithmetic gates.

Provenance. ids: D1501; proved in: definitions.md; tested by: positive arithmetic checker P1; reviewed in: positive-arithmetic-adjudication.md.

Definition 27.2 (Positive reference coefficients)**.** Let $A_{E,0} = P_{E,0}$ and, for $k \geq 1$, define

$$J_k(d) = \sum_{e|d} \mu(d/e)e^k = d^k \prod_{\ell | d \, \text{prime}} (1 - \ell^{-k}),$$

$$A_{E,k} = \sum_{d|r} \frac{J_k(d)}{k! b_d(p)} P_{E,d}^{\times}, \qquad B_E = A_{E,1}.$$

The empty product gives $J_k(1) = 1$. Set $\sigma_{E,0} = P_{E,0}$ and $\sigma_{E,k} = k! A_{E,k}/r^k$ for $k \geq 1$. Word coefficients are

$$A_{\text{word},k} = \sum_{k_1+\cdots+k_m=k} \bigotimes_j A_{E_j,k_j}.$$

For nonempty words define $\sigma_{\text{word},k} = k! A_{\text{word},k}/R^k$ when $k \geq 1$, and use the all-zero projection at grade zero.

Scope. These grades refer to powers of h, not to a Clifford hierarchy level. The full positive series will have a separate finite sufficient profile.

Provenance. ids: D1502; proved in: definitions.md; tested by: positive arithmetic checker P1, P5; reviewed in: positive-arithmetic-adjudication.md.

Proposition 27.3 (A unique positive reference and its Poisson decomposition)**.** PROVED
For every actual finite field $E = \mathbb{F}_{p^r}$, $\rho_E(h)$ is the unique period-uniform state with the prescribed masses. It is faithful for $h > 0$, equals I/p^r exactly at $h = \log p$, and tends to $P_{E,0}$ at zero. The unnormalized density has the entire trace-norm convergent expansion

$$D_E(h) = \sum_{k \geq 0} h^k A_{E,k}, \qquad \operatorname{Tr} A_{E,k} = \frac{r^k}{k!}, \qquad \rho_E(h) = e^{-rh} \sum_{k \geq 0} \frac{(rh)^k}{k!} \sigma_{E,k}.$$

For $k \geq 1$, $A_{E,k}$ is positive definite on the nonzero-label subspace. On a nonempty independent word,

$$\sigma_{\text{word},k} = \sum_{k_1+\cdots+k_m=k} \frac{k!}{\prod_j k_j!} \prod_j \left(\frac{r_j}{R}\right)^{k_j} \bigotimes_j \sigma_{E_j,k_j}.$$

Thus grades compose with multinomial weights, binomial for two factors. On the full matrix algebra the reference is tracial only at $h = \log p$.

Scope. The coefficient states and register dimensions retain p. Uniqueness uses the explicitly stated scalar action on each period block, not only agreement of the diagonal entries of an otherwise arbitrary density.

Provenance. ids: POS-STATE; proved in: reference-and-protocol.md, Sections 1–2; tested by: positive arithmetic checker P1, P7; reviewed in: positive-arithmetic-adjudication.md.

Derivation. Expanding the divisor sum for $b_d(e^h)$ gives $\sum_{k\geq 1} h^k J_k(d)/k!$, including $d=1$. Every $J_k(d)$ is positive by its prime-factor product, and $\sum_{d|r} J_k(d) = r^k$ by cancellation of the inner Moebius divisor sum. The nonzero period block has rank $b_d(p)$, so the displayed trace formulas follow. The masses fix the scalar on each block uniquely. Comparing the zero and one basis entries shows that their commutator test vanishes only when $e^h = p$, where the density is indeed scalar. Independent multiplication of the convergent series gives the multinomial weights. □

Definition 27.4 (Finite positive reference profile)**.** Put

$$L_E(h) = P_{E,0} + hB_E, \qquad L_{\mathrm{word}}(h) = \bigotimes_j L_{E_j}(h), \qquad \lambda_{\mathrm{word}}(h) = \frac{L_{\mathrm{word}}(h)}{\operatorname{Tr} L_{\mathrm{word}}(h)}.$$

A positive operator polynomial is a finite sum $A(h) = \sum_{k=0}^{m} h^k A_k$ with every coefficient positive in its finite block matrix algebra; zero coefficients are retained. For a subset S of word positions define

$$P_S = \bigotimes_j \begin{cases} I - P_{E_j,0} & j \in S, \\ P_{E_j,0} & j \notin S, \end{cases} \qquad B_S = \bigotimes_j \begin{cases} B_{E_j} & j \in S, \\ P_{E_j,0} & j \notin S. \end{cases}$$

Then $L_{\mathrm{word}} = \sum_S h^{|S|} B_S$ and $\operatorname{Tr} L_{\mathrm{word}} = \prod_j (1 + r_j h)$.

Scope. D and L are different families. The finite profile retains all data needed for the leading operational questions proved below, rather than all subleading probabilities at finite h.

Provenance. ids: D1503; proved in: definitions.md; tested by: positive arithmetic checker P5; reviewed in: positive-arithmetic-adjudication.md.

27.2 All named embeddings and arithmetic counting diagnostics

Proposition 27.5 (Reference restriction through every finite-field tower)**.** PROVED *For every named embedding $i : K \to E$, with absolute degrees s, r, in every characteristic and relative degree,*

$$J_i^* D_E(h) J_i = D_K(h), \qquad J_i^* A_{E,k} J_i = A_{K,k}, \qquad J_i^* L_E(h) J_i = L_K(h).$$

The retained inclusion decoder on $\rho_E(h)$ succeeds with probability $e^{(s-r)h}$ and returns $\rho_K(h)$ on success. These restrictions and successful probabilities compose exactly in all named towers. For the transported reference $F_E \rho_E(h) F_E^*$, the retained V_i decoder has the same success probability and returns $F_K \rho_K(h) F_K^*$. Its failure output is also the Fourier transport of the corresponding inclusion-decoder failure. Frobenius commutes with each reference coefficient. For λ_E the inclusion success probability is $(1+sh)/(1+rh)$ and the conditional output is λ_K.

Scope. No inverse relative degree is used. Canonical references are natural under named field embeddings and isomorphisms, not under arbitrary Hilbert-space coordinate identifications. Fourier transports a reference to its actual conjugate, which need not equal the canonical reference.

Provenance. ids: POS-DIAGRAM; proved in: reference-and-protocol.md, Section 3; tested by: positive arithmetic checker P2, P3; reviewed in: positive-arithmetic-adjudication.md.

Derivation. An embedding preserves and reflects $x^{p^d} = x$ by injectivity, so it preserves exact absolute Frobenius period and zero. The coefficient attached to a period uses $b_d(p)$ independently of the ambient field, proving all three restrictions. Trace normalization gives the success probability. The Fourier transfer identity gives the success square, while $(I - V_i V_i^*) F_E = F_E (I - J_i J_i^*)$ gives the failure square. □

Proposition 27.6 (Frobenius moments and active multiplication mass). PROVED *For every finite field $E = \mathbb{F}_{p^r}$ and integer k,*

$$\operatorname{Tr}(\rho_E(h)U_E^k) = e^{(\gcd(r,k)-r)h}, \qquad \gcd(r,0) = r.$$

On $d \geq 1$ independently prepared E controls and one E target, use the actual multiplication gate

$$M|x_1,\ldots,x_d,z\rangle = |x_1,\ldots,x_d,z+\textstyle\prod_j x_j\rangle.$$

The probability of all controls being nonzero, its gate trace, and squared-difference expectation are

$$w_d(h) = (1-e^{-rh})^d, \qquad \operatorname{Tr}(\rho_E(h)^{\otimes(d+1)}M) = 1 - w_d(h),$$
$$\operatorname{Tr}(\rho_E(h)^{\otimes(d+1)}(M-I)^*(M-I)) = 2w_d(h).$$

The active mass has order d with leading coefficient r^d. For $r \geq 2$, conditioning on nonfixed Frobenius basis labels and restricting to the marked common matrix action on every length-d orbit gives weights

$$\frac{b_d(e^h)}{e^{rh}-e^h} \longrightarrow \frac{\varphi(d)}{r-1}, \qquad d \mid r,\ d > 1,$$

with normalized block trace Tr/d. Here $\varphi(d) = J_1(d)$ counts integers from one through d coprime to d.

Scope. These are specified reference expectations, not a tracial-state assertion or a rule obtained by continuing every coupled arithmetic point count. At $r = 1$ the moving-label conditional sector is absent.

Provenance. ids: POS-COUNTS; proved in: reference-and-protocol.md, Section 4; tested by: positive arithmetic checker P1, P6; reviewed in: positive-arithmetic-adjudication.md.

The first formula sums precisely the period blocks fixed by the given power. Multiplication fixes a basis tuple precisely when a control is zero; the product reference is diagonal, so the same fixed-label test computes its trace. This uses no translation invariance of the target state. On an orbit block, the scalar reference coefficient cancels the actual number of orbits, giving the stated conditional moving-orbit trace.

27.3 Processes, profiles and conditional boundary records

Definition 27.7 (Positive-series process category). Objects are finite tagged direct sums of matrix algebras, with trace the sum of their ordinary matrix traces. A scalar normalizer is $Z(h) = \sum_{k\geq 0} z_k h^k$ with $z_k \geq 0$, $z_0 > 0$, and finite value for every real $h \geq 0$. A process representative $X \to Y$ is (Φ, Z) with $\Phi(h) = \sum_{k\geq 0} h^k \Phi_k$, each Φ_k completely positive, satisfying

$$\operatorname{Tr}_Y \Phi_k(a) \leq z_k \operatorname{Tr}_X(a) \quad \text{for every } a \geq 0 \text{ and every } k.$$

It is normalized when equality holds throughout. Instruments retain their external output tags. Equality is $Z'\Phi = Z\Phi'$ coefficientwise with identical tags. Composition and tensor use Cauchy products and multiply normalizers; the identity is $(\mathrm{id}, 1)$. Evaluation is $\Phi(h)/Z(h)$.

Scope. This is a semantic CP category, with ordinary trace normalization. It does not assert a faithful encoding of distinct source Kraus presentations. Leading normalized states are not the equality relation.

Provenance. ids: D1504; proved in: definitions.md; tested by: positive arithmetic checker P5; reviewed in: positive-arithmetic-adjudication.md.

Definition 27.8 (Arithmetic profile experiments and boundary records)**.** Retain every actual constant arithmetic CP process. Its action on a positive polynomial is $\Phi_* A = \sum_k h^k \Phi(A_k)$. A profile experiment object is (X, A), and an arrow $(X, A) \to (Y, C)$ is such a constant map with $C = \Phi_* A$; equality means equality of its typed CP map. Identity, composition and tensor use the actual matrix operations, with Cauchy tensor on profiles. Zero profiles are allowed but have no conditional state. The explicitly larger envelope may use all finite-dimensional constant CP maps. A finite reference protocol is a finite circuit of actual constant arithmetic maps and either ρ_E or λ_E preparations, with finitely many reference registers, named measurement placement and retained histories. Coherent amplitude compositions use the actual matrices before any prescribed CP instrument is applied. For a nonzero positive series $A(h) = \sum_k h^k A_k$, define

$$v(A) = \min\{k : A_k \neq 0\}, \quad M(A) = A_{v(A)}, \quad w(A) = \operatorname{Tr} M(A), \quad \eta(A) = M(A)/w(A).$$

The boundary record is (v, M), equivalently (v, w, η). With scalar normalizer having constant term z_0, its event probability has leading coefficient w/z_0, order v, and conditional limit η. Conditioning divides by the actual positive branch trace. The full profile is retained before later conditioning or continuation.

Scope. Canonical reference profiles are marked preparation data, not objects preserved by arbitrary arithmetic channels. Leading normalization alone is not an arrow or a functor on conditioned experiments.

Provenance. ids: D1505, D1506; proved in: definitions.md; tested by: positive arithmetic checker P4, P5; reviewed in: positive-arithmetic-adjudication.md.

Proposition 27.9 (Positive composition with the full arithmetic source)**.** PROVED *The positive-series processes form a symmetric monoidal category with continuous CP, ordinary-trace-nonincreasing evaluation for every $h \geq 0$. It contains every actual constant arithmetic CP map as $(\Phi, 1)$ and the normalized preparation representatives*

$$(z \mapsto z D_E(h), e^{rh}), \qquad (z \mapsto z L_E(h), 1 + rh).$$

All actual amplitude equations and coherent arithmetic compositions retain their original matrices. Profile experiments form a symmetric monoidal category under coefficientwise CP action; all named instrument outcomes and sequential histories can be retained.

Scope. The broader series category allows additional positive-series processes. The finite sufficient profile theorem below concerns fixed CP branches and finite protocols generated by the stated reference preparations and constant arithmetic processes. The two scopes are distinct.

Provenance. ids: POS-PROCESS; proved in: finite-operational-boundary.md, Sections 1–3; tested by: positive arithmetic checker P4, P5; reviewed in: positive-arithmetic-adjudication.md.

Positive coefficients remain CP under Cauchy composition and tensor. Their ordinary trace bounds multiply with the scalar normalizers and bound all matrix-entry series on compact intervals. Cross-multiplied equality respects composition because scalar series commute with the maps and a scalar series with nonzero constant term can be cancelled coefficientwise. The reference preparations satisfy coefficient normalization because $\operatorname{Tr} A_{E,k} = r^k/k!$.

Proposition 27.10 (A finite sufficient profile for every future CP branch)**.** PROVED *Let $E_1, \ldots, E_m$ be any finite reference word and let Φ be any nonzero fixed finite-dimensional CP map on its full register. Then $\Phi(D_{\text{word}})$ and $\Phi(L_{\text{word}})$ have the same first nonzero matrix coefficient and the same vanishing order, at most m. The conclusion remains valid after arbitrary later fixed CP continuation and independent tensor, including entangling maps. Consequently every finite reference protocol has identical leading event orders, probability coefficients and conditional output densities under the exact and finite-profile preparations. Keeping the full polynomial suffices for every such future branch question.*

Scope. Subleading probabilities and entire finite-h families are not asserted equal. The degree bound counts all reference registers allocated to a finite protocol, including those reserved for alternative adaptive histories. The zero branch has no conditional limit under either preparation.

Provenance. ids: POS-FINITE; proved in: finite-operational-boundary.md, Sections 4–6; tested by: positive arithmetic checker P5; reviewed in: positive-arithmetic-adjudication.md.

Derivation. On each nonzero support mask S, the exact word begins as $h^{|S|}B_S$ and has only positive higher coefficients supported on that mask. The matrix B_S is positive definite there. If $\Phi(B_S) = 0$, positivity forces Φ to annihilate every positive matrix on the mask: such a matrix is bounded above by a scalar multiple of B_S. Linearity then kills the whole supported algebra. Thus a killed mask contributes no later coefficient. The first surviving mask size supplies precisely the same leading sum $\sum_{|S|=v} \Phi(B_S)$ in the exact and polynomial families. Some mask survives when $\Phi \neq 0$, because $\sum_S B_S$ is faithful on the full input space; hence $v \leq m$. Any later branch is another fixed CP composite. A finite protocol may allocate its independent reference inputs at the beginning and discard those unused on a history, reducing every branch to this same argument. Both word normalizers have constant term one, so their event coefficients and normalized leading densities also agree. □

The retained finite grades have a direct probabilistic meaning. A factor λ_E is a mixture of $P_{E,0}$ and B_E/r with activation probability $rh/(1 + rh)$. On a word, the grade-k conditional mixture runs over subsets S of size k, with weights proportional to $\prod_{j \in S} r_j$. The maximal grade m is needed for the event that all reference labels are nonzero. A leading state alone fails even on one field: it is $P_{E,0}$, but a later nonzero-label test has a real order-one branch hB_E.

27.4 One protocol detects the arithmetic operations together

Definition 27.11 (Relative-Frobenius multiplication protocol)**.** For a named proper embedding $i : K \to E$, let $q = p^s = |K|$ and choose named $a, b \in E$ satisfying $T_i(ab) = 0$ and $T_i(a^q b) = 1$. Prepare the reference word E, E, K, retaining the outcome selecting its computational labels $a, b, 0$. Apply V_i to the target, U_E^s to the first control, the actual multiplication gate

$$M_E^{(2)}|x, y, z\rangle = |x, y, z + xy\rangle,$$

and then F_E to the target. Apply the retained J_i decoder, whose success amplitude is J_i^* and failure amplitude is $I - J_i J_i^*$. On success apply F_K^* and measure the target label. All failures remain tagged. An optional coherence test dephases the target in its E basis immediately after V_i. Either exact or finite reference preparations may be used, with their boundary records compared as above.

Scope. The elements a, b are named choices. Their existence for every proper extension and the observable protocol outcomes are the following result.

Provenance. ids: D1507; proved in: definitions.md; tested by: positive arithmetic checker P4, P6; reviewed in: positive-arithmetic-adjudication.md.

Proposition 27.12 (A uniform mixed arithmetic boundary distinction)**.** PROVED *Every proper named finite-field embedding admits the choices a, b above. Its rare preparation has order two and positive leading coefficient $w_a w_b$, where $w_x = \varphi(d)/b_d(p)$ for a nonzero label of period d. The full protocol has decoder success one and final target $|1_K\rangle$. Removing relative Frobenius or multiplication gives $|0_K\rangle$. Removing target Fourier gives decoder success $1/\kappa_i$ when the relative degree is invertible in p, and zero when it is divisible by p. Dephasing the trace-fibre target changes decoder success from one to $1/\kappa_i$, with final target still $|1_K\rangle$ on success. For $\mathbb{F}_2 \subset \mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$, the choices $a = \alpha, b = \alpha^2$ have preparation coefficient $1/4$; changing the multiplication entry $\alpha^2\alpha^2 = \alpha$ to zero changes the observed target from one to zero.*

Scope. The result covers every proper finite extension, including characteristic-dividing degrees. Output labels zero and one are uniform; the preparation coefficient, matrices and dimensions retain characteristic and field data. Coherent trace-fibre evolution is part of the experiment.

Provenance. ids: POS-MIXED; proved in: reference-and-protocol.md, Section 5; tested by: positive arithmetic checker P4, P6, P7; reviewed in: positive-arithmetic-adjudication.md.

Derivation. Choose a_0 outside K. If a_0, a_0^q are dependent, write $a_0^q = \lambda a_0$ with $\lambda \neq 1$ and use $a = a_0 + 1$; its two coefficients in the independent pair $a_0, 1$ show that a, a^q are independent. Otherwise use $a = a_0$. Nondegeneracy of the relative trace pairing makes $b \mapsto (T_i(ab), T_i(a^q b))$ surjective onto K^2, so the prescribed b exists. Multiplication translates $V_i|0\rangle$ to $V_i|1\rangle$, and Fourier transports that to $J_i F_K|1\rangle$. The decoder and final inverse Fourier therefore return one. Without Frobenius or multiplication the same computation returns zero. Without Fourier, the actual Gram column $J_i^* V_i|1\rangle$ has squared norm $1/\kappa_i$ in the invertible-degree branch and vanishes in the singular branch. A dephased trace fibre gives a mixture of basis vectors, each with code success $1/\kappa_i$, which verifies the coherence test. □

27.5 What the boundary retains

For a concrete positivity comparison, take $\mathbb{F}_2 \subset \mathbb{F}_8$. The complement of the joint inclusion/trace-fibre code space has rank four and kills the basis vectors $0, 1$. Each of the six other labels has period three and reference weight $(t^3 - t)/(6t^3)$. Hence the new expectation of this complementary projection is $\frac{2}{3}(1 - t^{-2})$, while the earlier formal tracial prescription was $1 - 2t^{-2}$. Both agree with the actual arithmetic trace at $t = 2$. The positive extension uses a different reference state, as illustrated in Figure 3.

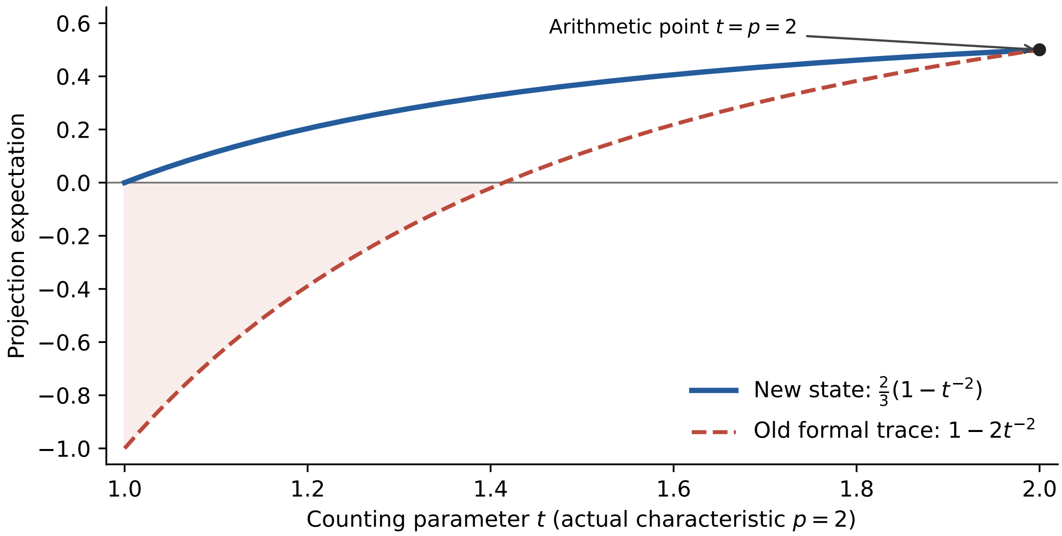

Figure 3: A full positive state replaces the failed tracial continuation. The arithmetic data are fixed at $\mathbb{F}_2 \subset \mathbb{F}_8$; only the counting reference varies. The shaded negative values cannot be projection probabilities.

The construction preserves actual arithmetic matrices and their relations, and supplies a new positive normalization and finite operational rule. One universal consequence is particularly simple. For a relative extension $\mathbb{F}_{p^s} \subset \mathbb{F}_{p^{sn}}$, the first nonzero-sector state satisfies, for every integer j,

$$\mathrm{Tr}\big(\sigma_{E,1} U_E^{sj}\big) = \frac{\gcd(n, j)}{n}.$$

Indeed the coefficient of h in $\mathrm{Tr}(D_E(h)U_E^{sj}) = e^{\gcd(sn,sj)h}$ is $\mathrm{Tr}(B_E U_E^{sj})$, and $\sigma_{E,1} = B_E/(sn)$. The relative Frobenius signature therefore depends only on the extension degree. The full construction does not impose characteristic independence. For a quadratic field, the nonzero boundary state $B_E/2$ has Frobenius first moment $1/2$ for both $p = 2$ and $p = 3$. At $p = 2$ its basis weights are $1/2$ on the nonzero prime-field label and $1/4$ on each other label; at $p = 3$ they are $1/4$ on each nonzero prime-field label and $1/12$ on each of the six other labels.

Nor does this rule continue every coupled point count by substituting t for p. The event $x = y \neq 0$ on two independently prepared E references has probability $\sum_{x\neq 0} \rho_E(x)^2$ and order two, consistently with its containment in the event that both labels are nonzero. The actual copy isometry $|x\rangle \mapsto |x, x\rangle$ transports one reference and gives that event order one. Its output is correlated and is not reset to two independent canonical references. Likewise a prime-field coordinate unitary transports a field reference to its actual, generally correlated image.

The earlier two-chart continuation varies angle matrices. The present construction keeps those actual matrices fixed and varies preparations. Agreement with its normalized tangent or weighted tower connector is not automatic. A comparison must specify its reference and channel maps and prove the relevant diagrams. The concrete surviving content here is the positive period normalization, a finite sufficient profile for every future arithmetic branch, and the mixed zero-versus-one protocol above.

28 Relative Frobenius in a composite quantum boundary

Arithmetic composition can retain quantum information that separate invariant reductions discard. The construction in this section assembles three actual field registers: two labels and their product. Simultaneous Frobenius is then removed by invariant projection, while Frobenius on one factor, transported through multiplication, acts on a relative coordinate. The relative coordinate is a quantum register of dimension equal to the extension degree. A specified preparation vanishes at the ordinary counting endpoint, but its first nonzero grade retains that register and an experiment detecting both Frobenius and coherence.

The six statements below passed the capped review with a repaired degree-one scope clause in the spectral addendum. They concern a conditional quantum boundary of already assembled arithmetic systems. They impose neither a characteristic-one arithmetic category nor tensor and direct-sum operations on the resulting collection of boundaries.

28.1 Arithmetic composition before specialization

For a named field $E = \mathbb{F}_{p^r}$, the Hilbert register is $H_E = \ell^2(E)$ with orthonormal basis $|x\rangle$. Fix a primitive pth root ζ_p and the additive character $\psi_E(x) = \zeta_p^{\mathrm{Tr}_{E/\mathbb{F}_p}(x)}$. The arithmetic amplitudes include zero preparations, shifts, phases, Frobenius, the Fourier transform, multiplication into a separate target, and the named field inclusion and trace-fibre isometries:

$$X_E(a)|x\rangle = |x + a\rangle, \qquad Z_E(b)|x\rangle = \psi_E(bx)|x\rangle,$$

$$U_E|x\rangle = |x^p\rangle, \qquad F_E|x\rangle = |E|^{-1/2} \sum_y \psi_E(-xy)|y\rangle,$$

$$M_E^{(d)}|x_1, \ldots, x_d, z\rangle = |x_1, \ldots, x_d, z + \prod_j x_j\rangle.$$

For a named embedding $i : K \to E$, let $T_i(y) = i^{-1}(\mathrm{Tr}_{E/i(K)}\, y)$ and $\kappa_i = |E|/|K|$. Then

$$J_i|a\rangle = |i(a)\rangle, \qquad V_i|a\rangle = \kappa_i^{-1/2} \sum_{T_i(y)=a} |y\rangle.$$

These maps act on ordered independent register words. All square roots are positive, and the Weyl ordering is $W(a,b) = Z(-b)X(a)$.

Provenance. ids: D8, D1301–D1304, D1306, D1308, FRB-FROB, FRB-TRANSFER; proved in: foundation.md; transfers-example.md; tested by: arithmetic Frobenius checkers; reviewed in: frobenius-hierarchy-adjudication.md.

Definition 28.1 (Completed concrete arithmetic processes)**.** Fix a finite named field diagram in characteristic p. Take the actual matrix image of its arithmetic amplitudes, with scalars in $\mathbb{Q}(\zeta_p, \sqrt{p})$ and equality of typed actual matrices. Its finite additive completion has finite lists of objects and rectangular matrices of arrows. Its self-adjoint projection completion has objects (X, e) with $e = e^* = e^2$ an actual endomorphism, represented on $\operatorname{ran} e$, and arrows $f : (X, e) \to (Y, g)$ satisfying $f = gfe$. Independent tensor uses the tensor of words, projections and arrows.

Retained processes are finite Kraus lists of these arrows with $\sum_j K_j^* K_j \le I$; normalized lists have equality. Classical output tags use block algebras and the sum of ordinary matrix traces. Coherent additive sums retain the source's rectangular Hom blocks. Additional reference states are the period references and finite profiles specified below, together with fixed zero preparations. Collective assembly means a specified actual isometry, arithmetic circuit and projection.

Scope. No surjectivity onto all complex CP maps or new faithfulness theorem for the older syntactic source is asserted. Coherent sums and classical tags are different constructions.

Provenance. ids: D1601; proved in: definitions.md; tested by: composite boundary checker C1, C6; reviewed in: composite-boundary-adjudication.md.

Proposition 28.2 (Positive processes on completed arithmetic systems)**.** PROVED *The completed concrete arithmetic source has finite coherent sums, independent tensor and self-adjoint retracts. Its Kraus lists realize completely positive, trace-nonincreasing maps; normalized lists realize CPTP maps. Their compositions and independent tensors retain these properties, with every classical outcome kept when specified.*

Scope. These are prelimit constructions. Their total descent to the selected boundary objects is not required or inferred.

Provenance. ids: CMP-SOURCE; proved in: relational-construction.md, Section 1; tested by: composite boundary checker C1, C6; reviewed in: composite-boundary-adjudication.md.

28.2 A relative coordinate inside a multiplication graph

Definition 28.3 (Relative multiplication-graph register)**.** Fix a named extension E/K with $|K| = q = p^s$, $|E| = q^n = p^r$ and $r = sn$. Put $\sigma(x) = x^q$ and $U|x\rangle = |\sigma(x)\rangle$. For every σ-orbit O of length $d \mid n$ and $k \in \mathbb{Z}/d$, put

$$v_{O,k} = \frac{1}{\sqrt{d}} \sum_{x \in O} |x, \sigma^k(x), x\sigma^k(x)\rangle.$$

Let $G_{d,k}$ be the diagonal projector onto precisely the tuples in these sums as O varies over length-d orbits. Define

$$G = \sum_{d|n} \sum_{k=0}^{d-1} G_{d,k}, \qquad \Delta = U \otimes U \otimes U, \qquad T = \frac{1}{n} \sum_{j=0}^{n-1} \Delta^j,$$

and set

$$P = TG, \qquad P_d = T \sum_k G_{d,k}, \qquad Q_{d,k} = TG_{d,k}.$$

The relational register is $H^{\mathrm{rel}}_{E/K} = \operatorname{ran} P$ with its inherited inner product. For every $n \geq 1$ define the copied graph map $C : H_E \to H_E^{\otimes 3}$ by $C|x\rangle = |x, x, x^2\rangle$.

Scope. No orbit origin is chosen. The averages use rational complex scalars; they never invert n in the finite field. Projector and range properties are established by the next proposition.

Provenance. ids: D1602; proved in: definitions.md; tested by: composite boundary checker C2, C3; reviewed in: composite-boundary-adjudication.md.

Definition 28.4 (Relative Frobenius and common observables)**.** For the preceding named extension use $M = M_E^{(2)}$ and define

$$R = M(I \otimes U \otimes I)M^*, \qquad E_{d;a,b} = Q_{d,a}R^{a-b}Q_{d,b}.$$

Let the common observable algebra be the complex star algebra on $\operatorname{ran} P$ generated by R and the $Q_{d,k}$. Its proposed abstract comparison is

$$B_n = \bigoplus_{d|n} M_d(\mathbb{C}), \qquad e_{d;a,b} \longmapsto E_{d;a,b}.$$

Write $S_d|k\rangle = |k+1 \pmod d\rangle$ and $u_n^{\mathrm{rel}} = \bigoplus_{d|n} S_d$. The primitive relative component is $d = n$; its common algebra is $M_n(\mathbb{C})$.

Scope. The common algebra does not include arbitrary operators on or between distinct orbit multiplicities. The arithmetic realization still retains those multiplicity spaces.

Provenance. ids: D1603; proved in: definitions.md; tested by: composite boundary checker C2, C3; reviewed in: composite-boundary-adjudication.md.

Proposition 28.5 (An arithmetic matrix algebra after collective descent)**.** **PROVED** *For every named finite extension E/K, the vectors $v_{O,k}$ form an orthonormal basis of $H^{\mathrm{rel}}_{E/K}$. The operators $P_d, Q_{d,k}$ are orthogonal projections. Simultaneous Frobenius fixes this register, while actual relative Frobenius satisfies*

$$\Delta v_{O,k} = v_{O,k}, \qquad R v_{O,k} = v_{O,k+1}.$$

The displayed comparison identifies the common observable algebra with B_n, represented with $c_d(q)/d$ copies of $M_d(\mathbb{C})$, where $c_d(q) = \sum_{a|d} \mu(d/a)q^a$ counts exact-period labels. On every $d > 1$ block, $\operatorname{Ad} R$ *is a nonidentity CPTP automorphism and the observables are noncommutative.*

Scope. Relative Frobenius is Frobenius on one factor transported through actual multiplication. It is different from the simultaneous Frobenius removed by invariant projection.

Provenance. ids: CMP-REL; proved in: relational-construction.md, Sections 2–3; tested by: composite boundary checker C2, C3; reviewed in: composite-boundary-adjudication.md.

Arithmetic calculation. On a tuple, multiplication and its inverse give

$$R(x, y, z) = (x, \sigma(y), z - xy + x\sigma(y)).$$

Thus the multiplication graph is preserved and its relative index advances. Simultaneous Frobenius permutes the summands of each orbit sum. Its average is the rank-one projector onto that sum, since the d distinct tuples repeat n/d times in the average. The graph tests select a relative index; conjugating them by R supplies every common matrix unit. In particular $[R, Q_{d,0}]v_{O,0} = v_{O,1} \neq 0$ when $d > 1$. $\square$

For one orbit, the separately fixed subspace is the line spanned by $f_O = d^{-1/2}\sum_{x\in O}|x\rangle$. Multiplying two separately fixed copies into a zero target produces only

$$\frac{1}{d}\sum_{x,y\in O}|x,y,xy\rangle = \frac{1}{\sqrt{d}}\sum_k v_{O,k}.$$

This is one vector in the d-dimensional jointly descended register. The comparison concerns orbitwise invariant reduction, not a claim that the full single-field observable algebra becomes scalar.

28.3 A vanishing preparation with a surviving quantum experiment

For the reference used here, let $h \geq 0$, $t = e^h$, and let $P^{\times}_{E,a}$ project onto nonzero labels of exact absolute Frobenius period $a \mid r$. Put

$$b_1(t) = t - 1, \qquad b_a(t) = \sum_{e\mid a}\mu(a/e)t^e \quad (a > 1), \qquad D_E(h) = |0\rangle\langle 0| + \sum_{a\mid r}\frac{b_a(e^h)}{b_a(p)}P^{\times}_{E,a}.$$

Then $\rho_E(h) = e^{-rh}D_E(h)$ is the period reference, with $\rho_E(\log p) = I/|E|$. Its finite profile is

$$L_E(h) = |0\rangle\langle 0| + hB_E, \qquad B_E = \sum_{a\mid r}\frac{\varphi(a)}{b_a(p)}P^{\times}_{E,a}, \qquad \lambda_E(h) = \frac{L_E(h)}{1+rh}.$$

For a nonzero positive output series $A(h) = \sum_k h^k A_k$, retain its first nonzero order v, coefficient $M = A_v$, trace $w = \operatorname{Tr} M$ and conditional state M/w. For normalizers with constant term one, w is the leading event-probability coefficient. Full profiles are retained before later conditioning.

Provenance. ids: D1501–D1503, D1506, POS-STATE, POS-FINITE; proved in: reference-and-protocol.md; finite-operational-boundary.md; tested by: positive arithmetic checker; reviewed in: positive-arithmetic-adjudication.md.

Definition 28.6 (Copied reference instrument)**.** For E/K of relative degree $n > 1$, let Π_n project onto labels of exact relative Frobenius period n. Prepare one $\rho_E(h)$ or $\lambda_E(h)$ and two fixed zero ancillas. Define the arithmetic isometry $C|x\rangle = |x, x, x^2\rangle$ by controlled addition and multiplication. The three preparation amplitudes are

$$K_{\text{cut}} = I - \Pi_n, \qquad K_{\text{fail}} = (I - T)C\Pi_n, \qquad K_{\text{ok}} = TC\Pi_n.$$

Their tags are primitive-cut failure in H_E, averaging failure in $H_E^{\otimes 3}$, and success in that same three-register word. After success apply R and measure $Q_{n,1}$ with its complement retained. In the identity comparison replace R by I. In the dephased comparison insert computational dephasing of all three slots immediately before R.

Scope. The two fixed zero ancillas are not independently prepared period references. The final effect includes the invariant projection; a diagonal graph test alone would not detect the stated coherence.

Provenance. ids: D1604; proved in: definitions.md; tested by: composite boundary checker C4–C6; reviewed in: composite-boundary-adjudication.md.

Definition 28.7 (Primitive relative quantum boundary)**.** Retain the leading event record of the copied reference instrument and its conditional state on the common $M_n(\mathbb{C})$ algebra. The proposed boundary realization has state $|0\rangle\langle 0|$, relative-index projections, the common matrix units and the channel $\operatorname{Ad} S_n$, where S_n is the single n-cycle. Its arithmetic processes are those induced by these displayed matrices and specified instruments. A larger category containing all CP maps is an optional envelope. Physical multiplicity states at different characteristics are not identified.

Scope. No tensor or direct-sum operation between boundary objects is stipulated. The full physical profile remains necessary for arbitrary later retained arithmetic continuations.

Provenance. ids: D1605; proved in: definitions.md; tested by: composite boundary checker C3–C6; reviewed in: composite-boundary-adjudication.md.

Theorem 28.8 (A quantum Frobenius witness in the first nonzero grade)**.** PROVED *For every finite E/K with $|K| = p^s$ and relative degree $n > 1$, in every characteristic, the copied reference instrument is CPTP with its stated histories. Its successful preparation has probability*

$$w_{s,n}(h) = \frac{c_n(t^s)}{nt^{sn}}, \qquad t = e^h, \qquad w_{s,n}(h) = \frac{s\varphi(n)}{n}h + O(h^2).$$

For $h > 0$, its conditional state on the common M_n block is $|0\rangle\langle 0|$. The final test has conditional probabilities

circuit	*relative Frobenius*	*identity*	*dephased*
probability	1	0	$1/n$

The finite-profile preparation has success $hs\varphi(n)/(n(1+snh))$ and the same leading event records.

Scope. The primitive branch is zero at the ordinary endpoint. The theorem concerns its specified graded conditional remnant. It preserves the arithmetic realization and its histories without identifying every physical state or CP map across characteristics.

Provenance. ids: CMP-BOUNDARY; proved in: boundary-and-spectrum.md, Sections 1–2; tested by: composite boundary checker C4–C6; reviewed in: composite-boundary-adjudication.md.

Derivation. The unnormalized reference mass on the fixed field of σ^d is t^{sd} by reference restriction. Divisor inversion therefore makes the primitive relative mass $c_n(t^s)$. For x in a primitive orbit, $TC|x\rangle = v_{O,0}/\sqrt{n}$, so the diagonal input succeeds with probability $1/n$ times that mass divided by t^{sn}. Differentiation at zero gives $s\varphi(n)/n$.

Relative Frobenius takes each prepared vector to $v_{O,1}$, while identity leaves it orthogonal to the final effect. Dephasing replaces each orbit sum by its uniform computational mixture. Each of its n tuples has squared overlap $1/n$ with the final invariant vector. The result persists under the physical mixture over orbits. □

The primitive-cut failure probability is $1 - c_n(t^s)/t^{sn}$; averaging failure has probability $(1 - 1/n)c_n(t^s)/t^{sn}$. The final test splits only the success history. Thus every omitted success is accounted for by a named physical output. The scalar $1/n$ concerns the diagonal reference: on coherent inputs $K^*_{\text{ok}}K_{\text{ok}}$ is an orbitwise fixed-vector projector, not Π_n/n.

28.4 The binary quadratic example

Take $E = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$ over $K = \mathbb{F}_2$. Its primitive orbit is $\{\alpha, \alpha^2\}$, and the two graph vectors are

$$v_0 = \frac{|\alpha, \alpha, \alpha^2\rangle + |\alpha^2, \alpha^2, \alpha\rangle}{\sqrt{2}},$$
$$v_1 = \frac{|\alpha, \alpha^2, 1\rangle + |\alpha^2, \alpha, 1\rangle}{\sqrt{2}}.$$

Their common algebra is a qubit algebra. Actual multiplication-conjugated relative Frobenius swaps them. The preparation probability is $w(h) = (1-e^{-h})/2$, hence has coefficient $1/2$ at first order. At the arithmetic point $t = 2$, preparation succeeds with probability $1/4$. The full final joint success is $1/4$, becomes zero when Frobenius is removed, and becomes $1/8$ when the orbit coherence is dephased. These distinctions are generated inside the three arithmetic registers.

Provenance. ids: CMP-REL, CMP-BOUNDARY; proved in: relational-construction.md; boundary-and-spectrum.md; tested by: composite boundary checker C2–C6; reviewed in: composite-boundary-adjudication.md.

28.5 Embedding comparisons and an actual Fourier return

Definition 28.9 (Relational transfers and retained Fourier return)**.** For a named embedding $i : E \to F$ over the same fixed K, restrict $J_i^{\otimes 3}$ to the all-period relational code and call it J_i^{rel}. Its decoder has success amplitude $(J_i^{\mathrm{rel}})^*$ and failure amplitude $I - J_i^{\mathrm{rel}}(J_i^{\mathrm{rel}})^*$, with different target tags. Towers retain every stopped history and length-d components remain length d in the upper ambient field.

On a primitive code of degree $n > 1$, put $F_{\mathrm{first}} = F_E \otimes I \otimes I$. The return and failure amplitudes are its restrictions followed by P_n and $I - P_n$, respectively; the failure retains the ambient word.

Scope. The embedding comparison fixes the base field K. No claim about changing the relative base or Fourier transforms on other slots is included.

Provenance. ids: D1606; proved in: definitions.md; tested by: composite boundary checker C7, C8; reviewed in: composite-boundary-adjudication.md.

Proposition 28.10 (Coherent transfers without orbit origins)**.** PROVED *Every such named embedding gives an isometry satisfying $J_i^{\otimes 3} v_{O,k} = v_{i(O),k}$. These isometries compose in every fixed-base tower and intertwine relative Frobenius and the component tests. Their retained decoders are CPTP, and success maps compose with every stopped failure history kept distinct.*

Scope. A lower primitive component is not replaced by the upper field's primitive component. No compatible orbit origins are needed, including for twisted field embeddings.

Provenance. ids: CMP-NATURAL; proved in: relational-construction.md, Section 4; tested by: composite boundary checker C7; reviewed in: composite-boundary-adjudication.md.

Proposition 28.11 (First-register Fourier compression)**.** PROVED *For every primitive graph code of relative degree $n > 1$,*

$$P_n F_{\mathrm{first}} P_n = |E|^{-1/2} D_{\mathrm{phase}} P_n, \qquad D_{\mathrm{phase}} v_{O,k} = \psi_E(-x^2) v_{O,k} \quad (x \in O).$$

The phase is independent of the selected x in O. Retained return has probability $1/|E|$ for every code state and conditional return is identity on the common M_n observables.

Scope. The physical multiplicity phases and the unconditioned return probability retain arithmetic data. The ambient failure branch is not discarded.

Provenance. ids: CMP-FOURIER; proved in: boundary-and-spectrum.md, Section 3; tested by: composite boundary checker C8; reviewed in: composite-boundary-adjudication.md.

Kernel calculation. Fourier in the first slot fixes the other two coordinates y, z. On the primitive graph $y \neq 0$ and $z = xy$ forces $x = z/y$. Only the diagonal Fourier kernel survives compression, namely $|E|^{-1/2}\psi_E(-x^2)$. Absolute trace is invariant along the relative Frobenius orbit. The compressed amplitude consequently has squared modulus $1/|E|$ on the entire code. Complementary projection supplies the other outcome. □

28.6 Which spectrum and which trace survive

Definition 28.12 (Finite relational spectral data)**.** Use ordinary Hilbert trace and $\det(I - zR)$ on the all-period relational code, with actual orbit multiplicities. Keep separately the normalized trace tr_d on each common M_d and the linear-map trace on its channel. To obtain a reference trace on the common blocks, apply the copied reference and invariant projection, retain success, then uniformly randomize over R^j, $0 \le j < n$, explicitly discarding the randomization label. The channel can use n copies of R^j/n for each j, so it needs no additional square root in the arithmetic scalar field. The copied map is defined for every $n \ge 1$; moving-sector conditioning requires $n > 1$. This preparation differs from the unrandomized witness.

Scope. These definitions concern finite systems. They posit no infinite trace, regularized determinant, or identification with a Riemann spectral operator.

Provenance. ids: D1607; proved in: definitions.md; tested by: composite boundary checker C9; reviewed in: composite-boundary-adjudication.md.

Proposition 28.13 (Finite spectral and reference identities)**.** PROVED *For every finite E/K and integer j, with* $\gcd(n, 0) = n$*,*

$$\mathrm{Tr}(R^j) = q^{\gcd(n,j)}, \qquad \det(I - zR) = \prod_{d|n} (1 - z^d)^{c_d(q)/d}.$$

On the full code algebra the channel's linear-map trace is instead $|\,\mathrm{Tr}(R^j)|^2$*. The primitive implementer has the nth roots of unity as eigenvalues, each once on its common block; its channel has their ratios, each root with multiplicity n.*

The randomized copied reference has unnormalized common-block functional

$$t^{-sn} \sum_{d|n} \frac{c_d(t^s)}{d} \, \mathrm{tr}_d(a_d).$$

For $n > 1$, conditioning its moving blocks $d > 1$ before taking the first nonzero grade gives positive normalized weights proportional to $\varphi(d)/d$. For $n = 1$ the moving sector is zero and has no conditional state.

Scope. Ordinary spectral trace and normalized reference expectation are distinct. The finite determinant identity supplies an arithmetic comparison for subsequent spectral work; no infinite completion is assumed here.

Provenance. ids: CMP-TRACE; proved in: boundary-and-spectrum.md, Section 4; tested by: composite boundary checker C9; reviewed in: composite-boundary-adjudication.md.

The finite trace identity follows by counting fixed vectors in each relative cycle. In particular, for $|z| < 1$,

$$\det(I - zR)^{-1} = \exp\left(\sum_{j \ge 1} \frac{z^j}{j} \, \mathrm{Tr}(R^j) \right).$$

The positive quantum remnant and this arithmetic spectral identity are already properties of an assembled system. Any later tower completion, rescaling of Frobenius iterates, or new boundary composition must be specified and justified independently.

29 Completing the arithmetic degree boundary

The relative quantum registers admit a concrete completion across extension degrees. Their finite comparison maps come from arithmetic field embeddings, and their unnormalized traces come from the first nonzero copied-reference coefficients. These two pieces of data determine the completion studied here. Its Frobenius action remains a legitimate quantum automorphism and becomes outer in the chosen minimal unitization. A separate positive degree regularization supplies normalized states and an explicit zeta quotient. The two statements in this section passed independent capped review; the tail cutoff is explicitly restricted to integers.

29.1 Divisor corners and their arithmetic provenance

For an extension of relative degree N, the moving common observable algebra consists of one matrix block $M_d(\mathbb{C})$ for each divisor $d > 1$ of N. Its relative Frobenius is the cyclic shift $S_d|k\rangle = |k+1 \pmod{d}\rangle$ in each block. The randomized copied reference has first coefficient $s\varphi(d)/d$ on that block, where s is the absolute degree of the base field. Dividing by this common positive factor leaves the weights used below.

Provenance. ids: D1602, D1603, D1607, CMP-REL, CMP-NATURAL, CMP-TRACE; proved in: relational-construction.md; boundary-and-spectrum.md; tested by: composite boundary checker; reviewed in: composite-boundary-adjudication.md.

Definition 29.1 (Divisor corners and a concrete degree completion)**.** For $N \geq 2$ put

$$A_N = \bigoplus_{\substack{d|N \\ d>1}} M_d(\mathbb{C}), \qquad \eta_d = \frac{\varphi(d)}{d}, \qquad \theta_N(a) = \sum_{\substack{d|N \\ d>1}} \eta_d \operatorname{tr}_d(a_d), \quad \operatorname{tr}_d = \frac{1}{d}\operatorname{Tr}.$$

Use the maximum of the block operator norms. For $N \mid M$, let $j_{N,M} : A_N \to A_M$ extend block coordinates by zero. On the Hilbert space

$$\mathcal{H}_{\mathrm{deg}} = \bigoplus_{d\geq 2}^{\ell^2} \mathbb{C}^d,$$

let A_{deg} be the block-diagonal algebra of sequences (a_d) with $\|a_d\| \to 0$, in the supremum norm, and let $B_{\mathrm{deg}} = A_{\mathrm{deg}} + \mathbb{C}I$ be its concrete minimal unitization. For $a \geq 0$ in A_{deg} define the extended-valued functional

$$\theta(a) = \sum_{d\geq 2} \eta_d \operatorname{tr}_d(a_d).$$

Scope. The inclusions are nonunital corners. This is a specified completion of the common arithmetic blocks; no total tensor or coherent direct-sum rule on the distinguished primitive systems is stipulated. The Hilbert-space symbol is not a Hamiltonian.

Provenance. ids: D1611; proved in: definitions.md; tested by: composite spectrum checker; reviewed in: composite-spectrum-adjudication.md.

The corner inclusion has an arithmetic interpretation. Fix a base $K = \mathbb{F}_q$ and $N \mid M$. Every Frobenius orbit of length $d \mid N$ in $\mathbb{F}_{q^M}$ is already contained in $\mathbb{F}_{q^N}$: its labels are fixed by the Nth Frobenius power. Thus the relational encoder carries the entire smaller period block to the entire same-period block upstairs. Conjugation by that encoder, extending by zero off its image, realizes $j_{N,M}$. In particular,

$$\theta_M \circ j_{N,M} = \theta_N.$$

After normalizing the finite traces, restriction instead has success $\theta_N(1)/\theta_M(1)$, with the smaller normalized reference as its conditional state.

29.2 Frobenius dynamics and positive degree states

Definition 29.2 (Completed Frobenius and degree regularization)**.** On $\mathcal{H}_{\deg}$ put

$$U_{\deg} = \bigoplus_{d\geq 2} S_d, \qquad \alpha(a)_d = S_d a_d S_d^*,$$

and let α fix the scalar unit of $B_{\deg}$. For a real $\beta > 1$, define

$$Z_{\deg}(\beta) = \sum_{d\geq 2} \frac{\varphi(d)}{d^{\beta+1}}, \qquad D_\beta = \bigoplus_{d\geq 2} d^{-\beta} I_d,$$

$$\rho_\beta = \bigoplus_{d\geq 2} \frac{\varphi(d)}{Z_{\deg}(\beta) d^{\beta+2}} I_d, \qquad \omega_\beta(a) = \mathrm{Tr}_{\mathcal{H}_{\deg}}(\rho_\beta a).$$

Define $\zeta(x) = \sum_{m\geq 1} m^{-x}$ for real $x > 1$, and for an integer j define the implementing-operator moment

$$M_\beta(j) = \mathrm{Tr}_{\mathcal{H}_{\deg}}(\rho_\beta U_{\deg}^j).$$

Scope. The regulator is additional central degree-weight data, not an arithmetic amplitude or a Frobenius generator. The moment uses the named Hilbert implementation; $U_{\deg}$ need not belong to $B_{\deg}$. No analytic continuation or unregularized infinite trace is stipulated.

Provenance. ids: D1612; proved in: definitions.md; tested by: composite spectrum checker; reviewed in: composite-spectrum-adjudication.md.

Theorem 29.3 (An arithmetic degree completion with outer Frobenius)**.** PROVED *The moving common algebras A_N with the fixed-base arithmetic corner maps form a directed system whose norm completion is $A_{\deg}$. The compatible finite traces extend to the faithful, lower-semicontinuous semifinite trace θ on $A_{\deg}$.*

Relative Frobenius extends to the unital CP automorphism α of $B_{\deg}$. This automorphism is outer in $B_{\deg}$ and is spatially implemented by $U_{\deg}$ on $\mathcal{H}_{\deg}$. The implementing unitary has all roots of unity as point spectrum, each with infinite multiplicity, and the whole unit circle as its spectrum.

Scope. Outerness refers specifically to the minimal unitization. The same action is inner in the larger product multiplier algebra. The trace is densely defined on the nonunital algebra, not asserted densely defined on its unitization. This is a discrete cycle dynamics, with no Riemann-zero identification or rescaled continuous-time generator asserted.

Provenance. ids: CMP-COMPLETE; proved in: degree-completion.md, Sections 1–2; tested by: composite spectrum checker; reviewed in: composite-spectrum-adjudication.md.

Construction and the source of outerness. Every finite collection of degrees lies among the divisors of a common multiple. The algebraic union is therefore all finitely supported matrix blocks, whose supremum-norm completion is precisely the sequence algebra with vanishing tails. Conjugating each block by its arithmetic cyclic shift preserves this norm and positivity at every matrix level.

If a unitary $w = \lambda I + a$ in the minimal unitization implemented α, then $\|a_d\| \to 0$ and $|\lambda| = 1$. For a rank-one block projection $e_d = |0\rangle\langle 0|$,

$$\|w_d e_d w_d^* - e_d\| \leq 2\|w_d - \lambda I_d\| \longrightarrow 0.$$

But cyclic Frobenius sends e_d to the orthogonal projection $|1\rangle\langle 1|$, whose difference from e_d has norm one for every $d \geq 2$. This contradicts inner implementation in the unitization.

The positive trace is the supremum of its finite block sums. Finite central cutoffs approximate positive elements in norm and in increasing trace, proving the stated semifiniteness and lower semicontinuity. Finally, finite cycle eigenvectors form an orthonormal basis of the Hilbert sum. Roots of unity are dense on the circle; off it the block resolvents have a uniform bound. This gives the stated spectrum. □

The regularized state is stationary, but the action is observable on other preparations. In each inherited finite block, a relative-index state at zero is carried to the orthogonal state at one. The coherent three-register arithmetic test from the preceding section remains its finite realization. No unrelated quantum block has been appended to produce this action.

Theorem 29.4 (A positive degree reference and its convergent zeta identity)**.** PROVED *For every real $\beta > 1$, the operator ρ_β is positive and trace class with ordinary trace one. Its state on $B_{\deg}$ is faithful and α-invariant. For every integer cutoff $D \geq 1$, the normalizer satisfies*

$$Z_{\deg}(\beta) = \frac{\zeta(\beta)}{\zeta(\beta+1)} - 1, \qquad \sum_{d>D} \frac{\varphi(d)}{d^{\beta+1}} \leq \frac{D^{1-\beta}}{\beta - 1} \quad (D \geq 1).$$

For a nonzero integer j,

$$M_\beta(j) = \frac{1}{Z_{\deg}(\beta)} \sum_{\substack{d \mid |j| \\ d \geq 2}} \frac{\varphi(d)}{d^{\beta+1}}, \qquad M_\beta(0) = 1.$$

Scope. The zeta identity is established in its convergent real domain. The degree regulator is specified rather than claimed unique. The moment is an implementing-operator trace paired with a trace-class state, not the trace of a conjugation channel or an unregularized infinite Frobenius trace.

Provenance. ids: CMP-MELLIN; proved in: degree-completion.md, Section 3; tested by: composite spectrum checker; reviewed in: composite-spectrum-adjudication.md.

Positive coefficients and divisor convolution. The bound $\varphi(d) \leq d$ compares the normalizer to the convergent series $\sum d^{-\beta}$ and gives the integral tail estimate. Each d-dimensional block contributes $\varphi(d)/(Z_{\deg} d^{\beta+1})$ to the ordinary trace of ρ_β, so these contributions sum to one. Every eigenvalue is positive and each block is scalar, giving faithfulness and Frobenius invariance.

Counting integers by their greatest common divisor with k gives $\sum_{d|k} \varphi(d) = k$. Absolute convergence then permits grouping the product of positive series:

$$\left(\sum_{d \geq 1} \frac{\varphi(d)}{d^{\beta+1}} \right) \zeta(\beta+1) = \sum_{k \geq 1} \frac{\sum_{d|k} \varphi(d)}{k^{\beta+1}} = \zeta(\beta).$$

Removing the $d = 1$ term gives the claimed subtraction. The underlying classical totient Dirichlet identity is also recorded in NIST DLMF, equation 27.4.6. The moment formula uses $\mathrm{Tr}(S_d^j) = d$ when $d \mid j$, and zero otherwise. The case $j = 0$ uses trace-one normalization separately. □

The infinite unregularized trace does not follow by dropping the regulator: the identity has infinite θ-mass, so the unitary powers are not θ-integrable. Also, if logarithmic degree weights are written as an operator with value $\log d$ on the dth block, that operator is central. Its conjugation flow fixes the block algebra and must not be identified with the nontrivial relative Frobenius action.

This completion gives one explicit positive quantum realization of the surviving degree data. It preserves the arithmetic cyclic action while leaving the fate of other composition laws open.

For example, two independent 2-cycles have two 2-cycles under simultaneous action; they are not a single 4-cycle. The corresponding matrix algebras can have the same dimension while their distinguished Frobenius actions differ. The existing greatest-common-divisor/least-common-multiple reblocking keeps that multiplicity information when such independent composites are retained.

Provenance. ids: FRL-COMP; proved in: orbit-composition.md; tested by: Frobenius boundary checker; reviewed in: frobenius-boundary-adjudication.md.

30 A uniform limit of finite arithmetic assemblies

The completed reference can itself be obtained from finite assembled experiments. Choose finitely many extension degrees, prepare the corresponding tagged arithmetic systems, and run the copied primitive protocol in each degree. Uniform relative randomization supplies a tracial common block on success. The counting boundary and the degree completion then converge together in trace norm. This theorem passed its own capped review, with the exact input reference made explicit in the definition.

Definition 30.1 (Finite degree mixture and success profiles)**.** Fix an integer base degree $s \geq 1$, a real $\beta > 1$, and named extensions E_d/K with $|K| = p^s$ and $[E_d : K] = d$ for $d \geq 2$. For an integer cutoff $D \geq 2$ put

$$L_D(\beta) = \sum_{d=2}^{D} d^{-\beta}, \qquad \pi_d = \frac{d^{-\beta}}{L_D(\beta)}.$$

Prepare the classical degree tag with these probabilities. In the selected degree prepare the exact period reference $\rho_{E_d}(\log t)$, then apply the three copied-reference preparation amplitudes. On success, uniformly randomize the relative index and explicitly discard that randomizer label. Every failed degree and stopped primitive/averaging history remains available.

For $t > 1$ define

$$w_d(t) = \frac{c_d(t^s)}{dt^{sd}}, \qquad v_d(t) = \frac{w_d(t)}{s(t-1)}, \qquad Z_D(\beta, t) = \sum_{d=2}^{D} d^{-\beta} v_d(t),$$

where $c_d(x) = \sum_{a|d} \mu(d/a) x^a$. Set $v_d(1) = \varphi(d)/d$ and use that value in $Z_D(\beta, 1)$. On the named Hilbert degree sum, let the common successful density have blocks

$$\rho_{D,t}\big|_{\mathbb{C}^d} = \frac{d^{-\beta} v_d(t)}{dZ_D(\beta, t)} I_d \quad (2 \leq d \leq D),$$

and zero elsewhere. Define L_∞, Z_∞ and $\rho_{\infty,t}$ by the corresponding infinite sums when they converge.

Scope. The degree prior is additional classical preparation data. These are common-observable outputs, not identifications of the physical field multiplicity states. At $t = 1$, $\rho_{D,1}$ designates the retained first grade; it is not normalization of a successful zero-probability event. Randomization makes this reference stationary. The unrandomized finite protocol remains the separate Frobenius/coherence witness.

Provenance. ids: D1621; proved in: definitions.md; tested by: composite joined-limit checker; reviewed in: composite-joined-adjudication.md.

The reference instrument used in each degree prepares one period-uniform field state and two fixed zero ancillas, maps $|x\rangle$ to $|x, x, x^2\rangle$, and projects onto simultaneous relative-Frobenius invariants after the primitive-period cut. Its common successful state is $|0\rangle\langle 0|$ and its success probability is $w_d(t)$. Relative randomization averages its d cyclic translates to I_d/d. For $a_d(t) = c_d(t^s)/t^{sd}$, the three preparation histories within that degree have probabilities

$$1 - a_d(t), \qquad (1 - 1/d)a_d(t), \qquad a_d(t)/d.$$

They sum to one before multiplication by π_d. The finite mixture therefore has total success

$$W_D(\beta, t) = \frac{s(t-1)Z_D(\beta, t)}{L_D(\beta)}.$$

Provenance. ids: D1604, D1607, CMP-BOUNDARY, CMP-TRACE; proved in: boundary-and-spectrum.md; tested by: composite boundary checker; reviewed in: composite-boundary-adjudication.md.

Theorem 30.2 (A joined counting and degree limit of assembled quantum states)**.** PROVED
For every integer $s \geq 1$ and real $\beta > 1$, the finite degree-mixture protocol is normalized with its stated histories. Its rescaled success factors satisfy $0 < v_d(t) \leq 1$ for all $d \geq 2, t > 1$ and extend continuously to $v_d(1) = \varphi(d)/d$.

For every fixed $T > 1$, the infinite normalizer converges uniformly on $1 \leq t \leq T$. Its density satisfies

$$\rho_{\infty,t} \longrightarrow \rho_\beta \quad \textit{in trace norm as } t \downarrow 1,$$

where

$$\rho_\beta\big|_{\mathbb{C}^d} = \frac{\varphi(d)}{Z_{\deg}(\beta)d^{\beta+2}} I_d, \qquad Z_{\deg}(\beta) = \frac{\zeta(\beta)}{\zeta(\beta+1)} - 1.$$

For every integer $D \geq 2$ and $1 \leq t \leq T$,

$$\|\rho_{D,t} - \rho_{\infty,t}\|_1 \leq \frac{2^{\beta+2}T^s}{\beta - 1} D^{1-\beta}.$$

Consequently both iterated limits and every path $D \to \infty$, $t \downarrow 1$ give the same ρ_β. With $h = \log t$, their first-order success rate tends to

$$\frac{sZ_{\deg}(\beta)}{\zeta(\beta) - 1}.$$

Probabilities for each fixed bounded common effect converge as well, including after any fixed number of relative Frobenius iterations.

Scope. The degree prior and common-observable comparison are specified choices. The statement includes no arbitrary characteristic-dependent physical continuation and requires no total endpoint tensor or direct-sum operation. The raw event at $t = 1$ still has probability zero. The cutoff bound is a uniform estimate and need not be sharp or smaller than the trivial bound two.

Provenance. ids: CMP-JOINT-LIMIT; proved in: joined-limit.md, Sections 1–4; tested by: composite joined-limit checker; reviewed in: composite-joined-adjudication.md.

The uniform estimate. The positive period polynomials satisfy $c_d(x) \leq x^d - 1$ for $d > 1, x \geq 1$, since their divisor sum is x^d and the degree-one term is x. Also

$$1 - t^{-sd} = (t-1)\sum_{j=1}^{sd} t^{-j} \leq sd(t-1).$$

Together these give $v_d(t) \leq 1$. Dividing the polynomial $c_d(t^s)$ by $t - 1$ gives its continuous endpoint value $\varphi(d)/d$. Thus the summands of Z_∞ are dominated uniformly by $d^{-\beta}$.

The degree-two block bounds every normalizer away from zero on the compact interval:

$$v_2(t) = \frac{1}{2s}\sum_{j=1}^{s} t^{-j} \geq \frac{1}{2}T^{-s}, \qquad Z_\infty(\beta, t) \geq 2^{-\beta-1}T^{-s}.$$

Trace norm between these scalar-block densities is the sum of the absolute differences of their block probabilities. Removing and renormalizing the tail gives exactly

$$\|\rho_{D,t} - \rho_{\infty,t}\|_1 = 2\frac{Z_\infty(\beta,t) - Z_D(\beta,t)}{Z_\infty(\beta,t)}.$$

The integer-cutoff bound for $\sum_{d>D} d^{-\beta}$ proves the displayed uniform estimate. A finite head is continuous at $t = 1$, and the tail is uniformly small, giving the remaining trace-norm limit. Finally, $(t-1)/\log t \to 1$ gives the retained success rate. Unitary Frobenius preserves trace norm, so fixed-effect convergence follows. □

This construction gives the completed state as the limit of actual finite assemblies performed before specialization. Its common randomized state is stationary under Frobenius, while the quantum automorphism remains nontrivial on the available finite-block preparations and observables. The representation has supplied an infinite quantum system without requiring a field structure or a conventional arithmetic composition law at characteristic one.

31 The Hunting of the Symplectic Phantasm

The direction adopted on 9 September 2026 begins with finite symplectic objects and their quantum realizations, then asks how arithmetic processes can act on systems assembled across primes. The procedure is sober, accretive, and careful: admitted results are reused at their exact scope, and new contracts state the extra choices and comparisons still required. The relation to the Riemann hypothesis remains a motivation.

The theorem for finite abelian configurations already provides matrix realization, irreducibility and uniqueness in arbitrary configuration size. The local Weyl task is its transport to symplectic coordinates and the half-form convention. The existing configuration-product theorem supplies the underlying Hilbert tensor comparison. Field traces, Frobenius permutations and ordinary-trace arithmetic processes likewise have admitted foundations.

Provenance. ids: F1-DUAL,F1-WEYL,F1-REAL,F1-FUNCT,FRB-TRACE,FRB-FROB,FRP-CP; proved in: Existing admitted proof shards; tested by: Existing checkers; finite bridge probes below; reviewed in: Original adjudications; no status change.

The Weyl realization, affine covariance, tensor comparison, relation composition, compact structure and odd-prime projective stabilizer comparison, scalar normalization, finite instrument laws and coherent/tagged sums, trace restriction, relative Frobenius and subsystem decoders below have passed their capped reviews. The remaining three propositions retain sketch status. Their finite probes check the interfaces with admitted constructions; the arbitrary-rank conclusions rest on the written proofs.

Provenance. ids: SP-WEYL,SP-EGOROV,SP-TENSOR,SP-LREL,SP-COMPACT,SP-STAB-REL,SP-SCALAR,SP-CP,SP-SUM,SP-TRACE,SP-FROB,SP-SUBSYS; proved in: Structured local, relation, process, sum and arithmetic proofs; tested by: Exact finite controls at their declared scope; reviewed in: Stage-1, relation, process, sum and arithmetic adjudications.

31.1 Choosing categorical structure

A continuing research aim, recorded on 10 September 2026, is to understand which structural properties the arithmetic and quantum categories can carry, how those properties interact, and how the comparison functors transport them. Each proposed property needs an exact category, its additional data and laws, and evidence at the stated scope. The purpose is to make choices of structure responsive to the arithmetic construction.

The first questions concern composition and exchange; daggers and duals; cups, caps and state/process correspondence; scalar coefficients and normalization; coherent sums and classical tags; and the structures available after completion. Properties of functors also require attention: the source and target, variance, coherence maps and meaning of preservation must be stated. A property known for one category is not assigned to a subcategory, quotient or completion without its comparison argument.

The question about cups and caps now has an explicit answer for affine Lagrangian relations. The opposite-form dual and diagonal cup and cap satisfy both snake equations and the stated compatibility with relational converse, including the empty relation and the zero-space unit. Naming an arrow retains its ordered affine relation as a state of the combined dual-input and output object. Over an odd prime field, the intertwiner construction below supplies the corresponding operator line and proves the dagger symmetric monoidal stabilizer equivalence modulo all nonzero complex scalars.

Provenance. ids: D1714,D1715,SP-LREL,SP-COMPACT,SP-STAB-REL; proved in: Structured relation, compact and intertwiner proofs; tested by: Exact F2/F3 relations and F3/F5 stabilizer controls; reviewed in: Relation-cluster adjudication.

The relational scalar calculation retains only the empty and nonempty scalar, and every closed cup–cap loop is the latter. The quantum comparison likewise identifies all nonzero scalar multiples. The scalar calculation below identifies the contraction condition for an actual successful branch. A class-invariant normalization rule can also specify a branch; the quotient itself selects no rule, and compatibility with composition is a separate obligation. The finite instrument theorem keeps ordinary traces and all named outcomes explicit. Its trace adjoint is an ambient reverse map and need not be a reverse branch. The coherent completion now realizes every linear map between its Hilbert sums, while the tagged algebra retains only diagonal blocks. The prescribed dephasing channel removes cross-block terms and preserves ordinary trace. The empty coherent sum is the zero Hilbert space, distinct from the quantum tensor unit. Trace restriction now identifies the named-character Weyl algebras, relative Frobenius has exact unitary covariance, and compatible subsystem encodings give observable inclusions with ordinary-trace decoders that compose in reverse. The remaining completion controls precede the global arithmetic action and state. Operator vectorization and the CP Choi construction still require their separate relation comparison. The structural record will grow with those results, without treating an uninvestigated property as false or a desired combination as established.

Provenance. ids: SP-COMPACT,SP-STAB-REL,SP-SCALAR,SP-CP,SP-SUM,SP-TRACE,SP-FROB,SP-SUBSYS; proved in: Admitted compact, projective, process, sum and arithmetic comparisons; tested by: Finite controls at their declared scope; reviewed in: Relation, process, sum and arithmetic adjudications.

31.2 Shared register and hierarchy conventions

Definition 31.1 (Trace-framed arithmetic registers, recalled)**.** Fix a prime p and a named primitive complex p-th root ζ_p. For a finite field $E/\mathbb{F}_p$ of cardinality p^r, retain E, $H_E = \ell^2(E)$ with basis δ_x, and $\psi_E(x) = \zeta_p^{\mathrm{Tr}_{E/\mathbb{F}_p}(x)}$. Its prime-field phase space is $S_E = E \oplus E$ with $\Omega_E((a,b),(a',b')) = \mathrm{Tr}_{E/\mathbb{F}_p}(ab' - a'b)$. Set $X_E(a)\delta_x = \delta_{x+a}$, $Z_E(b)\delta_x = \psi_E(bx)\delta_x$, and $W_E(a,b) = Z_E(-b)X_E(a)$. An ordered list has the tensor-product Hilbert space and orthogonal direct sum phase space; the empty list has Hilbert space $\mathbb{C}$. The field structures and ordered factors are retained data.

Scope. This recalls the existing fixed character family in all characteristics. A general chosen base character is distinguished below.

Provenance. ids: D1301; proved in: Existing definition; tested by: Existing arithmetic checks; reviewed in: Unchanged convention.

Definition 31.2 (Pauli group and Clifford hierarchy)**.** For an ordered list $A = (E_1, \ldots, E_n)$ of trace-framed registers put $P_A = \{\lambda \bigotimes_j W_{E_j}(a_j, b_j) : \lambda \in U(1)\}$. Set $C_1(A) = P_A$ and, for $r \geq 1$, set $C_{r+1}(A) = \{U \text{ unitary} : UPU^* \in C_r(A) \text{ for every } P \in P_A\}$. The exact level of U is the least positive r with $U \in C_r(A)$, if it exists. All scalar phases are included, also at $p = 2$.

Scope. Levels beyond the second are not stipulated to be groups. The pure stabilizer category below uses the existing second level.

Provenance. ids: D1307; proved in: Existing definition; tested by: Existing hierarchy checks; reviewed in: Unchanged hierarchy.

31.3 Definition layer

Definition 31.3 (finite symplectic objects and affine symmetries)**.** Fix a finite field k of characteristic p. A symplectic object is a finite-dimensional k-vector space V with a k-bilinear alternating form ω_V whose radical is zero. Its rank is n when $\dim_k V = 2n$. The zero space is allowed. Write $\overline{V} = (V, -\omega_V)$ and equip $V \oplus W$ with $\omega_V \oplus \omega_W$. A subspace $L \subseteq V$ is Lagrangian when $L = L^\perp$, where $L^\perp = \{v : \omega_V(v, L) = 0\}$. The linear symmetry groupoid S_k has these objects and form-preserving linear isomorphisms. The affine symmetry groupoid $\mathsf{S}_k^{\mathrm{aff}}$ has arrows $(t, g) : V \to W$, where $g : V \to W$ is such an isomorphism and $t \in W$, acting by $v \mapsto gv + t$. Composition is $(s, h) \circ (t, g) = (s + ht, hg)$ and the identity is $(0, 1_V)$. For affine arrows $(t, g) : V \to W$ and $(t', g') : V' \to W'$, prescribe

$$(t, g) \oplus (t', g') = ((t, t'), g \oplus g') : V \oplus V' \longrightarrow W \oplus W'.$$

The monoidal unit is the zero space with identity $(0, 1_0)$. For symplectic spaces U, V, W, prescribe the associator, left and right unitors, and symmetry to be the zero-translation affine arrows whose linear parts are

$$\begin{aligned}
a_{U,V,W} &: (U \oplus V) \oplus W \longrightarrow U \oplus (V \oplus W), & ((u, v), w) &\longmapsto (u, (v, w)), \\
\lambda_V &: 0 \oplus V \longrightarrow V, & (0, v) &\longmapsto v, \\
\rho_V &: V \oplus 0 \longrightarrow V, & (v, 0) &\longmapsto v, \\
\sigma_{V,W} &: V \oplus W \longrightarrow W \oplus V, & (v, w) &\longmapsto (w, v).
\end{aligned}$$

Scope. All characteristics are included in this classical datum. Rank one means dimension two; no quantum realization is part of the definition.

Provenance. ids: D1701; proved in: Stipulation; SP-GH07, SP-W09; tested by: See the scoped bridge checks; reviewed in: Reuses: D1,D14; no promotion.

Definition 31.4 (affine Lagrangian relations)**.** Fix a finite field k and finite-dimensional symplectic k-spaces V, W. An arrow $R : V \to W$ is either the empty relation or an affine translate of a Lagrangian linear subspace of $\overline{V} \oplus W$. For $R : V \to W$ and $S : W \to Z$ prescribe

$$S \circ R = \{(v, z) : \text{there exists } w \in W \text{ with } (v, w) \in R, (w, z) \in S\}.$$

The identity is $\Delta_V = \{(v, v) : v \in V\}$; the dagger is relational converse. The product is the direct sum on objects and the Cartesian product on relations, with coordinates reordered from $(\overline{V} \oplus W) \oplus (\overline{V'} \oplus W')$ to $\overline{V \oplus V'} \oplus (W \oplus W')$. The unit object is the zero symplectic space. Write $\mathsf{L}_k^{\mathrm{aff}}$ for this candidate.

Scope. Closure and category/coherence laws are proof obligations. Empty relations are retained; nonlinear subvarieties are outside this definition.

Provenance. ids: D1702; proved in: Stipulation; SP-W09, SP-LW14, SP-CK21; tested by: See the scoped bridge checks; reviewed in: Reuses: D1701; no promotion.

Definition 31.5 (odd-characteristic Weyl datum in arbitrary rank)**.** Fix a finite field k of odd characteristic, a nontrivial additive character $\psi : k \to U(1)$, and a finite-dimensional symplectic k-space (V, ω_V). Put $\beta_V = \omega_V/2$. Extend the existing Weyl notation to $A_{\psi,\beta_V}(V)$, with basis $\{W_{\beta_V}(v) : v \in V\}$ and prescribed operations

$$W_{\beta_V}(v)W_{\beta_V}(z) = \psi(\omega_V(v,z)/2)W_{\beta_V}(v+z), \qquad W_{\beta_V}(v)^* = W_{\beta_V}(-v), \qquad 1 = W_{\beta_V}(0).$$

Prescribe $\tau_V(\sum_v c_v W_{\beta_V}(v)) = c_0$. On $k \times V$ prescribe $(t,v)(t',z) = (t+t'+\omega_V(v,z)/2, v+z)$, with central character ψ. For $V = k^n \oplus k^n$ and $\omega_V((a,b),(a',b')) = a\cdot b' - a'\cdot b$, let $H_{k,n} = \ell^2(k^n)$ with counting inner product and its standard coordinate identification with $\ell^2(k)^{\otimes n}$. Use the distinct symmetrized-model notation

$$W^{\mathrm{s}}_{k,n}(a,b) = \psi(a\cdot b/2)\bigotimes_{j=1}^{n} W_{\beta_0}(a_j,b_j), \qquad (W^{\mathrm{s}}_{k,n}(a,b)f)(x) = \psi(-b\cdot x + a\cdot b/2)f(x-a).$$

Here $W_{\beta_0}(a_j,b_j)\delta_y = \psi(-b_j(y+a_j))\delta_{y+a_j}$ is the existing reference model. At $n=0$ use $H_{k,0} = \mathbb{C}$ and the empty tensor. Transporting this standard formula to an abstract symplectic space requires a named symplectic coordinate identification.

Independently of such a coordinate choice, extend the model-groupoid prescription to arbitrary rank. The category $\mathrm{Mod}_{\psi,\beta_V}(V)$ has objects (H,π), where H is a finite-dimensional complex Hilbert space and

$$\pi : A_{\psi,\beta_V}(V) \longrightarrow \mathrm{End}_{\mathbb{C}}(H)$$

is a unital $*$-representation whose underlying module is simple and for which every $\pi(W_{\beta_V}(v))$ is unitary. Its morphisms $U : (H,\pi) \to (H',\pi')$ are unitary intertwiners satisfying $U\pi(a) = \pi'(a)U$ for every $a \in A_{\psi,\beta_V}(V)$; identities and composition are the ordinary identities and composition of unitary maps. The category $\mathrm{PMod}_{\psi,\beta_V}(V)$ has the same objects and each morphism set is the quotient by the action $U \mapsto \lambda U$ of $\lambda \in U(1)$, with

$$[U'] \circ [U] = [U' \circ U].$$

When $V = V(k)$ has rank one and $\beta_V = \omega_V/2$, this is the existing rank-one model groupoid; the rank-one categories for other polarizing cocycles retain their existing meaning.

Scope. The half-form convention excludes characteristic two. Named symplectic coordinates belong only to the construction and transport of the standard formula; they are not data of an arbitrary object of $\mathrm{Mod}_{\psi,\beta_V}(V)$. The finite-abelian realization is already admitted; the Weyl-realization proposition records its coordinate, phase and central-quotient transport.

Provenance. ids: D1703; proved in: Stipulation; SP-GH07, SP-PRASAD09, SP-GROSS06; tested by: See the scoped bridge checks; reviewed in: Reuses: D3,D4,D5,D8,D9,D1001,D1002,D1003,D1301,D1701; no promotion.

Definition 31.6 (actual stabilizer amplitudes)**.** Fix an odd prime p and the reference root $\zeta_p = \exp(2\pi i/p)$ in the trace-framed field registers. For the ordered word $A = (\mathbb{F}_p, \ldots, \mathbb{F}_p)$ of length $n \geq 0$, use the existing full-scalar Pauli group P_A and second hierarchy level $C_2(A)$ on $H_{\mathbb{F}_p,n} = \ell^2(\mathbb{F}_p^n)$. Define $\mathsf{Stab}_p^{\mathrm{amp}}$ to have these Hilbert spaces as objects and actual linear maps generated under composition, tensor product and adjoint by the unitaries in $C_2(A)$, the computational preparation $\mathbb{C} \to H_{\mathbb{F}_p,1}$ sending 1 to δ_0, and every scalar map $c : \mathbb{C} \to \mathbb{C}$ with $c \in \mathbb{C}$. Use the coordinate identification $H_{\mathbb{F}_p,m+n} = H_{\mathbb{F}_p,m} \otimes H_{\mathbb{F}_p,n}$. Equality is equality of the resulting linear maps, including zero.

Scope. Scalars and amplitudes are retained. Such a map defines a physical successful branch only with the required norm or instrument normalization. No closure under addition is imposed here.

Provenance. ids: D1704; proved in: Stipulation; SP-GROSS06, SP-CK21; tested by: See the scoped bridge checks; reviewed in: Reuses: D1301,D1307,D1703; no promotion.

Definition 31.7 (stabilizer amplitudes modulo invertible scalars)**.** Fix an odd prime p and the actual stabilizer amplitude category with objects $\ell^2(\mathbb{F}_p^n)$. In each Hom-set identify two nonzero maps T, S precisely when $S = cT$ for some $c \in \mathbb{C}^\times$. Keep the zero map in a separate class. Write $\mathsf{Stab}_p^{\mathrm{proj}}$ for the quotient with the induced composition, tensor product and adjoint. This scalar quotient is different from quotienting unitary intertwiners only by $U(1)$.

Scope. The quotient records no success probability or norm. A coherent sum of independently rescaled representatives is not prescribed by this definition.

Provenance. ids: D1705; proved in: Stipulation; SP-CK21; tested by: See the scoped bridge checks; reviewed in: Reuses: D1704; no promotion.

Definition 31.8 (finite quantum branches and instruments)**.** Extend the existing ordinary-trace CP target to a finite nonempty family $X = (H_a)_{a\in A}$ of arbitrary nonzero finite-dimensional complex Hilbert spaces. Use its existing notation $B_X = \bigoplus_a \mathrm{End}(H_a)$ and $\mathrm{Tr}_X = \sum_a \mathrm{Tr}_{H_a}$. For another such family $Y = (K_b)_{b\in B}$, a branch $\Phi : B_X \to B_Y$ is a linear map admitting finite Kraus lists $K_{ba,j} : H_a \to K_b$ such that

$$\Phi(\rho)_b = \sum_{a,j} K_{ba,j}\rho_a K_{ba,j}^*, \qquad \sum_{b,j} K_{ba,j}^* K_{ba,j} \le 1_{H_a} \quad \text{for every } a.$$

Branch equality is equality of the linear maps. A channel has equality in every displayed inequality. An instrument has a finite nonempty outcome set O and branches $\Phi_o : B_X \to B_Y$ with a common source and target, whose sum is a channel. Its retained version has codomain $\bigoplus_{o\in O} B_Y$ and sends ρ to $(\Phi_o(\rho))_o$. Prescribe the outcome-first block identification

$$\bigoplus_{o\in O} B_Y = \bigoplus_{(o,b)\in O\times B} \mathrm{End}(K_b), \qquad (\rho_o)_o = (\rho_{o,b})_{(o,b)}.$$

For another such family $Z = (L_c)_{c\in C}$ and instruments $(\Phi_o)_{o\in O} : B_X \to B_Y$ and $(\Psi_r)_{r\in R} : B_Y \to B_Z$, prescribe their sequential family to be

$$(\Psi_r \circ \Phi_o)_{(o,r)\in O\times R};$$

the first component is the earlier outcome. For systems X', Y' of the same kind and an independent instrument $(\Theta_s)_{s\in S} : B_{X'} \to B_{Y'}$, prescribe the tensor family $(\Phi_o \otimes \Theta_s)_{(o,s)\in O\times S}$ in listed-factor order. For every complex-linear map $\Phi : B_X \to B_Y$, prescribe its ordinary-trace adjoint $\Phi^{\mathrm{tr}\,*} : B_Y \to B_X$ by

$$\sum_b \mathrm{Tr}(y_b^*\Phi(x)_b) = \sum_a \mathrm{Tr}\big(\Phi^{\mathrm{tr}\,*}(y)_a^* x_a\big) \quad (x \in B_X,\ y \in B_Y).$$

Use the existing state and outcome convention: $\rho \ge 0$, $\mathrm{Tr}_X \rho = 1$, outcome probability $\mathrm{Tr}_Y \Phi_o(\rho)$, and conditional normalization only at positive probability. Composition is composition of maps; tensor uses Cartesian products of tags and ordinary Hilbert tensor factors.

Scope. This is an ambient finite quantum process definition. It does not assert that every branch comes from the arithmetic or stabilizer generators. Traces are ordinary, not dimension-normalized. The ordinary-trace adjoint is an ambient reverse-direction map; even when it is completely positive, it is not required to be a trace-nonincreasing branch.

Provenance. ids: D1706; proved in: Stipulation; SP-WAT18; tested by: Exact finite process controls; reviewed in: Reuses: D1325,D1326,D1327; scalar and process properties admitted separately.

Definition 31.9 (coherent additive completion and classical tags)**.** Fix an odd prime p and its actual stabilizer amplitude category. First take the complex linear span of each Hom-set inside the ambient linear maps. Its matrix completion has objects finite lists $(H_{\mathbb{F}_p,n_1},\dots,H_{\mathbb{F}_p,n_s})$, including the empty list, and arrows matrices whose entries lie in these linear spans. Composition is matrix multiplication and dagger is adjoint transpose. Realize a list as $H=\bigoplus_{i=1}^{s} H_{\mathbb{F}_p,n_i}$. For finite lists $X=(H_i)_{i=1}^{s}$ and $Y=(K_j)_{j=1}^{t}$ of these model spaces, with $s,t\ge 0$, prescribe the realization of an arrow matrix $T=(T_{ji})$ by

$$(\xi_i)_i \longmapsto \left(\sum_{i=1}^{s} T_{ji}\xi_i\right)_j : \bigoplus_{i=1}^{s} H_i \longrightarrow \bigoplus_{j=1}^{t} K_j.$$

The empty list realizes the zero Hilbert space; the unique empty block matrix to or from that list has the zero action of its stated type. The coherent observable algebra of the realized list is $\mathrm{End}(H)$. For a nonempty list the tagged observable algebra is instead $\bigoplus_i \mathrm{End}(H_{\mathbb{F}_p,n_i})$, embedded as the block-diagonal operators on H. Both constructions and their chosen block decomposition are recorded; they are not identified with one another. For a nonempty list $X=(H_i)_{i=1}^{s}$ with realized space H, prescribe $P_i^X\in\mathrm{End}(H)$ by $(P_i^X\xi)_j=\delta_{ij}\xi_j$ on the ordered summands, and prescribe

$$\Delta_X:\mathrm{End}(H)\longrightarrow\mathrm{End}(H),\qquad \Delta_X(a)=\sum_{i=1}^{s} P_i^X a P_i^X.$$

Scope. This completion deliberately adds linear combinations. Whether it preserves a proposed Gaussian fragment, or comes from classical disjoint unions, is a separate question. The empty list is permitted in the linear completion; no channel on the zero Hilbert space is prescribed. The range, positivity and trace properties of the block-dephasing map remain proof obligations.

Provenance. ids: D1707; proved in: Stipulation; SP-CK21, SP-WAT18; tested by: Exact finite sum controls; reviewed in: Reuses: D1704,D1706; SP-SUM admitted separately.

Definition 31.10 (bosonic Fock space and bounded second quantization)**.** For a complex Hilbert space H, let $P_r=(r!)^{-1}\sum_{\pi\in S_r}U_\pi$, where $U_\pi(h_1\otimes\cdots\otimes h_r)=h_{\pi^{-1}(1)}\otimes\cdots\otimes h_{\pi^{-1}(r)}$, and set $\mathrm{Sym}^r H=\operatorname{ran}P_r$, with $\mathrm{Sym}^0 H=\mathbb{C}$. Define

$$\Gamma_s(H)=\widehat{\bigoplus}_{r\ge 0}\mathrm{Sym}^r H,\qquad \Omega=1\in\mathrm{Sym}^0 H.$$

The hat means Hilbert direct sum. The algebraic finite-particle space is $\bigoplus_{r\ge 0}^{\mathrm{alg}}\mathrm{Sym}^r H$. For a contraction $T:H\to K$, prescribe $\Gamma_s(T)=\bigoplus_{r\ge 0}T^{\otimes r}|_{\mathrm{Sym}^r H}$. Particle number is the operator $N_H\xi=(r\xi_r)_r$ on the domain $\{\xi:\sum_r r^2\|\xi_r\|^2<\infty\}$.

Scope. Boundedness and functor laws are obligations. No bounded second quantization of an arbitrary expansive map is asserted, and no unspecified categorical free-monoid property is imposed.

Provenance. ids: D1708; proved in: Stipulation; SP-DER06; tested by: See the scoped bridge checks; reviewed in: Reuses: D1010; no promotion.

Definition 31.11 (scalar restriction and a named Frobenius datum)**.** Let E/K be a finite extension of finite fields, with $|K|=p^s$, and let (V,ω_E) be a finite-dimensional symplectic E-space. On $\mathrm{Res}_{E/K}V$ prescribe $\omega_{\mathrm{Res}}(v,w)=\mathrm{Tr}_{E/K}(\omega_E(v,w))$. For a named nontrivial additive character χ_K, write $\chi_{E/K}=\chi_K\circ\mathrm{Tr}_{E/K}$. The notation ψ_E remains reserved for the fixed absolute-trace family; no equality with $\chi_{E/K}$ is stipulated for an arbitrary χ_K. Write $\sigma_{E/K}(x)=x^{|K|}=\sigma_E^s(x)$. An abstract Frobenius datum additionally names a bijective $\sigma_{E/K}$-semilinear map

$\varphi_V : V \to V$ satisfying $\omega_E(\varphi_V v, \varphi_V w) = \sigma_{E/K}(\omega_E(v, w))$. On $E^n \oplus E^n$ use coordinatewise $\sigma_{E/K}$. Its Hilbert permutation is $U_{E/K,n} = (U_E^s)^{\otimes n}$ on $\ell^2(E^n)$, with the empty tensor identity at $n = 0$. The absolute permutations σ_E, U_E keep their existing meanings.

Scope. The classical data allow characteristic two; the half-form quantum comparison is restricted to odd characteristic. Abstract symplectic spaces do not acquire descent data by omission. This does not define a field-inclusion functor.

Provenance. ids: D1709; proved in: Stipulation; SP-STFIELD, SP-GH07; tested by: See the scoped bridge checks; reviewed in: Reuses: D3,D1301,D1302,D1303,D1701; no promotion.

Definition 31.12 (a specified quantum subsystem decoder)**.** Fix an odd-characteristic finite field k, a nontrivial additive character ψ, and a symplectic linear injection $j : U \to V$ between finite-dimensional symplectic k-spaces. Put $W = j(U)^{\perp_V}$. A quantum subsystem datum additionally specifies finite-dimensional irreducible unitary Weyl model Hilbert spaces H_U, H_W, H_V, with W_U^{s}, W_W^{s}, W_V^{s} denoting the symmetrized operators transported through their named coordinates, and a unitary $J : H_U \otimes H_W \to H_V$ satisfying

$$J(W_U^{\mathrm{s}}(u) \otimes W_W^{\mathrm{s}}(w))J^* = W_V^{\mathrm{s}}(ju + w).$$

Define the observable inclusion $\iota_J(a) = J(a \otimes 1)J^*$ and the state decoder $\mathcal{D}_J(\rho) = \mathrm{Tr}_{H_W}(J^*\rho J)$, where the partial trace is specified by

$$\mathrm{Tr}(\mathcal{D}_J(\rho)a) = \mathrm{Tr}(\rho\iota_J(a)) \qquad (a \in \mathrm{End}(H_U)).$$

For compatible iterated tensor decompositions, take composable symplectic linear injections

$$U \xrightarrow{j} V \xrightarrow{\ell} X$$

and use the actual orthogonal complements

$$W_j = j(U)^{\perp_V}, \qquad W_\ell = \ell(V)^{\perp_X}, \qquad W_{\ell j} = (\ell j(U))^{\perp_X}.$$

The candidate comparison of complements is the linear map

$$W_j \oplus W_\ell \longrightarrow W_{\ell j}, \qquad (w, z) \longmapsto \ell(w) + z.$$

An iterated quantum subsystem datum consists of single-step and composite data sharing the model spaces H_U, H_V, H_X and additionally specifies the complement model spaces and unitaries

$$\begin{aligned} J_j &: H_U \otimes H_{W_j} \longrightarrow H_V, \\ J_\ell &: H_V \otimes H_{W_\ell} \longrightarrow H_X, \\ J_{\ell j} &: H_U \otimes H_{W_{\ell j}} \longrightarrow H_X, \\ J_{j,\ell}^\perp &: H_{W_j} \otimes H_{W_\ell} \longrightarrow H_{W_{\ell j}}. \end{aligned}$$

The first three unitaries satisfy the preceding Weyl compatibility equation for j, ℓ, and ℓj, respectively. The complement comparison satisfies

$$J_{j,\ell}^\perp(W_{W_j}^{\mathrm{s}}(w) \otimes W_{W_\ell}^{\mathrm{s}}(z))(J_{j,\ell}^\perp)^* = W_{W_{\ell j}}^{\mathrm{s}}(\ell(w) + z).$$

Call the iterated data compatible when there exists $\lambda \in U(1)$ such that, for every $\xi \in H_U$, $\eta \in H_{W_j}$ and $\zeta \in H_{W_\ell}$,

$$J_\ell\big(J_j(\xi \otimes \eta) \otimes \zeta\big) = \lambda J_{\ell j}\big(\xi \otimes J_{j,\ell}^\perp(\eta \otimes \zeta)\big).$$

The right-hand route uses the ordinary Hilbert reassociation $(\xi \otimes \eta) \otimes \zeta \mapsto \xi \otimes (\eta \otimes \zeta)$. The phase λ is allowed but is not retained as an additional choice. The associated maps have types

$$\begin{aligned} \iota_{J_j} &: \mathrm{End}(H_U) \to \mathrm{End}(H_V), & \mathcal{D}_{J_j} &: \mathrm{End}(H_V) \to \mathrm{End}(H_U), \\ \iota_{J_\ell} &: \mathrm{End}(H_V) \to \mathrm{End}(H_X), & \mathcal{D}_{J_\ell} &: \mathrm{End}(H_X) \to \mathrm{End}(H_V), \\ \iota_{J_{\ell j}} &: \mathrm{End}(H_U) \to \mathrm{End}(H_X), & \mathcal{D}_{J_{\ell j}} &: \mathrm{End}(H_X) \to \mathrm{End}(H_U). \end{aligned}$$

For every rank-zero symplectic factor use its existing model $H_0 = \mathbb{C}$, the identity Weyl operator and the ordinary Hilbert tensor unitors in the same displayed types and compatibility equation.

Scope. The orthogonal direct-sum decomposition, nondegeneracy of every complement, well-definedness into the combined complement and the symplecticity and bijectivity of the displayed complement comparison, existence of the compatible model unitaries, the observable/decoder laws, phase independence and composition of compatible decoders are proof obligations. Compatibility is an explicit condition on these chosen data; it is not assigned to arbitrary independently chosen model unitaries. This datum is not scalar restriction, a generic field trace, a field support code or a non-bijective interpretation of finite-field Frobenius.

Provenance. ids: D1710; proved in: Stipulation; SP-STFIELD, SP-WAT18; structured subsystem proofs; tested by: Exact finite arithmetic controls; reviewed in: Reuses: D1701,D1703,D1706; arithmetic adjudication.

Definition 31.13 (finite-prime tensor assembly with a chosen reference)**.** For every prime p, choose an integer $d_p \geq 1$, a Hilbert space $H_p = \mathbb{C}^{d_p}$ and a positive operator ρ_p of trace one. For a finite set of primes P, put $A_P = \bigotimes_{p \in P} \mathrm{End}(H_p)$, using increasing prime order, and $A_\varnothing = \mathbb{C}$. For $P \subseteq Q$ prescribe $\iota_{QP}(a) = a \otimes 1$ with the required coordinate reordering. Define the candidate norm completion $A_{\mathrm{pr}} = \overline{\varinjlim_P A_P}^{\|\cdot\|}$ using the finite matrix norms. On finite tensors prescribe $\varphi_{\mathrm{pr}}(\bigotimes_{p \in P} a_p) = \prod_{p \in P} \mathrm{Tr}(\rho_p a_p)$.

Scope. The inductive and state-extension properties belong to a lemma. The matrices are specified data, not yet an all-prime Weyl assignment. Neither inter-prime arithmetic maps nor a factor type is supplied.

Provenance. ids: D1711; proved in: Stipulation; SP-CM08, SP-WAT18; tested by: See the scoped bridge checks; reviewed in: Reuses: D1706; no promotion.

Definition 31.14 (a represented Bost–Connes control system)**.** On $H_{\mathrm{BC}} = \ell^2(\mathbb{N}_{>0})$ with basis $(\delta_m)_{m \geq 1}$, fix the usual embedding $\mathbb{Q}/\mathbb{Z} \to U(1)$ given by $r \mapsto \exp(2\pi i r)$, and prescribe

$$e(r)\delta_m = \exp(2\pi i m r)\delta_m, \qquad \mu_n \delta_m = \delta_{nm} \quad (n \geq 1).$$

Let $A_{\mathrm{BC}}^{\mathrm{rep}} = C^*(e(r), \mu_n : r \in \mathbb{Q}/\mathbb{Z}, n \geq 1) \subseteq B(H_{\mathrm{BC}})$. Define $H_{\log}\delta_m = (\log m)\delta_m$ on $\{\xi : \sum_m (\log m)^2 |\xi_m|^2 < \infty\}$ and prescribe $\sigma_u(a) = e^{iuH_{\log}} a e^{-iuH_{\log}}$. Define $\nu_n(a) = \mu_n a \mu_n^*$ and $L_n(a) = \mu_n^* a \mu_n$. For real $b > 1$ prescribe $Z_{\mathrm{BC}}(b) = \sum_{m \geq 1} m^{-b}$ and $\rho_{\mathrm{BC},b} = Z_{\mathrm{BC}}(b)^{-1} e^{-bH_{\log}}$.

Scope. This is a concrete represented control, not an assertion of faithfulness of a universal presentation or an identification with the sought symplectic system. Convergence, invariance and CP properties remain lemma obligations.

Provenance. ids: D1712; proved in: Stipulation; SP-BC95, SP-CM04, SP-CM08; tested by: See the scoped bridge checks; reviewed in: Reuses: none; no promotion.

Definition 31.15 (state, GNS and modular comparison data)**.** A C^*-dynamical datum is a unital C^*-algebra A with a point-norm continuous homomorphism $\sigma : \mathbb{R} \to \mathrm{Aut}(A)$. A state is a positive linear functional φ with $\varphi(1) = 1$. Its GNS Hilbert space H_φ is the prescribed completion of A/N_φ with $N_\varphi = \{a : \varphi(a^*a) = 0\}$ and $\langle [a], [b] \rangle = \varphi(a^*b)$, left action $\pi_\varphi(a)[b] = [ab]$, and cyclic vector $\Omega_\varphi = [1]$. Write $M_\varphi = \pi_\varphi(A)''$. For real $b > 0$, a KMS_b state means that for every $a, c \in A$ there is a bounded continuous function on $0 \le \operatorname{Im} z \le b$, holomorphic in the interior, whose boundary values are $F_{a,c}(u) = \varphi(a\sigma_u(c))$ and $F_{a,c}(u + ib) = \varphi(\sigma_u(c)a)$. A modular comparison datum additionally requires Ω_φ to be cyclic and separating for M_φ. With $S_\varphi(a\Omega_\varphi) = a^*\Omega_\varphi$ and polar decomposition $\overline{S}_\varphi = J_\varphi \Delta_\varphi^{1/2}$, its modular-flow prescription is $a \mapsto \Delta_\varphi^{iu} a \Delta_\varphi^{-iu}$.

Scope. Analytic existence and comparison theorems must be cited or proved before use. No equality between modular flow, the specified physical flow, or Frobenius is a stipulation.

Provenance. ids: D1713; proved in: Stipulation; SP-CM08, SP-CCM07; tested by: See the scoped bridge checks; reviewed in: Reuses: D1122,D1326; no promotion.

Definition 31.16 (compact data for affine Lagrangian relations)**.** Fix a finite field k and finite-dimensional symplectic k-spaces U, V, W. Use the affine Lagrangian relation prescription for arrows between these spaces. Prescribe the compact dual of V to be the existing opposite-form object $\overline{V}$. Prescribe the following coherence arrows to be the graphs of the displayed linear tuple maps:

$$\begin{aligned}
a_{U,V,W} &: (U \oplus V) \oplus W \longrightarrow U \oplus (V \oplus W), & ((u, v), w) &\longmapsto (u, (v, w)),\\
\lambda_V &: 0 \oplus V \longrightarrow V, & (0, v) &\longmapsto v,\\
\rho_V &: V \oplus 0 \longrightarrow V, & (v, 0) &\longmapsto v,\\
\sigma_{V,W} &: V \oplus W \longrightarrow W \oplus V, & (v, w) &\longmapsto (w, v).
\end{aligned}$$

With the factors in exactly the displayed order, prescribe

$$\eta_V : 0 \longrightarrow \overline{V} \oplus V, \qquad \eta_V = \{(0, (v, v)) : v \in V\},$$

and

$$\epsilon_V : V \oplus \overline{V} \longrightarrow 0, \qquad \epsilon_V = \{((v, v), 0) : v \in V\}.$$

For $R : V \to W$, prescribe its name by

$$\ulcorner R \urcorner = (1_{\overline{V}} \oplus R) \circ \eta_V : 0 \longrightarrow \overline{V} \oplus W.$$

For $Q : 0 \to \overline{V} \oplus W$, prescribe its unname by

$$\llcorner Q \lrcorner = \lambda_W \circ (\epsilon_V \oplus 1_W) \circ a^{-1}_{V,\overline{V},W} \circ (1_V \oplus Q) \circ \rho_V^{-1} : V \longrightarrow W.$$

For scalars $s, t : 0 \to 0$, prescribe their tensor as the transported endomorphism

$$\lambda_0 \circ (s \oplus t) \circ \lambda_0^{-1} : 0 \longrightarrow 0.$$

Scalar calculations use ordinary existential relational composition, with no additional scalar weight or middle-witness multiplicity.

Scope. The prescribed data are classical and include all finite fields, characteristic two, affine translates and empty relations. Their Lagrangian typing, compact laws, state/process correspondence and scalar properties are proof obligations. No quantum normalization, amplitude, probability, stabilizer equivalence or Choi theorem is prescribed.

Provenance. ids: D1714; proved in: Stipulation; SP-CK21, SP-LW14, SP-W09; tested by: Exact relation controls; reviewed in: Reuses: D1701,D1702; SP-COMPACT admitted separately.

Definition 31.17 (stabilizer intertwiner space)**.** Fix an odd prime p and, for integers $n \geq 0$, use the existing trace-framed character $\psi_{\mathbb{F}_p}$, standard symplectic spaces

$$V_n = \mathbb{F}_p^n \oplus \mathbb{F}_p^n, \qquad \omega_n((a,b),(a',b')) = a \cdot b' - a' \cdot b,$$

and the standard Hilbert models $H_{\mathbb{F}_p,n}$ with their symmetrized Weyl operators $W^{\mathrm{s}}_{\mathbb{F}_p,n}$. Put $\Omega_{m,n} = -\omega_m \oplus \omega_n$. For integers $m, n \geq 0$ and a nonempty affine Lagrangian relation $R : V_m \to V_n$, prescribe $\mathcal{E}_p(R)$ to consist of the complex linear maps $T : H_{\mathbb{F}_p,m} \to H_{\mathbb{F}_p,n}$ satisfying

$$W^{\mathrm{s}}_{\mathbb{F}_p,n}(w) T W^{\mathrm{s}}_{\mathbb{F}_p,m}(v)^* = \psi_{\mathbb{F}_p}(-\Omega_{m,n}(r,(v,w)))\, T$$

for every $r \in R$ and every $(v,w) \in R - R$. For the empty relation prescribe $\mathcal{E}_p(\varnothing) = \{0\}$.

Scope. Using a single origin in the equations, the line dimension, membership in the actual stabilizer-amplitude category, functoriality, dagger and tensor compatibility, and equivalence are proof obligations. The datum is restricted to standard spaces over an odd prime field and the fixed trace-framed character. It prescribes no representative normalization, success probability, CP map or arithmetic-source comparison.

Provenance. ids: D1715; proved in: Stipulation; SP-CK21, SP-GROSS06, SP-BC24; tested by: Exact stabilizer comparison; reviewed in: Reuses: D1301,D1307,D1701,D1702,D1703,D1704,D1705; SP-STAB-REL admitted separately.

31.4 Lemma contracts and remaining comparisons

The first contract is a finite-abelian corollary with an explicit phase and central-quotient comparison. The underlying tensor theorem is already available. The statements below distinguish these inherited results from the remaining affine naturality, relation semantics and analytic work. The first nine propositions are proved; the remaining six retain sketch status.

Proposition 31.18 (Weyl realization)**.** **PROVED** *For every odd-characteristic finite field k of cardinality q, nontrivial additive character ψ, and symplectic k-space V of dimension $2n$, at $\beta_V = \omega_V/2$ the Weyl algebra is a unital $*$-algebra with faithful normalized trace τ_V and is $*$-isomorphic to* $\mathrm{End}(\mathbb{C}^{q^n})$. *Its standard formula is a unitary irreducible realization; irreducible unitary representations of the prescribed Heisenberg group with central character ψ are unique up to unitary equivalence. Unitary intertwiners between irreducible unitary models are unique up to $U(1)$. Rank zero gives $\mathbb{C}$; rank one specializes to the existing Weyl algebra at $\beta = \omega/2$.*

Scope. Odd characteristic only; no preferred abstract-space basis or genuine symmetry lift is asserted.

Provenance. ids: SP-WEYL; proved in: Structured proof: reuse.md, sections 1–2; inherited results as cited; tested by: Exact finite probes at their declared scope; reviewed in: Single blind review and stage-1 adjudication.

Proof. If V is nonzero, choose a nonzero vector $e \in V$. Nondegeneracy supplies f with $\omega_V(e,f) = 1$, and the plane spanned by e, f is nondegenerate. Its orthogonal complement is again symplectic, so induction gives a symplectic coordinate isomorphism

$$V \simeq k^n \oplus k^n, \qquad \omega_V((a,b),(a',b')) = a \cdot b' - a' \cdot b.$$

For $V = 0$, use the unique coordinate map between zero spaces; this case is also recovered below from the empty configuration group. The image of ψ is the group of pth roots selected by ψ. For $A = (k^n, +)$, the characters

$$\chi_b(x) = \psi(-b \cdot x)$$

are all the characters of A: if $b \neq 0$, the functional $x \mapsto b \cdot x$ is onto, so χ_b is nontrivial, and the previously proved finite-character count turns this injection into a bijection.

In these coordinates the finite-abelian Weyl multiplier is $\psi(a \cdot b')$. Rephase its operators by

$$c(a, b) = \psi(a \cdot b/2), \qquad W^{\mathrm{s}}(a, b) = c(a, b) W_{\mathrm{ref}}(a, \chi_b).$$

Expanding the dot product gives

$$\begin{aligned} a \cdot b' + \frac{1}{2} a \cdot b + \frac{1}{2} a' \cdot b' - \frac{1}{2}(a + a') \cdot (b + b') \\ = \frac{1}{2}\big(a \cdot b' - a' \cdot b\big). \end{aligned}$$

Thus the rephased operators have exactly the half-form multiplier and satisfy $(W^{\mathrm{s}}(v))^* = W^{\mathrm{s}}(-v)$. On the computational basis,

$$W^{\mathrm{s}}(a, b)\delta_y = \psi\left(-b \cdot y - \frac{1}{2} a \cdot b\right) \delta_{y+a},$$

which is the stated wavefunction formula after writing the output coordinate as $x = y + a$.

Rephasing by nonzero scalars does not change the operator span. The finite Stone–von Neumann theorem already proved for A therefore gives all of $\mathrm{End}(\mathbb{C}^A)$, with

$$\mathrm{Tr}(W^{\mathrm{s}}(v)) = q^n \, \mathbf{1}_{v=0}, \qquad \tau_V = q^{-n} \,\mathrm{Tr}\,.$$

This trace is normalized and faithful because $\mathrm{Tr}(B^* B) > 0$ for every nonzero matrix B.

It remains to compare central coordinates. The homomorphism

$$(t, a, b) \longmapsto \big(\psi(t + a \cdot b/2), a, \chi_b\big)$$

from the prescribed k-central Heisenberg group onto the finite-abelian phase group has kernel $\{(t, 0, 0) : \psi(t) = 1\}$. A unitary representation with the stipulated central character kills this kernel, and conversely a unitary representation of the quotient pulls back with that central character. The previously proved finite Stone–von Neumann result now supplies dimension q^n, irreducibility, unitary equivalence, and uniqueness of unitary intertwiners up to $U(1)$. For $n = 0$ the construction is $\mathbb{C}$; for $n = 1$ it is the already defined rank-one half-form Weyl algebra. □

Proposition 31.19 (Affine symmetries and projective implementation). **PROVED** *For a fixed odd-characteristic finite field k and nontrivial additive character ψ, put $\beta_V = \omega_V/2$ on each symplectic space. An affine symplectic arrow $(t, g) : V \to W$ induces the $*$-isomorphism $\alpha_{t,g}(W_{\beta_V}(v)) = \psi(\omega_W(t, gv))W_{\beta_W}(gv)$. These assignments respect identities and affine composition exactly. On irreducible unitary models they have unique projective unitary implementers. Composition of those implementers agrees in the $U(1)$ quotient.*

Scope. This is a covariant algebra functor and a projective implementation statement. Choosing a genuine linear lift is a separate comparison.

Provenance. ids: SP-EGOROV; proved in: Structured proof: egorov.md; inherited results as cited; tested by: Exact finite probes at their declared scope; reviewed in: Single blind review and stage-1 adjudication.

Proof. It is enough first to check the prescribed map on the Weyl basis. For $v, z \in V$, symplecticity of g gives

$$\begin{aligned} \alpha_{t,g}(W_{\beta_V}(v))\alpha_{t,g}(W_{\beta_V}(z)) &= \psi\left(\omega_W(t, gv) + \omega_W(t, gz) + \frac{1}{2}\omega_W(gv, gz)\right) W_{\beta_W}(g(v + z)) \\ &= \psi\left(\omega_W(t, g(v + z)) + \frac{1}{2}\omega_V(v, z)\right) W_{\beta_W}(g(v + z)) \\ &= \alpha_{t,g}\big(W_{\beta_V}(v)W_{\beta_V}(z)\big). \end{aligned}$$

The unit is fixed. Since $\overline{\psi(x)} = \psi(-x)$, the same basis gives

$$\alpha_{t,g}(W_{\beta_V}(v)^*) = \alpha_{t,g}(W_{\beta_V}(v))^*.$$

The inverse affine arrow is $(-g^{-1}t, g^{-1})$, so this homomorphism is a $*$-isomorphism.

The semidirect-product law is exact, rather than projective. If $(t,g) : V \to W$ and $(s,h) : W \to Z$, then

$$\begin{aligned}(\alpha_{s,h}\alpha_{t,g})(W_{\beta_V}(v)) &= \psi(\omega_W(t, gv) + \omega_Z(s, hgv))\, W_{\beta_Z}(hgv) \\ &= \psi(\omega_Z(s + ht, hgv))\, W_{\beta_Z}(hgv) \\ &= \alpha_{s+ht,hg}(W_{\beta_V}(v)).\end{aligned}$$

The identity case follows by taking $(t,g) = (0, 1_V)$. Pure translation is inner:

$$W_{\beta_W}(t)W_{\beta_W}(v)W_{\beta_W}(t)^* = \psi(\omega_W(t,v))W_{\beta_W}(v),$$

so the affine formula is the Weyl translation followed by the linear Egorov map.

Let (H_V, π_V) and (H_W, π_W) be objects of the arbitrary-rank model groupoids just defined. Pulling π_W back along $\alpha_{t,g}$ gives another object of the source model groupoid. Finite Stone–von Neumann uniqueness supplies a unitary $U_{t,g} : H_V \to H_W$ with

$$U_{t,g}\pi_V(a)U_{t,g}^* = \pi_W(\alpha_{t,g}(a)), \qquad a \in A_{\psi,\beta_V}(V),$$

and any two such unitaries differ by a phase. For two composable affine arrows, $U_{s,h}U_{t,g}$ and $U_{s+ht,hg}$ implement the same exact algebra map calculated above. Their classes therefore agree after quotienting by $U(1)$. This proves projective composition without selecting a genuine phase lift. □

Proposition 31.20 (Symplectic sum and quantum tensor compatibility). **PROVED** *For a fixed odd-characteristic finite field k and nontrivial additive character ψ, put $\beta_V = \omega_V/2$, $\beta_W = \omega_W/2$. The assignment $W_{\beta_V \oplus \beta_W}(v,w) \mapsto W_{\beta_V}(v) \otimes W_{\beta_W}(w)$ gives a natural trace-preserving $*$-isomorphism $A_{\psi,\beta_V\oplus\beta_W}(V \oplus W) \to A_{\psi,\beta_V}(V) \otimes A_{\psi,\beta_W}(W)$. Together with the rank-zero unit these maps obey the associativity, unit and symmetry diagrams. The corresponding unitary model identifications obey these diagrams projectively.*

Scope. The source operation is symplectic direct sum. This does not construct a coherent additive completion of the classical category.

Provenance. ids: SP-TENSOR; proved in: Structured proof: tensor.md; inherited results as cited; tested by: Exact finite probes at their declared scope; reviewed in: Single blind review and stage-1 adjudication.

Proof. Temporarily denote the displayed generator map by $\Theta_{V,W}$. The direct-sum form gives

$$\begin{aligned}&\Theta_{V,W}(W_{\beta_V\oplus\beta_W}(v,w))\Theta_{V,W}(W_{\beta_V\oplus\beta_W}(v',w')) \\ &\quad = \psi\left(\frac{1}{2}\omega_V(v,v') + \frac{1}{2}\omega_W(w,w')\right) W_{\beta_V}(v+v') \otimes W_{\beta_W}(w+w'),\end{aligned}$$

which is the image of the product in the source algebra. The map also preserves the unit and involution. It takes the Weyl basis indexed by $V \oplus W$ bijectively to the tensor-product Weyl basis, and hence is a $*$-isomorphism. On these basis elements,

$$(\tau_V \otimes \tau_W)\big(W_{\beta_V}(v) \otimes W_{\beta_W}(w)\big) = \mathbf{1}_{v=0}\mathbf{1}_{w=0} = \tau_{V\oplus W}(W_{\beta_V\oplus\beta_W}(v,w)),$$

so it preserves the normalized trace.

In standard coordinates, reorder $((a_1, b_1), (a_2, b_2))$ as $((a_1, a_2), (b_1, b_2))$. The previously proved configuration-product unitary sends $\delta_{(x,y)}$ to $\delta_x \otimes \delta_y$. The half-form phase is compatible with this unitary because

$$\psi\left(\frac{1}{2}(a_1 \cdot b_1 + a_2 \cdot b_2)\right) = \psi(a_1 \cdot b_1/2)\,\psi(a_2 \cdot b_2/2).$$

Thus the standard Weyl operator on the direct sum is carried exactly to the tensor of the two standard Weyl operators.

Naturality for translations and linear symplectic maps is also exact. For affine arrows $(t, g) : V \to V'$ and $(s, h) : W \to W'$, both routes around the naturality square send the generator labelled by (v, w) to

$$\psi(\omega_{V'}(t, gv) + \omega_{W'}(s, hw))\, W_{\beta_{V'}}(gv) \otimes W_{\beta_{W'}}(hw).$$

Rebracketing three summands, adjoining the zero space, or exchanging two summands likewise sends every Weyl generator to the same ordered tensor by either route. The associativity, unit and symmetry diagrams therefore commute at the algebra level.

For arbitrary objects (H_V, π_V), (H_W, π_W), and $(H_{V\oplus W}, \pi_{V\oplus W})$ of the model groupoids just defined, finite Stone–von Neumann uniqueness supplies an implementing unitary

$$J_{V,W} : H_{V\oplus W} \longrightarrow H_V \otimes H_W$$

for the displayed algebra isomorphism, unique up to $U(1)$. In an affine naturality diagram, the two model-level routes implement the same exact algebra map just calculated, so they differ by one phase. The same argument applied to the exact algebra associativity, unit and symmetry diagrams shows that every corresponding model diagram commutes in the $U(1)$ quotient. This uses only projective classes and makes no coherent choice of unitary representatives. □

Proposition 31.21 (Composition of affine Lagrangian relations). PROVED *For every finite field k, the affine Lagrangian relation prescription is closed under relational composition and product and forms a dagger symmetric monoidal category. Its unit is the zero symplectic space. Graphs of affine symplectic isomorphisms define a faithful symmetric monoidal functor from the affine symmetry groupoid.*

Scope. This concerns finite linear/affine geometry, including empty relations. It does not assert a normalized quantization of those relations.

Provenance. ids: SP-LREL; proved in: Reviewed structured relation proofs; tested by: Exact F2/F3 relation and compact controls; reviewed in: Single blind review, sole repair and relation-cluster adjudication.

Proof. For linear arrows $R \subseteq \overline{V} \oplus W$ and $S \subseteq \overline{W} \oplus Z$, put

$$X = \overline{V} \oplus W \oplus \overline{W} \oplus Z, \quad L = R \oplus S, \quad C = \overline{V} \oplus \Delta_W \oplus Z.$$

Then L is Lagrangian, C is coisotropic, and

$$C^\perp = \{(0, w, w, 0) : w \in W\}.$$

The reduced space $C/C^\perp$ is symplectically identified with $\overline{V} \oplus Z$ by $[(v, w, w, z)] \mapsto (v, z)$. For completeness, if $K = C^\perp$, $\dim X = 2N$ and $\dim K = r$, then

$$(L + K)^\perp = L \cap C, \qquad \dim(L \cap C) = N - r + \dim(L \cap K).$$

It follows that the image of $L \cap C$ in C/K is isotropic of dimension $N - r$, half the reduced dimension, hence is Lagrangian. Under the displayed identification this image is exactly

$$\{(v, z) : \exists w,\ (v, w) \in R,\ (w, z) \in S\} = S \circ R.$$

For nonempty affine translates, intersection with C is either empty or, after choosing one point (v_0, w_0, w_0, z_0), is that point plus the intersection of the direction space with C. Its outer projection is therefore (v_0, z_0) plus the linear composite. This proves affine and empty closure without a transversality assumption and without division, so it includes characteristic two.

Associativity and the identity laws follow by expanding the existential middle variables. Converse exchanges the two endpoint factors; this map negates the ambient form and hence preserves Lagrangian subspaces, including in characteristic two. Direct sum preserves Lagrangian direction spaces, and the prescribed reorder is symplectic because

$$(-\omega_V + \omega_W) + (-\omega_{V'} + \omega_{W'}) = -(\omega_V + \omega_{V'}) + (\omega_W + \omega_{W'}).$$

The usual rebracketing, zero-space and exchange graphs give the coherence maps, whose diagrams reduce to equality of tuple maps. Finally, for an affine symplectic isomorphism $f(v) = gv + t$, its graph is the translate of $\{(v, gv)\}$; the latter is isotropic of half dimension. Graphs preserve identity, composition and direct sum and determine their functions, giving the faithful symmetric monoidal functor. $\square$

Proposition 31.22 (Dagger compact affine Lagrangian relations)**.** **PROVED** *For every finite field k, the prescribed compact data make $\mathsf{L}_k^{\mathrm{aff}}$ a dagger compact category with dual $\overline{V}$. The diagonal cup and cap are affine Lagrangian relations and satisfy*

$$\lambda_V \circ (\epsilon_V \oplus 1_V) \circ a^{-1}_{V,\overline{V},V} \circ (1_V \oplus \eta_V) \circ \rho_V^{-1} = 1_V,$$
$$\rho_{\overline{V}} \circ (1_{\overline{V}} \oplus \epsilon_V) \circ a_{\overline{V},V,\overline{V}} \circ (\eta_V \oplus 1_{\overline{V}}) \circ \lambda_{\overline{V}}^{-1} = 1_{\overline{V}}.$$

They obey $\eta_V^\dagger = \epsilon_V \circ \sigma_{\overline{V},V}$. Name and unname are inverse bijections $\mathsf{L}_k^{\mathrm{aff}}(V, W) \cong \mathsf{L}_k^{\mathrm{aff}}(0, \overline{V} \oplus W)$ that retain the same ordered affine relation and preserve the empty relation. The endomorphisms of 0 are exactly $\varnothing$ and Δ_0. Composition and tensor transported by λ_0 have the same multiplication, with $\varnothing$ absorbing and Δ_0 the identity. For every V, $\epsilon_V \circ \sigma_{\overline{V},V} \circ \eta_V = \Delta_0$.

Scope. Classical affine Lagrangian relations over every finite field, including characteristic two and empty relations. No quantum normalization, amplitude, probability, stabilizer equivalence or Choi theorem.

Provenance. ids: SP-COMPACT; proved in: Reviewed structured relation proofs; tested by: Exact F2/F3 relation and compact controls; reviewed in: Single blind review, sole repair and relation-cluster adjudication.

Proof. The diagonal in $\overline{V} \oplus V$ is isotropic because its form is $-\omega_V(v, w) + \omega_V(v, w) = 0$ and is Lagrangian because its dimension is $\dim V$. This types both η_V and ϵ_V with their displayed factor orders. Expanding the first snake, an input x is related through the cup to $(x, (u, u))$, the inverse associator sends this to $((x, u), u)$, and the cap forces $x = u$; the final output is x. Expanding the second snake similarly sends x through $((u, u), x)$ and the cap forces $u = x$. Thus the two fully unital composites in the proposition are the relevant identity relations. Converse turns the cup into the diagonal effect on $\overline{V} \oplus V$, which is $\epsilon_V \circ \sigma_{\overline{V},V}$.

For any $R : V \to W$, its name contains $(0, (v, w))$ exactly when $(v, w) \in R$. Conversely, the unname of Q relates v to w exactly when $(0, (v, w)) \in Q$: the cap forces the copied input coordinate to equal the first coordinate of the state. Hence the two maps are inverse and retain affine, nonfunctional and empty relations.

A relation $0 \to 0$ is a subset of a singleton, so the only two allowed scalars are $\varnothing$ and Δ_0. Direct expansion shows that composition and $\lambda_0 \circ (s \oplus t) \circ \lambda_0^{-1}$ have the two-element Boolean multiplication table. In the closed cup–cap composite the zero vector is always a middle witness, so existential relation composition gives Δ_0 for every V; no cardinality or amplitude is retained. $\square$

Proposition 31.23 (The scalar-quotient stabilizer comparison)**.** PROVED *For an odd prime p, restrict the affine Lagrangian relation category to the standard spaces $V_n = \mathbb{F}_p^n \oplus \mathbb{F}_p^n$, $n \geq 0$. For every nonempty arrow $R : V_m \to V_n$, $\mathcal{E}_p(R)$ is a one-dimensional subspace of* $\mathrm{Hom}_{\mathbb{C}}(H_{\mathbb{F}_p,m}, H_{\mathbb{F}_p,n})$ *contained in* $\mathsf{Stab}_p^{\mathrm{amp}}(H_{\mathbb{F}_p,m}, H_{\mathbb{F}_p,n})$*, while $\mathcal{E}_p(\varnothing) = \{0\}$. The assignments $V_n \mapsto H_{\mathbb{F}_p,n}$, $R \mapsto [T]$ for any nonzero $T \in \mathcal{E}_p(R)$, and $\varnothing \mapsto [0]$ define a dagger symmetric monoidal equivalence to* $\mathsf{Stab}_p^{\mathsf{proj}}$*.*

Scope. Standard-object props over odd prime fields, modulo all invertible complex scalars, retaining a separate zero. No norm, success probability, CP map or arithmetic-source exhaustion.

Provenance. ids: SP-STAB-REL; proved in: Reviewed intertwiner-line, functor and equivalence proofs; tested by: Exact stabilizer comparison controls; reviewed in: Single blind review, sole repair and relation-cluster adjudication.

Proof. On the Hilbert–Schmidt space $\mathrm{Hom}_{\mathbb{C}}(H_{\mathbb{F}_p,m}, H_{\mathbb{F}_p,n})$ put

$$\rho(v, w)T = W^{\mathrm{s}}_{\mathbb{F}_p,n}(w)TW^{\mathrm{s}}_{\mathbb{F}_p,m}(v)^*.$$

The half-form Weyl law gives

$$\rho(x)\rho(y) = \psi_{\mathbb{F}_p}(\Omega_{m,n}(x, y)/2)\rho(x + y),$$

and the trace of $\rho(x)$ is zero unless $x = 0$, when it is p^{m+n}. If $R = r + L$ is nonempty, isotropy makes $\rho|_L$ an ordinary unitary representation and

$$P_R = \frac{1}{|L|}\sum_{\ell \in L} \psi_{\mathbb{F}_p}(\Omega_{m,n}(r, \ell))\rho(\ell)$$

is the orthogonal projector onto the displayed intertwiner space. Changing r by an element of L leaves every coefficient unchanged. Since $|L| = p^{m+n}$, Weyl trace orthogonality gives $\mathrm{Tr}(P_R) = 1$. Thus every nonempty relation gives one nonzero operator line, while the empty relation gives only zero.

Composition also detects empty affine intersections. For $R : V_m \to V_n$, let

$$A_R = \{a : (0, a) \in L_R\};$$

the range of a nonzero $T \in \mathcal{E}_p(R)$ is exactly the simultaneous Weyl eigenspace for A_R with character $a \mapsto \psi_{\mathbb{F}_p}(-\omega_n(r_n, a))$. The analogous input eigenspace describes the initial support. For composable nonempty R, S, the two middle support projectors have nonzero product exactly when their characters agree on the intersection of their isotropic label groups. By symplectic annihilators, this is exactly the condition that the affine middle projections meet, hence exactly that $S \circ R$ is nonempty. In that case, choosing a common middle origin cancels the two middle form terms and proves

$$\mathcal{E}_p(S)\mathcal{E}_p(R) = \mathcal{E}_p(S \circ R).$$

When the middle projections do not meet, the projector product and every operator composite are zero.

For bare relational converse the endpoint form changes sign. Taking the Hilbert adjoint of the defining equation gives exactly

$$\mathcal{E}_p(R^\dagger) = \mathcal{E}_p(R)^*.$$

The direct-sum form adds and the admitted grouped Weyl comparison gives $\mathcal{E}_p(R \oplus S) = \mathcal{E}_p(R) \otimes \mathcal{E}_p(S)$, including the zero and rank-zero cases. The identity relation has the scalar line of the identity operator by the full-matrix Weyl commutant.

For target membership, use the explicit symplectic identification $c_m : \overline{V}_m \to V_m$, $c_m(a,b) = (a,-b)$, and

$$\operatorname{vec}(T) = \sum_x \delta_x \otimes T\delta_x.$$

Complex conjugation gives $\overline{W^{\mathrm{s}}(a,b)} = W^{\mathrm{s}}(a,-b)$, so vectorization sends $\mathcal{E}_p(R)$ to the stabilizer-state line of $(c_m \oplus 1)(\ulcorner R \urcorner)$. Every affine Lagrangian state is a Clifford image of $\delta_0^{\otimes(m+n)}$: extend a basis of its direction to a symplectic basis and use the admitted affine Egorov implementer. Unvectorization contracts with the stabilizer Bell effect, so every operator in the line is an actual stabilizer amplitude. For an empty arrow of type $V_m \to V_n$, let $e_j : \mathbb{C} \to H_{\mathbb{F}_p,j}$ be the tensor of j computational zero preparations, with $e_0 = 1_{\mathbb{C}}$. Its image is the typed actual zero map

$$e_n \circ (0 : \mathbb{C} \to \mathbb{C}) \circ e_m^* : H_{\mathbb{F}_p,m} \longrightarrow H_{\mathbb{F}_p,n},$$

using only the preparation, scalar, tensor, adjoint and composition generators.

Finally take an arbitrary $U \in C_2(A)$. Conjugation has the unique form

$$UW(v)U^* = \gamma(v)W(gv).$$

The Weyl product and commutator force g to be linear symplectic and γ to be an additive character. Hence uniquely $\gamma(v) = \psi_{\mathbb{F}_p}(\omega(t, gv))$, so U spans the line of the affine graph (t, g). This treats every second-level unitary. Together with the computational preparation, its adjoint, and the scalar generators, it proves fullness. The projective Weyl stabilizer of a nonzero line recovers its Lagrangian direction, and its eigencharacter recovers the affine coset, which proves faithfulness. The object assignment is surjective, completing the stated dagger symmetric monoidal equivalence, using the relation and compact laws proved above. □

Proposition 31.24 (Scalar normalization). **PROVED** *For nonzero finite-dimensional Hilbert spaces H, K and a linear map $T : H \to K$, the map $\Phi_T(\rho) = T\rho T^*$ is completely positive and is trace-nonincreasing exactly when $T^*T \le 1$. For every complex c, $\Phi_{cT} = |c|^2\Phi_T$. The scalar-quotient stabilizer prescription stipulates no actual successful branch or probability. A branch can be specified using an admissible representative or an admissible class-invariant normalization rule; a phase change alone does not change the branch. Compatibility of an added normalization rule with composition is a separate obligation.*

Scope. Zero outcomes are retained and never conditionally normalized. This describes the additional data for an operational lift, not a claimed canonical choice of that data.

Provenance. ids: SP-SCALAR; proved in: Inherited: FRP-CP; reviewed structured process proofs; tested by: Exact finite process controls; reviewed in: Single blind review, sole repair and process-cluster adjudication.

Proof. For every finite-dimensional auxiliary Hilbert space E and $R \ge 0$,

$$(1_E \otimes \Phi_T)(R) = (1_E \otimes T)R(1_E \otimes T)^* \ge 0,$$

including when $T = 0$, so Φ_T is completely positive. A direct rectangular matrix calculation gives

$$\operatorname{Tr}_K(T\rho T^*) = \operatorname{Tr}_H(T^*T\rho).$$

If $T^*T \le 1$, positivity of $1 - T^*T$ proves trace nonincrease. Conversely, applying trace nonincrease to every $\rho = |x\rangle\langle x|$ gives $\langle x, (1 - T^*T)x\rangle \ge 0$, hence $T^*T \le 1$.

For every $c \in \mathbb{C}$,

$$\Phi_{cT}(\rho) = c\bar{c}T\rho T^* = |c|^2\Phi_T(\rho).$$

Thus unit-modulus rescaling leaves the branch fixed. If $T \ne 0$ and $|c| \ne 1$, it changes the actual CP map and every nonzero success weight; the zero operator remains the zero branch for every scalar. The scalar quotient identifies all nonzero rescalings but prescribes no actual branch, probability, representative or normalization rule. One may choose an admissible representative. Alternatively, the class-invariant rule

$$\mathcal{N}_{[T]}(\rho) = \frac{T\rho T^*}{\|T\|^2}$$

is independent of the representative because numerator and denominator both scale by $|c|^2$, and it is TNI because $T^*T/\|T\|^2 \leq 1$. The zero class is handled separately by the zero branch. Neither method is stipulated by the scalar quotient, and compatibility of an added rule with composition is a separate obligation; no functor or no-go theorem is asserted. A zero-probability output is retained without conditional normalization. ☐

Proposition 31.25 (Closure and normalization of finite instruments)**.** PROVED *For finite quantum systems with ordinary block traces, the Kraus branch prescription is equivalent to complete positivity and trace nonincrease. Channels, branches and instruments compose and tensor with the stated normalizations. Instruments have a common source and target; retaining outcomes sends the output to the direct sum of the outcome blocks, without a factor equal to the number of outcomes. Sequential instruments retain the ordered pair of outcomes, and summing over those pairs gives a channel. Adjoint maps under trace duality reverse the direction; a channel has a unital adjoint.*

Scope. This is an ambient process theorem. It makes no claim that arithmetic or stabilizer generators exhaust these maps.

Provenance. ids: SP-CP; proved in: Inherited: FRP-CP; reviewed structured process proofs; tested by: Exact finite process controls; reviewed in: Single blind review, sole repair and process-cluster adjudication.

Proof. For a finite Kraus family $K_{ba,j} : H_a \to K_b$, every amplified output block is a finite sum

$$\sum_{a,j} (1 \otimes K_{ba,j}) R_a (1 \otimes K_{ba,j})^*,$$

and is positive. Conversely, if $\Phi : \bigoplus_a \mathrm{End}(H_a) \to \bigoplus_b \mathrm{End}(K_b)$ is CP, every component $\pi_b \Phi \iota_a$ is CP. Watrous's Theorem 2.22 supplies a finite Kraus list for each nonzero component; a zero component receives the empty list. Linearity reassembles these component lists into the displayed block map.

Put $A_a = \sum_{b,j} K^*_{ba,j} K_{ba,j}$. Ordinary rectangular trace gives

$$\mathrm{Tr}_Y \Phi(\rho) = \sum_a \mathrm{Tr}_{H_a}(A_a \rho_a).$$

Therefore trace nonincrease is equivalent to $A_a \leq 1_{H_a}$ for every input block: necessity follows by supporting a rank-one positive input on one chosen block. Trace preservation is similarly equivalent to $A_a = 1_{H_a}$ for every a. Kraus lists are presentations of an actual CP map; changing a list without changing that map does not change the ambient arrow, and does not identify the finer source-certified arithmetic arrows.

For a second branch with Kraus maps $L_{cb,t}$, composition has Kraus paths (b, j, t) and operators $L_{cb,t} K_{ba,j}$. Its effect is bounded by the first effect because the second effects are bounded by the identity. Tensor has operators $K_{ba,j} \otimes M_{b'a',u}$ on Cartesian block tags. Its effect is $A_a \otimes D_{a'} \leq 1$, and the matrix-unit spanning argument establishes the tensor formula on entangled as well as product inputs.

For an instrument $(\Phi_o)_{o \in O}$, the retained channel has blocks (o, b) and

$$\widehat{\Phi}(\rho)_{(o,b)} = \Phi_o(\rho)_b, \qquad \mathrm{Tr}(\widehat{\Phi}(\rho)) = \sum_o \mathrm{Tr}(\Phi_o(\rho)) = \mathrm{Tr}(\rho).$$

No outcome-count factor occurs. Sequential instruments have branches $\Psi_r\Phi_o$ indexed by the earlier/later pair (o,r); summing over those pairs is the composite of the two channel sums. Independent tensor outcomes are the listed-factor pairs (o,s).

For an arbitrary complex-linear block map, choose ordinary-trace-orthonormal matrix units $(E_p)_p$ and $(F_q)_q$ and write

$$\Phi(E_p) = \sum_q C_{q,p} F_q.$$

The prescription

$$\Phi^{\mathrm{tr}\,*}(F_q) = \sum_p \overline{C_{q,p}} E_p$$

extends linearly and satisfies $\langle \Phi^{\mathrm{tr}\,*}(y), x\rangle = \langle y, \Phi(x)\rangle$ on the matrix-unit bases and hence on every x, y. If two maps satisfy this identity, their difference is orthogonal to every matrix unit and is zero, proving existence and uniqueness for every linear map. For a CP map this general adjoint specializes to

$$(\Phi^{\mathrm{tr}\,*}(y))_a = \sum_{b,j} K^*_{ba,j} y_b K_{ba,j}.$$

It is CP and reverses direction. Its value on the output identity is A_a, so branches have subunital adjoints and channels have unital adjoints. This does not make the adjoint a reverse branch: the adjoint of ordinary discard $\mathrm{Tr} : M_2 \to \mathbb{C}$ sends $\lambda \mapsto \lambda I_2$ and increases the trace of 1 from one to two. The theorem therefore supplies ambient trace adjoints, without an automatic CP dagger on the trace-nonincreasing branch category. □

Proposition 31.26 (Coherent sums and tagged quantum systems)**.** PROVED *For an odd prime p, the complex span of the actual stabilizer maps $\ell^2(\mathbb{F}_p^n) \to \ell^2(\mathbb{F}_p^m)$ is the full space of linear maps. The matrix completion therefore realizes all linear maps between its coherent Hilbert sums. For a nonempty list of these model spaces $(H_i)_i$, the coherent algebra has dimension $(\sum_i \dim H_i)^2$ and the tagged algebra has dimension $\sum_i (\dim H_i)^2$; the tagged algebra is its block-diagonal subalgebra, with the corresponding trace-preserving block-dephasing channel.*

Scope. Adding coherent linear combinations enlarges the pure stabilizer fragment. A classical rig source is still to be constructed.

Provenance. ids: SP-SUM; proved in: Inherited: F1-REAL; reviewed matrix-unit and block proof; tested by: Exact finite sum controls; reviewed in: Single blind review, sole checker repair and sum adjudication.

Proof. For $x \in \mathbb{F}_p^n$, tensor the computational zero preparations and translate them to obtain an actual map

$$e_x : \mathbb{C} \longrightarrow \ell^2(\mathbb{F}_p^n), \qquad e_x(1) = \delta_x.$$

At $n = 0$ this is the identity on $\mathbb{C}$ indexed by the unique empty tuple. Adjoint closure gives e_x^*, so for $y \in \mathbb{F}_p^m$ the actual composite

$$E_{y,x} = e_y e_x^*$$

sends δ_z to $[x = z]\delta_y$. These are the p^{m+n} rectangular computational matrix units and form a basis of all linear maps between the two model spaces. Individual matrix units are actual amplitudes; their arbitrary complex sums enter at the prescribed linear completion. The projective scalar quotient does not supply this addition.

This enlargement is strict. For fixed ranks there are only finitely many affine Lagrangian relations over the finite field, so the admitted relation/stabilizer equivalence gives only finitely many nonzero projective stabilizer rays. In contrast, $\mathrm{End}(\ell^2(\mathbb{F}_p))$ has dimension $p^2 > 1$ and infinitely many rays: for independent matrix units E, F, the rays of $E + zF$, $z \in \mathbb{C}$, are pairwise distinct. The full complex span therefore contains maps outside the pure actual-amplitude fragment.

For ordered lists $X = (H_i)_{i=1}^s$ and $Y = (K_j)_{j=1}^t$, the prescribed block action is

$$(T_{ji})(\xi_i)_i = \left(\sum_i T_{ji}\xi_i\right)_j .$$

It is injective because inputs supported on one summand and projection to one output recover every T_{ji}. It is surjective because a linear map L has blocks $\pi_j L \iota_i$, which reconstruct L by linearity. Matrix composition and adjoint transpose realize ordinary composition and Hilbert adjoint. If either list is empty, its realization is the zero Hilbert space and the unique empty block matrix is the unique typed zero map. This empty list is distinct from the one-entry list $(\mathbb{C})$, and no channel on the zero space is asserted. Equal Hilbert spaces in repeated list positions remain separate summands.

For a nonempty list put $H = \bigoplus_i H_i$, $d_i = \dim H_i$, and $D = \sum_i d_i$. Block extraction gives

$$\mathrm{End}(H) = \bigoplus_{i,j} \mathrm{Hom}(H_j, H_i),$$

so its dimension is D^2. The tagged algebra retains only diagonal blocks $\bigoplus_i \mathrm{End}(H_i)$ and has dimension $\sum_i d_i^2$. With at least two recorded positions, any nonzero cross-block matrix unit exhibits the possible proper inclusion, including when the two underlying Hilbert spaces are equal.

The prescribed summand projections P_i^X satisfy $P_i^X P_j^X = [i = j] P_i^X$ and $\sum_i P_i^X = 1_H$. Therefore

$$\Delta_X(a) = \sum_i P_i^X a P_i^X$$

fixes each diagonal block and kills every off-diagonal block. It is unital, idempotent, and has exactly the tagged algebra as its range. The projections are a Kraus list with

$$\sum_i (P_i^X)^* P_i^X = 1_H,$$

so the admitted finite-block Kraus criterion makes Δ_X a channel. In an orthonormal basis adapted to the summands,

$$\mathrm{Tr}_H(\Delta_X(a)) = \sum_i \mathrm{Tr}_{H_i}(a_{ii}) = \mathrm{Tr}_H(a),$$

with no division by the number or dimensions of the blocks. This channel statement applies only to nonempty lists; no biproduct, fusion, rig, Gaussian-closure, or source-exhaustion conclusion is added. □

Proposition 31.27 (Fock completion and the exponential law)**.** SKETCH *On complex Hilbert spaces and contractions, the symmetric Fock prescription defines a functor and has a natural unitary exponential law $\Gamma_s(H \oplus K) \simeq \Gamma_s(H) \otimes \Gamma_s(K)$ with the standard normalized symmetric tensors. In particular $\Gamma_s(0) = \mathbb{C}$ and $\Gamma_s(\mathbb{C}) \simeq \ell^2(\mathbb{N}_0)$, with particle number acting diagonally on the latter.*

Scope. Infinite boundedness and the completion require written arguments. No equivalence between an unspecified classical free monoid and this Hilbert construction is asserted.

Provenance. ids: SP-FOCK; proved in: Inherited: No admitted earlier theorem claimed; remaining comparison in the DAG; tested by: Proposed falsifier only; reviewed in: SKETCH; promotion review pending.

Proposition 31.28 (Restriction of scalars preserves the Weyl datum). **PROVED** *For every finite extension E/K of odd-characteristic finite fields and symplectic E-space (V, ω_E), the trace form on $\mathrm{Res}_{E/K} V$ is nondegenerate and alternating. If χ_K is nontrivial, so is $\chi_{E/K} = \chi_K \circ \mathrm{Tr}_{E/K}$, and the identity on the underlying Weyl labels identifies the two half-form $*$-algebras exactly. These comparisons compose along towers by transitivity of trace.*

Scope. The extension degree may be divisible by the characteristic. This is restriction of scalars with a trace form, not the uncorrected inclusion of a subfield as a symplectic subsystem.

Provenance. ids: SP-TRACE; proved in: Inherited: FRB-TRACE; structured arithmetic proof; tested by: Exact arithmetic controls; existing bridge scope retained; reviewed in: theory/verdicts/phantasm-arithmetic-adjudication.md.

Restriction-of-scalars argument. Let $d = [E : K]$. The relative trace is K-linear and surjective, including when the characteristic divides d. Thus

$$\omega_{\mathrm{Res}}(v, w) = \mathrm{Tr}_{E/K}(\omega_E(v, w))$$

is alternating and K-bilinear. If $v \neq 0$, choose w_0 with $c = \omega_E(v, w_0) \neq 0$ and choose x with $\mathrm{Tr}_{E/K}(x) = 1$. Then $w = (x/c)w_0$ satisfies $\omega_{\mathrm{Res}}(v, w) = 1$, proving nondegeneracy in arbitrary rank. Trace surjectivity also proves that $\chi_{E/K} = \chi_K \circ \mathrm{Tr}_{E/K}$ is nontrivial, without identifying the named base character with the fixed absolute-trace character.

In odd characteristic, $1/2 \in K$ and

$$\mathrm{Tr}_{E/K}(\omega_E(v, w)/2) = \omega_{\mathrm{Res}}(v, w)/2.$$

Consequently

$$\chi_{E/K}(\omega_E(v, w)/2) = \chi_K(\omega_{\mathrm{Res}}(v, w)/2),$$

so the identity on the underlying labels preserves the half-form Weyl product. It also preserves negation, the zero label and the coefficient trace, and is therefore the exact unital $*$-isomorphism in the proposition. At rank zero it is the identity on $\mathbb{C}$.

For a named tower $L \subset K \subset E$ and characters induced from one named character of L, trace transitivity identifies the direct and iterated forms and characters. Both comparison routes are identity on the underlying labels and therefore compose literally. An unrelated named character of K is not silently identified with the induced one. This is restriction of scalars through trace, rather than uncorrected subfield inclusion or subsystem decoding. $\square$

Proposition 31.29 (Named arithmetic Frobenius covariance). **PROVED** *For a finite extension E/K of odd-characteristic finite fields, with $|K| = p^s$, $n \geq 0$ and a named nontrivial additive character χ_K, put $\chi_{E/K} = \chi_K \circ \mathrm{Tr}_{E/K}$. On the standard register $E^n \oplus E^n$ with the trace-form Weyl datum for $\chi_{E/K}$, coordinatewise $\#K$-power Frobenius is K-linear symplectic and its specified permutation unitary satisfies $U_{E/K,n} W^{\mathrm{s}}_{E,n}(a, b) U^*_{E/K,n} = W^{\mathrm{s}}_{E,n}(\sigma_{E/K}(a), \sigma_{E/K}(b))$. Its induced state channel is the invertible channel $\rho \mapsto U_{E/K,n} \rho U^*_{E/K,n}$ and its $[E : K]$-th power is identity.*

Scope. A smaller period is allowed. Abstract spaces require named semilinear data; Frobenius is not identified with partial trace.

Provenance. ids: SP-FROB; proved in: Inherited: FRB-FROB,FRB-TRACE; channel typing: SP-CP; structured arithmetic proof; tested by: Exact arithmetic controls; existing bridge scope retained; reviewed in: theory/verdicts/phantasm-arithmetic-adjudication.md.

Relative Frobenius argument. Put $q = |K| = p^s$, $d = [E : K]$ and $\sigma(x) = x^q$. This is a K-linear field automorphism and $\sigma^d = 1$. The conjugate-sum formula

$$\mathrm{Tr}_{E/K}(x) = \sum_{r=0}^{d-1} \sigma^r(x)$$

shows that relative trace is invariant under σ. Hence coordinatewise σ preserves the restricted symplectic form and every named relative character $\chi_K \circ \mathrm{Tr}_{E/K}$. This does not assert that an arbitrary base character is invariant under the one-step absolute p-power action.

The specified basis permutation satisfies $U_{E/K,n}\delta_y = \delta_{\sigma y}$. On basis vectors the symmetrized Weyl operator is

$$W^{\mathrm{s}}_{E,n}(a, b)\delta_y = \chi_{E/K}(-b \cdot y - a \cdot b/2)\delta_{y+a}.$$

Conjugating by the permutation, then using character invariance and applying σ to the phase argument, gives

$$U_{E/K,n} W^{\mathrm{s}}_{E,n}(a, b) U^*_{E/K,n} = W^{\mathrm{s}}_{E,n}(\sigma a, \sigma b).$$

Both phase coordinates are transformed. At $n = 0$ all operators are the identity on $\mathbb{C}$.

The single Kraus operator $U_{E/K,n}$ is unitary, so its conjugation is an ordinary-trace channel. Its inverse is conjugation by the adjoint, and its d-th power is identity because $U^d_{E/K,n} = 1$. No minimal-period claim, partial trace, characteristic-two half-form, or non-bijective Frobenius map is used. $\square$

Proposition 31.30 (Subsystem inclusion and its dual decoder)**.** PROVED *For an odd-characteristic finite field, a common nontrivial character and a symplectic injection $j : U \to V$, the orthogonal decomposition $V = j(U) \oplus j(U)^\perp$ admits the specified quantum subsystem realization. Its observable inclusion is a unital $*$-homomorphism and its trace-dual decoder is a channel. The decoder restricts Weyl characteristic functions to $j(U)$ and is unchanged by an overall phase of the implementing unitary. For compatible iterated tensor decompositions, the decoders compose as partial traces.*

Scope. The complement and Weyl model compatibility must be verified. Generic field embeddings and coordinate traces are not automatically such injections.

Provenance. ids: SP-SUBSYS; proved in: Inherited: FRP-CP; structured arithmetic proof; tested by: Exact finite arithmetic controls; reviewed in: theory/verdicts/phantasm-arithmetic-adjudication.md.

Subsystem geometry and model argument. For a symplectic injection $j : U \to V$, the image $j(U)$ is nondegenerate, so $j(U) \cap j(U)^\perp = 0$. The dimension identity $\dim A + \dim A^\perp = \dim V$ gives

$$V = j(U) \oplus j(U)^\perp.$$

The restricted form on the complement is nondegenerate because a vector in its radical also lies in the double perpendicular $j(U)$. Thus $(u, w) \mapsto ju + w$ is a symplectic isomorphism, including zero retained or zero complement factors.

The tensor of the chosen irreducible Weyl models for U and the complement, and the chosen model for V transported through this symplectic isomorphism, are irreducible unitary models of the same full matrix Weyl algebra. Admitted model uniqueness supplies a unitary J satisfying

$$J(W^{\mathrm{s}}_U(u) \otimes W^{\mathrm{s}}_W(w))J^* = W^{\mathrm{s}}_V(ju + w),$$

unique up to overall phase. This reuses the admitted realization rather than reproving finite Stone–von Neumann.

For composable $U \xrightarrow{j} V \xrightarrow{\ell} X$, the map

$$W_j \oplus W_\ell \longrightarrow W_{\ell j}, \qquad (w, z) \longmapsto \ell(w) + z$$

lands in the combined complement. Its two images are orthogonal, it is injective, dimensions agree, and it preserves the two forms; it is therefore symplectic. The same model comparison supplies the three registered injection unitaries and the complement unitary. The staged and direct routes implement the same Weyl label $\ell(ju)+\ell(w)+z$, so unitary-intertwiner uniqueness makes them differ by one allowed phase. This is exactly the registered pure-tensor compatibility, including the ordinary reassociation and rank-zero unitors. □

Observable and decoder argument. Conjugation gives

$$\iota_J(a) = J(a\otimes 1)J^*,$$

which is visibly linear, multiplicative, star-preserving and unital. For an orthonormal basis (e_r) of the complement model, put

$$K_r = (1\otimes \langle e_r|)J^* : H_V \to H_U.$$

Then

$$\mathcal{D}_J(\rho) = \sum_r K_r\rho K_r^* = \mathrm{Tr}_{H_W}(J^*\rho J), \qquad \sum_r K_r^*K_r = 1_{H_V}.$$

The decoder is therefore a completely positive ordinary-trace-preserving channel, with no division by the discarded dimension, and

$$\mathrm{Tr}(\mathcal{D}_J(\rho)a) = \mathrm{Tr}(\rho\iota_J(a)).$$

Putting the complement Weyl label equal to zero gives $\iota_J(W_U^{\mathrm{s}}(u)) = W_V^{\mathrm{s}}(ju)$ and hence

$$\mathrm{Tr}(\mathcal{D}_J(\rho)W_U^{\mathrm{s}}(u)) = \mathrm{Tr}(\rho W_V^{\mathrm{s}}(ju)).$$

Replacing J by a phase multiple changes neither the inclusion nor the decoder.

For the registered compatible iterated data, let A be the staged unitary and B the direct unitary after complement comparison and reassociation. Compatibility says $A=\lambda B$. The phase cancels from conjugation, and the complement unitary carries the tensor-product orthonormal basis to an orthonormal basis of the combined complement. Expanding both partial traces in these bases proves

$$\iota_{J_\ell}\iota_{J_j} = \iota_{J_{\ell j}}, \qquad \mathcal{D}_{J_j}\mathcal{D}_{J_\ell} = \mathcal{D}_{J_{\ell j}}.$$

This comparison is asserted only for the full registered compatibility data. It is a trace-preserving tensor-subsystem decoder, not the trace-nonincreasing success branch of a field support code. The full retained code decoder includes a separate failure outcome. □

Proposition 31.31 (Tensor assembly and product-state existence). **SKETCH** *For the specified finite matrix factors and trace-one positive reference operators indexed by primes, the finite tensor embeddings form an isometric unital inductive system. Its norm completion is a unital C^*-algebra, and the prescribed finite product states extend uniquely to a state on it. The GNS prescription gives its represented von Neumann algebra with a cyclic reference vector.*

Scope. No factor type, separating-vector property, inter-prime arithmetic process or Bost–Connes identification is asserted.

Provenance. ids: SP-PRIME; proved in: Inherited: No admitted earlier theorem claimed; remaining comparison in the DAG; tested by: Proposed falsifier only; reviewed in: SKETCH; promotion review pending.

Proposition 31.32 (Arithmetic semigroup and zeta control). **SKETCH** *In the specified represented Bost–Connes control, $\mu_m\mu_n = \mu_{mn}$, $\mu_n^*\mu_n = 1$, and $\mu_n^*e(r)\mu_n = e(nr)$, while $\mu_n e(r)\mu_n^* = n^{-1}\sum_{ns=r} e(s)$. The prescribed time evolution leaves the algebra invariant and satisfies $\sigma_u(\mu_n) = n^{iu}\mu_n$ and $\sigma_u(e(r)) = e(r)$. The maps L_n are unital completely positive and ν_n are corner $*$-endomorphisms. For real $b>1$, the Gibbs operator is trace class with* $\mathrm{Tr}(e^{-bH_{\log}}) = \zeta(b)$ *and the prescribed density has trace one.*

Scope. The Hamiltonian has logarithmic-integer eigenvalues. No zeta-zero spectrum, universal-representation faithfulness or full KMS classification is asserted.

Provenance. ids: SP-BC-CONTROL; proved in: Inherited: No admitted earlier theorem claimed; remaining comparison in the DAG; tested by: Proposed falsifier only; reviewed in: SKETCH; promotion review pending.

31.5 Decisions beyond the bootstrap

A normalized lift of the relation semantics must retain amplitudes and outcome probabilities. A classical additive construction must specify its objects, morphisms, two products and distributivity before it is called a rig quantization. Characteristic two needs its own central-extension data before any all-prime claim. Higher gates need phase precision and a composition law, rather than an unqualified polynomial-degree analogy.

The independent tensor assembly and the represented arithmetic control system do not yet define their desired combination. The next definition must supply actual arithmetic maps between the prime contributions, their relations and a reference state. A modular comparison then requires its analytic hypotheses and temperature convention. A spectral claim must name its operator, domain and trace before making an arithmetic comparison. These are decision gates for extending the graph, not unformulated theorems assigned a proof status.